Lattice Yang–Mills theory at strong coupling IV: the mass gap is $4\log(1/\beta)+O(1)$

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

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Abstract

For lattice Yang–Mills theory with Wilson's action in dimension $d\ge 3$ and $0<\beta<\beta_0(d)$ (the threshold of the cluster expansion of [F]), we prove that the mass gap of the infinite-volume theory of Part II, and the finite-volume gaps of [F] on spatial lattices with all sides ≥ 4, are at least $4\log(1/q(\beta))$, with $q(\beta)=e^3K(e^{4\beta}-1)$ and $K=4(2d-3)$. As $\beta\downarrow 0$ this is $4\log(1/\beta)-O(1)$. The leading coefficient 4 is the one predicted by the strong-coupling expansion (the lightest glueball is a tube with four plaquettes per unit of time, with predicted mass $-4\log u(\beta)+o(1)$, $u(\beta)\sim\beta/4$ for $SU(2)$), whereas the Dobrushin bounds of [F] and Part II give only $\log(1/\beta)-O(1)$. The proof combines three ingredients: duplicate variables, by which every term of the expansion of a covariance that does not connect the two observables cancels by the swap symmetry; the loop expansion and the ratio bound of Part III for the duplicated system; and a geometric lemma: a surface without free edges has at least four temporal plaquettes in each time layer it crosses, because these plaquettes form a graph of minimum degree two on the spatial lattice, whose girth is four (sides ≥ 4 are needed: with a side of length 3 the girth is three). In $d=2$ the girth is the spatial size, which gives an infinite mass gap in infinite volume and a finite-volume gap growing linearly with the spatial size, in agreement with the exact solution. As in [F], II and III nothing non-abelian is used, so the result belongs to the strong-coupling endpoint of the problem. We also prove the matching upper bound: for an irreducible non-trivial ρ of dimension N, $m\le 4\log(1/u(\beta))+D(\beta)$ with $u=c_\rho/(Nc_0)$ and an explicit $D(\beta)=O(\beta)$, on the whole range $0<\beta<\beta_0(d)$ when $N\le 4$ ($N\le 5$ if $\rho\cong\bar\rho$), with $D<0.32$ when $N\le 3$ (numerically $D<0.05$ for $U(1)$, $SU(2)$, $SU(3)$ in $d=3,4$), and for every ρ when β is small. The proof uses a positivity property of the transfer matrix that isolates the minimal tube (the four sides of a unit cube between two plaquettes), an exact evaluation of that tube by character orthogonality, a lemma that every other term of the resulting expansion has at least five further plaquettes, and the spectral inequality $\mathrm{Cov}\le e^{-m}\mathrm{Var}$ of the ground-state

chain of Part II. Together, for these ρ: $4\log(1/q(\beta)) \le m(\mu_\beta) \le 4\log(1/u(\beta)) + D(\beta)$ for all $\beta < \beta_0(d)$, so $m(\mu_\beta) = 4\log(1/\beta) + O(1)$, which is what principle 3 ("the strong-coupling test") of the research program requires. The additive constants of the two bounds still differ by about 35 for $SU(2)$ in $d = 4$. A transfer-matrix bound that weighs the temporal plaquettes of all but one time layer by the operator norm $u_* = \max_{r\neq 0} u_r$ of their kernel ($u_* = u$ for $\dim\rho \le 49$) improves the lower bound to $4\log(1/u(\beta)) - 4\log(eK) - O(\beta)$, so that for these ρ the two bounds differ by at most $4\log(eK) + O(\beta)$ (about 16.0 for $d = 4$ on the whole range). Counting the connecting surfaces by their fronts (the temporal plaquettes of one time layer, which can change only where a spatial plaquette or a support touches them) removes the remaining entropy and gives the sharp constant on the whole range: $m(\mu_\beta) \ge 4\log(1/u(\beta)) - \log(1 + \delta(\beta)) - O(\beta^5)$ with an explicit $\delta(\beta) = O(\beta)$, so that $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$. Grouping everything that meets at one time slice into a junction, the condition that no link of a connecting surface is free leaves a single junction of order β through which a front of four plaquettes can pass, the cap; this gives the first-order constant of the method, $\delta(\beta) = 2\beta(1 + O(\beta))$ when ρ has no trivial summand, i.e. $m(\mu_\beta) \ge 4\log(1/u(\beta)) - 2\beta + O(\beta^2)$. Evaluating the cap, and two sup-norm bounds of the upper bound, by characters gives the first-order term exactly: $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ with $\kappa(\beta) = \tau_0^{-1}\int(\operatorname{Re}\chi_\rho)^2 f_\beta\, dU = \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3\, dU + O(\beta^2)$, which vanishes to first order for $U(1)$ and for $SU(N)$, $N \neq 3$, with the fundamental representation, and is $\beta/6 + O(\beta^2)$ for $SU(3)$, where three fundamental fluxes can meet at a link. At second order the shifts and jumps of the tube drop out (in the duplicated system the covariance lives on the sector that is odd under the exchange of the copies, where their operators vanish), and $m(\mu_\beta) = 4\log(1/u(\beta)) - \log\Upsilon(\beta) + O(\beta^3)$, where $\Upsilon(\beta)$ is the largest variance of a real unit plaquette function in the span of $\chi_\rho, \overline{\chi_\rho}$ under the weight of one plaquette; for $SU(2)$, $m(\mu_\beta) = 4\log(1/u(\beta)) + \beta^2/8 + O(\beta^3)$. At third order nothing new enters: $m(\mu_\beta) = 4\log(1/u(\beta)) - \log\Upsilon(\beta) + O(\beta^4)$, so the term of order β^3 is that of $-\log\Upsilon$, which is $\beta^3/432$ for $SU(3)$, $-\beta^3/6000$ for $SU(5)$ (five fundamental fluxes meet at a link), and 0 for $U(1)$, $SU(2)$, $SU(4)$ and $SU(N)$, $N \ge 6$; for $SU(2)$, $m(\mu_\beta) = 4\log(4/\beta) + \frac{7}{24}\beta^2 + O(\beta^4)$. At fourth order the plaquette state moves: it hops to a neighbouring plaquette through a front widened to six edges, bulges by a unit cube, and for $d \ge 4$ turns or jumps through a unit cube of a time slice, each with relative weight u^4. Spreading the plaquette function over a finite set of plaquettes and applying the positivity of the transfer matrix to the fronts ∂q and $\partial q \triangle \partial c$ of these plaquettes in the crossing layer, we prove $\limsup_{\beta\downarrow 0} \beta^{-4}(m(\mu_\beta) - 4\log(1/u) + \log\Upsilon) \le -h\, u_1^4$ with $h = 26d - 70$ ($10d - 22$ in the C-odd sector, which carries the gap for $U(1)$) and $u_1 = \lim u/\beta$: the term of order β^4 is negative. We conjecture equality, $m(\mu_\beta) = 4\log(1/u) - \log\Upsilon - hu^4 + O(\beta^5)$. Numerically, exact transfer-matrix spectra of $\mathbb{Z}_2$ and $\mathbb{Z}_3$ on small tori and blocks of cubes follow the predicted band level by level. For $SU(2)$ and $SU(3)$ in $d = 4$ the conjecture reproduces the terms of order u^4 of the classical strong-coupling series ($-\frac{98}{3}u^4$ for the $SU(2)$ scalar glueball) in four of five channels. In the lower bound only the moves of the hopping matrix enter at order β^4: the junctions with four plaquettes are turns, jumps and U-turns through unit cubes, a six-edge front ends at leading order in one of two narrowings, and the shifts and jumps and the U-turns drop out. Each move goes through a junction whose plaquettes and fronts form a closed surface, on which the transfer operator is diagonal in the representations, with norm u per plaquette. So the lower bound has the same term for $C = +$, and for $SU(N)$, $2 \le N \le 5$, and every $d \ge 3$, $m(\mu_\beta) = 4\log(1/u) - \log\Upsilon - (26d - 70)u^4 + o(\beta^4)$: the conjecture holds for these groups (and for $U(1)$ in $d = 3$). For $SU(2)$ in $d = 4$ this is $m(\mu_\beta) = -4\log u + 2u^2 - \frac{98}{3}u^4 + o(u^4)$, the coefficient of the classical series. For $SU(2)$ the two bounds give $-1.3\cdot 10^{-4} \le m(\mu_\beta) - 4\log(1/u(\beta)) \le 4.8\cdot 10^{-4}$ for all $\beta < \beta_0(4)$ in $d = 4$, and $-2.5\cdot 10^{-4} \le m(\mu_\beta) - 4\log(1/u(\beta)) \le 2.4\cdot 10^{-3}$ for all $\beta < \beta_0(3)$ in $d = 3$ (a numerical evaluation of the explicit constants). The asymptotics itself, indeed the full expansion of the glueball mass, is classical (Schor; O'Carroll et al.); what is new here is a self-contained proof with explicit constants on the whole range, for general compact G and the infinite-volume theory of Part II.

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1 Introduction and main result

The foundations monograph [F] proves that the finite-volume mass gap is bounded below uniformly in the volume for small β, by Dobrushin's method; Part II [II] constructs the infinite-volume theory, identifies its transfer operator with a reversible Markov chain of time slices, and proves the same kind of bound, $m \geq \log(1/(6(d-1)\tanh\beta))$, for the infinite-volume mass gap $m(\mu_\beta)$. These bounds behave like $\log(1/\beta)$ as $\beta \downarrow 0$. The strong-coupling expansion predicts $m(\beta) = -4\log u(\beta) + O(u)$, where $u(\beta) = c_\rho(\beta)/(Nc_0(\beta))$ ($\approx \beta/4$ for $SU(2)$, $\approx \beta/(2N^2)$ for $N \geq 3$): the lightest state in the gauge-invariant sector propagates as a tube of plaquettes with four plaquettes per unit of Euclidean time. Principle 3 of the research program (`docs/03-research-program.md`, §3.1) asks every argument in the strong-coupling regime to reproduce this behaviour. Here we prove a lower bound with the correct leading coefficient for all $\beta < \beta_0(d)$ (Theorem 1.2) and the matching upper bound, on the same range for representations of small dimension and for every representation when β is small (Theorem 1.3); together, $m(\mu_\beta) = 4\log(1/\beta) + O(1)$.

Literature. The strong-coupling glueball spectrum was studied rigorously in the 1980s. According to the account of Faria da Veiga and O'Carroll [FO25, Sections II and V], Schor [Sch83, Sch84a, Sch84b] showed, for $SU(2)$ in three dimensions at fixed lattice spacing and small β, by the hyperplane decoupling method applied to the truncated plaquette–plaquette correlation, that the low-lying energy-momentum spectrum contains a single isolated one-glueball dispersion curve, whose mass is $-4\log\beta + r(\beta)$ with r analytic. By their abstracts,

O'Carroll, Braga and Schor [OBS85] ($2+1$ dimensions) and O'Carroll and Barbosa [OB85] ($3+1$ dimensions) studied the masses in the subspace generated by the time-zero plaquette fields (in [OB85]: at least one and at most four, with convergent expansions), and O'Carroll [OC85] the excited masses near $-6\log\beta$ in $2+1$ dimensions. We have read [FO25], the abstracts of [OB85, OC85], and the title and the first lines of the abstract of [OBS85], not the original papers (marked "verify"), and nothing below relies on them. These results are much sharper than ours: they determine the glueball mass to all orders in β, whereas the bounds proved here determine it up to $O(\beta^4)$ (Corollary 11.6). The title and the first lines of the abstract of [OBS85] mention a mass splitting for complex characters. For $SU(3)$ the cap of Section 9 gives a first-order splitting of the C-even and C-odd plaquette states, presumably the splitting found in [OBS85]; only the shift of the lowest (C-even) state is proved here (Corollary 9.6), that of the C-odd state is Heuristic (Remark 9.7(iii)). The strong-coupling series of the glueball masses were computed by Münster [Mün81] and Seo [Seo82]. We know them only as quoted by Langelage, Münster and Philipsen [LMP07, LMP08], which we have read. Their terms through order u^3 agree with Corollaries 10.9 and 11.6, and their terms of order u^4 agree with Conjecture 12.4 in four of five channels (Remark 12.10(iii)); for the lightest state of $SU(2)$ and of $SU(3)$ the term of order u^4 is proved here (Corollary 14.3). What this article adds for the project is a short self-contained proof, with explicit constants and the explicit range $\beta < \beta_0(d)$, for general compact G and for the infinite-volume theory of [II], together with a covariance bound for all bounded local observables.

Convention 1.1 (Labels). As in [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof here or in [F, II, III]; Remarks are commentary. We use the notation of [III]: $K := 4(2d-3)$, $\epsilon(\beta) = e^{2\beta} - 1$, $x(\beta) = e^2 K\epsilon(\beta)$, $\theta(\beta) = x/(1-x)$, and $\beta_0(d) = \frac{1}{2}\log(1 + \frac{1}{e^2K(K+2)})$, the threshold below which [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)] holds.

Theorem 1.2 (Main theorem). *Let G, ρ be as in [F, Convention 'Group'] with G metrisable, $d \geq 3$, $0 < \beta < \beta_0(d)$, and*

$$q(\beta) := e^3 K(e^{4\beta} - 1).$$

Then $q(\beta) < \frac{e(2+\epsilon_0)}{K+2} < 0.389$, where $\epsilon_0 = \frac{1}{e^2K(K+2)}$, and:

(i) for every periodic spatial lattice Λ_s with all sides $L_i \geq 4$, the finite-volume mass gap of [F, Definition 'Hamiltonian and mass gap' (`def:H`)] satisfies $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$;

(ii) the mass gap of the infinite-volume theory [II, Definition 'Mass gap of an infinite-volume lattice theory' (`iv:def:gap`)] satisfies $m(\mu_\beta) \geq 4\log(1/q(\beta))$;

(iii) for bounded measurable A, B on $G^{E(\mathbb{Z}^d)}$ depending only on finite link sets S_A, S_B, all of whose endpoints have time coordinates in $[a_1, a_2]$, resp. $[b_1, b_2]$, with $a_2 < b_1$,

$$|\mathrm{Cov}_{\mu_\beta}(A, B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q(\beta)}\, c(\beta)^{|S_A|+|S_B|}\, q(\beta)^{4(b_1-a_2)},$$

where $c(\beta) := \left(e^{2\theta(\beta)}(1 + \frac{1}{K})\right)^{2(d-1)}$.

As $\beta \downarrow 0$, $4\log(1/q(\beta)) = 4\log(1/\beta) - 4\log(4e^3K) + O(\beta)$.

Theorem 1.3 (Upper bound; the gap is $4\log(1/\beta) + O(1)$). *Let G, ρ be as in Theorem 1.2 (G metrisable), $d \geq 3$, and let ρ be irreducible and non-trivial, of dimension N; put $\iota := 1$ if $\rho \cong \bar{\rho}$ and $\iota := 0$ otherwise, and $u(\beta) := c_\rho(\beta)/(Nc_0(\beta))$. Suppose $N \leq 3$ (for instance $U(1)$, $SU(2)$, $SU(3)$). There is an explicit function $D(\beta)$ (Theorem 5.6) with $0 < D(\beta) < 0.32$ and $D(\beta) = O(\beta)$ such that for every $0 < \beta < \beta_0(d)$*

$$4\log\frac{1}{q(\beta)} \;\leq\; m(\mu_\beta) \;\leq\; 4\log\frac{1}{u(\beta)} + D(\beta);$$

the upper bound also holds for $m_{\Lambda_s}(\beta)$ on every periodic spatial lattice with all sides ≥ 3, and both bounds on every periodic spatial lattice with all sides ≥ 4. Hence

$$-4\log(4e^3K) - O(\beta) \;\leq\; m(\mu_\beta) - 4\log\frac{1}{\beta} \;\leq\; 4\log\frac{2N^2}{1+\iota} + O(\beta) \qquad (0 < \beta < \beta_0(d)).$$

Moreover, for every $0 < \beta < \beta_0(d)$, $m(\mu_\beta) \geq 4\log(1/u(\beta)) - 8(K+1)\theta_5(\beta) - \log(1+\delta_\infty(\beta))$ with $\theta_5 = O(\beta^5)$ and an explicit $\delta_\infty(\beta) = O(\beta)$, and the same holds for $m_{\Lambda_s}(\beta)$ when all spatial sides are ≥ 4, with a slightly larger δ (Theorem 7.8, Lemma 7.9, Corollary 7.11). So $m(\mu_\beta) - 4\log(1/u(\beta))$ lies in an interval of length at most $8(K+1)\theta_5 + \log(1+\delta_\infty) + D(\beta)$, and $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$ as $\beta \downarrow 0$. Grouping by junctions (Theorem 8.11, Corollary 8.13) improves δ_∞ to $\delta_J = 2\beta(1+O(\beta))$: $m(\mu_\beta) \geq 4\log(1/u(\beta)) - 2\beta + O(\beta^2)$. Evaluating the cap and the upper bound by characters (Theorems 9.4 and 9.5, Corollary 9.6) gives the first-order term, $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ with $\kappa(\beta) = \tau_0^{-1}\int(\operatorname{Re}\chi_\rho)^2 f_\beta\, dU$, which is $O(\beta^2)$ for $U(1)$ and $SU(2)$ and $\beta/6 + O(\beta^2)$ for $SU(3)$ (fundamental representations). At second order, $m(\mu_\beta) = 4\log(1/u(\beta)) - \log\Upsilon(\beta) + O(\beta^3)$, with $\Upsilon(\beta)$ the largest variance of a real unit plaquette function in the span of $\chi_\rho, \overline{\chi_\rho}$ under the weight w_β/c_0 (Theorems 10.2 and 10.8, Corollary 10.9); for $SU(2)$, $m(\mu_\beta) = 4\log(1/u(\beta)) + \beta^2/8 + O(\beta^3)$. At third order, $m(\mu_\beta) = 4\log(1/u(\beta)) - \log\Upsilon(\beta) + O(\beta^4)$ (Theorem 11.5, Corollary 11.6). At fourth order the plaquette state moves, and $m(\mu_\beta) \leq 4\log(1/u(\beta)) - \log\Upsilon(\beta) - (26d-70)u(\beta)^4 + o(\beta^4)$ for $SU(N)$, $2 \leq N \leq 5$ ($10d-22$ in place of $26d-70$ for $U(1)$) (Theorem 12.8, Corollary 12.9), and the lower bound has the same term for $SU(N)$, $2 \leq N \leq 5$, and for $U(1)$ in $d = 3$ (Theorem 14.2, Corollary 14.3): there $m(\mu_\beta) = 4\log(1/u) - \log\Upsilon - (26d-70)u^4 + o(\beta^4)$ ($8u^4$ for $U(1)$, $d = 3$), which is Conjecture 12.4 for these groups. The upper bound holds, with a larger explicit bound on D, whenever $N \leq 4$, or $N = 5$ and $\rho \cong \bar{\rho}$. Without any restriction on N, the upper bound holds for all sufficiently small β, so $m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$.

Here $\operatorname{Cov}(A,B) := \mu(AB) - \mu(A)\mu(B)$. Statement (iii) is the infinite-volume form of the covariance bound of Theorem 4.1; (i) follows from Theorem 4.1 by [F, Corollary 'Gap from Euclidean correlations' (`cor:gapfromcorr`)], and (ii) from (iii) by [II, Theorem 'OS reconstruction of the lattice theory' (`iv:thm:OS`)](iii). Theorem 1.3 is Corollary 5.8, together with Theorem 5.6(ii) and Lemma 5.7 for the upper bound on spatial lattices with sides ≥ 3, Corollary 7.11 (with Theorem 7.8 and Lemma 7.9) for the sharper lower bound and the asymptotics, Theorem 8.11 and Corollary 8.13 for the junctions, Theorems 9.4 and 9.5 and Corollary 9.6 for the first-order term, Theorems 10.2 and 10.8 and Corollary 10.9 for the second-order term, Theorem 11.5 and Corollary 11.6 for the third-order term, Theorem 12.8 and Corollary 12.9 for the fourth-order upper bound, and the last sentence of Theorem 5.6 for arbitrary N. For $SU(2)$ the interval of the last part has length at most $6.1 \cdot 10^{-4}$ ($d = 4$) and $2.7 \cdot 10^{-3}$ ($d = 3$) on the whole range (Remark 9.7; Numerical: an evaluation of the explicit constants).

Remark 1.4 (Comparison). For $d = 4$: the bound of Part II, $\log(1/(18\tanh\beta))$, is $\log(1/\beta) - 2.89 + O(\beta^2)$; the bound here is $4\log(1/\beta) - 29.5 + O(\beta)$. In the common range of validity $\beta < \beta_0(4) \approx 1.54 \cdot 10^{-4}$ the two are comparable near $\beta_0(4)$ (5.890 and 5.591 at $\beta_0(4)$), and the present one is larger for $\beta < 1.39 \cdot 10^{-4}$, by an additive amount that tends to infinity as $\beta \to 0$ (the ratio of the two bounds increases to 4; it is 1.92 at $\beta = 10^{-5}$). These values were computed with `python3`. The point of the theorem is the coefficient 4, not the numerical strength at fixed β; the additive constant -29.5 is far from the predicted $+4\log 4$ for $SU(2)$ (from $u \approx \beta/4$).

Strategy. (1) Let U and U' be two independent copies of the field, each with law μ. Then $2\operatorname{Cov}_\mu(A,B)$ is the integral of $(A-A')(B-B')$ against $\mu\otimes\mu$. The duplicated Wilson weight is $\prod_p(1+f_\beta(U_p))(1+f_\beta(U'_p)) = \prod_p(1+g_p)$ with $|g_p| \leq e^{4\beta} - 1$, so the loop expansion of [III]

applies to the duplicated system, with partition functions $\Xi(\cdot)^2$. (2) In this expansion every set of plaquettes that does not contain a component connecting the supports of A and B gives a product of two integrals, each odd under the exchange $U \leftrightarrow U'$, hence zero (Lemma 2.5). (3) A connecting component has no free edges away from the supports, so in every time layer between the supports it contains at least four temporal plaquettes (Lemma 3.2), hence at least 4×(number of layers) plaquettes (Corollary 3.4). (4) The sum over such sets is controlled by the ratio bound and the rooted counting of [III]. (5) The upper bound (Section 5) uses positivity instead of cancellation. Splitting the one-link operator of the transfer matrix into its trivial and non-trivial parts writes the covariance of two plaquettes at times 0 and 1 as a sum of non-negative terms; one of them is the expectation of a local function in which the four sides of the unit cube between the plaquettes carry non-trivial characters. Its loop expansion is evaluated exactly on the cube by character orthogonality, every other term has at least five further plaquettes, and the spectral inequality $\mathrm{Cov} \le e^{-m}\,\mathrm{Var}$ gives $m \le 4\log(1/u) + O(\beta)$, on the whole range $\beta < \beta_0(d)$ when $\dim\rho \le 4$ (≤ 5 if $\rho \cong \bar\rho$) and for every ρ when β is small. (6) Returning to the lower bound (Section 6): in the transfer-matrix representation of the terms of the duplicated expansion, a temporal plaquette is an operator between two time slices of norm $u_* = \max_{r\neq 0} u_r$, far below the sup norm $e^{4\beta} - 1$ used in (4); this gives $m \ge 4\log(1/u) - 4\log(eK) - O(\beta)$ (when $u_* = u$, e.g. $\dim\rho \le 49$). (7) Finally (Section 7) the connecting sets are encoded by their fronts: a front changes only where a marker (a spatial plaquette or a support) touches it, so the sum over connecting sets becomes a sum over paths of segments and clusters with decorations, in which an unchanged front of k edges costs a^k per layer, $a = e^{2(K+1)\theta_5} u_*$, and nothing in entropy. A generating function in which only the segments of one path carry the weight $z = 1/(a^4(1+\delta'))$ then gives $m \ge 4\log(1/u_*) - O(\beta)$ ($= 4\log(1/u) - O(\beta)$ when $u_* = u$). Counting at vertices, with the exact numbers of fronts with four and five edges, a sharper sup norm of g_p and a ratio bound that sees only closed polymers (which have at least five plaquettes), makes the constant effective on the whole range. (8) Grouping all markers and fronts that meet at a slice into junctions (Section 8) and splitting the weight of each segment between its two ends, the local form of the no-free-link condition shows that the only junction of order $b = e^{2(K+1)\theta_5}\|g_p\|_\infty$ (about 2β when ρ has no trivial summand) through which a four-edge front can pass is the cap, a plaquette closing the front at a slice where it continues; this gives $\delta = b(1 + O(\beta))$, the first-order constant of the method. (9) In Section 9 the transfer bound is applied junction by junction, with the gauge projections at the vertices of each junction; the cap then becomes an operator on the class functions of the holonomy of its plaquette, whose norm relative to the tube is $\kappa(\beta) + O(\beta^2)$, $\kappa(\beta) = \tau_0^{-1}\int(\mathrm{Re}\,\chi_\rho)^2 f_\beta\, dU$, and the same number replaces two sup-norm bounds of the upper bound; so $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$. (10) At second order (Section 10) the caps are summed into the tube, and the covariance is bounded on the sector of the duplicated system that is odd under the exchange of the copies. There the dressed tube has norm $u^4\Upsilon(\beta)(1 + O(\beta^6))$, the shifts and jumps of the tube vanish (their operators have rank one and are even), and the factor z of the generating function is charged only to the slices that the path reaches first; with the upper bound for the best plaquette function, $m(\mu_\beta) = 4\log(1/u(\beta)) - \log\Upsilon(\beta) + O(\beta^3)$. (11) At third order (Section 11) nothing new enters. The factor ϖ can be taken of order β^2; there are no junctions with three plaquettes and two four-edge fronts; and a junction with three front-pairs and one plaquette has a front of at least five edges. So the lower bound holds with $\delta' = O(\beta^4)$, and $m(\mu_\beta) = 4\log(1/u(\beta)) - \log\Upsilon(\beta) + O(\beta^4)$. (12) At fourth order (Section 12) the plaquette state moves: by hops and bulges through six-edge fronts, and by turns and jumps through unit cubes of a slice. For a plaquette function spread over a finite set $\mathcal{W}$ of plaquettes, positivity is applied with one term for every front ∂q and $\partial q \triangle \partial c$ in the crossing layer, $q \in \mathcal{W}$. Each term is evaluated exactly on one or two unit cubes. The rest is controlled because a set of plaquettes in which no link lies in exactly one plaquette has at least six plaquettes, and six only for a unit cube. This

gives $m \le 4\log(1/u) - \log\Upsilon - h_C u^4 + o(\beta^4)$, with h_C the top of the spectrum of the hopping matrix ($26d-70$; $10d-22$ in the C-odd sector, which carries the gap for $U(1)$); the conjecture is equality. (13) The junctions that enter the lower bound at order β^4 (Section 13) are only the shifts and jumps, the turns, jumps and U-turns through a unit cube ($Z_{4,4} \le 12(d-3)$, by the lemma on six plaquettes), and the bulges and hops through six-edge fronts, which end at leading order in one of two narrowings; three front-pairs force a front of at least eight edges (one plaquette) or six (two plaquettes). With $\varpi = b^{5/2}$ the shifts and jumps drop out, and so do the U-turns, whose exit runs back in time. Counted at the sup norm, the remaining moves, those of the hopping matrix, give the third-order lower bound with $\delta' = (\delta'_4 + o(1))\beta^4$, $\delta'_4 = 160(d-3) + 8K_S u_1^2$: 166 for $SU(2)$ in $d=4$ instead of $8.4\cdot 10^{10}$. (14) Each of these moves goes through a junction whose plaquettes, together with the fronts on either side, form a closed surface in which every link lies in exactly two faces (Section 14). On it the transfer operator, evaluated by characters, is diagonal in the representations, with norm u per plaquette instead of the sup norm $b \approx 2\beta$. So the constant becomes the row sum of the hopping matrix, $h_+ u_1^4$: the lower bound to fourth order matches the upper bound for $C=+$, and $m(\mu_\beta) = 4\log(1/u) - \log\Upsilon - (26d-70)u^4 + o(\beta^4)$ for $SU(N)$, $2 \le N \le 5$. The recursion follows the moving tube by itself, since the bound for a four-edge front does not depend on where the front is.

2 The duplicated system

Throughout this section and the next, $\Lambda = \prod_{\lambda=0}^{d-1} \mathbb{Z}/L_\lambda\mathbb{Z}$ is a periodic lattice with all $L_\lambda \ge 3$, with the conventions of [F, III]: $\mathcal{P} = \mathcal{P}(\Lambda)$, $E(\Lambda)$, $E(p)$, $E(X)$, adjacency and connectedness of sets of plaquettes, $f_\beta = w_\beta/c_0(\beta) - 1$, and for $\mathcal{Q} \subseteq \mathcal{P}$ the restricted partition function $\Xi(\mathcal{Q}) = \int \prod_{p\in\mathcal{Q}}(1 + f_\beta(U_p))\,dU$ of [III, Definition 'Restricted partition functions, loop polymers' (`al:def:Xi`)], so that $\Xi(\mathcal{P}) = Z_\Lambda(\beta)/c_0(\beta)^{|\mathcal{P}|}$.

Definition 2.1 (Duplicated system)**.** On $\Omega_\Lambda \times \Omega_\Lambda$ with the product Haar measure $dU\,dU'$ put, for $p \in \mathcal{P}$,

$$g_p(U,U') := \big(1 + f_\beta(U_p)\big)\big(1 + f_\beta(U'_p)\big) - 1.$$

For $X \subseteq \mathcal{P}$ and bounded measurable F on $\Omega_\Lambda \times \Omega_\Lambda$ let $\Phi_F(X) := \iint F \prod_{p\in X} g_p\,dU\,dU'$.

Lemma 2.2 (Properties of the duplicated system)**.** *(a)* $\|g_p\|_\infty \le \epsilon(2\beta) = e^{4\beta} - 1$.

(b) For $\mathcal{Q} \subseteq \mathcal{P}$, $\iint \prod_{p\in\mathcal{Q}}(1+g_p)\,dU\,dU' = \Xi(\mathcal{Q})^2$*; and* $\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}$ *has density* $\prod_{p\in\mathcal{P}}(1+g_p)/\Xi(\mathcal{P})^2$ *with respect to* $dU\,dU'$.

(c) Let $X \subseteq \mathcal{P}$*, let* ℓ *be a link lying in exactly one plaquette of* X*, and let* H *be bounded measurable and independent of* (U_ℓ, U'_ℓ)*. Then* $\iint H \prod_{p\in X} g_p\,dU\,dU' = 0$.

(d) If ρ *has no trivial summand (for instance if* ρ *is irreducible and non-trivial), then* $\|g_p\|_\infty \le \gamma_2(\beta) := e^{2\beta} - 1 + \beta^2 e^\beta$.

Proof. (a) By [III, Lemma `al:lem:facts`](a),(b), $e^{-\beta} \le w_\beta \le e^\beta$ and $1 \le c_0 \le e^\beta$, so $1+f_\beta = w_\beta/c_0 \in [e^{-2\beta}, e^\beta]$ and $(1+f_\beta(U_p))(1+f_\beta(U'_p)) \in [e^{-4\beta}, e^{2\beta}]$ and $g_p \in [e^{-4\beta}-1, e^{2\beta}-1]$; both $1-e^{-4\beta}$ and $e^{2\beta}-1$ are at most $e^{4\beta}-1$. (b) The integrand is $\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p))\cdot\prod_{p\in\mathcal{Q}}(1+f_\beta(U'_p))$; integrate over U and U' separately. For the density: $W_\beta(U)W_\beta(U') = c_0^{2|\mathcal{P}|}\prod_p(1+g_p)$ and $Z_\Lambda^2 = c_0^{2|\mathcal{P}|}\Xi(\mathcal{P})^2$. (c) Let p be the plaquette of X containing ℓ. Then $U_p = AU_\ell^{\pm1}B$ and $U'_p = A'U'^{\pm1}_\ell B'$ with A, B, A', B' independent of (U_ℓ, U'_ℓ), and no other factor of the integrand depends on (U_ℓ, U'_ℓ). By Fubini it suffices that $\iint g_p\,dU_\ell\,dU'_\ell = \int(1 + f_\beta(AU^{\pm1}B))\,dU \cdot \int(1 + f_\beta(A'U^{\pm1}B'))\,dU - 1 = 1\cdot 1 - 1 = 0$, by [III, Lemma `al:lem:facts`](c). (d) Put $y :=$

$\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}\rho(U) \in [-\beta,\beta]$. As in the proof of [III, Lemma `al:lem:facts`](b), $\int \operatorname{Tr}\rho(U)\,dU$ is the multiplicity of the trivial representation in ρ, here 0, so $\int y\,dU = 0$; and $e^y \le 1 + y + \frac{1}{2}y^2 e^{|y|}$. Hence $1 \le c_0 \le 1+\frac{1}{2}\beta^2 e^\beta$, so $1+f_\beta = w_\beta/c_0 \in [e^{-\beta}(1-\frac{1}{2}\beta^2 e^\beta), e^\beta]$, $(1+f_\beta(U_p))(1+f_\beta(U_p')) \in [e^{-2\beta}(1-\beta^2 e^\beta), e^{2\beta}]$, and $g_p \in [-(1-e^{-2\beta}) - \beta^2 e^{-\beta}, e^{2\beta}-1]$; finally $1-e^{-2\beta} \le e^{2\beta}-1$. □

Definition 2.3 (Loop polymers for a set of links)**.** For $S \subseteq E(\Lambda)$ let $\mathcal{P}_S := \{p : E(p)\cap S \neq \emptyset\}$, let $\mathcal{X}_S$ be the family of sets $X \subseteq \mathcal{P}$ each of whose connected components contains a plaquette of $\mathcal{P}_S$ (including $X = \emptyset$), and for $X \subseteq \mathcal{P}$ let $\overline{X} := \{p : E(p) \cap (E(X)\cup S) \neq \emptyset\}$. Put $J_S := 2(d-1)|S|$. Then $|\mathcal{P}_S| \le J_S$ and $|\overline{X}| \le (K+1)|X| + J_S$ (each link lies in $2(d-1)$ plaquettes, and each plaquette is adjacent to at most K plaquettes, [III, Lemma `al:lem:facts`](d)).

Lemma 2.4 (Duplicated loop expansion)**.** *Let $S \subseteq E(\Lambda)$ and let F be bounded, measurable, and dependent only on the variables (U_ℓ, U_ℓ'), $\ell \in S$. Then*

$$\int F\,d(\mu_{\Lambda,\beta}\otimes\mu_{\Lambda,\beta}) = \sum_{X\in\mathcal{X}_S} \Phi_F(X)\Big(\frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}\Big)^2.$$

Proof. This is the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] with f_β replaced by g, single link variables by pairs (U_ℓ, U_ℓ'), dU by $dU\,dU'$, and E_C by S. By Lemma 2.2(b), $\Xi(\mathcal{P})^2 \int F\,d(\mu\otimes\mu) = \sum_{T\subseteq\mathcal{P}} \int\!\!\int F \prod_{p\in T} g_p\,dU\,dU'$. For $T \subseteq \mathcal{P}$ let $X(T)$ be the union of the components of T meeting $\mathcal{P}_S$ and $Y(T) := T \setminus X(T)$. The proof of the claim in [III, Lemma `al:lem:loopexp`], which uses only the combinatorics of components, of $\mathcal{P}_S$ (there $\mathcal{P}_C$) and of $\overline{X}$, shows that $T \mapsto (X(T), Y(T))$ is a bijection onto the pairs with $X \in \mathcal{X}_S$ and $Y \subseteq \mathcal{P}\setminus\overline{X}$. For such a pair, $F\prod_{p\in X} g_p$ depends only on the variables of the links in $S\cup E(X)$ and $\prod_{p\in Y} g_p$ only on those in $E(Y)$, a disjoint set; so the integral factorises, and $\sum_{Y\subseteq\mathcal{P}\setminus\overline{X}} \int\!\!\int \prod_{p\in Y} g_p = \int\!\!\int \prod_{p\in\mathcal{P}\setminus\overline{X}}(1+g_p) = \Xi(\mathcal{P}\setminus\overline{X})^2$ by Lemma 2.2(b). Divide by $\Xi(\mathcal{P})^2$. □

Lemma 2.5 (Cancellation)**.** *Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B, put $S := S_A\cup S_B$ and $F(U,U') := (A(U)-A(U'))(B(U)-B(U'))$. Then:*

(i) $\int F\,d(\mu\otimes\mu) = 2\operatorname{Cov}_\mu(A,B)$ for every probability measure μ on Ω_Λ, and $\|F\|_\infty \le 4\|A\|_\infty\|B\|_\infty$;

(ii) if $X \in \mathcal{X}_S$ and no connected component of X contains both a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$, then $\Phi_F(X) = 0$;

(iii) if some link of $E(X)\setminus S$ lies in exactly one plaquette of X, then $\Phi_F(X) = 0$.

Proof. (i) Expand: $\int (A-A')(B-B')\,d(\mu\otimes\mu) = 2\mu(AB) - 2\mu(A)\mu(B)$. (ii) Let X_1 be the union of the components of X that contain a plaquette of $\mathcal{P}_{S_A}$, and $X_2 := X\setminus X_1$; by hypothesis no component of X_1 meets $\mathcal{P}_{S_B}$, and every component of X_2 meets $\mathcal{P}_{S_B}$ but not $\mathcal{P}_{S_A}$. A component containing a link of S_B contains a plaquette of $\mathcal{P}_{S_B}$; hence $E(X_1)\cap S_B = \emptyset$, and likewise $E(X_2)\cap S_A = \emptyset$; and $E(X_1)\cap E(X_2) = \emptyset$, since distinct components share no link. So $S_A\cup E(X_1)$ and $S_B\cup E(X_2)$ are disjoint, and since $dU\,dU'$ is a product measure,

$$\Phi_F(X) = \int\!\!\int (A-A') \prod_{p\in X_1} g_p \cdot \int\!\!\int (B-B') \prod_{p\in X_2} g_p.$$

The exchange $(U,U') \mapsto (U',U)$ preserves $dU\,dU'$ and every g_p (which is symmetric in U_p, U_p'), and changes the sign of $A-A'$; so the first factor equals its negative and vanishes. (iii) Apply Lemma 2.2(c) with $H = F\prod_{p'\in X\setminus\{p\}} g_{p'}$, p the plaquette containing the link. □

3 Geometry: four plaquettes per time layer

Time coordinates are taken in $\mathbb{Z}/L_0\mathbb{Z}$. A link $(x,0)$ is *temporal of layer* x_0 (it joins times x_0 and x_0+1); a plaquette $(x,0i)$, $1 \le i \le d-1$, is *temporal of layer* x_0. For $Y \subseteq \mathcal{P}$ and a layer s, Y_s denotes the set of temporal plaquettes of layer s in Y, and $T(p)$ the set of time coordinates of the corners of p (one time for a spatial plaquette, two consecutive times for a temporal one). Let $\Lambda_s := \prod_{i=1}^{d-1} \mathbb{Z}/L_i\mathbb{Z}$ be the spatial torus and g the *girth* of its nearest-neighbour graph (the length of a shortest cycle).

Lemma 3.1 (Girth)**.** *Let all $L_i \ge 3$. If $d \ge 3$, then $g = 4$ when all $L_i \ge 4$, and $g = 3$ when some $L_i = 3$. If $d = 2$, then $g = L_1$.*

Proof. Since all $L_i \ge 3$ the graph is simple. A cycle is a closed walk without backtracking, i.e. a sum of unit vectors $\pm e_i$ equal to 0 in Λ_s with no two consecutive steps opposite. For $d = 2$ the graph is the cycle $\mathbb{Z}/L_1\mathbb{Z}$ itself. For $d \ge 3$ a unit square is a cycle of length 4; a cycle of length 3 needs $\pm e_i \pm e_j \pm e_k = 0$ with no two opposite consecutive steps, which forces $i = j = k$ and $3e_i = 0$, i.e. $L_i = 3$; and conversely $L_i = 3$ gives the cycle $0, e_i, 2e_i$. □

Lemma 3.2 (Plaquettes per layer)**.** *Let all $L_i \ge 3$ $(1 \le i \le d-1)$. Let $S \subseteq E(\Lambda)$ and $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X. Let s be a layer none of whose temporal links belongs to S, and let Y be a connected component of X with $Y_s \neq \emptyset$. Then $|Y_s| \ge g$. In particular $|Y_s| \ge 4$ if $d \ge 3$ and all $L_i \ge 4$.*

Proof. The temporal links of layer s are $\ell_{\vec{x}} := ((s,\vec{x}),0)$, $\vec{x} \in \Lambda_s$, and the plaquettes containing $\ell_{\vec{x}}$ are the temporal plaquettes $((s,\vec{x}),0i)$ and $((s,\vec{x}-e_i),0i)$ of layer s. Let Γ be the graph on Λ_s whose edges are the plaquettes $((s,\vec{x}),0i) \in Y_s$, each joining $\vec{x}$ and $\vec{x}+e_i$ (the positions of its two temporal links). Since $L_i \ge 3$, distinct plaquettes give distinct edges and there are no loops, so Γ is a subgraph of the nearest-neighbour graph of Λ_s. If $\vec{x}$ is an endpoint of an edge, then $\ell_{\vec{x}} \in E(Y) \setminus S$, so $\ell_{\vec{x}}$ lies in at least two plaquettes of X; these are temporal plaquettes of layer s sharing the link $\ell_{\vec{x}}$ with a plaquette of Y, hence they belong to the component Y, i.e. to Y_s. So every non-isolated vertex of Γ has degree at least 2, and since $Y_s \neq \emptyset$, Γ contains a cycle (follow edges without immediately backtracking until a vertex repeats), which has at least g edges; they are distinct plaquettes of Y_s. The last statement is Lemma 3.1. □

Lemma 3.3 (Crossing layers)**.** *Let $Y \subseteq \mathcal{P}$ be connected and let $s \neq s'$ be layers with $Y_s = Y_{s'} = \emptyset$. Then either $T(p) \subseteq I_1$ for all $p \in Y$, or $T(p) \subseteq I_2$ for all $p \in Y$, where $I_1 := \{s+1,\dots,s'\}$ and $I_2 := \{s'+1,\dots,s\}$ are the two cyclic intervals of $\mathbb{Z}/L_0\mathbb{Z}$ determined by s, s'.*

Proof. I_1 and I_2 partition $\mathbb{Z}/L_0\mathbb{Z}$. A spatial plaquette has $T(p) = \{r\}$, contained in one of them. A temporal plaquette of layer $r \notin \{s,s'\}$ has $T(p) = \{r, r+1\}$, and $r, r+1$ lie in the same interval because the only consecutive pairs split between I_1 and I_2 are $\{s, s+1\}$ and $\{s', s'+1\}$. If $p, p' \in Y$ are adjacent, the time coordinates of the endpoints of a common link lie in $T(p) \cap T(p')$, so p, p' lie in the same class. By connectedness all plaquettes of Y lie in the same class. □

Corollary 3.4 (Size of connecting components)**.** *Let all $L_\lambda \ge 3$. Let $S_A, S_B \subseteq E(\Lambda)$ and integers $a_1 \le a_2 < b_1 \le b_2$ with $b_2 - a_1 < L_0$ be such that the endpoints of the links of S_A have time coordinates in $\{a_1,\dots,a_2\}$ and those of S_B in $\{b_1,\dots,b_2\}$ (modulo L_0). Put*

$$\tau_{\mathrm{in}} := b_1 - a_2, \qquad \tau_{\mathrm{out}} := L_0 - (b_2 - a_1), \qquad \tau := \min(\tau_{\mathrm{in}}, \tau_{\mathrm{out}}) \ \ (\ge 1).$$

Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus (S_A \cup S_B)$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \ge g\,\tau$. More precisely, with the inner layers $\{a_2,\dots,b_1-1\}$ and the outer layers $\{b_2,\dots,a_1-1+L_0\}$, Y contains at least g temporal plaquettes in every inner layer, or at least g temporal plaquettes in every outer layer.

Proof. Call the layers $s \in \{a_2, \dots, b_1 - 1\}$ inner (τ_{in} of them) and the layers $s' \in \{b_2, \dots, a_1 - 1 + L_0\}$ outer (τ_{out} of them); as integers, $s < s' < s + L_0$ for every inner s and outer s', so inner and outer layers are distinct. *These layers are free.* A temporal link of an inner layer s joins s and $s+1$, where $s+1 \in \{a_2+1, \dots, b_1\}$ is not in $\{a_1, \dots, a_2\}$ and $s \in \{a_2, \dots, b_1 - 1\}$ is not in $\{b_1, \dots, b_2\}$ modulo L_0 (because $b_2 - a_1 < L_0$); so it belongs neither to S_A nor to S_B. The same argument applies to outer layers. *Times of the plaquettes of Y at the supports.* Let $p \in Y \cap \mathcal{P}_{S_A}$, containing a link $\ell \in S_A$. If p is spatial, $T(p) = \{r\}$ with $r \in \{a_1, \dots, a_2\}$; if p is temporal of layer r, then ℓ is either its temporal link of layer r or a spatial link at time r or $r+1$, so $r \in \{a_1 - 1, \dots, a_2\}$. Now let s be inner and s' outer with $Y_s = Y_{s'} = \emptyset$, and I_1, I_2 as in Lemma 3.3. Then $T(p) \subseteq I_2 = \{s'+1, \dots, s\}$: the times $a_1, \dots, a_2$ lie in I_2 because $s' + 1 \le a_1 + L_0$ and $s \ge a_2$; the time $a_2 + 1$ occurs in $T(p)$ only if p is temporal of layer a_2, which is excluded when $s = a_2$ (as $Y_s = \emptyset$) and lies in I_2 when $s \ge a_2 + 1$; similarly the time $a_1 - 1$ occurs only if p is temporal of layer $a_1 - 1$, which is excluded when $s' \equiv a_1 - 1$ and lies in I_2 otherwise. In the same way $T(p) \subseteq I_1 = \{s+1, \dots, s'\}$ for every $p \in Y \cap \mathcal{P}_{S_B}$. *Conclusion.* If $Y_s = Y_{s'} = \emptyset$ for some inner s and some outer s', Lemma 3.3 puts all of Y into one of the disjoint classes I_1, I_2, which contradicts the previous paragraph, since Y meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Hence $Y_s \neq \emptyset$ for all inner layers, or $Y_{s'} \neq \emptyset$ for all outer layers. In either case Lemma 3.2 (with $S = S_A \cup S_B$) gives at least g plaquettes of Y in each of these layers, and plaquettes of different layers are different. □

4 The covariance bound and the mass gap

Theorem 4.1 (Covariance bound in finite volume)**.** *Let $d \ge 2$, $0 < \beta < \beta_0(d)$, and let Λ be a periodic lattice with all $L_\lambda \ge 3$; let g be the girth of Lemma 3.1. Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B as in Corollary 3.4, with $\tau = \min(\tau_{\text{in}}, \tau_{\text{out}})$. Then, with $q = q(\beta) < 1$ (Lemma 4.2) and $c = c(\beta)$ as in Theorem 1.2,*

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, B)| \le \frac{2\|A\|_\infty \|B\|_\infty}{1 - q}\, c^{|S_A| + |S_B|}\, q^{g\tau}.$$

Proof. If S_A or S_B is empty, A or B is constant and the covariance vanishes; so let both be non-empty. Let $S := S_A \cup S_B$ and F as in Lemma 2.5. By Lemmas 2.4 and 2.5, $2\,\mathrm{Cov}(A, B) = \sum_X \Phi_F(X)(\Xi(\mathcal{P} \setminus \overline{X})/\Xi(\mathcal{P}))^2$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X) \setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$; by Corollary 3.4 these have $|X| \ge n_0 := g\tau$. For each of them: $|\Phi_F(X)| \le 4\|A\|_\infty \|B\|_\infty\, \epsilon(2\beta)^{|X|}$ by Lemmas 2.5(i) and 2.2(a); and by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)] and Definition 2.3,

$$\Big(\frac{\Xi(\mathcal{P} \setminus \overline{X})}{\Xi(\mathcal{P})}\Big)^2 \le e^{2\theta|\overline{X}|} \le e^{2\theta(K+1)|X|} e^{2\theta J_S} \le e^{2|X|} e^{2\theta J_S},$$

using $(K+1)\theta < 1$ [III, Lemma `al:lem:constants`]. The number of $X \in \mathcal{X}_S$ with $|X| = n$ is at most $\binom{J_S + Kn}{n}$: this is [III, Corollary 'Counting loop polymers' (`al:cor:countloop`)] with E_C replaced by S (its proof uses only $|\mathcal{P}_S| \le J_S$ and the adjacency bound K; it requires $J_S \ge 1$, i.e. $S \neq \emptyset$). With $\binom{m}{n} t^n \le (1+t)^m$ at $t = 1/K$, $\binom{J_S + Kn}{n} \le (1 + \frac{1}{K})^{J_S} (1 + \frac{1}{K})^{Kn} K^n \le (1 + \frac{1}{K})^{J_S} (eK)^n$. Altogether

$$2|\,\mathrm{Cov}(A, B)| \le 4\|A\|_\infty \|B\|_\infty\, e^{2\theta J_S} \Big(1 + \frac{1}{K}\Big)^{J_S} \sum_{n \ge n_0} \big(eK \cdot e^2 \epsilon(2\beta)\big)^n = \frac{4\|A\|_\infty \|B\|_\infty\, c^{|S|}}{1 - q}\, q^{n_0},$$

since $e^3 K \epsilon(2\beta) = q$ and $(e^{2\theta}(1 + \frac{1}{K}))^{J_S} = c^{|S|}$. □

Lemma 4.2 (Constants)**.** *For $d \geq 2$ and $0 < \beta < \beta_0(d)$: $q(\beta) < \frac{e(2+\epsilon_0)}{K+2}$, which is < 0.909 for $d = 2$ and < 0.389 for $d \geq 3$; and $4 \log \frac{1}{q(\beta)} = 4 \log \frac{1}{\beta} - 4 \log(4e^3 K) + O(\beta)$ as $\beta \downarrow 0$. Moreover $\beta_0(d) < \beta_*(d) = \operatorname{artanh} \frac{1}{6(d-1)}$, the threshold of [II].*

Proof. With $\epsilon = e^{2\beta} - 1 < \epsilon_0$ (which is the condition $\beta < \beta_0(d)$), $e^{4\beta} - 1 = \epsilon(2+\epsilon) < \epsilon_0(2+\epsilon_0)$ and $e^3 K \epsilon_0 = \frac{e}{K+2}$; for $d = 2$, $K = 4$ and $\frac{e(2+\epsilon_0)}{6} < 0.909$; for $d \geq 3$, $K \geq 12$ and $\frac{e(2+\epsilon_0)}{14} < 0.389$. The asymptotics follow from $e^{4\beta} - 1 = 4\beta + O(\beta^2)$. The last claim is [II, Corollary 'Area law in infinite volume' (`iv:cor:arealaw`)] (first line of its proof). □

Proof of Theorem 1.2. The bound on q is Lemma 4.2. (iii) Since $\beta < \beta_*(d)$, the infinite-volume Gibbs measure μ_β of [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)] exists and is the limit of the periodised measures $\tilde{\mu}_{\Lambda_k,\beta}$ on every bounded measurable local function, for every sequence of periodic lattices with all sides tending to infinity. Take such Λ_k with spatial sides ≥ 4. For k large, the reduction map is injective on the links of $S_A \cup S_B$, the functions $A \circ \iota$, $B \circ \iota$ on Ω_{Λ_k} (in the notation of [II]) depend on the images of S_A, S_B, which satisfy the hypotheses of Theorem 4.1 with the same a_i, b_i, $g = 4$, $\tau_{\text{in}} = b_1 - a_2$ and $\tau_{\text{out}} \to \infty$. Theorem 4.1 bounds $\operatorname{Cov}_{\tilde{\mu}_{\Lambda_k,\beta}}(A, B)$ uniformly in k, and the left side converges to $\operatorname{Cov}_{\mu_\beta}(A, B)$.

(ii) Let $F \in \mathcal{A}_+$ in the notation of [II, Definition 'Reflection, shifts, half-space algebras' (`iv:def:reflection`)], depending only on a finite set of links whose endpoints have time coordinates in $[0, n_F]$. Then ΘF depends only on links with endpoint times in $[-n_F, 0]$ and $S_t F$ on links with endpoint times in $[t, t + n_F]$, with the same number of links. For $t \geq 1$ the two sets are disjoint and (iii) gives, by θ- and translation invariance of μ_β (so that $\mu_\beta(\Theta F)\mu_\beta(S_t F) = |\mu_\beta(F)|^2$),

$$\left|\mu_\beta(\Theta F \cdot S_t F) - |\mu_\beta(F)|^2\right| = \left|\operatorname{Cov}_{\mu_\beta}(\Theta F, S_t F)\right| \leq C_F\, q^{4t} \qquad (t \geq 1),$$

with C_F independent of t (here $b_1 - a_2 = t$); for $t = 0$ the left side is at most $2\|F\|_\infty^2$. So the hypothesis of [II, Theorem `iv:thm:OS`](iii) holds with $m = 4 \log(1/q)$ and the constant $\max(C_F, 2\|F\|_\infty^2)$, and $m(\mu_\beta) \geq 4 \log(1/q)$.

(i) Let A be a bounded gauge-invariant function on $\Omega_s = G^{E(\Lambda_s)}$ and $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ with $L_0 \geq 3$. The functions $\overline{A}(t)$ and $A(0)$ depend on the spatial links of the slices t and 0: in Corollary 3.4 take $a_1 = a_2 = 0$, $b_1 = b_2 = t$; for $1 \leq t \leq L_0/2$, $\tau = \min(t, L_0 - t) = t$, and $g = 4$ since all $L_i \geq 4$. Theorem 4.1 gives $|\langle \overline{A}(t) A(0) \rangle - \langle \overline{A}(t) \rangle \langle A(0) \rangle| \leq C_A q^{4t}$ with C_A depending on $\|A\|_\infty$ and $|E(\Lambda_s)|$ only, not on L_0; for $t = 0$ the left side is at most $2\|A\|_\infty^2$. [F, Corollary `cor:gapfromcorr`] (whose proof lets $L_0 \to \infty$ at fixed t, so that the bound is needed only for large L_0) gives $m_{\Lambda_s}(\beta) \geq 4 \log(1/q)$. □

5 The matching upper bound

Throughout this section $d \geq 3$, G is metrisable (as in Theorem 1.2), and ρ is irreducible and non-trivial (automatic for an irreducible faithful ρ when $G \neq \{1\}$); $N = d_\rho$, $\iota := 1$ if $\rho \cong \bar{\rho}$ and $\iota := 0$ otherwise, $\tau_0 := \frac{1+\iota}{2}$, and

$$u_r(\beta) := \frac{c_r(\beta)}{d_r\, c_0(\beta)} \quad (r \text{ irreducible}), \qquad u(\beta) := u_\rho(\beta) = \frac{c_\rho(\beta)}{N c_0(\beta)},$$

$$\varphi_\beta := 1 - \frac{c_0(\beta)}{w_\beta} = \frac{f_\beta}{1 + f_\beta}.$$

Since w_β is real and invariant under inversion, $c_r = \int \overline{\chi_r}\, w_\beta\, dU = \int \chi_r\, w_\beta\, dU = c_{\bar{r}}$ is real (and $c_r > 0$ by [F, Lemma 'Positivity of the character coefficients' (`lem:cr`)]); so $u_{\bar{r}} = u_r > 0$, and

$f_\beta = \sum_{r\neq 0} d_r u_r \chi_r$, absolutely and uniformly convergent ([F, Corollary `cor:uniform`]). Fix two spatial directions, labelled $1, 2$. On a periodic lattice Λ (all $L_\lambda \geq 3$) let $p_0 := (0, 12)$ be the spatial plaquette at the origin and time 0, $p_1 := (e_0, 12)$ its translate at time 1, Q the unit cube with bottom p_0 and top p_1, $\mathcal{Q}$ the set of its six faces, $X_0 \subseteq \mathcal{Q}$ its four sides (the temporal plaquettes $(y, 0i)$ of layer 0 with $(y, i) \in E(p_0)$), $S_Q := E(\mathcal{Q})$ its twelve edges and $S_A := E(p_0)$; and

$$A := \operatorname{Re}\chi_\rho(U_{p_0}), \qquad A_1 := \operatorname{Re}\chi_\rho(U_{p_1}).$$

The same notation is used on $\mathbb{Z}^d$.

The proof has three steps. (a) A positivity property of the transfer matrix (Lemma 5.2) bounds $\operatorname{Cov}(A, A_1)$ from below by the expectation of one function of the twelve edges of Q, in which the four sides carry the factor φ_β, i.e. are forced to carry non-trivial characters. (b) The loop expansion of [III], applied to this local expectation, gives an exact integral over the cube, evaluated by character orthogonality (Lemma 5.1), plus terms with at least five further plaquettes (Lemma 5.4). (c) The spectral inequality $\operatorname{Cov} \leq e^{-m} \operatorname{Var}$. Unlike the duplicated expansion of Section 2, nothing here compares the leading term with the other surfaces connecting p_0 and p_1: the remainder is relative to the minimal tube itself. This is why the result holds on the whole range $\beta < \beta_0(d)$.

Lemma 5.1 (Character computations)**.** *For every $\beta > 0$:*

(i) $\int A\, dU = 0$ *and* $\int A^2\, dU = \tau_0 = \frac{1+\iota}{2}$.

(ii) Let a_0, a_1 be real continuous class functions on G, invariant under $U \mapsto U^{-1}$, with $\sum_r |\hat a_{k,r}| d_r < \infty$, where $\hat a_{k,r} := \int a_k \overline{\chi_r}\, dU$. Then $\hat a_{k,r} = \hat a_{k,\bar r} \in \mathbb{R}$ and

$$\int a_0(U_{p_0})\, a_1(U_{p_1}) \prod_{s\in X_0} f_\beta(U_s)\, dU = \sum_{r\neq 0} \hat a_{0,r}\, \hat a_{1,r}\, u_r^4.$$

In particular $\int A A_1 \prod_{s\in X_0} f_\beta(U_s)\, dU = \tau_0 u^4$.

(iii) $\frac{(1+\iota)\beta e^{-\beta}}{2N^2} \leq u(\beta) \leq \frac{(1+\iota)\beta + \beta^2 e^\beta}{2N^2}$.

(iv) $\int |\varphi_\beta|\, dU \leq e^{3\beta}\beta\sqrt{\tau_0}/N \leq e^{4\beta} N\, u(\beta)/\sqrt{\tau_0}$.

Proof. (i) $\int \chi_\rho = 0$ since ρ is irreducible and non-trivial. With $\operatorname{Re}\chi = \frac12(\chi + \bar\chi)$, $\int (\operatorname{Re}\chi_\rho)^2 = \frac14(\int \chi_\rho^2 + 2\int |\chi_\rho|^2 + \int \overline{\chi_\rho}^2)$. By Schur orthogonality [F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](ii), $\int |\chi_\rho|^2 = 1$ and, since $\chi_{\bar\rho} = \overline{\chi_\rho}$, $\int \chi_\rho^2 = \int \chi_\rho \overline{\chi_{\bar\rho}} = \iota$; likewise $\int \overline{\chi_\rho}^2 = \iota$.

(ii) Substituting $U \mapsto U^{-1}$ (Haar measure is inversion invariant) and using $a_k(U^{-1}) = a_k(U)$, $\chi_r(U^{-1}) = \overline{\chi_r(U)}$: $\hat a_{k,r} = \int a_k \chi_r\, dU$, which is $\overline{\hat a_{k,r}}$ (a_k is real) and equals $\int a_k \overline{\chi_{\bar r}} = \hat a_{k,\bar r}$. Orient the six faces of Q as its boundary. The proof of [F, Proposition 'Cube activity' (`prop:cube`)] establishes, in its line "Altogether $\int \prod_p \chi_{r_p}(U_p)\, dU = \delta_{\text{all } r_p = r}\, d_r^{-4}$" (whose derivation does not use $r_p \neq 0$): for irreducible representations attached to the six faces, $\int \prod_{\text{faces}} \chi_{r_{\text{face}}}(U_{\text{face}})\, dU$ equals d_r^{-4} if all six are equivalent to the same r, and 0 otherwise. Each boundary-oriented face holonomy is conjugate to U_p or U_p^{-1}, and a_0, a_1, f_β are class functions invariant under inversion, so they are the same functions of the boundary-oriented holonomies. Insert $a_k = \sum_r \hat a_{k,r}\chi_r$ for the bottom and the top and $f_\beta = \sum_{r\neq 0} d_r u_r \chi_r$ for the four sides. The resulting six-fold series converges absolutely, because $|\int \prod \chi_{r_{\text{face}}}| \leq \prod d_{r_{\text{face}}}$, $\sum_r |\hat a_{k,r}| d_r < \infty$ and $\sum_r d_r^2 u_r = \sum_r d_r c_r / c_0 = e^\beta/c_0$ ([F, eq. `eq:trace`]); so it may be integrated term by term. Only the terms in which all six faces carry the same r survive, and $r \neq 0$ because f_β has no trivial component; such a term contributes $\hat a_{0,r}\hat a_{1,r}(d_r u_r)^4 d_r^{-4}$. For $a_0 = a_1 = \operatorname{Re}\chi_\rho$: if $\iota = 1$, χ_ρ is real, $\hat a_{k,\rho} = 1$, and the sum is u^4; if $\iota = 0$, $\hat a_{k,\rho} = \hat a_{k,\bar\rho} = \frac12$ and the sum is $2 \cdot \frac14 u^4$. In both cases it is $\tau_0 u^4$.

(iii) By the proof of [F, Lemma `lem:cr`], $c_\rho = \sum_{n,m\geq 0} \frac{t^{n+m}}{n!\,m!} m_\rho(n,m)$ with $t = \beta/(2N)$ and $m_\rho(n,m)$ the multiplicity of ρ in $\rho^{\otimes n} \otimes \bar\rho^{\otimes m}$. Here $m_\rho(0,0) = 0$, $m_\rho(1,0) = 1$, $m_\rho(0,1) = \iota$, and $m_\rho(n,m) \leq N^{n+m-1}$ (dimension count). Hence $(1+\iota)t \leq c_\rho \leq (1+\iota)t + N^{-1}(e^{2Nt} - 1 - 2Nt) \leq \frac{(1+\iota)\beta}{2N} + \frac{\beta^2 e^\beta}{2N}$, using $e^y - 1 - y \leq \frac{y^2}{2} e^y$ with $y = 2Nt = \beta$. Divide by Nc_0 and use $1 \leq c_0 \leq e^\beta$ ([III, Lemma `al:lem:facts`](b)).

(iv) As $1 + f_\beta = w_\beta/c_0 \geq e^{-2\beta}$ ([III, Lemma `al:lem:facts`](a),(b)), $|\varphi_\beta| \leq e^{2\beta}|f_\beta|$, and by the Cauchy–Schwarz inequality $\int |\varphi_\beta| \leq e^{2\beta}(\int f_\beta^2)^{1/2}$. Now $\int f_\beta^2 = c_0^{-2} \int (w_\beta - c_0)^2 \leq \int (w_\beta - 1)^2$, because $c_0 \geq 1$ and $c = c_0 = \int w_\beta$ minimises $\int (w_\beta - c)^2$. With $y := \frac{\beta}{N} \mathrm{Re}\, \chi_\rho$ we have $|y| \leq \beta$, $|e^y - 1| \leq |y| e^{|y|}$, and so $\int (w_\beta - 1)^2 \leq e^{2\beta} \int y^2 = e^{2\beta} \beta^2 \tau_0 / N^2$ by (i). This gives the first inequality; the second follows from $\beta \leq e^\beta N^2 u/\tau_0$, which is (iii). □

Lemma 5.2 (Positivity)**.** *Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ be periodic with all sides ≥ 3, $\beta > 0$, B a real bounded measurable function on Ω_s, $b \in \mathbb{R}$, and $E_0 \subseteq E_s$ a set of spatial links; for $\ell \in E_s$ let s_ℓ be the temporal plaquette of layer 0 (between the slices 0 and 1) based on ℓ. Then, with $\Delta_B := (B(0) - b)(B(1) - b)$,*

$$\langle \Delta_B \rangle_{\Lambda,\beta} \;\geq\; \Big\langle \Delta_B \prod_{\ell \in E_0} \varphi_\beta(U_{s_\ell}) \Big\rangle_{\Lambda,\beta}.$$

In particular, if $b = \langle B(0) \rangle_{\Lambda,\beta}$, which equals $\langle B(1) \rangle_{\Lambda,\beta}$ by translation invariance, the left side is $\mathrm{Cov}_{\Lambda,\beta}(B(0), B(1))$. More precisely, $\langle \Delta_B \rangle_{\Lambda,\beta} = \sum_{F \subseteq E_0} T_F$, where every

$$T_F := \Big\langle \Delta_B \prod_{\ell \in F} \varphi_\beta(U_{s_\ell}) \prod_{\ell \in E_0 \setminus F} (1 - \varphi_\beta(U_{s_\ell})) \Big\rangle_{\Lambda,\beta}$$

is non-negative; the right side above is T_{E_0}.

Proof. We use the transfer matrix $\hat T = V K_1 P_{\mathcal G} V$ of [F, Definition `def:T`] on $L^2(\Omega_s)$, where $P_{\mathcal G}$ is the projection onto the gauge-invariant subspace (denoted P in [F]). By [F, Lemma 'The one-link operator' (`lem:kop`)], the one-link operator is $k_\beta = c_0 \Pi_0 + k'_\beta$ with $k'_\beta := \sum_{r \neq 0} (c_r/d_r) \Pi_r$; both summands are positive operators ($c_r > 0$), with kernels c_0 and $w_\beta(UV^{-1}) - c_0$. For $F \subseteq E_0$ let $K_1^{(F)}$ be the tensor product over $\ell \in E_s$ of k'_β ($\ell \in F$), $c_0 \Pi_0$ ($\ell \in E_0 \setminus F$) and k_β ($\ell \notin E_0$). Expanding $\bigotimes_{\ell \in E_0} (c_0 \Pi_0 + k'_\beta)$ gives $K_1 = \sum_{F \subseteq E_0} K_1^{(F)}$. Each $K_1^{(F)}$ is positive, as a tensor product of positive operators, and its kernel is a product over the links of class functions of $U_\ell U_\ell'^{-1}$; so, as in the proof of [F, Lemma `lem:K`](i), it commutes with every R_g and hence with $P_{\mathcal G}$. Put $\hat T_F := V K_1^{(F)} P_{\mathcal G} V = P_{\mathcal G} (V K_1^{(F)} V) P_{\mathcal G}$. Then $\hat T_F \geq 0$ and $\hat T = \sum_F \hat T_F$, so $0 \leq \hat T_F \leq \hat T$ and $\hat T_F$ is trace class.

Let $\tilde B$ be multiplication by $B - b$. By [F, Theorem 'Transfer matrix formalism' (`thm:tm`)],

$$\langle \Delta_B \rangle = \frac{\mathrm{Tr}(\hat T^{L_0 - 1} \tilde B \hat T \tilde B)}{\mathrm{Tr}\, \hat T^{L_0}} = \sum_{F \subseteq E_0} \frac{\mathrm{Tr}(\hat T^{L_0 - 1} \tilde B \hat T_F \tilde B)}{\mathrm{Tr}\, \hat T^{L_0}}.$$

Every term is non-negative: by cyclicity of the trace ($\hat T_F$ is trace class, the other factors are bounded), $\mathrm{Tr}(\hat T^{L_0-1} \tilde B \hat T_F \tilde B) = \mathrm{Tr}(\hat T_F^{1/2}\, \tilde B \hat T^{L_0 - 1} \tilde B\, \hat T_F^{1/2})$, the trace of a positive operator. Finally, the term of F is T_F. Indeed, the proof of [F, Theorem `thm:tm`] applies verbatim when, in layer 0, the weights $w_\beta(U_{s_\ell})$ are replaced by $w_\beta(U_{s_\ell}) \varphi_\beta(U_{s_\ell}) = w_\beta(U_{s_\ell}) - c_0$ for $\ell \in F$ and by $w_\beta(U_{s_\ell})(1 - \varphi_\beta(U_{s_\ell})) = c_0$ for $\ell \in E_0 \setminus F$, class functions invariant under inversion: integrating out the temporal links of layer 0 then gives, by the computation in the proof of [F, Lemma `lem:K`](ii), the kernel of $K_1^{(F)} P_{\mathcal G}$ instead of that of $K = K_1 P_{\mathcal G}$, i.e. $\hat T_F$ instead of $\hat T$ between the slices 0 and 1. (Sides ≥ 2 would suffice here.) □

Remark 5.3. Lemma 5.2 is a form of reflection positivity for the reflection through the plane $x_0 = \frac{1}{2}$, which bisects links ([F, Remark 'Other reflections'](ii)), applied to each piece of a decomposition of the weights of the crossing plaquettes into positive-definite parts. It holds for every $\beta > 0$. Its point is that it discards, with the right sign, every contribution to the covariance except the one in which the four sides of Q carry non-trivial characters.

Lemma 5.4 (Five plaquettes)**.** *Let Λ be periodic with all sides ≥ 3, or $\Lambda = \mathbb{Z}^d$. Let $\mathcal{L} \subseteq E(\Lambda)$, and let $Y \neq \emptyset$ be a finite set of plaquettes such that every plaquette of Y contains at most one link of $\mathcal{L}$ and every link of $E(Y) \setminus \mathcal{L}$ lies in at least two plaquettes of Y. Then $|Y| \geq 5$.*

Proof. Two facts. (a) Distinct plaquettes share at most one link. (b) For a link m and a vertex $v \notin m$, at most one plaquette contains both. For (a): two links of a plaquette either meet at a corner, and two perpendicular links at a common vertex lie in exactly one plaquette; or they are opposite, (z, μ) and $(z + e_\nu, \mu)$, and then the plaquette is $(z, \mu\nu)$, determined because $z + e_\nu \neq z - e_\nu$ when $L_\nu \geq 3$. For (b): with $m = (z, \mu)$, the plaquettes containing m are $(z, \mu\nu)$ and $(z - e_\nu, \mu\nu)$, $\nu \neq \mu$; their vertices off m are $z \pm e_\nu$, $z + e_\mu \pm e_\nu$, and these $4(d-1)$ points are distinct when all sides are ≥ 3, so v determines the plaquette.

Let $q \in Y$. It has at least three links outside $\mathcal{L}$, each of which lies in a further plaquette of Y, and by (a) these further plaquettes are distinct; so $|Y| \geq 4$. Suppose $|Y| = 4$. Applying this to every plaquette of Y: each has exactly three links outside $\mathcal{L}$ (four would need four further plaquettes) and exactly one in $\mathcal{L}$, and it shares each of its three outside links with a different one of the other three plaquettes; by (a), any two plaquettes of Y share exactly one link. Label the links of q cyclically $m_{\mathcal{L}}, \ell_1, \ell_2, \ell_3$ with $m_{\mathcal{L}} \in \mathcal{L}$ (so ℓ_1 and ℓ_3 are opposite), let q_i be the plaquette of $Y \setminus \{q\}$ containing ℓ_i, and let $v := \ell_1 \cap \ell_2$, $v' := \ell_2 \cap \ell_3$. The plaquettes q_1, q_2 both contain v and share a link m; by (b), $v \in m$, since otherwise $q_1 = q_2$. Likewise q_2, q_3 share a link $m' \ni v'$. As q_1 shares only ℓ_1 with q, $m \neq \ell_2$, and likewise $m' \neq \ell_2$. So q_2 contains ℓ_2 and a further link at each endpoint of ℓ_2: it is spanned by ℓ_2 and a unit vector $e \perp \ell_2$, with m the link from v in direction e and m' the link from v' in direction e. The vector e is not parallel to ℓ_1: if $m = \ell_1$, then $q_2 \ni \ell_1$, but q_2 shares only ℓ_2 with q; and if m and ℓ_1 point in opposite directions from v, they are collinear and no plaquette contains both, whereas q_1 does. Hence e is perpendicular to ℓ_1 and ℓ_2, q_1 is the plaquette spanned at v by ℓ_1 and e, and q_3 the plaquette spanned at v' by ℓ_3 and e. Since $\ell_3 = \ell_1 + (v' - v)$, $q_3 = q_1 + (v' - v)$ is the translate of q_1 by a unit vector perpendicular to its plane, so q_1 and q_3 have no common vertex and share no link: a contradiction. Hence $|Y| \geq 5$. □

The bound is sharp: the five faces other than p_0 of a unit cube containing p_0 satisfy the hypotheses with $\mathcal{L} = E(p_0)$.

Lemma 5.5 (Two-point bounds in finite volume)**.** *Let $0 < \beta < \beta_0(d)$ and let Λ be periodic with all sides ≥ 3. Put*

$$q_* := eK\, e^{(K+1)\theta}\epsilon \;\leq\; x < 1,$$

$$\mathcal{T}_Q := (1+\epsilon)^6 e^{(24d-42)\theta}\Big(1+\frac{1}{K}\Big)^{24(d-2)}\frac{q_*^5}{1-q_*}, \qquad \mathcal{T}_p := (1+\epsilon)e^{(8d-11)\theta}\Big(1+\frac{1}{K}\Big)^{K}\frac{q_*^5}{1-q_*},$$

$$\bar{u} := \frac{(1+\iota)\beta + \beta^2 e^{\beta}}{2N^2}, \qquad a_* := N\bar{u}(1+\mathcal{T}_p) + \sqrt{\tau_0}\,\mathcal{T}_p, \qquad \eta := \frac{\epsilon}{\sqrt{\tau_0}} + \frac{a_* N\bar{u}}{\tau_0},$$

$$L(\beta) := \frac{(1-\eta)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1+\frac{a_*^2}{\tau_0}\Big)e^{16\beta}\,\frac{N^4}{\tau_0^2}\,\mathcal{T}_Q,$$

$$V(\beta) := (1+\epsilon)(1+\mathcal{T}_p) + \mathcal{T}_p,$$

with ϵ, x, θ, K as in the Convention. Then $a := \langle A\rangle_{\Lambda,\beta}$ satisfies $|a| \leq a_*$ and $\mathrm{Var}_{\Lambda,\beta}(A) \leq \tau_0\, V(\beta)$; and if $\eta \leq 1$,

$$\mathrm{Cov}_{\Lambda,\beta}(A, A_1) \;\geq\; \tau_0\, u(\beta)^4\, L(\beta).$$

Proof. The expansion. For $S \subseteq E(\Lambda)$ and a bounded measurable H depending only on the links of S, the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] applies verbatim (with E_C replaced by S, as in Definition 2.3) and gives

$$\langle H\rangle_{\Lambda,\beta} = \sum_{X\in\mathcal{X}_S} \Phi_H(X)\, r_X, \qquad \Phi_H(X) := \int H \prod_{p\in X} f_\beta(U_p)\, dU, \qquad r_X := \frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}.$$

If a link of $E(X)\setminus S$ lies in exactly one plaquette of X, then $\Phi_H(X)=0$ ([III, Lemma `al:lem:facts`](c)). Always $|\Phi_H(X)| \le \epsilon^{|X|}\int |H|\, dU$, by $\|f_\beta\|_\infty \le \epsilon$.

Let $(S,\mathcal{C})$ be $(S_Q,\mathcal{Q})$ or $(S_A,\{p_0\})$. Every plaquette not in $\mathcal{C}$ contains at most one link of S: two links of S in one plaquette either meet at a corner of Q (resp. p_0), and are then two perpendicular edges which span a face of Q (resp. p_0 itself), or are opposite links of the plaquette, i.e. parallel edges whose base points differ by a unit vector, which (sides ≥ 3) are edges of a common face. Each link lies in $2(d-1)$ plaquettes ([III, Lemma `al:lem:facts`](d)), and each edge of Q (resp. p_0) in two faces of Q (resp. in p_0); hence $|\mathcal{P}_{S_Q}| = 6+12(2d-4) = 24d-42$, $|\mathcal{P}_{S_Q}\setminus\mathcal{Q}| = 24(d-2)$, $|\mathcal{P}_{S_A}| = 1+4(2d-3) = 8d-11$ and $|\mathcal{P}_{S_A}\setminus\{p_0\}| = K$.

For $X\subseteq\mathcal{C}$ we have $E(X)\subseteq S$, hence $\overline{X}=\mathcal{P}_S$ and $r_X = r_\emptyset =: R_S$; and $\sum_{X\subseteq\mathcal{C}}\Phi_H(X) = \int H\prod_{p\in\mathcal{C}}(1+f_\beta(U_p))\,dU =: M_S(H)$. Let $X\in\mathcal{X}_S$ with $X\not\subseteq\mathcal{C}$ and $\Phi_H(X)\neq 0$, and put $Y := X\setminus\mathcal{C}$. A link of $E(Y)\setminus S$ lies in at least two plaquettes of X, none of which is in $\mathcal{C}$ (whose plaquettes have all their links in S); so Lemma 5.4 applies with $\mathcal{L}=S$ and gives $|Y|\ge 5$. Every component of Y contains a plaquette of $\mathcal{P}_S\setminus\mathcal{C}$: it lies in a component of X, which contains a plaquette of $\mathcal{P}_S$; if the component of Y is not itself a component of X, one of its plaquettes shares a link with a plaquette of $X\cap\mathcal{C}$, a link of S. Hence, by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] applied to the graph on $(\mathcal{P}\setminus\mathcal{C})\cup\{o\}$ whose root o is joined to the J' plaquettes of $\mathcal{P}_S\setminus\mathcal{C}$, there are at most $\binom{J'+Kn}{n} \le (1+\frac{1}{K})^{J'}(eK)^n$ possible Y with $|Y|=n$. Moreover $\overline{X} = \mathcal{P}_S\cup\{p : E(p)\cap E(Y)\neq\emptyset\}$, so $|\overline{X}| \le |\mathcal{P}_S|+(K+1)|Y|$ and, by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)], $r_X \le e^{\theta|\mathcal{P}_S|+\theta(K+1)|Y|}$. Summing over $X\cap\mathcal{C}$ (at most $(1+\epsilon)^{|\mathcal{C}|}$ in total weight) and over Y,

$$\big|\langle H\rangle_{\Lambda,\beta} - R_S\, M_S(H)\big| \;\le\; \int |H|\,dU\cdot(1+\epsilon)^{|\mathcal{C}|}e^{\theta|\mathcal{P}_S|}\Big(1+\frac{1}{K}\Big)^{J'}\sum_{n\ge 5} q_*^n, \tag{1}$$

which is $\int|H|\cdot\mathcal{T}_Q$ for $(S_Q,\mathcal{Q})$ and $\int|H|\cdot\mathcal{T}_p$ for $(S_A,\{p_0\})$. (Here $q_*\le x$ because $(K+1)\theta<1$, [III, Lemma `al:lem:constants`].)

The plaquette p_0. With $H=1$: $M_{S_A}(1) = \int(1+f_\beta) = 1$, so $|R_{S_A}-1|\le\mathcal{T}_p$. With $H=A$: $\int A(1+f_\beta(U_{p_0}))\,dU = \mathrm{Re}\int\chi_\rho w_\beta\,dU/c_0 = c_{\bar\rho}/c_0 = Nu$ (as $\int\chi_\rho\chi_r = \delta_{r\bar\rho}$), and $\int|A| \le (\int A^2)^{1/2} = \sqrt{\tau_0}$; so $|a|\le(1+\mathcal{T}_p)Nu+\sqrt{\tau_0}\,\mathcal{T}_p \le a_*$, by Lemma 5.1(iii). With $H=A^2$: $M_{S_A}(A^2)\le(1+\epsilon)\tau_0$, so $\mathrm{Var}(A)\le\langle A^2\rangle \le (1+\mathcal{T}_p)(1+\epsilon)\tau_0+\tau_0\mathcal{T}_p = \tau_0 V$.

Positivity. By Lemma 5.2 with $B=\mathrm{Re}\,\chi_\rho$ evaluated on p_0 (so that $B(0)=A$, $B(1)=A_1$), $b=a$ and $E_0=E(p_0)$,

$$\mathrm{Cov}(A,A_1) \;\ge\; \langle H_Q\rangle, \qquad H_Q := (A-a)(A_1-a)\prod_{s\in X_0}\varphi_\beta(U_s),$$

a function of the twelve edges of Q.

The cube. With $H=1$: $M_{S_Q}(1) = \int\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p))\,dU = 1+\int\prod_{p\in\mathcal{Q}}f_\beta(U_p)\,dU =: M_1$. (For $\emptyset\neq X\subsetneq\mathcal{Q}$: the face-adjacency graph of the cube is connected, so some $p\in X$ and $p'\in\mathcal{Q}\setminus X$ share an edge of Q; that edge lies in exactly these two faces of Q, hence in exactly one plaquette of X, and $\int\prod_X f_\beta = 0$.) By Lemma 5.1(ii) with $a_0=a_1=f_\beta$, $M_1 = 1+\sum_{r\neq0} d_r^2 u_r^6 \in [1,1+\epsilon^6]$. So $R_{S_Q}M_1 \ge 1-\mathcal{T}_Q$ by (1). With $H=H_Q$: since $\varphi_\beta(1+f_\beta)=f_\beta$,

$$M_Q := M_{S_Q}(H_Q) = \int\tilde a(U_{p_0})\,\tilde a(U_{p_1})\prod_{s\in X_0} f_\beta(U_s)\,dU, \qquad \tilde a := (\mathrm{Re}\,\chi_\rho - a)(1+f_\beta).$$

The function $\tilde{a}$ is a real class function invariant under inversion, and its character expansion converges absolutely (that of f_β does, and multiplication by $\operatorname{Re}\chi_\rho$ maps χ_r to a combination of characters with total dimension-weighted coefficient $\le N d_r$). By Lemma 5.1(ii), $M_Q = \sum_{r\neq 0} \hat{a}_r^2 u_r^4 \ge \frac{2}{1+\iota}\hat{a}_\rho^2 u^4$ (the terms $r = \rho, \bar{\rho}$, equal), where

$$\hat{a}_\rho = \int \tilde{a}\,\overline{\chi_\rho}\,dU = \tau_0 + \int \operatorname{Re}\chi_\rho\, f_\beta\,\overline{\chi_\rho}\,dU - a\,Nu \;\ge\; \tau_0 - \epsilon\sqrt{\tau_0} - a_* N\bar{u} = \tau_0(1-\eta) \ge 0,$$

using $\int \operatorname{Re}\chi_\rho\overline{\chi_\rho} = \tau_0$, $\int(1+f_\beta)\overline{\chi_\rho} = Nu$ and $|\int \operatorname{Re}\chi_\rho f_\beta \overline{\chi_\rho}| \le \epsilon(\int(\operatorname{Re}\chi_\rho)^2)^{1/2}(\int|\chi_\rho|^2)^{1/2}$. Since $\frac{2}{1+\iota}\tau_0^2 = \tau_0$, $M_Q \ge \tau_0(1-\eta)^2u^4$.

Next, $\int |H_Q|\,dU \le (\tau_0 + a^2)(\int|\varphi_\beta|)^4$. Indeed, write b_ℓ, t_ℓ for the bottom and top edges of Q over $\ell \in E(p_0)$ and v_y for its temporal edges. The holonomy of the side over $\ell = (y,i)$ is conjugate to $U_{b_\ell}U_{v_{y+e_i}}U_{t_\ell}^{-1}U_{v_y}^{-1}$ or its inverse, so for fixed bottom and temporal edges the map from $(U_{t_\ell})_\ell$ to $W := (U_{s_\ell})_\ell$ is a bijection of G^4 preserving Haar measure. By the Cauchy–Schwarz inequality, $\int|H_Q| \le (\int(A-a)^2\prod_\ell|\varphi_\beta(W_\ell)|)^{1/2}(\int(A_1-a)^2\prod_\ell|\varphi_\beta(W_\ell)|)^{1/2}$. In the first integral, A depends only on the bottom edges and W is Haar distributed independently of them, so it equals $\int(A-a)^2\cdot(\int|\varphi_\beta|)^4 = (\tau_0+a^2)(\int|\varphi_\beta|)^4$ by Lemma 5.1(i); the second is the same, with bottom and top exchanged. By Lemma 5.1(iv), $\int|H_Q| \le (\tau_0 + a_*^2)e^{16\beta}N^4u^4/\tau_0^2$.

Finally, by (1), $M_Q \ge 0$ and $R_{S_Q} \ge (1-\mathcal{T}_Q)/M_1 \ge (1-\mathcal{T}_Q)/(1+\epsilon^6)$,

$$\langle H_Q\rangle \;\ge\; R_{S_Q}M_Q - \mathcal{T}_Q\int|H_Q| \;\ge\; \tau_0u^4\Big[\frac{(1-\eta)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1+\frac{a_*^2}{\tau_0}\Big)e^{16\beta}\frac{N^4}{\tau_0^2}\mathcal{T}_Q\Big] = \tau_0u^4L(\beta).$$

□

Theorem 5.6 (Upper bound)**.** *Let $d \ge 3$, G metrisable, ρ irreducible and non-trivial, $0 < \beta < \beta_0(d)$, and with the notation of Lemma 5.5 suppose that $\eta(\beta) \le 1$ and $L(\beta) > 0$. Put $D(\beta) := \log(V(\beta)/L(\beta))$. Then*

(i) $m(\mu_\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$;

(ii) $m_{\Lambda_s}(\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$ for every periodic spatial lattice Λ_s with all sides ≥ 3.

As $\beta \downarrow 0$, $D(\beta) = 2(1+2/\sqrt{\tau_0})\beta + O(\beta^2)$; in particular the hypotheses hold for all sufficiently small β.

Proof. Write Γ_1 and Γ_0 for the covariance of A_1 and A and for the variance of A in the relevant theory. By Lemma 5.5, $\Gamma_1 \ge \tau_0u^4L > 0$ and $\Gamma_0 \le \tau_0 V$, so $\log(\Gamma_0/\Gamma_1) \le 4\log(1/u) + D(\beta)$, provided these bounds hold for the theory in question. It remains to show $m \le \log(\Gamma_0/\Gamma_1)$.

(ii) Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z})\times\Lambda_s$ with $L_0 \ge 3$, and A the corresponding gauge-invariant function on the spatial configurations, so that $A_1 = A(1)$ in the notation of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)]. Since A is gauge invariant, [F, Theorem 'Ground-state expectations' (`thm:gs`)] applies: its first formula with $t = 1$ for the pair (A, A), and its second formula for A and for A^2, together with $\langle A(1)\rangle_\Lambda = \langle A(0)\rangle_\Lambda$ (translation invariance, [F, Theorem `thm:basic`](iv)), give, as $L_0 \to \infty$, $\operatorname{Cov}_\Lambda(A_1, A) \to \langle Q_\Omega\hat{A}\Omega, e^{-H}Q_\Omega\hat{A}\Omega\rangle$ and $\operatorname{Var}_\Lambda(A) \to \|Q_\Omega\hat{A}\Omega\|^2$, where Q_Ω is the projection orthogonal to the vacuum, as in the proof of [F, Theorem 'Gap ⇔ clustering in time' (`thm:gapclust`)](i); and $\langle Q_\Omega\hat{A}\Omega, e^{-H}Q_\Omega\hat{A}\Omega\rangle \le e^{-m_{\Lambda_s}(\beta)}\|Q_\Omega\hat{A}\Omega\|^2$. The bounds of Lemma 5.5 are uniform in L_0, so they hold for the limits Γ_1, Γ_0, and $e^{-m_{\Lambda_s}(\beta)} \ge \Gamma_1/\Gamma_0$.

(i) By [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)](iii) (note $\beta < \beta_0(d) < \beta_*(d)$, Lemma 4.2), the covariance and the variance under μ_β are limits of those on periodic lattices with all sides tending to infinity, so the bounds of Lemma 5.5 hold for them. With ν and P as in [II, Corollary 'The time-slice chain' (`iv:cor:chain`)] and $\tilde{A} := A - \nu(A)$, a function of the time-zero slice, [II, Corollary `iv:cor:chain`](iv) gives $\Gamma_1 = \langle\tilde{A}, P\tilde{A}\rangle_\nu$ and

$\Gamma_0 = \|\tilde{A}\|_\nu^2$. Here $\tilde{A} \perp \mathbf{1}$; and $P \geq 0$ [II, Theorem 'Transfer operator, ground-state chain and mass gap' (`iv:thm:gap`)](ii), so $\|P|_{\mathbf{1}^\perp}\| = \sup \operatorname{spec}(P|_{\mathbf{1}^\perp}) = 1 - \operatorname{gap} P = e^{-m(\mu_\beta)}$ by [II, Theorem `iv:thm:gap`](iii). Hence $\Gamma_1 \leq e^{-m(\mu_\beta)}\Gamma_0$.

Finally, as $\beta \downarrow 0$: $\epsilon = 2\beta + O(\beta^2)$, $x, \theta, q_* = O(\beta)$, so $\mathcal{T}_Q, \mathcal{T}_p = O(\beta^5)$; $\bar{u}, a_* = O(\beta)$ and $\eta = 2\beta/\sqrt{\tau_0} + O(\beta^2)$. Hence $\log V = 2\beta + O(\beta^2)$ and $\log L = -2\eta + O(\beta^2)$. □

Lemma 5.7 (The constants on the whole range)**.** *Let $d \geq 3$ and $0 < \beta < \beta_0(d)$. Then $\epsilon < 8.06 \cdot 10^{-4}$, $\mathcal{T}_Q < 8.2 \cdot 10^{-4}$, $\mathcal{T}_p < 1.5 \cdot 10^{-5}$, $\eta < 1.14 \cdot 10^{-3}$, $V(\beta) < 1.00085$ and*

$$L(\beta) \;>\; 0.9969 - 8.2 \cdot 10^{-4}\, \frac{N^4}{\tau_0^2}.$$

Hence, if $N^4 \leq 1200\,\tau_0^2$ (i.e. $N \leq 4$, or $N = 5$ and $\rho \cong \bar{\rho}$), then $L(\beta) > 0.0129$ and $D(\beta) < \log(1.00085/(0.9969 - 8.2 \cdot 10^{-4} N^4/\tau_0^2))$; if $N \leq 3$ (e.g. $G = U(1)$, $SU(2)$, $SU(3)$), then $L(\beta) > 0.73$ and $D(\beta) < 0.32$. Moreover all the constants $\epsilon, x, \theta, q_*, \mathcal{T}_Q, \mathcal{T}_p, \bar{u}, a_*, \eta$ increase with β, so V increases, L decreases, and D increases with β.

Proof. $\beta_0(d)$ decreases with d, and $\beta < \beta_0(d)$ means $\epsilon < \epsilon_0(d) := \frac{1}{e^2 K(K+2)} \leq \frac{1}{168 e^2} < 8.056 \cdot 10^{-4}$. By [III, Lemma `al:lem:constants`], $x < \frac{1}{K+2}$ and $\theta < \frac{1}{K+1}$; so $q_* \leq x < \frac{1}{K+2}$ and $\frac{q_*^5}{1-q_*} < \frac{1}{(K+2)^4(K+1)}$. With $K = 8d - 12 \geq 12$: $(24d-42)\theta < \frac{24d-42}{8d-11} < 3$, $(8d-11)\theta < 1$, $(1+\frac{1}{K})^{24(d-2)} = (1+\frac{1}{K})^{3K-12} < e^3$ and $(1+\frac{1}{K})^K < e$. Hence $\mathcal{T}_Q < 1.00485\, e^6/(14^4 \cdot 13) < 8.12 \cdot 10^{-4}$ and $\mathcal{T}_p < 1.00081\, e^2/(14^4 \cdot 13) < 1.49 \cdot 10^{-5}$. Next $\beta < \beta_0(3) < 4.027 \cdot 10^{-4}$ gives $N\bar{u} < 4.03 \cdot 10^{-4}/N$, $a_* < 4.18 \cdot 10^{-4}$ (as $\sqrt{\tau_0} \leq 1$), and, with $\tau_0 \geq \frac{1}{2}$, $\eta < \sqrt{2} \cdot 8.056 \cdot 10^{-4} + 2 \cdot 4.18 \cdot 10^{-4} \cdot 4.03 \cdot 10^{-4} < 1.14 \cdot 10^{-3}$. So $(1-\eta)^2(1-\mathcal{T}_Q)/(1+\epsilon^6) > 0.99691$, $(1 + a_*^2/\tau_0) e^{16\beta} < 1.00647$, which gives the bound on L; and $V < 1.000806 \cdot 1.0000149 + 0.0000149 < 1.00085$. If $N^4 \leq 1200\tau_0^2$, then $L > 0.9969 - 0.984 = 0.0129$; since $\tau_0 \in \{\frac{1}{2}, 1\}$, this condition means $N^4 \leq 300$ ($\iota = 0$) or $N^4 \leq 1200$ ($\iota = 1$). For $N \leq 3$, $N^4/\tau_0^2 \leq 4 \cdot 81 = 324$, so $L > 0.9969 - 0.2657 > 0.73$ and $D < \log(1.00085/0.73) < 0.32$. Monotonicity is clear from the definitions. □

Corollary 5.8 (Two-sided bounds on the whole range)**.** *Let $d \geq 3$, G metrisable and ρ irreducible and non-trivial of dimension N, with $N^4 \leq 1200\,\tau_0^2$ (i.e. $N \leq 4$, or $N = 5$ and $\rho \cong \bar{\rho}$; e.g. $U(1)$, $SU(2)$, $SU(3)$, $SU(4)$). For every $0 < \beta < \beta_0(d)$,*

$$4 \log \frac{1}{q(\beta)} \;\leq\; m(\mu_\beta) \;\leq\; 4 \log \frac{1}{u(\beta)} + D(\beta), \qquad D(\beta) = O(\beta),$$

with $0 < D(\beta) < \log(1.00085/(0.9969 - 8.2 \cdot 10^{-4} N^4/\tau_0^2))$, and $D(\beta) < 0.32$ if $N \leq 3$; explicitly,

$$4 \log \frac{1}{\beta} - 4 \log(4e^3 K) - 4 \log \frac{e^{4\beta} - 1}{4\beta} \;\leq\; m(\mu_\beta) \;\leq\; 4 \log \frac{1}{\beta} + 4 \log \frac{2N^2}{1+\iota} + 4\beta + D(\beta).$$

The same bounds hold for $m_{\Lambda_s}(\beta)$ for every periodic spatial lattice with all sides ≥ 4.

If ρ is any irreducible non-trivial representation (no condition on N), the upper bounds hold for all sufficiently small β (Theorem 5.6); in particular $m(\mu_\beta) = 4 \log(1/\beta) + O(1)$ as $\beta \downarrow 0$, and the same holds uniformly for the finite-volume gaps on spatial lattices with all sides ≥ 4.

Proof. The lower bounds are Theorem 1.2(ii) and (i), with $q(\beta) = 4e^3 K\beta \cdot \frac{e^{4\beta}-1}{4\beta}$. The upper bounds are Theorem 5.6 and Lemma 5.7 (which also gives $\eta \leq 1$), with $\log \frac{1}{u} \leq \log \frac{2N^2}{(1+\iota)\beta} + \beta$ (Lemma 5.1(iii)). $D > 0$ because $V \geq 1 \geq L$. □

Remark 5.9 (Numbers, the first version, and checks). (i) Numerically (`python3` with `mpmath`), the supremum of $D(\beta)$ on $(0, \beta_0(d))$, its limit as $\beta \uparrow \beta_0(d)$ (D is increasing, Lemma 5.7), is

$4.8\cdot10^{-3}$, $4.9\cdot10^{-2}$ and $3.8\cdot10^{-3}$ for $SU(2)$, $SU(3)$ and $U(1)$ in $d=3$, and $1.4\cdot10^{-3}$, $1.0\cdot10^{-2}$ and $1.3\cdot10^{-3}$ in $d=4$; and $D(\beta)/\beta\to 6$ for $SU(2)$, $\to 2(1+2\sqrt{2})\approx 7.66$ when $\iota=0$. So on the whole range the upper bound is $4\log(1/u)$ up to a few per cent. The two-sided window of Corollary 5.8 still has width about $4\log(4e^3K)+4\log\frac{2N^2}{1+\iota}$ (about 35.1 for $SU(2)$ in $d=4$). Since the strong-coupling expansion predicts $m=4\log(1/u)+O(u)$, it is expected (heuristic) to come entirely from the constant of the lower bound (Remark 15.3(ii)).

(ii) The first version of this section (9 November 2026; see the repository history and `journal/2026-11-09b.md`) obtained the upper bound from the duplicated expansion of Section 2: the four sides of Q form the unique minimal connecting set, with the exactly computable term $\Phi_F(X_0)=2\tau_0u^4$, and every other connecting set has at least five plaquettes. There the remainder, with sup-norm weights $\approx 4\beta\cdot eK$ per plaquette, had to be compared with the leading term τ_0u^4, $u\approx(1+\iota)\beta/(2N^2)$, which restricted the result to $\beta\lesssim 10^{-16}$. Lemma 5.2 removes this comparison: the remainder in Lemma 5.5 is relative to the tube itself, and the only price is the factor $(\int|\varphi_\beta|/u)^4\le e^{16\beta}N^4/\tau_0^2$.

(iii) Checks (a compass, not part of the proofs). Exact transfer-matrix computations for $\mathbb{Z}_2$ and $\mathbb{Z}_3$ in $d=3$ on a 3×3 spatial torus (physical spaces of dimension 2^{10} and 3^{10}): all sixteen terms of the decomposition in the proof of Lemma 5.2 are non-negative, the covariance dominates $\langle H_Q\rangle$, $\langle H_Q\rangle/(\tau_0u^4)\to1$, and for $\beta<\beta_0(3)$ the bounds of Lemma 5.5 hold with room (e.g. $\langle H_Q\rangle/(\tau_0u^4)=0.9999997\ge L=0.998$ at $\beta=4\cdot10^{-4}$ for $\mathbb{Z}_2$). On this lattice the true gap is $\approx 3\log(1/u)$ (a torelon around a side of length 3), so Theorem 5.6(ii) is valid but not sharp there, as Remark 15.3(iii) expects. For S_3 with its two-dimensional representation, on a spatial graph consisting of one square, the decomposition was checked against brute-force summation over gauge-fixed configurations. Lemma 5.4 (with $\mathcal{L}=S_Q$ and $\mathcal{L}=S_A$) was checked by exhaustive enumeration on the tori $(3,3,3)$ and $(4,4,4)$ up to five plaquettes (the only five-plaquette sets are the caps: the five faces other than F of a unit cube $\neq Q$ sharing a face F with Q, resp. of a unit cube containing p_0) and on $(3,3,3,3)$ and $(4,3,3,3)$ up to four plaquettes.

6 A sharper lower bound: temporal plaquettes at their operator norm

In the proof of Theorem 4.1 every plaquette of a connecting set is bounded by the sup norm $\|g_p\|_\infty\le e^{4\beta}-1\approx4\beta$. For the temporal plaquettes this is far from sharp: viewed as the kernel of an operator between two time slices, $f_\beta(UV^{-1})$ has operator norm $\max_{r\neq0}u_r(\beta)$, which is $u(\beta)\approx(1+\iota)\beta/(2N^2)$ when $\dim\rho\le49$ (Corollary 6.5). The transfer-matrix representation of the terms of the duplicated loop expansion lets every temporal plaquette but those of one time layer carry this weight. The result is a lower bound of the form $4\log(1/u)-4\log(eK)-O(\beta)$ (when $u_*=u$), whose additive constant does not depend on G. Throughout this section G is metrisable when the infinite-volume theory of [II] is used (as in Theorem 1.2), u_r is as in Section 5, and

$$u_*(\beta):=\sup_{r\neq0}u_r(\beta)\ \le\ \epsilon(\beta),$$

the inequality because $u_r=d_r^{-1}\int f_\beta\overline{\chi_r}\,dU$ and $\int|\chi_r|\le1$. The ratios of partition functions are controlled by the following sharpening of [III, Proposition 'Removing plaquettes' (`al:prop:remove`)]: only closed polymers have non-zero activity.

Lemma 6.1 (Removing plaquettes: closed polymers)**.** *Let Λ be periodic with all sides ≥3, $0<\beta<\beta_0(d)$ and $\theta_5(\beta):=x(\beta)^5/(1-x(\beta))$, so that $\theta_5\le\theta$ and $\theta_5=O(\beta^5)$. Then for every $\mathcal{Q}\subseteq\mathcal{P}$*

$$e^{-\theta_5(\beta)|\mathcal{P}\setminus\mathcal{Q}|}\ \le\ \frac{\Xi(\mathcal{Q})}{\Xi(\mathcal{P})}\ \le\ e^{\theta_5(\beta)|\mathcal{P}\setminus\mathcal{Q}|}.$$

Proof. Let γ be a polymer and ℓ a link lying in exactly one plaquette p of γ. Then $U_p = AU_\ell^{\pm 1}B$ with A, B independent of U_ℓ, no other factor of $\prod_{p' \in \gamma} f_\beta(U_{p'})$ depends on U_ℓ, and $\int f_\beta(AU^{\pm 1}B)\, dU = 0$ ([III, Lemma `al:lem:facts`](c)); so $z(\gamma) = 0$. Hence $z(\gamma) \neq 0$ only if every link of $E(\gamma)$ lies in at least two plaquettes of γ, and then $|\gamma| \geq 5$ by Lemma 5.4 with $\mathcal{L} = \emptyset$. In the proof of [III, Lemma 'The Kotecký–Preiss condition for plaquette polymers' (`al:lem:KPsums`)] the sum over n may therefore start at $n = 5$, which gives $\sum_{\gamma \cap S \neq \emptyset} |z(\gamma)| e^{|\gamma|} \leq \theta_5 |S|$ for every $S \subseteq \mathcal{P}$; and [III, Lemma `al:lem:ratio`] applies with $a(\gamma) = |\gamma|$, because $(K+1)\theta_5 \leq (K+1)\theta < 1$. The proof of [III, Proposition `al:prop:remove`] then gives the claim. □

Lemma 6.2 (Layer operators)**.** *Let $\beta > 0$, let Π_0 be the projection onto the constants in $L^2(G)$, and let F_f be the integral operator on $L^2(G)$ with kernel $f_\beta(UV^{-1})$. Then $F_f = \sum_{r \neq 0} u_r \Pi_r$ (notation of [F, Lemma 'The one-link operator' (`lem:kop`)]), so $F_f \geq 0$, $\|F_f\| = u_*$ and $\operatorname{Tr} F_f = e^\beta/c_0 - 1$. The integral operator G_g on $L^2(G \times G)$ with kernel $g(UV^{-1}, U'V'^{-1}) := (1 + f_\beta(UV^{-1}))(1 + f_\beta(U'V'^{-1})) - 1$ is*

$$G_g = (\Pi_0 + F_f) \otimes (\Pi_0 + F_f) - \Pi_0 \otimes \Pi_0 = F_f \otimes \Pi_0 + \Pi_0 \otimes F_f + F_f \otimes F_f,$$

and $G_g \geq 0$, $\|G_g\| = u_*$, $\operatorname{Tr} G_g = (e^\beta/c_0)^2 - 1 \leq e^{2\beta} - 1$.

Proof. The kernel of $f_\beta = \sum_{r \neq 0} (c_r/c_0)\chi_r$ is the kernel of k_β/c_0 minus that of Π_0, so [F, Lemma `lem:kop`] gives $F_f = \sum_{r \neq 0} \frac{c_r}{d_r c_0} \Pi_r = \sum_{r \neq 0} u_r \Pi_r$, with $u_r > 0$; its norm is $\sup_{r \neq 0} u_r$ and its trace $\sum_{r \neq 0} u_r d_r^2 = (\operatorname{Tr} k_\beta - c_0)/c_0 = e^\beta/c_0 - 1$ by [F, eq. `eq:trace`]. The kernel of Π_0 is 1, which gives the formula for G_g. Its three summands are positive and act on the mutually orthogonal subspaces $\Pi_0^\perp \otimes \Pi_0$, $\Pi_0 \otimes \Pi_0^\perp$ and $\Pi_0^\perp \otimes \Pi_0^\perp$, so $\|G_g\| = \max(u_*, u_*, u_*^2) = u_*$, because $u_* \leq 1$: $|u_r| \leq \int |\chi_r| w_\beta\, dU/(d_r c_0) \leq 1$. Finally $\operatorname{Tr} G_g = (1 + \operatorname{Tr} F_f)^2 - 1$, and $c_0 \geq 1$. □

Lemma 6.3 (Transfer bound for duplicated terms)**.** *Let Λ be periodic, let A, B be bounded measurable functions of the spatial links of the time slices 0 and t, $0 < t < L_0$, and $F := (A - A')(B - B')$. For $X \subseteq \mathcal{P}$ let $T_s(X)$ be the set of temporal plaquettes of X in layer s, $T(X) := \bigcup_s T_s(X)$, $S(X) := X \setminus T(X)$ (its spatial plaquettes), $S_s(X)$ the spatial plaquettes of X at slice s, and $m(X) := \min_s |T_s(X)|$. Then*

$$|\Phi_F(X)| \;\leq\; 4\|A\|_\infty \|B\|_\infty\, \gamma_g^{|S(X)|}\, \big(e^{2\beta} - 1\big)^{m(X)}\, u_*^{|T(X)| - m(X)},$$

where $\gamma_g := \|g_p\|_\infty$ (the same for all p; $\gamma_g \leq e^{4\beta} - 1$ by Lemma 2.2(a)).

Proof. As in the proof of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)], now for the duplicated system: label the spatial links of slice s in both copies by $(U^{(s)}, U'^{(s)}) \in \Omega_s \times \Omega_s$ and integrate out the temporal links of each layer. For layer s the integrand contains g_p for the temporal plaquettes $p \in T_s(X)$ and no factor for the others; the temporal links of the two copies are integrated independently, and every factor is a class function of the plaquette holonomy in each copy, invariant under inversion. So the computation in the proof of [F, Lemma `lem:K`](ii) shows that the integral over the temporal links of layer s is the kernel of

$$\mathcal{K}_s := \Big(\bigotimes_{\ell \in T_s(X)} G_g \otimes \bigotimes_{\ell \notin T_s(X)} (\Pi_0 \otimes \Pi_0) \Big)(P_{\mathcal{G}} \otimes P_{\mathcal{G}}),$$

where temporal plaquettes are identified with the spatial links on which they are based and $P_{\mathcal{G}}$ is the projection onto the gauge-invariant subspace (denoted P in [F]). Let $\mathbb{M}_s$ be multiplication by the product of the g_p, $p \in S(X)$ at slice s, times $A(U^{(0)}) - A(U'^{(0)})$ if $s = 0$ and times $B(U^{(t)}) - B(U'^{(t)})$ if $s = t$. Then, exactly as in [F, Theorem `thm:tm`], $\Phi_F(X) = \operatorname{Tr}\big(\mathbb{M}_0 \mathcal{K}_0 \mathbb{M}_1 \mathcal{K}_1 \cdots \mathbb{M}_{L_0 - 1} \mathcal{K}_{L_0 - 1}\big)$; every $\mathcal{K}_s$ is trace class. By Hölder's inequality for

traces, $|\Phi_F(X)| \le \|\mathcal{K}_{s_0}\|_1 \prod_{s\neq s_0}\|\mathcal{K}_s\| \prod_s \|\mathbb{M}_s\|$ for any s_0. By Lemma 6.2, $\|\mathcal{K}_s\| \le u_*^{|T_s(X)|}$ and $\|\mathcal{K}_s\|_1 \le (e^{2\beta}-1)^{|T_s(X)|}$ (and $\|\Pi_0\otimes\Pi_0\| = \mathrm{Tr}(\Pi_0\otimes\Pi_0) = 1$, $\|P_{\mathcal{G}}\otimes P_{\mathcal{G}}\| = 1$); by Lemma 2.2(a), $\|\mathbb{M}_s\| \le \gamma_g^{|S_s(X)|}$ times $2\|A\|_\infty$ or $2\|B\|_\infty$ at $s = 0, t$. Choose s_0 with $|T_{s_0}(X)| = m(X)$. □

Theorem 6.4 (Sharper covariance bound and gap)**.** *Let $d \ge 2$, $0 < \beta < \beta_0(d)$, let Λ be periodic with all sides ≥ 3 and girth g (Lemma 3.1), and let A, B be bounded measurable functions of finitely many spatial links of the time slices 0 and t, with $1 \le t \le L_0/2$; S is the set of these links. Put $\hat q(\beta) := eK\, e^{2(K+1)\theta_5(\beta)}(e^{4\beta}-1) \le q(\beta)$ and, with θ_5 as in Lemma 6.1,*

$$q_u(\beta) := eK\, e^{2(K+1)\theta_5(\beta)}\, u_*(\beta).$$

Then $\hat q < 1$ and, with $c = c(\beta)$ as in Theorem 1.2,

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A,B)| \;\le\; \frac{2\|A\|_\infty\|B\|_\infty}{1-\hat q}\, c^{|S|}\, q_u^{gt}.$$

Consequently, for $d \ge 3$: (i) $m_{\Lambda_s}(\beta) \ge 4\log(1/q_u(\beta))$ for every periodic spatial lattice Λ_s with all sides ≥ 4; (ii) $m(\mu_\beta) \ge 4\log(1/q_u(\beta))$.

Proof. $\hat q \le q < 1$ because $(K+1)\theta < 1$ ([III, Lemma `al:lem:constants`], Lemma 4.2). As in the proof of Theorem 4.1, $2\,\mathrm{Cov}(A,B) = \sum_X \Phi_F(X) r_X^2$, $r_X := \Xi(\mathcal{P}\setminus\overline{X})/\Xi(\mathcal{P})$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X)\setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Fix such an X. Temporal links are never in S, so Lemma 3.2, applied to a component of X that meets layer s, shows that $T_s(X)$ is empty or has at least g elements, for every layer s. Put $n(X) := |T(X)| - m(X)$. If some layer contains no temporal plaquette of X, then $m(X) = 0$, and by Corollary 3.4 the connecting component has at least g temporal plaquettes in every inner layer or in every outer layer, so $n(X) \ge g\min(t, L_0 - t) = gt$. Otherwise every layer has at least $m(X) \ge g$ temporal plaquettes and $n(X) \ge (L_0-1)m(X) \ge (L_0-1)g \ge gt$. Let $\gamma := e^{4\beta}-1$. Since $u_* \le \epsilon \le \gamma$, $e^{2\beta}-1 \le \gamma$ and $\gamma_g \le \gamma$, Lemma 6.3 gives

$$|\Phi_F(X)| \le 4\|A\|_\infty\|B\|_\infty\, \gamma^{|X|-n(X)} u_*^{n(X)} \le 4\|A\|_\infty\|B\|_\infty\, \gamma^{|X|}\,(u_*/\gamma)^{gt},$$

and $|X| \ge n(X) \ge gt$. The rest is the computation in the proof of Theorem 4.1, with $r_X^2 \le e^{2\theta_5|\overline{X}|} \le e^{2\theta J_S} e^{2\theta_5(K+1)|X|}$ (Lemma 6.1) in place of $e^{2\theta|\overline{X}|}$, keeping the factor $e^{2\theta_5(K+1)|X|}$:

$$2|\,\mathrm{Cov}(A,B)| \le 4\|A\|_\infty\|B\|_\infty \Big(\frac{u_*}{\gamma}\Big)^{gt} c^{|S|} \sum_{n\ge gt} \big(eK\, e^{2\theta_5(K+1)}\gamma\big)^n = \frac{4\|A\|_\infty\|B\|_\infty\, c^{|S|}}{1-\hat q}\Big(\frac{u_*}{\gamma}\Big)^{gt}\hat q^{\,gt},$$

and $\hat q u_*/\gamma = q_u$.

(i) As in the proof of Theorem 1.2(i): for gauge-invariant bounded A on Ω_s, the bound with $g = 4$ (all spatial sides ≥ 4, Lemma 3.1) is uniform in L_0, and [F, Corollary `cor:gapfromcorr`] gives $m_{\Lambda_s}(\beta) \ge 4\log(1/q_u)$.

(ii) Let ψ be a real bounded measurable function of finitely many spatial links of the time-zero slice, and $\tilde\psi := \psi - \nu(\psi)$ with ν and P as in [II, Corollary 'The time-slice chain' (`iv:cor:chain`)]. By [II, Corollary `iv:cor:chain`](iv), $\langle\tilde\psi, P^t\tilde\psi\rangle_\nu = \mathrm{Cov}_{\mu_\beta}(\psi(U^{(0)}), \psi(U^{(t)}))$, and by [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)](iii) this is the limit of the covariances of the periodised observables on periodic lattices with all sides tending to infinity (and spatial sides ≥ 4). The bound above, with $g = 4$, gives $\langle\tilde\psi, P^t\tilde\psi\rangle_\nu \le C_\psi q_u^{4t}$ for all $t \ge 1$. Since $P \ge 0$ [II, Theorem 'Transfer operator, ground-state chain and mass gap' (`iv:thm:gap`)](ii), the spectral measure σ_ψ of $\tilde\psi$ with respect to P satisfies $\int \lambda^t\, d\sigma_\psi \le C_\psi q_u^{4t}$, hence $\sigma_\psi((q_u^4, 1]) = 0$. Such ψ are dense in the real $L^2(\nu)$, by [II, Lemma `iv:lem:measures`](iv) with the link set $E^{(0)}$ and the unitary map $B \mapsto B(U^{(0)})$ (as in the proof of [II, Theorem

iv:thm:gap](i)), so the $\tilde{\psi}$ are dense in the real part of $\mathbf{1}^{\perp}$; since P maps real functions to real functions, the spectral projection of $P|_{\mathbf{1}^{\perp}}$ for $(q_u^4, 1]$ vanishes, $\sup\operatorname{spec}(P|_{\mathbf{1}^{\perp}}) \le q_u^4$, and $m(\mu_\beta) = -\log\sup\operatorname{spec}(P|_{\mathbf{1}^{\perp}}) \ge 4\log(1/q_u)$ by [II, Theorem iv:thm:gap](iii). □

Corollary 6.5 (The window on the whole range)**.** *Let $d \ge 3$, $0 < \beta < \beta_0(d)$, and let ρ be irreducible and non-trivial with $N \le 49$. Then $u_*(\beta) = u(\beta)$, so*

$$-4\log(eK) - 8(K+1)\theta_5(\beta) \;\le\; m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \;\le\; D(\beta),$$

the upper inequality when $N^4 \le 1200\tau_0^2$ (Corollary 5.8); the lower one also holds for $m_{\Lambda_s}(\beta)$ on spatial lattices with all sides ≥ 4. As $\beta \downarrow 0$ the width of the window tends to $4\log(eK)$, which is 13.9 for $d = 3$ and 16.0 for $d = 4$, for every such G and ρ.

Proof. For irreducible $r \notin \{0, \rho, \bar{\rho}\}$, the multiplicity $m_r(n, m)$ of r in $\rho^{\otimes n} \otimes \bar{\rho}^{\otimes m}$ vanishes for $n+m \le 1$ and satisfies $d_r m_r(n, m) \le N^{n+m}$; the series of the proof of [F, Lemma lem:cr] gives, with $t = \beta/(2N)$, $c_r \le d_r^{-1} \sum_{k\ge 2}(2Nt)^k/k! = (e^\beta - 1 - \beta)/d_r$, so $u_r \le e^\beta - 1 - \beta \le \frac{1}{2}\beta^2 e^\beta$. By Lemma 5.1(iii), $u_\rho = u_{\bar{\rho}} = u \ge (1+\iota)\beta e^{-\beta}/(2N^2)$. Hence $u_* = u$ as soon as $\beta e^{2\beta} \le (1+\iota)/N^2$, which holds for $\beta < \beta_0(d) \le \beta_0(3) < 4.03\cdot 10^{-4}$ when $N \le 49$. The lower bound is Theorem 6.4 with $\log(1/q_u) = \log(1/u) - \log(eK) - 2(K+1)\theta_5$; the upper bound is Corollary 5.8. Finally $\theta_5(\beta) \to 0$ and $D(\beta) \to 0$ (Theorem 5.6). □

Remark 6.6 (Comparison and what remains). (i) Numerically (python3 with mpmath; u computed exactly), at $\beta = \beta_0(4) \approx 1.54 \cdot 10^{-4}$ for $SU(2)$ the lower bounds are $4\log(1/q_u) = 24.68$ (Theorem 6.4), $4\log(1/q) = 5.59$ (Theorem 1.2) and 5.89 (Part II), and the upper bound is 40.67; the window between the two bounds of Corollary 6.5 is $4\log(eK) + O(\beta)$, about 16.0 ($d = 4$; 13.9 for $d = 3$) on the whole range, against 35.1 for Theorem 1.2 and Corollary 5.8 ($SU(2)$, $d = 4$). (The first version of this section had θ in place of θ_5, and its window at $\beta_0(d)$ was 24.0 and 21.9; now $8(K+1)\theta_5 < 2.1 \cdot 10^{-4}$.) Section 7 raises the lower bound at $\beta_0(4)$ to 39.29 (Remark 7.10).

(ii) What remains is polymer entropy: the factor eK per required temporal plaquette comes from the rooted counting of all connected sets ([III, Lemma al:lem:rooted]), whereas a connecting surface with exactly four temporal plaquettes in each layer is essentially rigid: if T_s and T_{s+1} denote its temporal plaquettes in two consecutive layers, the symmetric difference of the corresponding sets of spatial links is contained in the edges of the spatial plaquettes of X at slice $s+1$ (a horizontal link lying in no such plaquette lies in 0 or 2 plaquettes of X). So a tube can change position only by paying for spatial plaquettes; Section 7 turns this into a proof that the loss tends to 0 with β (Theorem 7.8). The strong-coupling expansion predicts $m = 4\log(1/u) + O(u)$; the classical results [Sch83, Sch84a, Sch84b] (as reported in [FO25]) determine m to all orders.

(iii) The structural point is that the transfer-matrix view weighs the temporal plaquettes of all but one layer by the operator norm of their kernel, u_* ($= u \approx (1+\iota)\beta/(2N^2)$ for $\dim\rho \le 49$), instead of the sup norm $\approx 2\beta$ (or $\approx 4\beta$ in the duplicated system); only the spatial plaquettes, which a minimal tube does not need, are bounded by sup norms.

(iv) Checks (a compass): for S_3 with its two-dimensional representation on (time $\mathbb{Z}/2$) × (one square), the terms $\Phi_F(X)$ of the duplicated expansion for five sets X were computed by brute-force summation over gauge-fixed configurations and by the transfer representation of the proof of Lemma 6.3; they agree to all printed digits and satisfy the bound of Lemma 6.3 (theory/checks/lower-bound/).

7 The sharp constant: counting by fronts

The remaining loss $4\log(eK)$ in Theorem 6.4 is entropy: the rooted counting of all connected sets charges every temporal plaquette a factor eK, although a surface with exactly four temporal plaquettes in each layer is rigid. In this section the connecting sets are organised by their *fronts*, the temporal plaquettes of a single layer. A front can change only where a marker (a spatial plaquette or a support) touches it, so an unchanged front of k edges costs a^k per layer, with $a = e^{2(K+1)\theta_5}u_*$, and carries no entropy. The outcome is $m \geq 4\log(1/u_*) - O(\beta)$, the sharp constant: together with Section 5, $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$ (here $u_* = u$, Corollary 6.5). Counting the fronts and clusters that meet a given front or cluster at a vertex, and using the exact numbers of fronts with four and five edges, makes the bound effective on the whole range: for $SU(2)$ the bound for $m(\mu_\beta)$ lies within 1.4 ($d = 4$) and 0.7 ($d = 3$) of $4\log(1/u)$ for all $\beta < \beta_0(d)$ (Remark 7.10).

Throughout this section $d \geq 3$, $0 < \beta < \beta_0(d)$, G is metrisable whenever the infinite-volume gap $m(\mu_\beta)$ is concerned, Λ is periodic with all sides ≥ 3 and all spatial sides ≥ 4 (so $g = 4$), A, B are bounded measurable functions of finitely many spatial links of the slices 0 and t, $1 \leq t \leq L_0/2$, with link sets S_A, S_B, $S := S_A \cup S_B$, and $F := (A - A')(B - B')$. Let Γ be the nearest-neighbour graph of Λ_s; we identify a temporal plaquette of layer s with the edge of Γ on which it is based, a spatial link at time s with an edge of Γ, and S_A, S_B with sets of edges of Γ. For a set E of edges, $V(E)$ is the set of their endpoints. A *front in* Γ is a non-empty set of edges, connected (any two of its edges are linked by a chain of edges of the set, consecutive ones sharing an endpoint), in which every vertex has degree ≥ 2. Put

$$P_v := 2(d-1)(d-2), \qquad K_S := 4(2d-5), \qquad N_k^v := \max_v \#\{\varphi : \ |\varphi| = k, \ v \in V(\varphi)\},$$

φ ranging over the fronts in Γ. P_v is the number of plaquettes of Λ_s containing a given vertex; a link of Λ_s lies in $2(d-2)$ plaquettes of Λ_s; and a plaquette of Λ_s shares a link with K_S others.

Definition 7.1 (Fronts, clusters, markers, segments)**.** Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X, and such that some layer s_* contains no temporal plaquette of X; times are represented in $\{s_* + 1, \dots, s_* + L_0\}$.

(a) For a layer s, $T_s \subseteq E(\Gamma)$ is the set of temporal plaquettes of X in layer s. A *front* of layer s is a connected component of T_s.

(b) A *cluster* at slice s is a connected component (two plaquettes being connected when they share a link) of the set of spatial plaquettes of X at time s; $E(C) \subseteq E(\Gamma)$ is its set of links and $|C|$ its number of plaquettes. The *markers* at slice s are the clusters at slice s, together with S_A if $s \equiv 0$ and S_B if $s \equiv t$; $E(\mathfrak{m})$ is the edge set of a marker $\mathfrak{m}$.

(c) A marker $\mathfrak{m}$ at slice s *touches* a front φ of layer $s-1$ or s if $E(\mathfrak{m})$ and φ have a common vertex, i.e. some edge of $E(\mathfrak{m})$ equals or shares an endpoint with an edge of φ.

(d) A *segment* is a pair $\sigma = (\varphi, [s_1, s_2))$, $s_1 < s_2$, such that φ is a front of every layer $s \in [s_1, s_2)$, no marker at a slice $s \in (s_1, s_2)$ touches φ, and some marker at slice s_1 and some marker at slice s_2 touch φ. We write $\ell(\sigma) := s_2 - s_1$ and $|\sigma| := |\varphi|$.

Lemma 7.2 (Fronts and segments)**.** *Let X be as in Definition 7.1.*

(i) Every vertex of Γ has degree 0 or at least 2 in T_s; every front of a layer is a front in Γ and has at least 4 edges.

(ii) For every slice s, $T_{s-1} \triangle T_s \subseteq \bigcup_{\mathfrak{m}} E(\mathfrak{m})$, the union over the markers $\mathfrak{m}$ at slice s.

(iii) *If a front φ of layer $s-1$ is touched by no marker at slice s, then φ is a front of layer s; and if a front of layer s is touched by no marker at slice s, it is a front of layer $s-1$.*

(iv) *Every front of every layer belongs to exactly one segment, i.e. for every front φ of a layer s there is exactly one segment $(\varphi, [s_1, s_2))$ with $s \in [s_1, s_2)$.*

(v) *$|X| = \sum_\sigma |\sigma|\,\ell(\sigma) + \sum_C |C|$, the sums over the segments and the clusters of X.*

Proof. (i) This is the argument of Lemma 3.2 (temporal links are never in S), and a non-empty graph of minimum degree ≥ 2 contains a cycle, of length ≥ 4 (Lemma 3.1). (ii) A link at time s lies in the temporal plaquettes based on it in layers $s-1$ and s, and in the spatial plaquettes of time s containing it. If it lies in no marker at slice s, it is not in S and in no spatial plaquette of X, so it lies in 0 or 2 plaquettes of X: in both of T_{s-1}, T_s or in neither. (iii) Let φ be a front of layer $s-1$ touched by no marker at slice s. By (ii), $\varphi \subseteq T_s$. If an edge $e \in T_s \setminus \varphi$ shared an endpoint with an edge of φ, then $e \notin T_{s-1}$ (φ is a component of T_{s-1}), so $e \in E(\mathfrak{m})$ for a marker $\mathfrak{m}$ at slice s by (ii), and $\mathfrak{m}$ would touch φ. So φ is a component of T_s. The second statement is symmetric. (iv) Let φ be a front of layer s. Going up, let $s_2 > s$ be the first slice at which a marker touches φ; by (iii) and induction, φ is a front of every layer in $[s, s_2)$. Such a slice exists: otherwise φ would be a front of every later layer, including s_*. Likewise let $s_1 \leq s$ be the last slice $\leq s$ at which a marker touches φ. Then $(\varphi, [s_1, s_2))$ is a segment, and it is the only one containing (φ, s), because a segment cannot contain a slice touching φ in its interior. (v) Every plaquette of X lies in exactly one cluster or in exactly one pair (φ, s), hence by (iv) in exactly one segment. □

Definition 7.3 (Incidence graph)**.** For X as in Definition 7.1, $\mathcal{I}(X)$ is the graph whose vertices are the segments and the markers of X; a segment $(\varphi, [s_1, s_2))$ is joined to every marker at slice s_1 or s_2 that touches φ, and a cluster at slice 0 (resp. t) that contains an edge of S_A (resp. S_B) is joined to S_A (resp. S_B).

Lemma 7.4 (Connectivity and covering)**.** *Let X be as in Definition 7.1.*

(i) *If two plaquettes of X share a link, the vertices of $\mathcal{I}(X)$ containing them are equal or joined by a path of length at most two. A plaquette containing a link of S_A (resp. S_B) lies in a vertex equal or joined to S_A (resp. S_B).*

(ii) *If $X \in \mathcal{X}_S$, every vertex of $\mathcal{I}(X)$ is joined by a path to S_A or to S_B; if moreover a component of X contains a plaquette of $\mathcal{P}_{S_A}$ and one of $\mathcal{P}_{S_B}$, then $\mathcal{I}(X)$ contains a path from S_A to S_B.*

(iii) *If π is a path in $\mathcal{I}(X)$ from S_A to S_B, then $\sum_{\sigma\in\pi} \ell(\sigma) \geq \min(t, L_0 - t) = t$.*

Proof. (i) Two temporal plaquettes of the same layer sharing a (temporal) link lie in the same front, hence in the same segment. A temporal plaquette and a spatial plaquette of time s sharing a link e: the cluster C of the latter contains e, so it touches the front $\varphi \ni e$ (of layer $s-1$ or s), and the segment of φ has s as an end slice by Lemma 7.2(iv); so they are joined. Two spatial plaquettes sharing a link lie at the same time, in the same cluster. Two temporal plaquettes based on the same edge e in layers $s-1$ and s: let $\varphi \ni e$ and $\varphi' \ni e$ be their fronts. If $\varphi = \varphi'$ and no marker at slice s touches φ, they lie in the same segment; if some marker touches $\varphi = \varphi'$ at slice s, both segments end at s and are joined to it. Let $\varphi \neq \varphi'$. If $\varphi \subseteq T_s$, then $\varphi \subseteq \varphi'$ (it is connected and meets φ'), and if also $\varphi' \subseteq T_{s-1}$ then $\varphi = \varphi'$; so, by symmetry, we may assume $\varphi \not\subseteq T_s$. Follow a path of edges $e = e_0, e_1, \ldots, e_m$ in φ (consecutive edges sharing an endpoint) ending outside T_s, and let e_j be its first edge outside φ'. Then $e_j \notin T_s$ (otherwise it would lie in the component φ' of its neighbour e_{j-1}), so $e_j \in T_{s-1} \setminus T_s$, and by Lemma 7.2(ii) $e_j \in E(\mathfrak{m})$ for a marker $\mathfrak{m}$ at slice s. This $\mathfrak{m}$ touches φ ($e_j \in \varphi$) and

φ' (e_j shares an endpoint with $e_{j-1} \in \varphi'$), so both segments end at s and are joined to $\mathfrak{m}$. Finally, a temporal plaquette based on an edge of S_A lies in layer -1 or 0, its front is touched by S_A, and its segment ends at the slice 0; a spatial plaquette containing a link of S_A lies at time 0, in a cluster joined to S_A. (ii) Each component of $X \in \mathcal{X}_S$ contains a plaquette with a link in S_A or S_B; apply (i) along chains of adjacent plaquettes. (iii) Assign to a marker at slice s the point s and to a segment $(\varphi, [s_1, s_2))$ the interval $[s_1, s_2]$, in the representation of Definition 7.1. Joined vertices get intersecting sets, so the union of the sets along π is an interval containing the points assigned to S_A and S_B. These points are L_0 and $L_0 + t$ when s_* is an outer layer, and t and L_0 when s_* is an inner one; the layers between them are the t inner layers in the first case and the $L_0 - t$ outer layers in the second. Every such layer, a unit interval between consecutive integers, lies in $[s_1, s_2]$ for some segment $\sigma = (\varphi, [s_1, s_2))$ of π (markers are points), hence in its set of layers. □

Lemma 7.5 (Small fronts)**.** *Let φ be a front in Γ (all sides of Λ_s are ≥ 4).*

(i) $|V(\varphi)| \leq |\varphi|$.

(ii) If $|\varphi| = 4$, then φ is the boundary of a plaquette of Λ_s or a straight cycle $\{\{v + je_\lambda, v + (j+1)e_\lambda\} : 0 \leq j < 4\}$ around a side of length $L_\lambda = 4$.

(iii) If $|\varphi| = 5$, then φ is a straight cycle around a side of length 5.

Consequently, if n_j is the number of spatial directions in which Λ_s has side j,

$$N_4^v = P_v + n_4 \leq (d-1)(2d-3), \qquad N_5^v = n_5 \leq d-1, \qquad N_k^v \leq (d-1)\binom{(4d-6)k}{k-1} \quad (k \geq 4).$$

Proof. A closed walk of length m in Γ lifts to a walk in $\mathbb{Z}^{d-1}$ from 0 to a point w with $w_\lambda \in L_\lambda \mathbb{Z}$ for every λ. If m_λ is its number of steps in direction λ, then $m = \sum_\lambda m_\lambda$, $m_\lambda \geq |w_\lambda|$ and $m_\lambda \equiv w_\lambda \pmod 2$. So if $w = 0$, m is even; and if $w_\lambda \neq 0$, then $m_\lambda \geq L_\lambda \geq 4$. Hence for $m \leq 5$ either $w = 0$, or exactly one component w_λ is non-zero, $w_\lambda = \pm L_\lambda$, and then $m - m_\lambda$ is even. (a) Γ has no cycle of length 3: 3 is odd and < 4. (b) Let C be a cycle of length 4. If $w = 0$, every m_λ is even; the four steps are not all in one direction (such a walk visits at most three vertices), so there is one step in each sense in each of two directions λ, λ', and the four vertices are the corners of a unit square of the plane $\{\lambda, \lambda'\}$: C is the boundary of a plaquette. If $w \neq 0$, then $m_\lambda = 4 \geq L_\lambda = |w_\lambda| \geq 4$, so all four steps go in direction λ, in the same sense: C is a straight cycle and $L_\lambda = 4$. (c) A cycle of length 5 has $w \neq 0$; then $5 - m_\lambda$ is even, so $m_\lambda = 5$, and $4 \leq L_\lambda \leq m_\lambda$ with $L_\lambda \equiv m_\lambda \pmod 2$ gives $L_\lambda = 5 = |w_\lambda|$: a straight cycle.

(i) $2|V(\varphi)| \leq \sum_v \deg_\varphi(v) = 2|\varphi|$. (ii) Γ is a simple graph (all sides are ≥ 3), so three vertices span at most three edges; hence $|V(\varphi)| = 4$, every vertex has degree 2, and the connected graph φ is a cycle; apply (b). (iii) If $|V(\varphi)| = 5$, φ is a cycle, as in (ii); apply (c). If $|V(\varphi)| = 4$, φ is a simple graph with four vertices and five edges, the complete graph minus one edge, which contains a triangle, contrary to (a). Three vertices span at most three edges.

The plaquettes of Λ_s containing a vertex v are the four plaquettes of each coordinate plane that have v as a corner, P_v in all, and the straight cycles through v are one for each direction with side 4 (resp. 5); this gives N_4^v and N_5^v. For the last bound: a front containing v contains at least two of the $2(d-1)$ edges at v, and for each edge e the connected sets of k edges containing e number at most $\binom{(4d-6)k}{k-1}$, by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] for the graph on $E(\Gamma)$ in which two edges are adjacent when they share an endpoint (root e, $J = K = 4d-6$, $n = k-1$); summing over e counts every front through v at least twice. □

Lemma 7.6 (Local counts)**.** *Let X be as in Definition 7.1.*

(i) Given a marker $\mathfrak{m}$ at slice s and integers $k \geq 4$, $\ell \geq 1$, at most $2|V(E(\mathfrak{m}))|N_k^v$ candidate segments $(\varphi, [s_1, s_2))$ (that is, φ a front in Γ with $|\varphi| = k$, $s_2 - s_1 = \ell$ and $s \in \{s_1, s_2\}$) have a vertex of φ in $V(E(\mathfrak{m}))$.

(ii) Given a segment σ and $n \geq 1$, at most $2|\sigma|P_v\binom{K_S n}{n-1}$ clusters C with $|C| = n$, at an end slice of σ, touch its front; at most $2(d-2)|S_A|\binom{K_S n}{n-1}$ clusters C with $|C| = n$ at slice 0 contain an edge of S_A, and likewise for S_B.

(iii) $|V(E(C))| \leq 4|C|$ for every cluster C, and $|V(S_A)| \leq 2|S_A|$, $|V(S_B)| \leq 2|S_B|$.

Proof. (i) The front contains one of the $|V(E(\mathfrak{m}))|$ vertices of $E(\mathfrak{m})$; given this vertex there are at most N_k^v choices of φ; and $s \in \{s_1, s_2\}$ (two choices) together with ℓ determines the interval. (ii) The cluster lies at one of the two end slices; it contains a vertex of the front φ of σ (at most $|V(\varphi)| \leq |\sigma|$ choices, Lemma 7.5(i)) and a plaquette of that slice containing this vertex (P_v choices), and it is a connected set of n plaquettes of the slice containing that plaquette, of which there are at most $\binom{K_S n}{n-1}$ by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] (root the plaquette, $J = K = K_S$). For S_A: an edge of S_A, a plaquette of slice 0 containing it ($2(d-2)$ choices), and the same count. (iii) A plaquette has four vertices, a link two. □

Lemma 7.7 (Tree sums)**.** *Let $\mathcal{V}$ be a set, $w : \mathcal{V} \to [0, \infty)$, and $\mathcal{N}(v) \subseteq \mathcal{V}$ for $v \in \mathcal{V}$ (the possible children of v). For $v \in \mathcal{V}$ let $\Psi(v)$ be the sum, over all finite rooted trees whose root is labelled v and in which the children of a vertex labelled v' carry distinct labels in $\mathcal{N}(v')$, of the product of w over the non-root labels. If $h : \mathcal{V} \to [0, \infty)$ satisfies $\sum_{c \in \mathcal{N}(v)} w(c)e^{h(c)} \leq h(v)$ for all v, then $\Psi(v) \leq e^{h(v)}$.*

Proof. Let Ψ_n be the same sum restricted to trees of depth $\leq n$; $\Psi_0 = 1$. A tree of depth $\leq n$ is a set of distinct children of the root with subtrees of depth $\leq n-1$, so $\Psi_n(v) \leq \prod_{c \in \mathcal{N}(v)}(1 + w(c)\Psi_{n-1}(c)) \leq \exp \sum_c w(c)\Psi_{n-1}(c)$. By induction $\Psi_n \leq e^h$, and $\Psi = \lim_n \Psi_n$. □

Theorem 7.8 (The sharp constant)**.** *Let $\gamma_g := \|g_p\|_\infty$ (so $\gamma_g \leq e^{4\beta} - 1$, and $\gamma_g \leq \gamma_2(\beta)$ if ρ has no trivial summand, Lemma 2.2(a),(d)), let θ_5 be as in Lemma 6.1, and put $a := e^{2(K+1)\theta_5}u_*$ and $b := e^{2(K+1)\theta_5}\gamma_g$. For $\lambda_C, \lambda_T \geq 0$ put*

$$\eta_C := 2P_v \sum_{n \geq 1} \binom{K_S n}{n-1} (be^{\lambda_C})^n, \qquad \kappa_C := 8P_v \sum_{n \geq 1} n \binom{K_S n}{n-1} (be^{\lambda_C})^n,$$

$$\eta_T := 8 \sum_{k \geq 4} N_k^v \, e^{\lambda_T k} \frac{a^k}{1 - a^k}, \qquad \nu := 2 \sum_{k \geq 4} k N_k^v \, e^{\lambda_T k} a^{k-4}, \qquad \delta := \kappa_C \, \nu,$$

and suppose that the four series converge, $\eta_C \leq \lambda_T$, $\eta_T \leq \lambda_C$ and $a^4(1 + \delta) < 1$. Then for every $\delta' > \delta$ with $a^4(1 + \delta') \leq 1$ there is $C_{A,B}$, independent of t and L_0, with

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, B)| \leq C_{A,B}\big(a^4(1 + \delta')\big)^t + \frac{2\|A\|_\infty \|B\|_\infty c^{|S|}}{1 - \hat{q}} \, \hat{q}^{\,4L_0}.$$

Consequently $m_{\Lambda_s}(\beta) \geq 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1 + \delta)$, with δ and the hypotheses taken for the spatial lattice Λ_s; and

$$m(\mu_\beta) \;\geq\; 4\log\frac{1}{u_*} - 8(K+1)\theta_5 - \log(1 + \delta_\infty),$$

where δ_∞, and the hypotheses, are taken with $N_4^v = P_v$, $N_5^v = 0$ and $N_k^v = (d-1)\binom{(4d-6)k}{k-1}$ for $k \geq 6$ (which bound the counts when all spatial sides are ≥ 6, Lemma 7.5). As $\beta \downarrow 0$ the hypotheses hold, with $\lambda_T = O(\beta)$ and $\lambda_C = O(\beta^4)$, and $\delta = 64P_v N_4^v \gamma_g(1 + O(\beta))$; if ρ has no trivial summand, $\delta_\infty \leq 128P_v^2\beta(1 + O(\beta))$.

Proof. As in the proof of Theorem 6.4, $2\operatorname{Cov}(A,B) = \sum_X \Phi_F(X) r_X^2$ over the sets $X \in \mathcal{X}_S$ without free links outside S that have a component meeting $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Those X with a temporal plaquette in every layer have $|X| \geq 4L_0$ (Lemma 3.2), and the computation in the proof of Theorem 4.1, with $\hat{q}$ in place of q (as in the proof of Theorem 6.4), bounds their contribution by the second term. For the other X, Lemma 6.3 (with $m(X) = 0$), $r_X^2 \leq e^{2\theta_5 J_S} e^{2\theta_5 (K+1)|X|}$ (Lemma 6.1 and $|\overline{X}| \leq (K+1)|X| + J_S$) and Lemma 7.2(v) give

$$|\Phi_F(X)| r_X^2 \leq 4\|A\|_\infty \|B\|_\infty e^{2\theta_5 J_S}\, W(X), \qquad W(X) := \prod_\sigma a^{|\sigma|\ell(\sigma)} \prod_C b^{|C|}.$$

Fix δ' as in the statement and put $z := 1/(a^4(1+\delta')) \geq 1$. For each such X choose a path $\pi(X)$ in $\mathcal{I}(X)$ from S_A to S_B of minimal length (Lemma 7.4(ii)) and a spanning forest $\mathcal{F}(X)$ of $\mathcal{I}(X)$ each of whose trees contains exactly one vertex of $\pi(X)$ (possible by Lemma 7.4(ii)). By Lemma 7.4(iii), $z^t \leq z^{\sum_{\sigma\in\pi(X)} \ell(\sigma)}$, so

$$z^t W(X) \leq \prod_{\sigma\in\pi(X)} \left(z a^{|\sigma|}\right)^{\ell(\sigma)} \prod_{\sigma\notin\pi(X)} a^{|\sigma|\ell(\sigma)} \prod_C b^{|C|}.$$

The set X is the union of the plaquette sets of the vertices of $\mathcal{I}(X)$, so it is determined by the pair $(\pi(X), \mathcal{F}(X))$, the vertices being described geometrically (a segment by (φ, s_1, s_2), a cluster by its slice and its plaquettes). Hence $z^t \sum_X W(X)$ is at most the sum, over all sequences $\pi = (S_A, v_1, \dots, v_r, S_B)$ of segments and clusters in which consecutive terms are joined in the sense of Definition 7.3, of $\prod_{v\in\pi} w_\pi(v)\Psi(v)$. Here $w_\pi(\sigma) = (za^{|\sigma|})^{\ell(\sigma)}$, $w_\pi(C) = b^{|C|}$ and $w_\pi(S_A) = w_\pi(S_B) = 1$, and Ψ is the tree sum of Lemma 7.7 with $w(\sigma) = a^{|\sigma|\ell(\sigma)}$, $w(C) = b^{|C|}$ and children given by Definition 7.3 (supports excluded).

Decorations. Let $h(C) := \lambda_C|C|$, $h(\sigma) := \lambda_T|\sigma|$, and $h(S_A) := |S_A|(\lambda_T' + \eta_T/2)$ with $\lambda_T' := 2(d-2)\sum_n \binom{K_S n}{n-1}(be^{\lambda_C})^n$, and likewise for S_B. By Lemma 7.6: for a cluster C, $\sum_\sigma a^{|\sigma|\ell(\sigma)} e^{\lambda_T|\sigma|} \leq 2|V(E(C))| \sum_k N_k^v e^{\lambda_T k} \sum_{\ell\geq 1} a^{k\ell} \leq |C|\eta_T \leq \lambda_C|C|$; for a segment σ, $\sum_C b^{|C|} e^{\lambda_C|C|} \leq |\sigma|\eta_C \leq \lambda_T|\sigma|$; for S_A, the clusters contribute at most $|S_A|\lambda_T'$ and the segments at most $2|V(S_A)| \sum_k N_k^v e^{\lambda_T k} a^k/(1-a^k) \leq |S_A|\eta_T/2$. By Lemma 7.7, $\Psi \leq e^h$.

The path. Since $za^4 = 1/(1+\delta')$ and $a \leq 1$, $\sum_{\ell\geq 1}(za^k)^\ell = za^k/(1-za^k) \leq a^{k-4}/\delta'$ for $k \geq 4$. Hence, by Lemma 7.6(i), for a marker $\mathfrak{m}$ the segments σ joined to $\mathfrak{m}$, weighted by $|\sigma| w_\pi(\sigma) e^{h(\sigma)}$, sum to at most $|V(E(\mathfrak{m}))|\nu/\delta'$; and by Lemma 7.6(ii),(iii), for a segment σ the clusters C joined to σ, weighted by $|V(E(C))| b^{|C|} e^{h(C)}$, sum to at most $|\sigma|\kappa_C$. A minimal path has the form S_A, possibly a cluster C_0 at slice 0, then $\sigma_1, C_1, \sigma_2, \dots, C_{r-1}, \sigma_r$ (clusters are not joined to clusters), possibly a cluster C_r at slice t, and S_B. For a segment σ let $R(\sigma)$ be the sum, over the possible continuations of a path after σ (namely S_B; or C, S_B; or C, σ' followed by a continuation of σ'), of the product of $w_\pi(v)e^{h(v)}$ over the vertices v of the continuation. Then

$$R(\sigma) \leq e^{h(S_B)}(1 + |\sigma|\eta_C) + \sum_{C\sim\sigma} b^{|C|} e^{h(C)} \sum_{\sigma'\sim C} w_\pi(\sigma') e^{h(\sigma')} R(\sigma'),$$

and by induction on the length of the continuations $R(\sigma) \leq M|\sigma|$ with $M := e^{h(S_B)}(1+\eta_C)/(1-\delta/\delta')$: the double sum is at most $M\sum_{C\sim\sigma} b^{|C|} e^{h(C)} |V(E(C))|\nu/\delta' \leq M|\sigma|\kappa_C\nu/\delta' = (\delta/\delta')M|\sigma|$, and $1 + |\sigma|\eta_C \leq (1+\eta_C)|\sigma|$. The beginning of the path is treated in the same way:

$$z^t \sum_X W(X) \leq e^{h(S_A)} M \Big(|V(S_A)| + \sum_{C_0} b^{|C_0|} e^{h(C_0)} |V(E(C_0))| \Big) \frac{\nu}{\delta'} =: C_0 < \infty,$$

the sum over the clusters C_0 at slice 0 containing an edge of S_A, which is at most $8(d-2)|S_A| \sum_n n\binom{K_S n}{n-1}(be^{\lambda_C})^n$. Hence $\sum_X W(X) \leq C_0(a^4(1+\delta'))^t$, which is the covariance bound with $C_{A,B} = 2C_0\|A\|_\infty\|B\|_\infty e^{2\theta_5 J_S}$.

Gaps. Let A be bounded, measurable and gauge invariant on Ω_s. Applying the covariance bound to the real and imaginary parts of $\overline{A}(t)$ and $A(0)$, letting $L_0 \to \infty$ at fixed t (the second term tends to 0) and using [F, Theorem 'Ground-state expectations' (`thm:gs`)], we get $|\langle \Omega, \hat{A}^* e^{-tH} \hat{A}\Omega\rangle - |\langle A\rangle_\nu|^2| \le C_A(a^4(1+\delta'))^t$ for all $t \ge 0$, with C_A independent of t ($t = 0$ is trivial), hence $m_{\Lambda_s}(\beta) \ge -\log(a^4(1+\delta'))$ by [F, Theorem 'Gap $\Leftrightarrow$ clustering in time' (`thm:gapclust`)](ii). (The bound is not uniform in L_0 for t up to $L_0/2$, as [F, Corollary `cor:gapfromcorr`] would require, unless β is small: near $\beta_0(d)$ the term $\hat{q}^{4L_0}$ decays more slowly than $(a^4(1+\delta'))^{L_0/2}$.) For $m(\mu_\beta)$ argue as in the proof of Theorem 6.4(ii), with q_u^{4t} replaced by $(a^4(1+\delta'))^t$: the infinite-volume covariance is the limit over periodic lattices with all sides tending to infinity, on which (sides ≥ 6) the counts are bounded as in the definition of δ_∞, so the covariance bound holds with any $\delta' > \delta_\infty$ and the second term vanishes in the limit. Finally let $\delta' \downarrow \delta$ (resp. δ_∞), and note $4\log(1/a) = 4\log(1/u_*) - 8(K+1)\theta_5$.

Small β. Here $a, b = O(\beta)$. Take $\lambda_T := \eta_C|_{\lambda_C=1} = O(\beta)$ and then $\lambda_C := \max(\eta_T, \beta^4) \le 1$; since η_C increases with λ_C, both conditions hold, $\lambda_C = O(\beta^4)$ (as $\eta_T = O(a^4)$), $\kappa_C = 8P_v b(1+O(\beta))$, $\nu = 8N_4^v(1+O(\beta))$, and $\delta = 64P_vN_4^v\gamma_g(1+O(\beta))$, $a^4(1+\delta) \to 0$. If ρ has no trivial summand, $\gamma_g \le \gamma_2 = 2\beta + O(\beta^2)$, and $N_4^v = P_v$ in δ_∞. □

Lemma 7.9 (The hypotheses on the whole range)**.** *For every $d \ge 3$ and $0 < \beta < \beta_0(d)$, every G and ρ, and every spatial lattice with all sides ≥ 4, the hypotheses of Theorem 7.8 hold with $\lambda_C = \lambda_T = \frac{1}{10}$, both for δ (with the bounds of Lemma 7.5 for N_k^v) and for δ_∞; moreover $\eta_C < 0.023$ and $\eta_T < 10^{-9}$.*

Proof. The series increase with a, b and the N_k^v, and we use the largest bounds for the latter, $N_4^v \le (d-1)(2d-3)$, $N_5^v \le d-1$. Since $u_* \le \epsilon$ and $\gamma_g \le e^{4\beta} - 1 = \epsilon(2+\epsilon)$, $a \le \bar{a} := e^{2(K+1)\theta_5}\epsilon$ and $b \le \bar{b} := e^{2(K+1)\theta_5}\epsilon(2+\epsilon)$, which increase with β; so it suffices to bound the series at $\beta = \beta_0(d)$, where $\epsilon = \epsilon_0 := 1/(e^2K(K+2))$ and $x = 1/(K+2)$. Then $2(K+1)\theta_5 \le 2(K+1)x^5/(1-x) = 2(K+2)^{-4} < 6\cdot 10^{-5}$, so $\bar{a} \le 1.0001\,\epsilon_0$ and $y := \bar{b}e^{1/10} \le 2.212\,\epsilon_0 \le 0.3/(K(K+2))$. We write $K = 8d-12 \ge 12$, so that $K_S = K-8$, $4d-6 = K/2$, $2P_v = (K^2-16)/16$ and $(d-1)(2d-3) = K(K+4)/32$, and use $\binom{Jn}{n-1} \le e(eJ)^{n-1}$ for $n \ge 2$ (from $\binom{m}{j} \le (em/j)^j$). With $w := eK_Sy \le 0.82/(K+2) \le 0.06$,

$$\eta_C \le 2P_vy\Big(1 + \frac{ew}{1-w}\Big) \le \frac{K^2-16}{16} \cdot \frac{0.3}{K(K+2)} \cdot 1.18 < 0.023.$$

With $r := e^{1.1}(K/2)\bar{a} \le 0.21/(K+2) \le 0.015$, the terms $k \ge 6$ of η_T are at most $8(d-1)e^{1.1}\bar{a}\sum_{k\ge6} r^{k-1}(1-\bar{a}^6)^{-1} < 10^{-10}$, and the terms $k = 4, 5$ at most $8\frac{K(K+4)}{32}e^{0.4}\bar{a}^4(1+10^{-6}) + 8(d-1)e^{0.5}\bar{a}^5 < 10^{-10}$, using $\bar{a} \le 1.0001/(e^2K(K+2)) \le 8.1\cdot 10^{-4}$; so $\eta_T < 10^{-9}$. All series converge ($w, r < 1$). Finally $\kappa_C \le 8P_vy\big(1 + e((1-w)^{-2}-1)\big) < 0.11$ and, bounding every N_k^v by $(d-1)e(eK/2)^{k-1}$, $\bar{a}^4\nu \le 2(d-1)e^{1.1}\bar{a}\sum_{k\ge4} kr^{k-1} < 10^{-6}$; so $a^4(1+\delta) \le \bar{a}^4 + 0.11\,\bar{a}^4\nu < 10^{-6}$. □

Remark 7.10 (Numbers, the first version, and checks). (i) For $SU(2)$ (ρ has no trivial summand and $u_* = u$, computed exactly), with λ_C, λ_T the smallest solution of the two conditions (obtained by iterating them from $\lambda_C = 0$), the lower bound of Theorem 7.8 for $m(\mu_\beta)$ lies below $4\log(1/u)$ by 0.0018, 0.018, 0.17, 0.90 and 1.38 at $\beta = 10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(4)/2$ and $\beta_0(4)$ for $d = 4$, and by 0.0002, 0.0020, 0.020, 0.35 and 0.62 at $\beta = 10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(3)/2$ and $\beta_0(3)$ for $d = 3$. The loss increases with β: a and b do (as θ_5, $u = I_2/I_1$ and γ_2 do), $\eta_C, \eta_T, \kappa_C, \nu$ increase with a, b and the λ's, and the smallest solution, the limit of the iteration of an increasing map started at 0, increases with (a, b); so its supremum over the range is its limit at $\beta_0(d)$. For the finite-volume gap on spatial lattices with all sides ≥ 4 the loss is at most 1.56 ($d = 4$) and 0.83 ($d = 3$). (With the choice $\lambda_C = \lambda_T = \frac{1}{10}$ of Lemma 7.9 instead of

the smallest solution, the loss at $\beta_0(d)$ is 1.75 and 0.87.) With the upper bound (Corollary 5.8; $D < 1.4 \cdot 10^{-3}$ and $D < 4.9 \cdot 10^{-3}$, D increasing, Remark 5.9), for all $0 < \beta < \beta_0(d)$

$$-1.38 \le m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \le 0.0014 \quad (d = 4),$$
$$-0.63 \le m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \le 0.005 \quad (d = 3)$$

(Numerical: an evaluation of the explicit constants, `python3` with `mpmath`). Theorem 6.4 loses $4\log(eK)$, i.e. 16.0 and 13.9, throughout. As $\beta \downarrow 0$, $\delta_\infty/\beta \to 128P_v^2$, which is 18432 for $d = 4$ and 2048 for $d = 3$.

(ii) The first version of this section (commit `1766bb3`) counted the fronts and clusters meeting a marker or a front through an edge rather than through a vertex, bounded N_k by $e(e(4d-6))^{k-1}$ for every k, and used $e^{4\beta} - 1$ and θ in place of γ_g and θ_5; its δ/β tended to $3.4 \cdot 10^6$ ($d = 4$) and $4.1 \cdot 10^5$ ($d = 3$), and near $\beta_0(d)$ it was weaker than Theorem 6.4. That version said that its hypotheses fail at $\beta_0(4)$; an independent referee found this to be false (the iteration had been started at $\lambda_C = 1$ instead of 0): they hold, with a loss of 27.7.

(iii) The constants are still far from optimal. The dominant term of δ comes from clusters of one plaquette at an end of a segment with a four-edge front; most of these are impossible, since such a plaquette at slice s needs each of its links in $T_{s-1} \cup T_s$. For the family of tubes whose fronts are boundaries of plaquettes, with at most three spatial plaquettes per slice, an exact transfer-matrix sum (first done by the referee) gives the growth factor $a^4(1+b)$ per layer up to $O(b^2)$, i.e. an effective δ of about $b \approx 2\beta$ rather than $128P_v^2\beta$ (Numerical). Section 8 proves $\delta = b(1 + O(\beta))$.

(iv) Checks (a compass): the constants of (i), and those of Lemma 7.9 for $3 \le d \le 12$; an exhaustive enumeration of the fronts with at most six edges through a vertex on all spatial tori with sides in $\{4, \dots, 7\}$ ($d = 3, 4$), which confirms Lemma 7.5 and the value of P_v, and gives also $N_6^v = 12\binom{d-1}{2} + 96\binom{d-1}{3} + 18(d-2)n_4 + n_6$; the transfer-matrix sum of (iii); a test of Lemmas 7.2 and 7.4 on 400 random sets X on the periodic lattice $8 \times 5 \times 5$ (the referee's independent test covered 4513 pairs (X, s_*) on five lattices); and the enumeration of K_S and $2(d-2)$. The scripts are in `theory/checks/lower-bound/`.

Corollary 7.11 (The gap is $4\log(1/u) + O(\beta)$, on the whole range)**.** *Let G be metrisable and ρ irreducible and non-trivial with $N \le 49$. Then for every $0 < \beta < \beta_0(d)$*

$$-8(K+1)\theta_5(\beta) - \log(1 + \delta_\infty(\beta)) \;\le\; m(\mu_\beta) - 4\log\frac{1}{u(\beta)},$$

with δ_∞ as in Theorem 7.8 for any λ_C, λ_T satisfying its hypotheses (for instance $\frac{1}{10}$); the same holds for $m_{\Lambda_s}(\beta)$, with δ in place of δ_∞, on every periodic spatial lattice with all sides ≥ 4. If $N^4 \le 1200\tau_0^2$, then also $m(\mu_\beta) - 4\log(1/u(\beta)) \le D(\beta)$. As $\beta \downarrow 0$, $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$, and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the spatial lattices with all sides ≥ 4.

Proof. Theorem 7.8 and Lemma 7.9, with $u_* = u$ (Corollary 6.5). The upper bound is Corollary 5.8 when $N^4 \le 1200\tau_0^2$, and for small β the last sentence of Theorem 5.6 (which has no condition on N; for the finite-volume gaps, Theorem 5.6(ii)). As $\beta \downarrow 0$, $\theta_5 = O(\beta^5)$, $\delta, \delta_\infty = O(\beta)$ (Theorem 7.8) and $D(\beta) = O(\beta)$. □

8 The first-order constant: junctions

In Theorem 7.8 the constant δ is about $2 \cdot 10^4\beta$ ($d = 4$). Its dominant term counts a cluster of one plaquette at an end of a segment with a four-edge front, together with any front that touches it, although almost all of these configurations violate the condition that no link of

X is free (Remark 7.10(iii)). In this section all markers and fronts that meet at a slice are grouped into a *junction*; the path of Section 7 becomes a path of junctions and segments, and the weight of each segment is split between its two end junctions. The local form of the no-free-link condition then leaves exactly one junction of order b through which a path can pass: the *cap*, one plaquette closing a four-edge front at a slice where the front continues. Every other junction is of order b^2, or carries a further front and with it a factor a^2, or lets the path continue with a front of at least six edges, which the next segment suppresses by a factor a^2. The outcome is $\delta = b\,(1 + O(\beta))$ (Theorem 8.11), the first-order constant of the method; $b \approx 2\beta$ when ρ has no trivial summand. For $SU(2)$ the lower bound for $m(\mu_\beta)$ lies within $5 \cdot 10^{-4}$ ($d = 4$) and $1.1 \cdot 10^{-3}$ ($d = 3$) of $4\log(1/u)$ on the whole range (Remark 8.12).

The setting is that of Section 7; in addition we put $T_v := 4(P_v - 1)$, an upper bound for the number of plaquettes of Λ_s other than a given one that share a vertex with it.

Definition 8.1 (Junctions)**.** Let X be as in Definition 7.1 and s a slice. A *front-pair* at slice s is a pair $p = (\varphi, \lambda)$ with $\lambda \in \{s-1, s\}$ and φ a front of layer λ touched by a marker at slice s; we write $|p| := |\varphi|$. Front-pairs at different slices are different objects, even when the pairs coincide (as for a segment of length 1, whose front φ of layer s gives (φ, s) at the slices s and $s+1$). Let H_s be the graph whose vertices are the markers and the front-pairs at slice s, two of them being adjacent when their edge sets ($E(\mathfrak{m})$, resp. φ) have a common vertex. A *junction* at slice s is a connected component J of H_s. We write $M(J)$ and $\Phi(J)$ for its markers and front-pairs, $E(J)$ for the union of their edge sets, $E_\mu(J)$ for the union of the fronts of its front-pairs of layer μ, $E_M(J) := \bigcup_{\mathfrak{m}\in M(J)} E(\mathfrak{m})$, and $|J|_C$ for the number of plaquettes of the clusters in $M(J)$.

Lemma 8.2 (Junctions)**.** *Let X be as in Definition 7.1.*

(i) Every junction contains a marker, and every front-pair is adjacent to a marker of its junction. For every segment $\sigma = (\varphi, [s_1, s_2))$, (φ, s_1) is a front-pair at slice s_1 and $(\varphi, s_2 - 1)$ one at slice s_2, and every front-pair arises in this way from exactly one segment. The junctions containing these two front-pairs are the end junctions *of σ; they lie at different slices.*

(ii) If J is a junction at slice s and $e \in E(J)$, then every marker at slice s whose edge set contains e belongs to J, and for every front ψ of layer $\mu \in \{s-1, s\}$ with $e \in \psi$, (ψ, μ) is a front-pair and belongs to J.

(iii) If J contains no support, then for every $e \in E(J)$

$$n_J(e) := [e \in E_{s-1}(J)] + [e \in E_s(J)] + \pi_J(e) \;\geq\; 2,$$

where $\pi_J(e)$ is the number of plaquettes of the clusters of J that contain e.

(iv) $|X| = \sum_\sigma |\sigma|\,\ell(\sigma) + \sum_J |J|_C$, the sums over the segments and the junctions of X.

Proof. (i) A front-pair is touched by a marker at its slice, which is adjacent to it. A front φ of layer s touched by a marker at slice s lies in a segment $(\varphi, [s_1, s_2))$ with $s \in [s_1, s_2)$ (Lemma 7.2(iv)); no marker at a slice in (s_1, s_2) touches φ, so $s = s_1$. Likewise a front-pair $(\varphi, s-1)$ is the end of its segment, $s = s_2$. Uniqueness is Lemma 7.2(iv), and $s_1 < s_2$. (ii) A marker whose edge set contains e shares the endpoints of e with the vertex of J whose edge set contains e, so it is adjacent to it. Let $\psi \ni e$ be a front of layer μ. If ψ is touched by a marker at slice s, (ψ, μ) is a front-pair adjacent to that vertex of J. Otherwise no marker at slice s contains e, so $e \in \varphi$ for some $(\varphi, \nu) \in \Phi(J)$. If $\nu = \mu$, then $\psi = \varphi$, the component of T_μ containing e; if $\nu \neq \mu$, then ψ is a front of layer ν as well (Lemma 7.2(iii)), so again $\psi = \varphi$. In both cases ψ is touched by a marker at slice s after all, a contradiction. (iii) If the

link e of time s were in S, then $s \equiv 0$ and $e \in S_A$, or $s \equiv t$ and $e \in S_B$, and that support would belong to J by (ii). So the link e of time s is not in S, and it lies in a plaquette of X (a plaquette of a cluster, or the temporal plaquette based on e in the layer of a front-pair). Hence it lies in at least two plaquettes of X. These are the temporal plaquettes based on e in layers $s-1$ and s, which by (ii) are counted by the first two terms, and the spatial plaquettes of time s containing e, which lie in clusters containing e, hence by (ii) in clusters of J. (iv) Lemma 7.2(v); every cluster lies in exactly one junction. □

Definition 8.3 (Junction graph)**.** For X as in Definition 7.1, $\mathcal{J}(X)$ is the bipartite graph whose vertices are the junctions and the segments of X, every segment being joined to its two end junctions; J_A and J_B are the junctions containing S_A and S_B.

Lemma 8.4 (Connectivity and covering for junctions)**.** *Let $X \in \mathcal{X}_S$ be as in Definition 7.1, with a component that contains a plaquette of $\mathcal{P}_{S_A}$ and one of $\mathcal{P}_{S_B}$. Then every vertex of $\mathcal{J}(X)$ is joined by a path to J_A or to J_B, there is a path from J_A to J_B, and every such path $J_A = J_0, \sigma_1, J_1, \dots, \sigma_r, J_r = J_B$ satisfies $\sum_i \ell(\sigma_i) \geq t$.*

Proof. Send a segment of $\mathcal{I}(X)$ to itself and a marker to its junction. If a segment and a marker are joined in $\mathcal{I}(X)$, the marker touches the front of the segment at an end slice s, i.e. it is adjacent in H_s to the corresponding front-pair, so the segment is joined to the image of the marker; a cluster joined to S_A shares an edge with it, hence lies in J_A. So paths of $\mathcal{I}(X)$ become walks of $\mathcal{J}(X)$, and Lemma 7.4(ii) gives the first two statements. The last is proved as Lemma 7.4(iii), a junction at slice s being assigned the point s. □

Weights. For a junction J and a segment σ put

$$w(J) := b^{|J|_C} \prod_{p \in \Phi(J)} a^{|p|/2}, \qquad w(\sigma) := a^{|\sigma|(\ell(\sigma)-1)}.$$

Every segment has exactly two end front-pairs (Lemma 8.2(i)), so by Lemma 8.2(iv) the weight $W(X)$ of the proof of Theorem 7.8 factorises as $W(X) = \prod_J w(J) \prod_\sigma w(\sigma)$.

Lemma 8.5 (Fronts with six edges)**.** *If all sides of Λ_s are ≥ 4, every front with six edges is a cycle, and $N_6^v \leq (2(d-1))^5/2 = 16(d-1)^5$.*

Proof. Let φ be a front with six edges. Γ has no triangles (proof of Lemma 7.5), and a graph on four vertices with at least five edges contains one, so $|V(\varphi)| \in \{5, 6\}$. If $|V(\varphi)| = 6$, all degrees are 2 and φ is a cycle. If $|V(\varphi)| = 5$, the degrees are $(4, 2, 2, 2, 2)$ or $(3, 3, 2, 2, 2)$. Following the edges at the vertices of degree ≥ 3 through the vertices of degree 2, in the first case φ is the union of two cycles through the vertex of degree 4, of lengths ≥ 4 each: too many edges. In the second case it is either two cycles joined by a path (at least $4+4+1$ edges) or two vertices x, y joined by three internally disjoint paths of lengths $l_1 + l_2 + l_3 = 6$, with $l_i + l_j \geq 4$ (no cycles of length ≤ 3) and at most one $l_i = 1$ (Γ is simple); so $l_1 = l_2 = l_3 = 2$ and x, y have three common neighbours. But a common neighbour z of $x \neq y$ gives $y - x \equiv e + e'$ with unit vectors $e = z - x$, $e' = y - z$, and as all sides are ≥ 4 every class $y - x \not\equiv 0$ is represented by at most two ordered pairs (e, e'): by (e, e') and (e', e) if $y - x \equiv e + e'$ with $e \neq \pm e'$, and by (e, e) and $(-e, -e)$ if $y - x \equiv 2e$ (the second only when the side in that direction is 4). Hence φ is a 6-cycle. A 6-cycle through v is traversed by two closed walks of length 6 from v, and there are at most $(2(d-1))^5$ such walks. □

Lemma 8.6 (Junctions with two front-pairs)**.** *Let J be a junction without support at slice s whose front-pairs are exactly $p = (\varphi, \lambda)$ and $p' = (\varphi', \lambda')$, with $|\varphi| = 4$, and let $n := |J|_C$.*

(i) *If $\lambda \neq \lambda'$, then $\varphi \triangle \varphi' \subseteq E_M(J)$ and $\varphi' \subseteq E_M(J) \cup \varphi$; if $\lambda = \lambda'$, then $\varphi \cup \varphi' \subseteq E_M(J)$. Every plaquette of J has at least $5 - n$ edges in $\varphi \cup \varphi'$. Every plaquette of J is joined to $V(\varphi)$ by a chain of plaquettes of J, consecutive ones sharing a vertex and the first having a vertex in $V(\varphi)$.*

(ii) *If $|\varphi'| = 4$ and $n = 1$, then $\lambda \neq \lambda'$, $\varphi = \varphi'$, and the plaquette of J is the one bounded by φ (a* cap*).*

(iii) *If $|\varphi'| = 4$ and $n = 2$, then $\lambda \neq \lambda'$, φ and φ' are the boundaries of two distinct plaquettes q, q' that share a vertex, and q, q' are the plaquettes of J (a* shift *of the tube if q, q' share a link, a* jump *if they share only a vertex).*

(iv) *If $|\varphi'| = 4$ and $n = 3$, then $\varphi' \neq \varphi$, every plaquette of J is the plaquette bounded by φ, the one bounded by φ', or the unique plaquette containing two given distinct edges $e \in \varphi$, $e' \in \varphi'$, and at least one plaquette of J is of the last kind.*

(v) *If $|\varphi'| = 5$, then $n \geq 4$.*

Proof. (i) By Lemma 8.2(iii), an edge of $E(J)$ outside $E_M(J)$ has $n_J(e) \geq 2$ from the two layers alone; this gives the first statements. An edge of a plaquette of J outside $\varphi \cup \varphi'$ lies in no front of J, so it lies in two plaquettes of J; two plaquettes share at most one link (proof of Lemma 5.4), so a plaquette has at most $n - 1$ such edges. For the chains: let $\mathcal{A}$ be the set of plaquettes of J joined to $V(\varphi)$ by such chains, $\mathcal{B}$ the others, and suppose $\mathcal{B} \neq \emptyset$. Plaquettes sharing a link share a vertex, so each cluster lies in $\mathcal{A}$ or in $\mathcal{B}$, and a cluster in $\mathcal{B}$ shares no vertex with φ or with a cluster in $\mathcal{A}$. By the first statement every edge of φ' lies in $E(\mathcal{A}) \cup \varphi$ or in $E(\mathcal{B})$. If every edge of φ' lies in $E(\mathcal{A}) \cup \varphi$, the clusters in $\mathcal{B}$ are adjacent in H_s only to each other (a common vertex with φ' would be one with $\mathcal{A}$ or φ), and J is disconnected. If every edge lies in $E(\mathcal{B})$, then p' and the clusters in $\mathcal{B}$ are adjacent only to each other (p and the clusters in $\mathcal{A}$ share no vertex with $E(\mathcal{B})$), and again J is disconnected. Otherwise, φ' being connected, some vertex is an endpoint of an edge of each kind, and then lies in $V(E(\mathcal{B})) \cap V(E(\mathcal{A}) \cup \varphi)$, a contradiction. (ii) If $\lambda = \lambda'$, φ and φ' are distinct components of T_λ, so vertex-disjoint, and (i) gives $8 \leq 4n$. So $\lambda \neq \lambda'$, the four edges of the plaquette lie in $\varphi \cup \varphi'$, and $\varphi \triangle \varphi'$ lies in its boundary. Two distinct cycles of length 4 in Γ share at most one edge (two plaquette boundaries by the proof of Lemma 5.4; a plaquette contains no two collinear edges, so a straight cycle shares at most one edge with a plaquette boundary; distinct straight cycles are edge-disjoint), so $\varphi \neq \varphi'$ would give $|\varphi \triangle \varphi'| \geq 6$. Hence $\varphi = \varphi'$ contains the boundary of the plaquette, and they are equal. (iii) If $\lambda = \lambda'$, (i) shows that the two plaquettes have all their edges in $\varphi \cup \varphi'$, so they are the plaquettes bounded by φ and φ' (a cycle of length 4 contained in the union of two vertex-disjoint ones is one of them); but these have disjoint vertex sets, and J would be disconnected. So $\lambda \neq \lambda'$. Every edge of $E_M(J)$ outside $\varphi \cup \varphi'$ lies in both plaquettes, so there is at most one such edge e_0. If $\varphi = \varphi'$, each plaquette has three edges in φ and is bounded by φ, which is impossible for two distinct plaquettes. So $\varphi \neq \varphi'$; a plaquette whose boundary is neither φ nor φ' has at most one edge in each and at most one further edge e_0, three in all, which is impossible. So the boundaries of the two plaquettes are φ and φ', and J is connected only if they share a vertex. (iv) By (i) every plaquette has two edges in $\varphi \cup \varphi'$. A plaquette with two edges in one cycle of length 4 is the plaquette that cycle bounds (two edges of a plaquette at a corner, or opposite ones, determine it, and a straight cycle contains at most one edge of a plaquette). If $\varphi' = \varphi$, all three plaquettes would be the one bounded by φ. Otherwise a plaquette not bounded by φ or φ' has exactly one edge $e \in \varphi$ and one edge $e' \in \varphi'$, $e \neq e'$, and is the only plaquette containing both. As there are three plaquettes, at least one is of this kind. (v) A front with five edges is a straight cycle (Lemma 7.5), with five collinear edges, around a side of length 5. By (i) these

lie in $E_M(J) \cup \varphi$. A plaquette has at most one edge on a given line; φ has at most one if it bounds a plaquette, and none if it is straight (its direction has side 4). So $5 \le n+1$. □

Lemma 8.7 (Counting junctions with two front-pairs)**.** *Let $p = (\varphi, \lambda)$ be a front-pair candidate at slice s with $|\varphi| = 4$ (a front in Γ and a layer). For $n \ge 1$ and $k \ge 4$ let $Z_{n,k}$ be the number of pairs (J, p') with J a junction without support at slice s of some set X as in Definition 7.1, with $\Phi(J) = \{p, p'\}$, $|J|_C = n$ and $|p'| = k$. Then $Z_{1,4} \le 1$, $Z_{2,4} \le 4(P_v - 1) - K_S$, $Z_{3,4} \le 48\binom{18}{2}(d-2)(2d-3)$, and for all n, k*

$$Z_{n,k} \le 2\binom{4P_v + T_v n}{n} X_{n,k},$$

where $X_{n,4} := (n+1)(2d-3)$, $X_{n,5} := 0$ if $n \le 3$ or if no side of Λ_s equals 5 and $X_{n,5} := \frac{4}{5}(n+1)$ otherwise, and $X_{n,k} := \min\{\binom{4n+4}{k}, (4n+4)N_k^v\}$ for $k \ge 6$.

Proof. $Z_{1,4}$ and $Z_{2,4}$: Lemma 8.6(ii),(iii); for (iii), q' is a plaquette other than q with a vertex in $V(q)$, and counting the pairs (v, q') with $v \in V(q) \cap V(q')$, at most $4(P_v - 1)$, counts the K_S plaquettes that share a link with q twice. $Z_{3,4}$: by Lemma 8.6(iv), choose $e \in \varphi$ (4 ways), a plaquette $P \ni e$ ($2(d-2)$ ways), an edge $e' \ne e$ of P (3 ways), the cycle $\varphi' \ni e'$ of length 4 (at most $2d-3$ ways: the plaquettes containing e' and the straight cycle through it), the layer λ' (2 ways), and two further plaquettes among the at most $2+16$ allowed by Lemma 8.6(iv). For the general bound: by Lemma 8.6(i) the set of the n plaquettes of J, together with φ, is connected in the graph in which φ is joined to the at most $4P_v$ plaquettes with a vertex in $V(\varphi)$ and plaquettes sharing a vertex are joined; by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] there are at most $\binom{4P_v + T_v n}{n}$ such sets, and they determine the clusters. Given them, φ' is a front contained in $F := E_M(J) \cup \varphi$, with $|F| \le 4n+4$ and $|V(F)| \le 4n+4$, at one of two layers. Fronts with $k \ge 6$ edges in F number at most $\binom{|F|}{k}$ and at most $|V(F)|N_k^v$; fronts with four edges in F are plaquette boundaries (at most $2(d-2)|F|/4$, each edge of F lying in $2(d-2)$ plaquettes of the slice) or straight cycles, which are edge-disjoint and so at most $|F|/4$; straight cycles with five edges are edge-disjoint, at most $|F|/5$, and occur only for $n \ge 4$, and only if a side of Λ_s equals 5 (Lemma 8.6(v)). □

Lemma 8.8 (Marked tree sums)**.** *In the setting of Lemma 7.7, let h satisfy its hypothesis, and let $\omega_1, \omega_2 : \mathcal{V} \to [0, \infty)$. Let $\Psi_i(v)$ be the same sum as $\Psi(v)$ over trees with one marked non-root vertex u, with $w(u)$ replaced by $\omega_i(u)$, and $\Psi_{12}(v)$ the sum over trees with two distinct marked non-root vertices u_1, u_2, with $w(u_j)$ replaced by $\omega_j(u_j)$. If $g_1, g_2, g_{12} : \mathcal{V} \to [0, \infty)$ satisfy*

$$G_i(v) := \sum_{c \in \mathcal{N}(v)} (\omega_i(c) + w(c) g_i(c)) e^{h(c)} \le g_i(v) \qquad (i = 1, 2),$$
$$G_1(v) G_2(v) + \sum_{c \in \mathcal{N}(v)} (\omega_1(c) g_2(c) + \omega_2(c) g_1(c) + w(c) g_{12}(c)) e^{h(c)} \le g_{12}(v)$$

for all v, then $\Psi_i \le g_i e^h$ and $\Psi_{12} \le g_{12} e^h$.

Proof. By induction on the depth n, as in Lemma 7.7, with $\Psi_{\cdot,n}$ the sums over trees of depth $\le n$. A tree with one mark is a set of children of the root of which exactly one carries the mark or has it in its subtree, so

$$\Psi_{i,n}(v) \le \Psi_n(v) \sum_c (\omega_i(c) \Psi_{n-1}(c) + w(c) \Psi_{i,n-1}(c)) \le e^{h(v)} G_i(v).$$

With two marks, either one child carries or contains both, or two distinct children carry or contain one each:

$$\frac{\Psi_{12,n}(v)}{\Psi_n(v)} \le \sum_c \big(\omega_1(c)\Psi_{2,n-1}(c) + \omega_2(c)\Psi_{1,n-1}(c) + w(c)\Psi_{12,n-1}(c)\big) + \prod_{i=1,2}\sum_c \big(\omega_i(c)\Psi_{n-1}(c) + w(c)\Psi_{i,n-1}(c)\big),$$

which is at most $g_{12}(v)$ by the induction hypothesis. □

Definition 8.9 (Constants of the junction expansion)**.** Let a, b be as in Theorem 7.8, and take N_4^v, N_5^v from Lemma 7.5, $N_6^v \le 16(d-1)^5$ (Lemma 8.5) and $N_k^v \le (d-1)\binom{(4d-6)k}{k-1}$ for $k \ge 7$. For $\xi_F, \xi_C, h_J, F \ge 0$ put $c_J := e^{h_J}/(1-a^4)$ and, for $j = 0, 1, 2$,

$$Y_j := \sum_{n\ge1} n^j \binom{K_S n}{n-1} (be^{\xi_C})^n, \qquad A_j := e^{c_J}\sum_{k\ge4} k^j N_k^v a^{k/2} e^{\xi_F k}, \qquad B_j := \sum_{k\ge4} k^j N_k^v f(k) e^{\xi_F k},$$
$$f(4) := 1, \qquad f(k) := Fk^2 e^{\xi_F k} a^{k-4} \quad (k \ge 5),$$

and let $g_F, g_C, g_F', g_C', \kappa_F, \kappa_C \ge 0$ satisfy

$$\begin{aligned} P_v g_C Y_1 + B_0 + g_F A_1 &\le g_F, \qquad & 4P_v g_C Y_1 + 8B_0 + 8g_F A_1 &\le g_C,\\ P_v g_C' Y_1 + A_0 + g_F' A_1 &\le g_F', \qquad & 4P_v g_C' Y_1 + 8A_0 + 8g_F' A_1 &\le g_C',\\ g_F g_F' + P_v \kappa_C Y_2 + g_F' B_1 + g_F A_1 + \kappa_F A_2 &\le \kappa_F,\\ g_C g_C' + 4P_v \kappa_C Y_2 + 8(g_F' B_1 + g_F A_1 + \kappa_F A_2) &\le \kappa_C. \end{aligned}$$

Finally let

$$\Sigma_4 := b + (4P_v - 4 - K_S)b^2 + 48\binom{18}{2}(d-2)(2d-3)\,b^3 + 2\sum_{n\ge4}(n+1)(2d-3)\binom{4P_v+T_v n}{n} b^n + 2\sum_{n\ge1}\binom{4P_v+T_v n}{n} b^n \sum_{k\ge5} X_{n,k} f(k) + 4P_v e^{4\xi_F}\big(\kappa_C Y_2 + 4(g_C g_F' + g_C' g_F)Y_1 + 16\kappa_F Y_0\big),$$

with $X_{n,k}$ as in Lemma 8.7.

Lemma 8.10 (Junction sums)**.** *In the setting of Definition 8.9 suppose that the series converge and that*

$$P_v Y_0 + A_0 \le \xi_F, \qquad 4P_v Y_0 + 8A_0 \le \xi_C, \qquad kP_v Y_0 a^{k/2} e^{\xi_F k} \le h_J \quad (k \ge 4).$$

Let p be a front-pair candidate at slice s, $k := |p|$, and let J range over the junctions without support at slice s containing p, of sets X as in Definition 7.1. Then:

(i) $\displaystyle\sum_J b^{|J|_C} \prod_{\psi\in\Phi(J)\setminus\{p\}} (a^{|\psi|/2} e^{c_J}) \le kP_v Y_0 e^{\xi_F k}$;

(ii) with $\mathcal{S}(p) := \sum_J \sum_{p'\in\Phi(J)\setminus\{p\}} b^{|J|_C} \prod_{\psi\in\Phi(J)\setminus\{p,p'\}} (a^{|\psi|/2} e^{c_J}) f(|p'|)$*:* $\mathcal{S}(p) \le \Sigma_4$ *if* $k = 4$*, and* $\mathcal{S}(p) \le kP_v e^{\xi_F k}(g_C Y_1 + g_F k Y_0)$ *for all* k*.*

Proof. Apply Lemmas 7.7 and 8.8 with $\mathcal{V}$ the set of candidate clusters at slice s (connected sets of plaquettes of the slice) and candidate front-pairs, $w(C) := b^{|C|}$, $w(\psi) := a^{|\psi|/2} e^{c_J}$, the children of a cluster being the clusters and the front-pairs (of either layer) with which it shares a vertex, and the children of a front-pair being the clusters and the front-pairs of

the other layer with which it shares a vertex; $h(C) := \xi_C|C|$, $h(\psi) := \xi_F|\psi|$. Counting as in Lemma 7.6 (a cluster through a vertex: a plaquette at the vertex and a connected set through it; a front-pair of a given layer through a vertex: $N^v_{k'}$ fronts of each size k'), the children of a front-pair ψ contribute at most $|\psi|(P_vY_0 + A_0)$ to the condition of Lemma 7.7 and those of a cluster C at most $4|C|P_vY_0 + 8|C|A_0$, since $|V(\psi)| \le |\psi|$ and $|V(E(C))| \le 4|C|$; so $\Psi \le e^h$. In the same way the six inequalities of Definition 8.9 are the hypotheses of Lemma 8.8 with $\omega_1(\psi) := f(|\psi|)$, $\omega_2(\psi) := w(\psi)$, $\omega_i(C) := 0$, $g_1 = g_F|\cdot|$, $g_2 = g'_F|\cdot|$, $g_{12} = \kappa_F|\cdot|^2$ on front-pairs and $g_1 = g_C|\cdot|$, $g_2 = g'_C|\cdot|$, $g_{12} = \kappa_C|\cdot|^2$ on clusters (using $|v| \le |v|^2$).

A junction $J \ni p$ without support is a connected subgraph of this graph, and p is adjacent to a cluster C_0 of J (Lemma 8.2(i)); so J is the label set of a tree rooted at p in which C_0 is a child of p, and a pair (J, p') is such a tree with a marked vertex. The trees in which C_0 is a child of the root contribute at most $w(C_0)\Psi(C_0)\cdot\Psi(p)$, and those with marks at most the corresponding sums of the decompositions in the proof of Lemma 8.8. Summing over the at most $kP_v\binom{K_S n}{n-1}$ clusters C_0 with n plaquettes touching p gives (i) and the second bound in (ii). For the first bound in (ii), split the pairs (J, p'): if $\Phi(J) = \{p, p'\}$ and $|p'| = 4$, they are counted by $Z_{n,4}$ (Lemma 8.7) with weight b^n; if $\Phi(J) = \{p, p'\}$ and $|p'| \ge 5$, by $Z_{n,k}$ with weight $b^n f(k)$; if $|\Phi(J)| \ge 3$, choose a further front-pair p'' and mark it with $\omega_2 = w$: these terms are bounded by the doubly marked sum, at most $\sum_{C_0} w(C_0) e^{\xi_C|C_0|} e^{4\xi_F} (\kappa_C|C_0|^2 + g_C|C_0|\,4g'_F + g'_C|C_0|\,4g_F + 16\kappa_F)$. □

Theorem 8.11 (The first-order constant)**.** *In the setting of Definition 8.9 and Lemma 8.10, let $\delta' > 0$ and suppose that*

$$\Sigma_4 < \delta', \qquad P_v(g_C Y_1/5 + g_F Y_0) < F\delta', \qquad a^4(1+\delta') \le 1.$$

Then there is $C_{A,B}$, independent of t and L_0, with

$$\left|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A,B)\right| \le C_{A,B}\big(a^4(1+\delta')\big)^t + \frac{2\|A\|_\infty\|B\|_\infty c^{|S|}}{1-\hat q}\,\hat q^{\,4L_0},$$

and $m_{\Lambda_s}(\beta) \ge 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1+\delta')$. With the values $N^v_4 = P_v$, $N^v_5 = 0$, valid when all spatial sides are ≥ 6, the same holds for $m(\mu_\beta)$. Let δ_J be the infimum of the admissible δ' (for the respective values of N^v_4, N^v_5). As $\beta \downarrow 0$ the hypotheses hold for every $\delta' > b(1+c\beta)$ with a constant c, so $\delta_J = b\,(1+O(\beta))$; if ρ has no trivial summand, $\delta_J \le 2\beta(1+O(\beta))$ and

$$m(\mu_\beta) \;\ge\; 4\log\frac{1}{u_*(\beta)} - 2\beta + O(\beta^2).$$

Proof. As in the proof of Theorem 7.8, the sets X with a temporal plaquette in every layer give the second term, and for the others $|\Phi_F(X)|r_X^2 \le 4\|A\|_\infty\|B\|_\infty e^{2\theta_5 J_S} W(X)$ with $W(X) = \prod_J w(J)\prod_\sigma w(\sigma)$. Put $z := 1/(a^4(1+\delta')) \ge 1$. For each such X choose a path $\pi(X)$ of minimal length from J_A to J_B in $\mathcal{J}(X)$ and a spanning forest $\mathcal{F}(X)$ of $\mathcal{J}(X)$ each of whose trees contains exactly one vertex of $\pi(X)$ (Lemma 8.4); then $z^t \le z^{\sum_{\sigma\in\pi(X)}\ell(\sigma)}$, and X is determined by $(\pi(X), \mathcal{F}(X))$. On $\pi(X)$ we combine each segment with the factors $a^{|\sigma|/2}$ of its two end junctions: a path segment carries $(za^{|\sigma|})^{\ell(\sigma)}$, and a path junction J the weight $b^{|J|_C}\prod a^{|\psi|/2}$ over its front-pairs ψ that are not ends of path segments. Every intermediate junction of the path contains no support, since $\pi(X)$ is simple and J_A, J_B are its ends; and junctions off the path contain no support, since every support lies in J_A or J_B.

Decorations. The children of a junction in $\mathcal{F}(X)$ are segments with an end in it, those of a segment the junction at its other end. A path segment has no children. In Lemma 7.7 we label a junction off the path by the pair (J, p_0), p_0 the front-pair at which it is reached from its parent, with possible children the segments with an end at the other front-pairs of J, and a segment by the pair (σ, end) of the end at which it is reached, with possible child the junction at its other

end; so the possible children depend only on the label. Give $(\sigma,\cdot)$ the weight $w(\sigma)$ and $h := h_J$, and (J,p_0) the weight $w(J)$ and $h := c_J(|\Phi(J)|-1)$. The segments with an end at a given front-pair ψ of a junction contribute at most $\sum_{\ell\ge1} a^{|\psi|(\ell-1)}e^{h_J} \le c_J$, which gives the condition of Lemma 7.7 at junctions; at a segment ending at a front-pair p^* with $|p^*|=k$, the junctions $J' \ni p^*$ contribute $\sum w(J')e^{h(J')} = a^{k/2}\sum b^{|J'|_C}\prod_{\psi\neq p^*}(a^{|\psi|/2}e^{c_J}) \le a^{k/2}kP_vY_0e^{\xi_F k} \le h_J$ by Lemma 8.10(i). Hence the decorations of a path junction J are bounded by e^{c_J} for each of its front-pairs that is not an end of a path segment.

The path. For a front-pair p at a path junction that is the start of the next path segment, let $\mathcal{V}(p)$ be the sum over the continuations of the path (the segment, of any length ℓ, the junction J' at its far end with its decorations, and then either $J'=J_B$ or a further start $p'\in\Phi(J')$ and its continuation) of the product of the weights just described. Since $\sum_{\ell\ge1}(za^k)^\ell \le a^{k-4}/\delta'$ for $k\ge4$,

$$\mathcal{V}(p) \le \frac{a^{|p|-4}}{\delta'}\Big(E_B(|p|) + \sup_{p^*}\sum_{J'\ni p^*}\sum_{p'} b^{|J'|_C}\prod_{\psi\neq p^*,p'}(a^{|\psi|/2}e^{c_J})\,\mathcal{V}(p')\Big),$$

where p^* ranges over the front-pairs with the front of p at the far end and $E_B(k)$ bounds the contribution of the junctions $J'=J_B$. A junction $J'\ni p^*$ containing S_B is the label set of a tree rooted at p^* in which S_B occurs; so $E_B(k)$ is at most the singly marked sum of Lemma 8.8 with $\omega(S_B) := e^{h(S_B)}$, $h(S_B) := |V(S_B)|(P_vY_0+2A_0)$ (which bounds the subtrees of S_B, whose children are the clusters and front-pairs touching it), and $\omega := 0$ on all other vertices, S_B being excluded from the unmarked sums. The function $g := G|\cdot|$ satisfies the hypothesis of that lemma for a constant G depending on $|S_B|$ (the term $\omega(S_B)e^{h(S_B)}$ appears only at the neighbours of S_B, and $|v|\ge1$), so $E_B(k) \le Gke^{\xi_F k} \le E_Bk^2e^{\xi_F k}$ with $E_B := G$. We show by induction on the length of the continuations that $\mathcal{V}(p) \le Mf(|p|)$ for a constant M. If $|p|=4$, Lemma 8.10(ii) bounds the right-hand side by $(E_B(4)+M\Sigma_4)/\delta' \le M$ when $M \ge E_B(4)/(\delta'-\Sigma_4)$. If $|p|=k\ge5$, it is at most $a^{k-4}(E_Bk^2e^{\xi_F k} + MkP_ve^{\xi_F k}(g_CY_1+g_FkY_0))/\delta' \le Mf(k)$ when $M \ge E_B/(F\delta' - P_v(g_CY_1/5+g_FY_0))$, since $(g_CY_1+g_FkY_0)/k \le g_CY_1/5+g_FY_0$. The beginning of the path is treated in the same way: the junctions $J_A\ni S_A$ with an exit p_1 are trees rooted at S_A with a marked front-pair p_1 (weight $f(|p_1|)$), and the singly marked bound gives a factor $C_A<\infty$ depending on $|S_A|$. So $z^t\sum_X W(X) \le MC_A$, which is the covariance bound.

Gaps: as in the proof of Theorem 7.8.

Small β. Here $a,b = O(\beta)$. Take $h_J := a^2$, $\xi_F := 2P_vb$, $\xi_C := 8P_vb$: for small β, $Y_0 \le b(1+O(\beta))$, $A_j = O(\beta^2)$, so the three conditions of Lemma 8.10 hold. Take $F := 3P_vN_4^v$. Then $B_0 = N_4^v+O(\beta)$, $B_1 = 4N_4^v+O(\beta)$, and the six inequalities hold with $g_F = N_4^v(1+O(\beta))$, $g_C = 8N_4^v(1+O(\beta))$, $g_F', g_C' = O(\beta^2)$ and $\kappa_F,\kappa_C = O(\beta^2)$. Then $\Sigma_4 = b+O(\beta^2)$ (the terms after the first are $O(b^2)$, $O(b\,a^2F)$ and $O(\beta^3)$), and $P_v(g_CY_1/5+g_FY_0) = \frac{13}{5}P_vN_4^vb(1+O(\beta)) < F\delta'$ for δ' near b. Finally $b = e^{2(K+1)\theta_5}\gamma_g$, $\theta_5 = O(\beta^5)$, and $\gamma_g \le \gamma_2 = 2\beta+O(\beta^2)$ if ρ has no trivial summand. □

Remark 8.12 (Numbers and checks). (i) For $SU(2)$, with the parameters of Definition 8.9 found by iterating the conditions, the lower bound of Theorem 8.11 for $m(\mu_\beta)$ lies below $4\log(1/u)$ by $2.0\cdot10^{-7}$, $2.0\cdot10^{-6}$, $2.00\cdot10^{-5}$, $1.7\cdot10^{-4}$ and $4.9\cdot10^{-4}$ at $\beta = 10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(4)/2$ and $\beta_0(4)$ ($d=4$; $\delta_J/b = 1.00001$, 1.0001, 1.0012, 1.08, 1.49), and by $2.0\cdot10^{-7}$, $2.0\cdot10^{-6}$, $2.00\cdot10^{-5}$, $4.2\cdot10^{-4}$ and $1.09\cdot10^{-3}$ for $d=3$; on spatial lattices with all sides ≥4 the loss is at most $5.3\cdot10^{-4}$ ($d=4$) and $1.12\cdot10^{-3}$ ($d=3$). These values extend to the whole range: every left side in the hypotheses of Lemma 8.10, Definition 8.9 and Theorem 8.11 is non-decreasing in (a,b), and $a(\beta)$, $b(\beta)$ increase with β (as θ_5, $u=I_2/I_1$ and γ_2 do), so parameters admissible at $\beta_0(d)$ remain admissible for every smaller β. With the upper bound (Corollary 5.8), for all

$0 < \beta < \beta_0(d)$

$$-5.0 \cdot 10^{-4} \le m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \le 1.4 \cdot 10^{-3} \quad (d = 4),$$
$$-1.1 \cdot 10^{-3} \le m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \le 4.9 \cdot 10^{-3} \quad (d = 3)$$

(Numerical: an evaluation of the explicit constants, `python3` with `mpmath`). As $\beta \downarrow 0$ the two bounds are $-2\beta(1 + O(\beta))$ and $6\beta(1 + O(\beta))$ (Remark 5.9); Section 9 improves both to $-\kappa(\beta) + O(\beta^2)$, and Section 10 to $-\log \Upsilon(\beta) + O(\beta^3)$, and Section 11 to $-\log \Upsilon(\beta) + O(\beta^4)$. Near $\beta_0(4)$ the largest corrections to $\Sigma_4 \approx b$ come from junctions with two front-pairs and at least four plaquettes, from exits with at least six edges, and from junctions with three or more front-pairs (each about $0.15\, b$ at $\beta_0(4)$); all are crude, and the first-order term b is the cap. The largest single source of the last two is the crude bound $N_7^v \le (d-1)\binom{7(4d-6)}{6}$ (about $3.9 \cdot 10^8$ for $d = 4$), whereas on large tori the fronts with seven edges are the 108 ($d = 4$) and 12 ($d = 3$) pairs of plaquettes sharing a link through a given vertex (an independent referee's enumeration); with these values δ_J/b at $\beta_0(4)$ would be about 1.40. Without the special structure of $SU(2)$, i.e. with $u_* \le \epsilon$ and $\|g_p\|_\infty \le e^{4\beta} - 1$, the hypotheses still hold at $\beta_0(d)$ for $3 \le d \le 6$, with losses 0.051, 0.21, 0.44, 0.70 (and 0.023, 0.094, 0.21, 0.36 when ρ has no trivial summand), so for every G and ρ on the whole range (Numerical, values for spatial sides ≥ 6).

(ii) The cap is a plaquette q at a slice where a tube with front ∂q continues, and its weight b is the sup norm of g_q. Its true contribution is an integral of characters; Section 9 evaluates it, together with the corresponding terms of the upper bound.

(iii) Checks (`theory/checks/lower-bound/`, a compass): the constants of (i); an exhaustive enumeration of the junctions with two front-pairs, an entry with four edges and at most two plaquettes (and three for $d = 3$) on spatial tori 6×6, 7×7, 4×6, $6 \times 6 \times 6$ and $4 \times 6 \times 6$, which finds exactly the cap, the 8 ($d = 3$) and 32 ($d = 4$) junctions of Lemma 8.6(iii) with four-edge exits ($= 4(P_v - 1) - K_S$), no junction with three plaquettes and a four-edge exit in $d = 3$, and none with a straight entry and a four-edge exit; and a test of Lemmas 8.2 and 8.4 on 400 random sets X on the periodic lattice $8 \times 5 \times 5$ (2228 junctions). An independent referee's checks, with separate code, confirmed the numbers of (i), Lemma 8.6 on nine tori in $d = 3, 4$ (with up to four plaquettes in $d = 3$, three in $d = 4$; also $Z_{3,4} = 0$ in $d = 4$, so the b^3 term of Σ_4 is an overestimate), the junction lemmas on about 23600 junctions of random connecting sets in $d = 3, 4$, and Lemma 8.8 on random weighted systems.

Corollary 8.13 (The gap to first order, on the whole range)**.** *Let G be metrisable and ρ irreducible and non-trivial with $N \le 49$. For every $0 < \beta < \beta_0(d)$ for which the hypotheses of Theorem 8.11 hold (by Remark 8.12, a numerical evaluation, all $\beta < \beta_0(d)$ when $3 \le d \le 6$),*

$$-8(K+1)\theta_5(\beta) - \log(1 + \delta_J(\beta)) \;\le\; m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \;\le\; D(\beta),$$

the upper bound when $N^4 \le 1200\tau_0^2$; and $-2\beta + O(\beta^2) \le m(\mu_\beta) - 4\log(1/u(\beta)) \le D(\beta) = O(\beta)$ as $\beta \downarrow 0$.

Proof. Theorem 8.11 with $u_* = u$ (Corollary 6.5) and Corollary 5.8, or Theorem 5.6 for small β. □

9 The cap by characters: the first-order term

In Theorem 8.11 the first-order constant b of the lower bound is the cap, a plaquette q at a slice where a tube with front ∂q continues, weighted by the sup norm of g_q (Remark 8.12(ii));

and the first-order term $2(1+2/\sqrt{\tau_0})\beta$ of $D(\beta)$ in Theorem 5.6 comes from two sup-norm bounds in the proof of Lemma 5.5. In this section both are evaluated by characters. In the transfer-matrix representation the cap is an operator on the class functions of the holonomy of q in the two copies; on the plaquette states $\chi_\rho(U_q)$, $\chi_{\bar\rho}(U_q)$ its norm, relative to the tube, is

$$\kappa(\beta) := \frac{1}{\tau_0}\int (\operatorname{Re}\chi_\rho)^2 f_\beta \, dU = \frac{\beta}{N\tau_0}\int (\operatorname{Re}\chi_\rho)^3 \, dU + O(\beta^2),$$

and the same number appears in the upper bound. The outcome is $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ (Corollary 9.6). The first-order term vanishes when the centre of G acts on ρ by a scalar that is not a cube root of unity: for $U(1)$, and for $SU(N)$ with $N \neq 3$ and the fundamental representation, $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta^2)$. For $SU(3)$ it does not vanish: $\rho\otimes\rho\otimes\rho$ contains the trivial representation, so three sheets of fundamental flux can meet along the edges of the cap, and the cap can reverse the orientation of the flux through the tube; here $\int(\operatorname{Re}\chi_\rho)^3\,dU = \frac14$ and $m(\mu_\beta) = 4\log(1/u(\beta)) - \beta/6 + O(\beta^2)$. Two further changes make the remainder of order β^2: the transfer bound is applied junction by junction, with the gauge projections at the vertices of each junction (Lemma 9.2), and path segments whose fronts have at least five edges are summed without the factor $1/\delta'$ of the proof of Theorem 8.11.

Throughout this section ρ is irreducible and non-trivial, with N, ι, τ_0, u_r, u as in Section 5; $L^2_{\mathrm{cl}}(G)$ is the space of square-integrable class functions on G, with orthonormal basis $(\chi_r)_r$; and

$$u_2(\beta) := \sup_{r\notin\{0,\rho,\bar\rho\}} u_r(\beta), \qquad \sigma_6(\beta) := \sum_{r\neq 0} d_r^2 u_r(\beta)^6.$$

The number $\kappa(\beta)$ is unrelated to the constants κ_F, κ_C of Definition 8.9.

Lemma 9.1 (The cap matrix element)**.** *For every $\beta > 0$:*

(i) For $a, b \in \{\rho, \bar\rho\}$ the numbers $\varphi_{ab} := \int \overline{\chi_a}\, f_\beta\, \chi_b\, dU$ are real and non-negative, with $\varphi_{\bar\rho\bar\rho} = \varphi_{\rho\rho}$ and $\varphi_{\bar\rho\rho} = \varphi_{\rho\bar\rho}$. If P_ρ is the orthogonal projection of $L^2_{\mathrm{cl}}(G)$ onto the span of χ_ρ and $\chi_{\bar\rho}$, and M_f is multiplication by f_β, then $\|P_\rho M_f P_\rho\| = \kappa(\beta)$; and $\int \operatorname{Re}\chi_\rho\, f_\beta\, \overline{\chi_\rho}\, dU = \tau_0\kappa(\beta)$.

(ii) $0 \le \kappa(\beta) \le \epsilon(\beta)$; $\int(\operatorname{Re}\chi_\rho)^3\,dU \ge 0$; and $\bigl|\kappa(\beta) - \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3\,dU\bigr| \le \frac12\beta^2 e^\beta(1+\beta)$.

(iii) If $\rho(z) = \zeta\mathbf{1}$ for a central element z of G with $\zeta^3 \neq 1$, then $\int(\operatorname{Re}\chi_\rho)^3\,dU = 0$, so $\kappa(\beta) = O(\beta^2)$. This applies to $U(1)$ with $\rho(e^{i\theta}) = e^{i\theta}$, to $\mathbb{Z}_n$ ($n \neq 1, 3$) with a faithful character, and to $SU(N)$ with the fundamental representation for $N = 2$ and $N \ge 4$. For $SU(3)$ with the fundamental representation, $\int(\operatorname{Re}\chi_\rho)^3\,dU = \frac14$, so $\kappa(\beta) = \frac{\beta}{6} + O(\beta^2)$.

(iv) $u_2(\beta) \le e^\beta - 1 - \beta$ and $\sigma_6(\beta) \le (2-\iota)N^2 u(\beta)^6 + u_2(\beta)^5(e^\beta - 1)$.

Proof. (i) Inserting the absolutely and uniformly convergent expansion $f_\beta = \sum_{r\neq0} d_r u_r \chi_r$ gives $\varphi_{ab} = \sum_r d_r u_r \int \overline{\chi_a}\chi_r\chi_b\,dU$, where $\int\overline{\chi_a}\chi_r\chi_b\,dU$ is the multiplicity of a in $r\otimes b$; as $u_r > 0$, $\varphi_{ab} \ge 0$. Since f_β is real and $\overline{\chi_r} = \chi_{\bar r}$, $\varphi_{\bar a\bar b} = \overline{\varphi_{ab}} = \varphi_{ab}$, which gives the two equalities. If $\iota = 1$, P_ρ is the projection onto the real function χ_ρ ($\|\chi_\rho\| = 1$), so $\|P_\rho M_f P_\rho\| = \varphi_{\rho\rho}$, which is $\int(\operatorname{Re}\chi_\rho)^2 f_\beta = \kappa$ as $\tau_0 = 1$. If $\iota = 0$, $\chi_\rho \perp \chi_{\bar\rho}$, and in this basis $P_\rho M_f P_\rho$ has diagonal entries $\alpha_\rho := \varphi_{\rho\rho}$ and off-diagonal entries $\gamma_\rho := \varphi_{\rho\bar\rho}$, so its eigenvalues are $\alpha_\rho \pm \gamma_\rho$, with eigenvectors $\chi_\rho \pm \chi_{\bar\rho}$; its norm is $\alpha_\rho + \gamma_\rho$, attained at $\chi_\rho + \chi_{\bar\rho} = 2\operatorname{Re}\chi_\rho$, and therefore equals the Rayleigh quotient of $\operatorname{Re}\chi_\rho$, which is κ because $\|\operatorname{Re}\chi_\rho\|^2 = \tau_0$ (Lemma 5.1(i)). Finally $\int \operatorname{Re}\chi_\rho f_\beta\overline{\chi_\rho}$ equals $\varphi_{\rho\rho} = \kappa$ if $\iota = 1$, and $\frac12(\varphi_{\rho\rho} + \varphi_{\rho\bar\rho}) = \frac12\kappa$ if $\iota = 0$.

(ii) $\kappa = \|P_\rho M_f P_\rho\| \ge 0$, and $\kappa \le \|M_f\| = \|f_\beta\|_\infty \le \epsilon$, because $1 + f_\beta \in [e^{-2\beta}, e^\beta]$ (proof of Lemma 2.2(a)). Expanding $(\operatorname{Re}\chi_\rho)^3 = \frac18(\chi_\rho + \overline{\chi_\rho})^3$ writes $\int(\operatorname{Re}\chi_\rho)^3$ as a combination with positive coefficients of the integrals $\int \chi_\rho^j \overline{\chi_\rho}^{3-j}$, $0 \le j \le 3$, which are multiplicities of the

trivial representation, hence non-negative. Put $y := \frac{\beta}{N}\mathrm{Re}\,\chi_\rho \in [-\beta,\beta]$ and $R := e^y - 1 - y \in [0, \frac{1}{2}\beta^2 e^\beta]$. As $\int y = 0$, $c_0 = 1 + \int R$, and $f_\beta = e^y/c_0 - 1$ gives $f_\beta - y = (R - \int R - y\int R)/c_0$; so $|f_\beta - y| \le \frac{1}{2}\beta^2 e^\beta(1+\beta)$, since $c_0 \ge 1$ and $|R - \int R| \le \frac{1}{2}\beta^2 e^\beta$. Hence $|\kappa - \tau_0^{-1}\int(\mathrm{Re}\,\chi_\rho)^2 y| \le \frac{1}{2}\beta^2 e^\beta(1+\beta)$, and $\tau_0^{-1}\int(\mathrm{Re}\,\chi_\rho)^2 y = \frac{\beta}{N\tau_0}\int(\mathrm{Re}\,\chi_\rho)^3$.

(iii) Haar measure is invariant under $U \mapsto zU$, and $\chi_\rho(zU) = \zeta\chi_\rho(U)$, $|\zeta| = 1$; so $\int \chi_\rho^j \overline{\chi_\rho}^{3-j} = \zeta^{2j-3}\int \chi_\rho^j \overline{\chi_\rho}^{3-j}$, and $\zeta^{2j-3} \in \{\zeta^{-3},\zeta^{-1},\zeta,\zeta^3\}$ is not 1. Take $z = -1$ for $U(1)$, a generator of $\mathbb{Z}_n$, and $z = e^{2\pi i/N}\mathbf{1}$ for $SU(N)$, for which $\zeta = e^{2\pi i/N}$ and $\zeta^3 = 1$ only if $N \in \{1,3\}$. For $SU(3)$ the same argument removes the terms with $j = 1,2$, and $\int \chi_\rho^3 = \int \overline{\chi_\rho}^3$ is the dimension of the space of $SU(3)$-invariant tensors $T = \sum T_{ijk} e_i \otimes e_j \otimes e_k \in (\mathbb{C}^3)^{\otimes 3}$ (the trace of the projection $\int U^{\otimes 3}\,dU$). The diagonal matrices $\mathrm{diag}(z_1,z_2,z_3)$, $z_1z_2z_3 = 1$, multiply T_{ijk} by $z_iz_jz_k$, so $T_{ijk} = 0$ unless $\{i,j,k\} = \{1,2,3\}$. Let $H \cong SU(2)$ act on the span of e_1, e_2 and fix e_3. As $\mathbb{C}e_1 \oplus \mathbb{C}e_2$ and $\mathbb{C}e_3$ are H-stable, $(\mathbb{C}^3)^{\otimes 3}$ is the H-stable direct sum of the eight tensor products of these two summands, so the components of an invariant tensor in them are invariant. Then $T = T^{(1)} + T^{(2)} + T^{(3)}$, where $T^{(m)}$ collects the terms with e_3 in the m-th factor; H preserves each of these three subspaces and acts on each as on $\mathbb{C}^2 \otimes \mathbb{C}^2$, whose H-invariants are the multiples of $e_1 \otimes e_2 - e_2 \otimes e_1$ (their dimension is $\int \chi_{1/2}^2 = 1$, and the determinant of $\mathbb{C}^2$ is invariant). So T is antisymmetric under the exchange of the values 1 and 2, and likewise for the other pairs: $T_{\sigma(1)\sigma(2)\sigma(3)} = \mathrm{sgn}(\sigma)T_{123}$, and T is a multiple of the determinant tensor, which is invariant. So $\int \chi_\rho^3 = 1$, $\int(\mathrm{Re}\,\chi_\rho)^3 = \frac{1}{8}(1+1) = \frac{1}{4}$, and (ii) with $N = 3$, $\tau_0 = \frac{1}{2}$ gives $\kappa = \frac{\beta}{6} + O(\beta^2)$.

(iv) The first bound is in the proof of Corollary 6.5. In σ_6 the terms $r \in \{\rho, \bar\rho\}$ contribute $(2-\iota)N^2u^6$, and the others at most $u_2^5 \sum_{r\neq 0} d_r^2 u_r = u_2^5(e^\beta/c_0 - 1) \le u_2^5(e^\beta - 1)$ (Lemma 6.2). □

Lemma 9.2 (Transfer bound by pieces)**.** *Let X be as in Definition 7.1, and s a slice. The* pieces *at slice s are the junctions at slice s and the fronts of layer $s-1$ touched by no marker at slice s (which are fronts of layer s as well, Lemma 7.2(iii)). For a junction J at slice s, let M_J be multiplication by the product of the g_p over the plaquettes p of the clusters of J, times $A(U^{(0)}) - A(U'^{(0)})$ if $S_A \in M(J)$ and times $B(U^{(t)}) - B(U'^{(t)})$ if $S_B \in M(J)$; for $\lambda \in \{s-1, s\}$ let $\tilde{\mathcal{K}}_{J,\lambda}^{1/2}$ be the tensor product over $e \in E(J)$ of $G_g^{1/2}$ if $e \in E_\lambda(J)$ and of $\Pi_0 \otimes \Pi_0$ otherwise (notation of Lemma 6.2); and let $O_J := \tilde{\mathcal{K}}_{J,s-1}^{1/2} M_J \tilde{\mathcal{K}}_{J,s}^{1/2}$, an operator on $L^2(G^{E(J)} \times G^{E(J)})$. For an untouched front φ let $E(\varphi) := \varphi$ and $O_\varphi := G_g^{\otimes\varphi}$. For a piece π let P^π be the orthogonal projection onto the functions of $(U_e, U'_e)_{e\in E(\pi)}$ that are invariant, in each copy, under the gauge transformations at the vertices of $V(E(\pi))$ (acting on the links of $E(\pi)$). Then*

$$|\Phi_F(X)| \;\le\; \prod_s \prod_{\pi\ at\ s} \|P^\pi O_\pi P^\pi\|,$$

and $\|P^\varphi O_\varphi P^\varphi\| \le u_*^{|\varphi|}$, $\|P^J O_J P^J\| \le \|M_J\| \prod_{p\in\Phi(J)} u_*^{|p|/2}$.

Proof. In the proof of Lemma 6.3, $\Phi_F(X) = \mathrm{Tr}(\mathbb{M}_0\mathcal{K}_0 \cdots \mathbb{M}_{L_0-1}\mathcal{K}_{L_0-1})$ and $\mathcal{K}_s = \tilde{\mathcal{K}}_s\mathbf{P}$, where $\mathbf{P} := P_{\mathcal{G}} \otimes P_{\mathcal{G}}$ and $\tilde{\mathcal{K}}_s \ge 0$ is the tensor product over the spatial links of G_g (on $T_s(X)$) and $\Pi_0 \otimes \Pi_0$ (elsewhere). It commutes with $\mathbf{P}$, so $\mathcal{K}_s = \tilde{\mathcal{K}}_s^{1/2}\mathbf{P}\tilde{\mathcal{K}}_s^{1/2}$, where $\tilde{\mathcal{K}}_s^{1/2}$ is the tensor product of the operators $G_g^{1/2}$ and $\Pi_0 \otimes \Pi_0$. By cyclicity of the trace and $\mathbf{P}^2 = \mathbf{P}$,

$$\Phi_F(X) = \mathrm{Tr}\Big(\prod_s \mathbf{P}\,\mathcal{O}_s\mathbf{P}\Big), \qquad \mathcal{O}_s := \tilde{\mathcal{K}}_{s-1}^{1/2}\,\mathbb{M}_s\,\tilde{\mathcal{K}}_s^{1/2}$$

(indices modulo L_0). For a layer s_* without temporal plaquettes, $\tilde{\mathcal{K}}_{s_*}^{1/2}$ is the rank-one projection Π onto the constants, and $\mathbf{P}\Pi = \Pi$; so $\mathbf{P}\mathcal{O}_{s_*+1}\mathbf{P} = \Pi(\mathbf{P}\mathcal{O}_{s_*+1}\mathbf{P})$ has trace norm at most its operator norm, and Hölder's inequality for traces gives $|\Phi_F(X)| \le \prod_s \|\mathbf{P}\mathcal{O}_s\mathbf{P}\|$.

Fix s. Distinct pieces at slice s have no common vertex: junctions are components of H_s; an untouched front shares no vertex with a marker at slice s; two untouched fronts are

distinct components of T_{s-1}; and an untouched front, a component of T_{s-1} and of T_s, sharing a vertex with the front of a front-pair would be that front, which is touched. A spatial link of slice s that lies in T_{s-1}, in T_s or in the edge set of a marker at slice s belongs to exactly one piece (by Lemma 8.2(ii), and Lemma 7.2(iii)), and on the links of a junction J the factor $\tilde{\mathcal{K}}_\lambda^{1/2}$ is $\tilde{\mathcal{K}}_{J,\lambda}^{1/2}$ (an edge of $E(J)$ lies in T_λ exactly when it lies in $E_\lambda(J)$, by Lemma 8.2(ii)), while on an untouched front φ the product $\tilde{\mathcal{K}}_{s-1}^{1/2}\tilde{\mathcal{K}}_s^{1/2}$ is $G_g^{\otimes\varphi}$. Moreover no link outside $E(\pi)$ at a vertex v of a piece π lies in T_{s-1}, in T_s or in a marker. Indeed, a marker at slice s with the vertex v touches π if π is a front, and is adjacent in H_s to a vertex of J whose edge set has the endpoint v if $\pi = J$. A front ψ of layer $\lambda \in \{s-1, s\}$ through v is π itself if π is a front (fronts of one layer are vertex-disjoint). If $\pi = J$ and ψ is touched at slice s, then (ψ, λ) is a front-pair adjacent to a vertex of J with the endpoint v, hence in J; if ψ is not touched, it is a front of both layers, so it is the front of every front-pair of J through v, which is touched, and no front-pair of J passes through v; then a marker of J does, and touches ψ: a contradiction. Hence $\mathcal{O}_s = \bigotimes_\pi O_\pi \otimes (\Pi_0 \otimes \Pi_0)^{\otimes\mathcal{R}}$, with $\mathcal{R}$ the remaining links. Now $\mathbf{P} = \prod_v P_v \otimes P_v$, where $P_v = \int R_h^{(v)}\, dh$ is the projection for the gauge transformations $R_h^{(v)}$ at the vertex v of the slice; as Π_0 is invariant under left and right translations, $R_h^{(v)}$ acts on $\mathcal{O}_s$, from either side, only through the links of the piece containing v, and trivially if v lies in no piece. So $\mathbf{P}\mathcal{O}_s\mathbf{P} = \bigotimes_\pi P^\pi O_\pi P^\pi \otimes (\Pi_0 \otimes \Pi_0)^{\otimes\mathcal{R}}$, whose norm is $\prod_\pi \|P^\pi O_\pi P^\pi\|$. The last two bounds follow from $\|G_g^{1/2}\| = u_*^{1/2}$ (Lemma 6.2), $\|\Pi_0 \otimes \Pi_0\| = 1$ and $|E_\lambda(J)| = \sum |p|$ over the front-pairs p of J of layer λ. □

With $\|M_J\| \le \gamma_g^{|J|_C}$ (times $2\|A\|_\infty$, $2\|B\|_\infty$ for J_A, J_B) this is the bound of Lemma 6.3 for sets with $m(X) = 0$, split into the factors $w(J)$ and $w(\sigma)$ of Section 8 (before the factors $e^{2(K+1)\theta_5}$). The point of the splitting is that a single factor can now be computed exactly.

Lemma 9.3 (The cap by characters)**.** *Let $\beta > 0$ with $u_*(\beta) = u(\beta)$, and let J be a cap at slice s (Lemma 8.6(ii)): $M(J) = \{\{q\}\}$ and $\Phi(J) = \{(\partial q, s-1), (\partial q, s)\}$. Then $\|P^J O_J P^J\| \le u^4 \bar\kappa_{\mathrm{cap}}$, where*

$$\bar\kappa_{\mathrm{cap}} := \kappa + \frac{u_2^2}{u^2}\Big(1 + \frac{u_2^2}{u^2}\Big)\epsilon + \frac{\sigma_6}{u^4} + \sqrt{2}\, e^\beta \sigma_6^{1/2} + u^4\gamma_g,$$

all at β. As $\beta \downarrow 0$, $\bar\kappa_{\mathrm{cap}} = \kappa(\beta) + (2-\iota)N^2 u(\beta)^2 + O(\beta^3)$.

Proof. Here $E(J) = \partial q = \{e_1, \dots, e_4\}$, $O_J = (G_g^{1/2})^{\otimes 4} M_{g_q} (G_g^{1/2})^{\otimes 4}$, and $g_q = (1 + f_\beta(U_q))(1 + f_\beta(U_q')) - 1$. A function of the four links of the square ∂q that is invariant under the gauge transformations at its corners is a class function of the holonomy U_q (fix the gauge on three links), and U_q is Haar distributed when the U_{e_i} are; so $(U, U') \mapsto (U_q, U_q')$ identifies the range of P^J with $\mathcal{H} := L^2_{\mathrm{cl}}(G) \otimes L^2_{\mathrm{cl}}(G)$, with orthonormal basis $e_{aa'} := \chi_a(U_q)\chi_{a'}(U_q')$. The factors of O_J commute with P^J (gauge transformations act by translations, which commute with G_g, and g_q is gauge invariant). As a function of U_{e_i}, $\chi_a(U_q)$ is a matrix coefficient of a or of $\bar a$, and $u_{\bar a} = u_a$; so by Lemma 6.2, $(G_g^{1/2})^{\otimes 4}$ multiplies $e_{aa'}$ by $\hat u_{aa'}^2$, where $\hat u_{a0} = u_a$, $\hat u_{0a'} = u_{a'}$, $\hat u_{aa'} = u_a u_{a'}$ $(a, a' \neq 0)$ and $\hat u_{00} = 0$. Hence $P^J O_J P^J$ is $\Lambda M_g \Lambda$ on $\mathcal{H}$, with $\Lambda e_{aa'} = \hat u_{aa'}^2 e_{aa'}$ and $\langle e_{aa'}, M_g e_{bb'}\rangle = m_{ab}m_{a'b'} - \delta_{ab}\delta_{a'b'}$, where $m_{ab} := \int \overline{\chi_a}(1+f_\beta)\chi_b\, dU = \delta_{ab} + \varphi_{ab}$ (the notation of Lemma 9.1, for all a, b). Since $\Lambda e_{00} = 0$, only $\mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_{12}$ matters, the spans of the e_{a0}, the $e_{0a'}$ and the $e_{aa'}$ with $a, a' \neq 0$. With $m_{00} = 1 + \int f_\beta = 1$, $\varphi_{a0} = d_a u_a$ and $\varphi_{0b} = d_b u_b$ $(a, b \neq 0)$, $v_a := d_a u_a^3$, $\mathsf{D} := \mathrm{diag}(u_a^2)_{a\neq 0}$ (of norm u^2, as $u_* = u$), $\mathsf{B} := (\varphi_{ab})_{a,b\neq 0}$ and $\mathfrak{m} := (m_{ab})_{a,b\neq 0}$ (compressions of M_f and M_{1+f}, so $\|\mathsf{B}\| \le \|f_\beta\|_\infty \le \epsilon$ and $\|\mathfrak{m}\| \le e^\beta$), the blocks of $\Lambda M_g \Lambda$ are: DBD from $\mathcal{H}_1$ to $\mathcal{H}_1$ and from $\mathcal{H}_2$ to $\mathcal{H}_2$; the rank-one operator $e_{0b'} \mapsto v_{b'} \sum_a v_a e_{a0}$, of norm $\sum_a v_a^2 = \sigma_6$, from $\mathcal{H}_2$ to $\mathcal{H}_1$, and its adjoint; $e_{bb'} \mapsto v_{b'} \sum_a (\mathsf{D}\mathfrak{m}\mathsf{D})_{ab} e_{a0}$, of norm at most $u^4 e^\beta \sigma_6^{1/2}$, from $\mathcal{H}_{12}$ to $\mathcal{H}_1$, likewise to $\mathcal{H}_2$, and the adjoints; and $(\mathsf{D}\otimes\mathsf{D})(\mathfrak{m}\otimes\mathfrak{m} - \mathbf{1})(\mathsf{D}\otimes\mathsf{D})$, of norm at most $u^8\gamma_g$, on $\mathcal{H}_{12}$, because $\mathfrak{m}\otimes\mathfrak{m} - \mathbf{1}$ is a compression of M_g. For DBD write

$\mathsf{D} = u^2 P_\rho + \mathsf{D}'$ with $\|\mathsf{D}'\| \le u_2^2$; since P_ρ and D' act on orthogonal subspaces, $\|\mathsf{DBD}\| \le \max(u^4\kappa, u_2^4\epsilon) + u^2u_2^2\epsilon \le n_1 := u^4\kappa + u^2u_2^2\epsilon + u_2^4\epsilon$ (Lemma 9.1(i)). The norm of a block operator is at most the largest eigenvalue of the matrix of the norms of its blocks, here

$$\begin{pmatrix} n_1 & \sigma_6 & n_y \\ \sigma_6 & n_1 & n_y \\ n_y & n_y & n_z \end{pmatrix}, \qquad n_y := u^4 e^\beta \sigma_6^{1/2}, \quad n_z := u^8\gamma_g.$$

Its eigenvectors are symmetric or antisymmetric in the first two coordinates; the antisymmetric one has the eigenvalue $n_1 - \sigma_6$, and on the symmetric ones it acts as $\left(\begin{smallmatrix} n_1+\sigma_6 & \sqrt{2}n_y \\ \sqrt{2}n_y & n_z \end{smallmatrix}\right)$, with eigenvalues at most $n_1 + \sigma_6 + n_z + \sqrt{2}n_y = u^4\bar\kappa_{\mathrm{cap}}$. As $\beta \downarrow 0$: $u \ge (1+\iota)\beta e^{-\beta}/(2N^2)$ (Lemma 5.1(iii)) and $u_2 = O(\beta^2)$, so $(u_2/u)^2\epsilon = O(\beta^3)$, $\sigma_6/u^4 = (2-\iota)N^2u^2 + O(\beta^7)$ and $\sqrt{2}e^\beta\sigma_6^{1/2} + u^4\gamma_g = O(\beta^3)$ (Lemma 9.1(iv)). □

Theorem 9.4 (The lower bound to first order)**.** *Suppose that $u_* = u$. In the setting of Definition 8.9 and Lemma 8.10 put $b_{\mathrm{cap}} := e^{2(K+1)\theta_5}\min(\bar\kappa_{\mathrm{cap}}, \gamma_g)$ and $\Sigma_4^\chi := \Sigma_4 - b + b_{\mathrm{cap}}$. Let $\delta' > 0$ satisfy*

$$\Sigma_4^\chi < \delta', \qquad P_v(g_C Y_1/5 + g_F Y_0) < F(1+\delta'-a), \qquad a^4(1+\delta') \le 1.$$

Then the conclusions of Theorem 8.11 hold for this δ'. Let δ_χ be the infimum of the admissible δ' (with $N_4^v = P_v$, $N_5^v = 0$ for $m(\mu_\beta)$). As $\beta \downarrow 0$ the hypotheses hold for every $\delta' > b_{\mathrm{cap}} + (4P_v - 4 - K_S)b^2 + c\beta^3$, with a constant c; so

$$\delta_\chi \le \kappa(\beta) + (2-\iota)N^2u(\beta)^2 + (4P_v - 4 - K_S)\gamma_g^2 + O(\beta^3), \qquad m(\mu_\beta) \ge 4\log\frac{1}{u(\beta)} - \kappa(\beta) - O(\beta^2).$$

Proof. Weights. By Lemmas 9.2 and 9.3, and $r_X^2 \le e^{2\theta_5 J_S} e^{2(K+1)\theta_5|X|}$, every set X with an empty layer satisfies

$$|\Phi_F(X)|\, r_X^2 \le 4\|A\|_\infty\|B\|_\infty e^{2\theta_5 J_S}\, W^\chi(X), \qquad W^\chi(X) := \prod_J w^\chi(J) \prod_\sigma w(\sigma),$$

where $w^\chi(J) := w(J)$ for junctions other than caps and $w^\chi(J) := a^4 b_{\mathrm{cap}} \le w(J)$ for caps; the factor $e^{2(K+1)\theta_5}$ of each plaquette is distributed as in Section 8, one for the plaquette of the cap and the factors of the half layers with its two front-pairs. (A cap contains no support, so the bound applies to every cap of X.)

The path. The proof of Theorem 8.11 applies to W^χ, with two changes. First, the decorations are bounded as there, as $w^\chi \le w$, but a cap on the path now carries b_{cap}: in the bound of Lemma 8.10(ii) for $k = 4$ the contribution $Z_{1,4}b$ of the cap becomes b_{cap}, so Σ_4 becomes Σ_4^χ. Second, for a front-pair p with $k := |p| \ge 5$ we have $za^k = a^{k-4}/(1+\delta') \le a/(1+\delta')$, so $\sum_{\ell\ge1}(za^k)^\ell \le a^{k-4}/(1+\delta'-a)$ instead of a^{k-4}/δ', and the induction step for $|p| \ge 5$ holds when $M \ge E_B/(F(1+\delta'-a) - P_v(g_C Y_1/5 + g_F Y_0))$.

Small β. Take h_J, ξ_F, ξ_C as in the proof of Theorem 8.11, but $F := 3P_vN_4^vb$. Then $f(k) = O(\beta a^{k-4})$ for $k \ge 5$, so $B_0 = N_4^v(1+O(\beta))$ and $B_1 = 4N_4^v(1+O(\beta))$, and the six inequalities of Definition 8.9 hold with $g_F = N_4^v(1+O(\beta))$, $g_C = 8N_4^v(1+O(\beta))$ and $g_F', g_C', \kappa_F, \kappa_C = O(\beta^2)$; so $P_v(g_C Y_1/5 + g_F Y_0) = \frac{13}{5}P_vN_4^vb(1+O(\beta)) < F(1+\delta'-a)$. In Σ_4^χ the term of $Z_{2,4}$ is $(4P_v - 4 - K_S)b^2$; the terms of three plaquettes and of $n \ge 4$ plaquettes are $O(b^3)$; those with exits of $k \ge 5$ edges are $O(b\,Fa) = O(\beta^3)$ (the five-edge exits occur only for $n \ge 4$); and the last term is $O((\kappa_C + \kappa_F + g_F' + g_C')b) = O(\beta^3)$. So $\Sigma_4^\chi = b_{\mathrm{cap}} + (4P_v - 4 - K_S)b^2 + O(\beta^3)$, and $b_{\mathrm{cap}} \le \bar\kappa_{\mathrm{cap}}(1+O(\beta^5))$, $b = \gamma_g(1+O(\beta^5))$ give the bound on δ_χ by Lemma 9.3. Finally $m(\mu_\beta) \ge 4\log(1/u) - 8(K+1)\theta_5 - \log(1+\delta')$ with $\theta_5 = O(\beta^5)$, and $\gamma_g \le \gamma_2 = O(\beta)$, $u = O(\beta)$. □

Theorem 9.5 (The upper bound by characters)**.** *In the setting of Lemma 5.5 put* $\eta_2 := a_* N\bar{u}/\tau_0$, $\kappa := \kappa(\beta)$, *and*

$$V^\chi := (1+\kappa)(1+\mathcal{T}_p)+\mathcal{T}_p, \qquad L^\chi := \frac{(1+\kappa-\eta_2)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1+\frac{a_*^2}{\tau_0}\Big)e^{16\beta}\,\frac{N^4}{\tau_0^2}\,\mathcal{T}_Q.$$

Then $V^\chi \le V$ *and, if* $\eta \le 1$, $L^\chi \ge L$. *If* $\eta_2 \le 1$ *and* $L^\chi > 0$, *then (i) and (ii) of Theorem 5.6 hold with* $D^\chi(\beta) := \log(V^\chi/L^\chi)$ *in place of* $D(\beta)$. *As* $\beta \downarrow 0$, $D^\chi = -\log(1+\kappa)+2\eta_2+O(\beta^3) = -\kappa(\beta)+O(\beta^2)$; *for* $SU(2)$, $D^\chi(\beta) = \frac{3}{8}\beta^2+O(\beta^3)$. *Moreover* $D^\chi \le -\log(1+\kappa)+D_0$, *where* $D_0 := \log((1+2\mathcal{T}_p)/L_0)$ *and* L_0 *is* L^χ *with* κ *replaced by* 0; $L_0 \ge L$ *if* $\eta \le 1$ *(as* $\eta_2 \le \eta$*), so* $L_0 > 0$ *under the hypotheses of Theorem 5.6, and* D_0 *increases with* β.

Proof. In the proof of Lemma 5.5 two estimates are replaced by identities. First, $M_{S_A}(A^2) = \int A^2(1+f_\beta)\,dU = \tau_0 + \int(\operatorname{Re}\chi_\rho)^2 f_\beta\,dU = \tau_0(1+\kappa)$ (Lemma 5.1(i) and the definition of κ), instead of $M_{S_A}(A^2) \le (1+\epsilon)\tau_0$; so $\operatorname{Var}(A) \le \langle A^2\rangle \le (1+\mathcal{T}_p)\tau_0(1+\kappa)+\tau_0\mathcal{T}_p = \tau_0 V^\chi$. Second, by Lemma 9.1(i), $\hat{a}_\rho = \tau_0 + \int \operatorname{Re}\chi_\rho f_\beta\overline{\chi_\rho}\,dU - aNu = \tau_0(1+\kappa) - aNu \ge \tau_0(1+\kappa-\eta_2) \ge 0$, as $|a| \le a_*$ and $u \le \bar{u}$; so $M_Q \ge \frac{2}{1+\iota}\hat{a}_\rho^2 u^4 \ge \tau_0(1+\kappa-\eta_2)^2u^4$. The rest of the proofs of Lemma 5.5 and Theorem 5.6 is unchanged and gives $\operatorname{Cov}(A,A_1) \ge \tau_0 u^4 L^\chi$, $\operatorname{Var}(A) \le \tau_0 V^\chi$ and the theorem. As $0 \le \kappa \le \epsilon$ (Lemma 9.1(ii)) and $\eta_2 \le \eta$, $V^\chi \le V$ and $L^\chi \ge L$. As $\beta \downarrow 0$: $\mathcal{T}_Q, \mathcal{T}_p = O(\beta^5)$, $a_* = O(\beta)$, $\eta_2 = O(\beta^2)$, so $\log V^\chi = \log(1+\kappa)+O(\beta^5)$ and $\log L^\chi = 2\log(1+\kappa) - 2\eta_2 + O(\beta^3)$. For $SU(2)$, $\kappa = O(\beta^2)$ by Lemma 9.1(iii); more precisely $\kappa = d_1u_1 = 3I_3(\beta)/I_1(\beta) = \frac{1}{8}\beta^2+O(\beta^4)$ ($u_j = I_{2j+1}/I_1$ for the spin-j representation, as in Remark 7.10), and $\eta_2 = \frac{1}{4}\beta^2+O(\beta^3)$, so $D^\chi = \frac{3}{8}\beta^2+O(\beta^3)$. Finally, $V^\chi \le (1+\kappa)(1+2\mathcal{T}_p)$ and $L^\chi \ge (1+\kappa)^2 L_0$, because $(1+\kappa-\eta_2)/(1+\kappa) \ge 1-\eta_2$; and $\mathcal{T}_p$, $\mathcal{T}_Q$, ϵ, a_*, η_2 increase with β (Lemma 5.7). □

Corollary 9.6 (The first-order term)**.** *Let* G *be metrisable and* ρ *irreducible and non-trivial. As* $\beta \downarrow 0$,

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \kappa(\beta) + O(\beta^2) = 4\log\frac{1}{u(\beta)} - \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3\,dU + O(\beta^2),$$

and the same holds for $m_{\Lambda_s}(\beta)$, *uniformly in the periodic spatial lattices with all sides* ≥ 4. *In particular* $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta^2)$ *for* $U(1)$, *and for* $SU(N)$ *with* $N \ne 3$ *and the fundamental representation; and for* $SU(3)$ *with the fundamental representation*

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \frac{\beta}{6} + O(\beta^2).$$

Proof. For small β, $u_* = u$ (proof of Corollary 6.5: as soon as $\beta e^{2\beta} \le (1+\iota)/N^2$). The lower bound is Theorem 9.4, with $\log(1+\delta_\chi) \le \delta_\chi$, and in finite volume the same theorem with the values of N_4^v, N_5^v of the lattice (bounded by Lemma 7.5, uniformly). The upper bound is Theorem 9.5, since $L^\chi \to 1$ and $\eta_2 \to 0$. The rest is Lemma 9.1(ii),(iii). □

Remark 9.7 (Numbers, meaning, and checks). (i) For $SU(2)$ ($\kappa = 3I_3/I_1$, $u = I_2/I_1$; u_2 and σ_6 bounded by Lemma 9.1(iv)), with the parameters found by iterating the conditions, the lower bound of Theorem 9.4 for $m(\mu_\beta)$ lies below $4\log(1/u)$ by $1.3\cdot10^{-12}$, $1.3\cdot10^{-10}$, $1.5\cdot10^{-8}$, $6.8\cdot10^{-6}$ and $1.28\cdot10^{-4}$ at $\beta = 10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(4)/2$ and $\beta_0(4)$ ($d=4$; $\delta_\chi/\beta^2 = 128.5$, 129.8, 151.7, 979, 3971), and by $3.2\cdot10^{-13}$, $3.3\cdot10^{-11}$, $3.4\cdot10^{-9}$, $1.0\cdot10^{-5}$ and $2.44\cdot10^{-4}$ for $d=3$ ($\delta_\chi/\beta^2 = 32.4$, 32.6, 34.5, 95, 219; at $\beta_0(3)$ most of the loss is $8(K+1)\theta_5$). On spatial lattices with all sides ≥ 4 the loss is at most $1.4\cdot10^{-4}$ ($d=4$) and $2.51\cdot10^{-4}$ ($d=3$). As $\beta \downarrow 0$, $\delta_\chi/\beta^2 \to \frac{1}{8}+\frac{1}{4}+4(4P_v-4-K_S)$, i.e. 128.375 ($d=4$) and 32.375 ($d=3$): the shifts and jumps of the tube, still at their sup norm, are now the leading term. The upper bound of Theorem 9.5 has $D^\chi = \frac{3}{8}\beta^2(1+O(\beta))$, and $D_0(\beta_0(d))$ is $4.8\cdot10^{-4}$ ($d=4$) and $2.4\cdot10^{-3}$ ($d=3$), where

$\mathcal{T}_Q$, the five-plaquette remainder, dominates. As before (Remark 8.12(i)) the hypotheses of Theorem 9.4 are monotone in (a, b, b_{cap}), the right side of the second one decreases with a, and a, b increase with β. In these evaluations γ_g is bounded by γ_2, and u_2, σ_6 by Lemma 9.1(iv); that the resulting b_{cap} increases with β was checked on a grid of 60 points of $(0, \beta_0(d)]$ and is part of the numerical evaluation. So for all $0 < \beta < \beta_0(d)$

$$-1.3 \cdot 10^{-4} \le m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \le 4.8 \cdot 10^{-4} \quad (d = 4),$$
$$-2.5 \cdot 10^{-4} \le m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \le 2.4 \cdot 10^{-3} \quad (d = 3),$$

and $-128.4\beta^2 \le m(\mu_\beta) - 4\log(1/u(\beta)) \le 0.375\beta^2$ up to factors $1 + O(\beta)$ ($d = 4$; $-32.4\beta^2$ for $d = 3$) (Numerical: an evaluation of the explicit constants, `python3` with `mpmath`). For $SU(3)$ (κ and u by Weyl integration) $\kappa/\beta = 0.1667$ on the whole range, and the two bounds are $-\kappa(1 + O(\beta))$ for small β (at $\beta = 10^{-7}$: $\delta_\chi = 1.6668 \cdot 10^{-8}$ and $D^\chi = -1.6667 \cdot 10^{-8}$, $\kappa = 1.6667 \cdot 10^{-8}$); near $\beta_0(4)$ the lower bound loses $1.1 \cdot 10^{-4}$ and $D^\chi \approx 9.0 \cdot 10^{-3}$, dominated by the factor $N^4/\tau_0^2 = 324$ in front of $\mathcal{T}_Q$.

(ii) Without the second change in the proof of Theorem 9.4 the method would not reach the second order: with $\sum_\ell (za^k)^\ell \le a^{k-4}/\delta'$ for $k \ge 5$ the constant F must be of order b/δ', and with $\delta' = O(\beta^2)$ the exits with six edges contribute $O(bFa^2) = O(\beta^2)$ with a large constant; numerically $\delta/\beta^2 \to 4.2 \cdot 10^3$ instead of 128.4 ($SU(2)$, $d = 4$).

(iii) Meaning. In the transfer-matrix picture the cap acts on the plaquette states $\chi_\rho(U_q)$ and $\chi_{\bar\rho}(U_q)$ by the matrix of Lemma 9.1(i); its eigenvectors are $\operatorname{Re}\chi_\rho$ and $\operatorname{Im}\chi_\rho$, which are even and odd under charge conjugation $U \mapsto \bar U$, with eigenvalues $\kappa = \alpha_\rho + \gamma_\rho$ and $\alpha_\rho - \gamma_\rho$. So to first order the C-even plaquette state is lowered by κ and, if $\iota = 0$, the C-odd one is shifted by $-(\alpha_\rho - \gamma_\rho)$ (Heuristic; only the lowest mass is bounded here). For $SU(3)$, $\alpha_\rho = O(\beta^2)$ and $\gamma_\rho = 3u + O(\beta^2)$: the two plaquette glueballs split at first order, $m_\pm \approx 4\log(1/u) \mp 3u$. The title and the first lines of the abstract of [OBS85] (complex characters, the subspace generated by the plaquette function and its complex conjugate, "mass splitting") suggest that this splitting was found there, in $2 + 1$ dimensions; we have not seen the paper.

(iv) Checks (`theory/checks/first-order/`, a compass): an exact computation for $\mathbb{Z}_3$ in $d = 3$ on the 3×3 spatial torus, in the sector of winding zero (dimension 3^8; there $\kappa = u$ exactly; this torus has sides $3 < 4$, outside the hypotheses), gives $(m - 4\log(1/u))/u = -0.99875$ for the lowest C-even state and $+1.00025$ for the lowest C-odd state at $\beta = 10^{-3}$, tending to ∓1; an independent referee repeated the computation on the 4×4 torus, which satisfies the hypotheses (3^{15} states), and found the same values to six or seven digits at $\beta = 0.05$ and 0.02; for $\mathbb{Z}_2$ ($\kappa = 0$) on the 4×4 torus, winding zero, $(m - 4\log(1/u))/u^2 \to 1$, so there is no first-order term. A brute-force computation of $\|P^J O_J P^J\|$ on the full two-copy space of the four links of a cap ($\mathbb{Z}_3$: dimension 3^8; S_3 with its two-dimensional representation, for which $\int \chi_\rho^3 = 1$ and $\kappa \approx \beta/2$: dimension 6^8) agrees with the class-function formula of the proof of Lemma 9.3 to all printed digits, lies below $u^4\bar\kappa_{\mathrm{cap}}$, and agrees with $u^4(\kappa + (2 - \iota)N^2u^2)$ to six digits at $\beta = 0.05$ and seven at $\beta = 0.01$. Weyl integration gives $\int(\operatorname{Re}\chi)^3 = 0$ for $SU(2)$, $SU(4)$, $SU(5)$ and $\frac14$ for $SU(3)$, and $\kappa_{SU(3)}(\beta)/(\beta/6) = 1 + O(\beta)$. An independent referee's checks, with separate code, reproduced every number of (i), the brute-force cap operator for $\mathbb{Z}_2$, $\mathbb{Z}_3$, $\mathbb{Z}_4$, S_3 and the quaternion group (whose centre acts by -1, so that $\kappa = O(\beta^2)$), the class-function cap for $SU(2)$ and $SU(3)$, and the exact $\mathbb{Z}_2$ and $\mathbb{Z}_3$ spectra.

(v) What remains at second order: the shifts and jumps of Lemma 8.6(iii), $4(P_v - 1) - K_S$ junctions of two plaquettes at their sup norm $b^2 \approx 4\beta^2$, whose character values should be of order N^2u^2, about $\beta^2/4$ for $SU(2)$ (Heuristic); the copy switch σ_6/u^4 and the term η_2, which come from the duplicated system and from the mean a; and, near $\beta_0(d)$, the crude counts of Remark 8.12(i) and $\mathcal{T}_Q$. Section 10 treats the first three: the shifts and jumps vanish on the

odd sector of the duplicated system, and the copy switch and the mean combine with κ into $-\log\Upsilon$.

10 The second-order term

Corollary 9.6 determines $m(\mu_\beta)$ up to $O(\beta^2)$. At the next order three effects enter: the second-order part of the cap; the mixing of the plaquette state with the vacuum, which in the duplicated system is the copy switch in the proof of Lemma 9.3; and the shifts and jumps of the tube (Lemma 8.6(iii)), at their sup norm the leading term of δ_χ (Remark 9.7(i)). The shifts and jumps drop out: the two plaquettes of such a junction have independent holonomies, so its operator has rank one and factorises through the vacuum; it vanishes on the sector of the duplicated system that is odd under the exchange of the two copies, and the covariance lives on that sector (Lemmas 10.4 and 10.5). What remains is one number, the largest variance of a plaquette function under the weight of a single plaquette:

$$\Upsilon(\beta) := \max\Big\{ \int \psi^2 \, \frac{w_\beta}{c_0}\, dU - \Big(\int \psi \, \frac{w_\beta}{c_0}\, dU \Big)^2 \;:\; \psi \in \operatorname{span}_{\mathbb{R}}\{\operatorname{Re}\chi_\rho, \operatorname{Im}\chi_\rho\},\ \int \psi^2\, dU = 1 \Big\}.$$

The outcome is $m(\mu_\beta) = 4\log(1/u(\beta)) - \log\Upsilon(\beta) + O(\beta^3)$ (Corollary 10.9). For $SU(2)$, $-\log\Upsilon(\beta) = \beta^2/8 + O(\beta^3)$: the gap has no first-order term above $4\log(1/u)$, and the second-order term $+\beta^2/8$. With the fundamental representations, $-\log\Upsilon = -\beta/6 + \beta^2/72 + O(\beta^3)$ for $SU(3)$, $-\beta^2/128 + O(\beta^3)$ for $SU(4)$ and $O(\beta^3)$ for $SU(N)$, $N \geq 5$; for $U(1)$ the maximum is attained by the C-odd function $\sqrt{2}\sin\theta$, and $-\log\Upsilon = \beta^2/8 + O(\beta^3)$.

The upper bound (Theorem 10.2) is the argument of Section 5 for the best plaquette function, with the mean subtracted exactly; its error is $O(\beta^5)$. The lower bound (Theorem 10.8) has four ingredients. (a) The caps are summed into the tube, which they do not change as a set of links (Lemma 10.3). (b) Between the two observables only the swap-odd sector of the duplicated system contributes (Lemma 10.4); there the dressed tube has norm $u^4\Upsilon(1+O(\beta^6))$ (Lemma 10.6), and the shifts and jumps vanish. (c) A slice can be bounded on the odd sector through any one of its pieces, the others being taken at their full norm times a factor C_ϖ of order $1/\beta$ (Lemma 10.7). (d) Along the path of junctions, the factor z of the generating function is charged only to the slices that the path reaches for the first time; at a slice it reaches again, a piece of the tube costs $C_\varpi u^4(1+\gamma_g) = O(\beta^3)$, which pays for the factors C_ϖ. The constant in the $O(\beta^3)$ of the lower bound is large (Remark 10.10(iii)).

The setting is that of Section 9: ρ is irreducible and non-trivial, with the notation φ_{ab}, α_ρ, γ_ρ, u_2, σ_6, $e_{aa'}$, $\hat{u}_{aa'}$, D, B, $\mathfrak{m}$, v there. We write $\langle h\rangle := \int h\, dU$ and

$$\mathsf{V} := P_\rho (M_{1+f_\beta} - |f_\beta\rangle\langle f_\beta|) P_\rho,$$

an operator on the span of χ_ρ and $\chi_{\bar\rho}$ in $L^2_{\mathrm{cl}}(G)$.

Lemma 10.1 (The plaquette variance). *(i) For real ψ in the span of $\chi_\rho, \chi_{\bar\rho}$, $\langle \psi, \mathsf{V}\psi\rangle = \operatorname{Var}_\beta(\psi) := \langle \psi^2(1+f_\beta)\rangle - \langle \psi(1+f_\beta)\rangle^2$, the variance of $\psi(U)$ when U has the density $w_\beta/c_0 = 1 + f_\beta$. The operator V is positive and commutes with complex conjugation, and its largest eigenvalue is $\Upsilon(\beta)$, with a real eigenvector. If $\iota = 1$, the eigenvector is χ_ρ and $\Upsilon = 1 + \kappa - N^2u^2$. If $\iota = 0$, the eigenvectors are $\sqrt{2}\operatorname{Re}\chi_\rho$ and $\sqrt{2}\operatorname{Im}\chi_\rho$, with the eigenvalues $\Upsilon_+ := 1 + \kappa - 2N^2u^2$ and $\Upsilon_- := 1 + \alpha_\rho - \gamma_\rho$, and $\Upsilon = \max(\Upsilon_+, \Upsilon_-)$.*

(ii) With $y := \frac{\beta}{N}\operatorname{Re}\chi_\rho$, uniformly in the real unit vectors ψ of this span, $\operatorname{Var}_\beta(\psi) = 1 + \langle y\psi^2\rangle + \frac{1}{2}\langle y^2\psi^2\rangle - \frac{1}{2}\langle y^2\rangle - \langle y\psi\rangle^2 + O(\beta^3)$.

(iii) For the fundamental representations: $SU(2)$: $-\log\Upsilon = \frac{\beta^2}{8} + O(\beta^3)$. $U(1)$: $-\log\Upsilon_+ = \frac{3}{8}\beta^2 + O(\beta^3)$ and $-\log\Upsilon_- = \frac{\beta^2}{8} + O(\beta^3)$, so $\Upsilon = \Upsilon_-$ for small β. $SU(3)$: $-\log\Upsilon =$

$-\log\Upsilon_+ = -\frac{\beta}{6} + \frac{\beta^2}{72} + O(\beta^3)$. *SU(4):* $-\log\Upsilon = -\log\Upsilon_+ = -\frac{\beta^2}{128} + O(\beta^3)$. *SU(N), $N \geq 5$:* $-\log\Upsilon = O(\beta^3)$.

Proof. (i) As $\langle\psi\rangle = 0$, $\langle f_\beta\psi\rangle = \langle(1+f_\beta)\psi\rangle$, and $\langle\psi, \mathsf{V}\psi\rangle = \langle(1+f_\beta)\psi^2\rangle - \langle f_\beta\psi\rangle^2$ is the variance. The three factors of V commute with complex conjugation (f_β is real), so the real and imaginary parts of an eigenvector are eigenvectors, and $\mathsf{V} \geq 0$ because it is non-negative on real vectors. If $\iota = 1$ the span is that of the real function χ_ρ, and $\mathrm{Var}_\beta(\chi_\rho) = 1 + \langle\chi_\rho^2 f_\beta\rangle - \langle f_\beta\chi_\rho\rangle^2 = 1 + \kappa - N^2u^2$ (Lemma 9.1(i); $\langle f_\beta\chi_\rho\rangle = d_{\bar\rho}u_{\bar\rho} = Nu$). If $\iota = 0$, in the orthonormal basis $\chi_\rho, \chi_{\bar\rho}$ the operator $P_\rho M_{1+f}P_\rho$ has the diagonal entries $1+\alpha_\rho$ and the off-diagonal entries γ_ρ (Lemma 9.1(i)), and $P_\rho f_\beta = Nu(\chi_\rho + \chi_{\bar\rho})$; so V has the diagonal entries $1 + \alpha_\rho - N^2u^2$ and the off-diagonal entries $\gamma_\rho - N^2u^2$, the eigenvectors $\chi_\rho \pm \chi_{\bar\rho}$ and the eigenvalues $1 + \alpha_\rho + \gamma_\rho - 2N^2u^2 = 1 + \kappa - 2N^2u^2$ and $1 + \alpha_\rho - \gamma_\rho$.

(ii) $w_\beta = e^y$, $|y| \leq \beta$, $\langle y\rangle = \langle\psi\rangle = 0$, $\langle\psi^2\rangle = 1$ and $|\psi| \leq \sqrt{2}N$. Taylor's formula with $|e^y - 1 - y - \frac{1}{2}y^2| \leq \frac{1}{6}\beta^3 e^\beta$ gives $\langle e^y\psi^2\rangle = 1 + \langle y\psi^2\rangle + \frac{1}{2}\langle y^2\psi^2\rangle + O(\beta^3)$, $\langle e^y\psi\rangle = \langle y\psi\rangle + O(\beta^2)$ and $c_0 = \langle e^y\rangle = 1 + \frac{1}{2}\langle y^2\rangle + O(\beta^3)$; insert these in $\mathrm{Var}_\beta(\psi) = \langle e^y\psi^2\rangle/c_0 - \langle e^y\psi\rangle^2/c_0^2$.

(iii) We use (ii) and, for $SU(N)$ with $N \geq 2$ and $\chi := \chi_\rho$: $\langle\chi^j\bar\chi^k\rangle = 0$ unless $j \equiv k$ modulo N (the centre, as in the proof of Lemma 9.1(iii)); $\langle|\chi|^2\rangle = 1$; $\langle|\chi^2|^2\rangle = 2$, since $\rho\otimes\rho$ is the sum of its symmetric and antisymmetric parts, which are irreducible and inequivalent; and $\langle\chi^N\rangle = 1$. For the last, the argument of Lemma 9.1(iii) for $SU(3)$ applies to N tensor factors: the diagonal subgroup forces every invariant tensor T to have $T_{i_1\cdots i_N} = 0$ unless $\{i_1, \dots, i_N\} = \{1, \dots, N\}$, the subgroups $SU(2)$ acting on two basis vectors force T to be antisymmetric under the exchange of any two values, and so T is a multiple of the determinant tensor, which is invariant. Then: $SU(2)$, $\psi = \chi$, $y = \frac{\beta}{2}\chi$: $\langle y\psi^2\rangle = 0$, $\frac{1}{2}\langle y^2\psi^2\rangle = \frac{\beta^2}{8}\langle\chi^4\rangle = \frac{\beta^2}{4}$, $\frac{1}{2}\langle y^2\rangle = \frac{\beta^2}{8}$ and $\langle y\psi\rangle^2 = \frac{\beta^2}{4}$, so $\Upsilon = 1 - \frac{\beta^2}{8} + O(\beta^3)$. $U(1)$, $y = \beta\cos\theta$: for $\psi = \sqrt{2}\cos\theta$, $\frac{1}{2}\langle y^2\psi^2\rangle = \beta^2\langle\cos^4\theta\rangle = \frac{3}{8}\beta^2$, $\frac{1}{2}\langle y^2\rangle = \frac{\beta^2}{4}$, $\langle y\psi\rangle^2 = \frac{\beta^2}{2}$ and $\langle y\psi^2\rangle = 0$; for $\psi = \sqrt{2}\sin\theta$, $\frac{1}{2}\langle y^2\psi^2\rangle = \beta^2\langle\cos^2\theta\sin^2\theta\rangle = \frac{\beta^2}{8}$ and $\langle y\psi\rangle = \langle y\psi^2\rangle = 0$. $SU(N)$, $N \geq 3$, $y = \frac{\beta}{N}\mathrm{Re}\,\chi$: expanding the powers of $\mathrm{Re}\,\chi = \frac{1}{2}(\chi + \bar\chi)$ and $\mathrm{Im}\,\chi = \frac{1}{2i}(\chi - \bar\chi)$ gives $\langle(\mathrm{Re}\,\chi)^2\rangle = \frac{1}{2}$, $\langle\mathrm{Re}\,\chi\,\mathrm{Im}\,\chi\rangle = 0$, $\langle(\mathrm{Re}\,\chi)^3\rangle = \frac{1}{4}[N=3]$, $\langle\mathrm{Re}\,\chi(\mathrm{Im}\,\chi)^2\rangle = -\frac{1}{4}[N=3]$, $\langle(\mathrm{Re}\,\chi)^4\rangle = \frac{3}{4} + \frac{1}{8}[N=4]$ and $\langle(\mathrm{Re}\,\chi)^2(\mathrm{Im}\,\chi)^2\rangle = \frac{1}{4} - \frac{1}{8}[N=4]$; so

$$\Upsilon_+ = 1 + \tfrac{\beta}{6}[N=3] + \tfrac{\beta^2}{128}[N=4] + O(\beta^3), \qquad \Upsilon_- = 1 - \tfrac{\beta}{6}[N=3] - \tfrac{\beta^2}{128}[N=4] + O(\beta^3),$$

which gives the values, with $-\log(1+\frac{\beta}{6}) = -\frac{\beta}{6} + \frac{\beta^2}{72} + O(\beta^3)$. □

The upper bound uses the cube identity of Lemma 5.1(ii) for real class functions that need not be invariant under inversion: for such an h with $\sum_r |\hat h_r| d_r < \infty$, $\hat h_r := \int h\,\overline{\chi_r}\,dU$,

$$\int h(U_{p_0})\,h(U_{p_1}) \prod_{s\in X_0} f_\beta(U_s)\,dU = \sum_{r\neq 0} |\hat h_r|^2 u_r^4. \tag{2}$$

Indeed, in the proof of Lemma 5.1(ii) the boundary-oriented holonomies of the bottom and the top face are conjugate to U_{p_0} and U_{p_1}, with exactly one of them inverted (the two faces carry the same orientation in space and have opposite outward normals), and for real h, $h(U^{-1}) = \sum_r \hat h_r\overline{\chi_r(U)} = \sum_r \overline{\hat h_r}\chi_r(U)$, because $\hat h_{\bar r} = \overline{\hat h_r}$. The six-fold sum of that proof then gives $\sum_{r\neq 0}\overline{\hat h_r}\hat h_r u_r^4$.

Theorem 10.2 (The upper bound to second order)**.** *In the setting of Lemma 5.5, let ψ be a real eigenvector of V for the eigenvalue $\Upsilon = \Upsilon(\beta)$ with $\langle\psi^2\rangle = 1$, $a_1 := |\langle f_\beta\psi\rangle|$, $a_2 := (1+\mathcal{T}_p)a_1 + \mathcal{T}_p$, and*

$$V_2 := (1+\mathcal{T}_p)(\Upsilon + a_1^2) + \mathcal{T}_p - \big((1-\mathcal{T}_p)a_1 - \mathcal{T}_p\big)_+^2,$$
$$L_2 := \frac{1-\mathcal{T}_Q}{1+\epsilon^6}\Big(\Upsilon - \mathcal{T}_p(1+a_1)\sqrt{2-\iota}\,Nu\Big)_+^2 - (1+a_2^2)\,e^{16\beta}\frac{N^4}{\tau_0^2}\,\mathcal{T}_Q,$$

where $x_+ := \max(x,0)$. *If* $L_2 > 0$, *then (i) and (ii) of Theorem 5.6 hold with* $D_2(\beta) := \log(V_2/L_2)$ *in place of* $D(\beta)$. *As* $\beta \downarrow 0$, $D_2(\beta) = -\log \Upsilon(\beta) + O(\beta^5)$.

Proof. Follow the proofs of Lemma 5.5 and Theorem 5.6 with $A := \psi(U_{p_0})$ and $A_1 := \psi(U_{p_1})$ in place of $\operatorname{Re}\chi_\rho(U_{p_0})$ and $\operatorname{Re}\chi_\rho(U_{p_1})$; A is a real, bounded, continuous and gauge-invariant function of the links of p_0. By (1), $|R_{S_A} - 1| \le \mathcal{T}_p$, $\langle\psi\rangle = 0$ and $\langle|\psi|\rangle \le 1$, the mean $a := \langle A\rangle_{\Lambda,\beta}$ satisfies $|a - R_{S_A}\langle(1+f_\beta)\psi\rangle| \le \mathcal{T}_p$, where $\langle(1+f_\beta)\psi\rangle = \langle f_\beta\psi\rangle$; hence $(1-\mathcal{T}_p)a_1 - \mathcal{T}_p \le |a| \le a_2$ and $|a - \langle f_\beta\psi\rangle| \le \mathcal{T}_p(1+a_1)$. Likewise $\langle A^2\rangle_{\Lambda,\beta} \le (1+\mathcal{T}_p)\langle(1+f_\beta)\psi^2\rangle + \mathcal{T}_p = (1+\mathcal{T}_p)(\Upsilon + a_1^2) + \mathcal{T}_p$, by Lemma 10.1(i). So $\operatorname{Var}(A) = \langle A^2\rangle - a^2 \le V_2$. By Lemma 5.2, $\operatorname{Cov}(A, A_1) \ge \langle H_Q\rangle$ with $H_Q := (A-a)(A_1-a)\prod_{s\in X_0}\varphi_\beta(U_s)$, and by (2), with $\tilde a := (\psi - a)(1+f_\beta)$,

$$M_Q = \sum_{r\neq 0} |\hat{\tilde a}_r|^2 u_r^4 \;\ge\; u^4 \|P_\rho \tilde a\|^2, \qquad P_\rho \tilde a = P_\rho M_{1+f_\beta}\psi - a\,P_\rho f_\beta = \mathsf{V}\psi + (\langle f_\beta\psi\rangle - a)P_\rho f_\beta,$$

because $P_\rho\psi = \psi$ and $P_\rho \mathbf{1} = 0$. As $\mathsf{V}\psi = \Upsilon\psi$ and $\|P_\rho f_\beta\| = \sqrt{2-\iota}\,Nu$, $\|P_\rho\tilde a\| \ge \Upsilon - \mathcal{T}_p(1+a_1)\sqrt{2-\iota}\,Nu$. As in the proof of Lemma 5.5, $\int|H_Q| \le \langle(\psi-a)^2\rangle(\int|\varphi_\beta|)^4 \le (1+a_2^2)e^{16\beta}N^4u^4/\tau_0^2$ and $R_{S_Q} \ge (1-\mathcal{T}_Q)/(1+\epsilon^6)$, so $\langle H_Q\rangle \ge R_{S_Q}M_Q - \mathcal{T}_Q\int|H_Q| \ge u^4 L_2$. The rest of the proof of Theorem 5.6 gives $m \le \log(\operatorname{Var}(A)/\operatorname{Cov}(A,A_1)) \le 4\log(1/u) + D_2$. As $\beta\downarrow 0$: $\mathcal{T}_p, \mathcal{T}_Q = O(\beta^5)$, $\epsilon^6 = O(\beta^6)$ and $a_1 \le \sqrt{2-\iota}\,Nu = O(\beta)$, so $V_2 = \Upsilon + O(\beta^5)$ and $L_2 = \Upsilon^2 + O(\beta^5)$. □

We turn to the lower bound. Let S be the exchange of the two copies, $(Sh)(U,U') := h(U',U)$, on $L^2(\Omega_s\times\Omega_s)$, and $P_\pm := \frac12(1\pm S)$; for a set E of spatial links, S_E and $P_\pm^E$ are defined in the same way on $L^2(G^E\times G^E)$. A *cap* of X is a junction J as in Lemma 8.6(ii): $M(J) = \{\{q\}\}$ and $\Phi(J) = \{(\partial q, s-1), (\partial q, s)\}$. For a plaquette q of a slice let

$$D_q := (G_g^{1/2})^{\otimes\partial q}\, M_{1+g_q}\, (G_g^{1/2})^{\otimes\partial q},$$

the *dressed tube*, on $L^2(G^{\partial q}\times G^{\partial q})$; it is the sum of the operator $G_g^{\otimes\partial q}$ of an untouched front ∂q and the operator O_J of a cap on q (Lemma 9.2).

Lemma 10.3 (Summing the caps)**.** *Let* X *be as in Definition 7.1, and* X° *the set obtained by removing the plaquettes of all caps of* X. *Then* X° *is as in Definition 7.1, has no cap, the same temporal plaquettes as* X, *and* $r_{X^\circ} = r_X$; *it is in* $\mathcal{X}_S$ *with a component meeting* $\mathcal{P}_{S_A}$ *and* $\mathcal{P}_{S_B}$ *if and only if* X *is. For a set* X' *as in Definition 7.1 without caps, let* $\mathcal{U}_s(X')$ *be the set of plaquettes* q *of slice* s *whose boundary* ∂q *is a front of* X' *untouched at* s. *The sets* X *with* $X^\circ = X'$ *are the sets* $X'\cup C$ *with* $C \subseteq \bigcup_s \mathcal{U}_s(X')$, *and*

$$\sum_{X:\ X^\circ = X'} \Phi_F(X) = \operatorname{Tr}\prod_s \mathbf{P}\,\mathcal{O}_s^d\,\mathbf{P}, \qquad \mathcal{O}_s^d := \tilde{\mathcal{K}}_{s-1}^{1/2}\,\mathbb{M}_s \prod_{q\in\mathcal{U}_s(X')}(1+g_q)\,\tilde{\mathcal{K}}_s^{1/2},$$

with $\tilde{\mathcal{K}}_s^{1/2}$ *and* $\mathbb{M}_s$ *those of* X' *(proof of Lemma 9.2). In the factorisation of Lemma 9.2,* $\mathbf{P}\mathcal{O}_s^d\mathbf{P}$ *is the tensor product of the operators of the pieces of* X' *at* s, *in which the untouched fronts* ∂q, $q\in\mathcal{U}_s(X')$, *carry* $P^qD_qP^q$ *instead of* $P^qG_g^{\otimes\partial q}P^q$.

Proof. Let J be a cap at s on q. Every link of ∂q at time s lies in the temporal plaquettes based on it in the layers $s-1$ and s, which belong to X and share that link; so removing q leaves no free link, does not change $E(X)$ (hence $\overline{X}$ and r_X) and does not change the components of X except by removing q from one of them. As J is a component of H_s, no other marker at slice s, in particular no support, shares a vertex with ∂q; so $q\notin\mathcal{P}_S$, the junctions other than J do not change (those at other slices do not depend on the plaquettes of slice s), and after the removal ∂q is a front of both layers untouched at s. Hence removing all caps creates no new cap, and X° has the stated properties. Conversely, let X' be without caps and $q\in\mathcal{U}_s(X')$. No

marker of X' at slice s shares a vertex with ∂q, and the fronts of the layers $s-1$ and s other than ∂q are vertex-disjoint from it; so in $X' \cup \{q\}$ the plaquette q is a cluster by itself, and $\{\{q\}, (\partial q, s-1), (\partial q, s)\}$ is a junction, a cap, while all other junctions are those of X'. Adding several such plaquettes at once acts in the same way, one slice and one front at a time. So $C \mapsto X' \cup C$ is a bijection onto $\{X : X^\circ = X'\}$. In the representation $\Phi_F(X) = \operatorname{Tr} \prod_s \mathbf{P}\mathcal{O}_s\mathbf{P}$ of the proof of Lemma 9.2, $X' \cup C$ differs from X' only by the factors g_q, $q \in C$, in $\mathbb{M}_s$; summing over C gives the factors $1 + g_q$. On the links of ∂q the factors $\tilde{\mathcal{K}}^{1/2}_{s-1}$ and $\tilde{\mathcal{K}}^{1/2}_s$ of X' are $(G_g^{1/2})^{\otimes \partial q}$, which gives D_q, and the factorisation of the proof of Lemma 9.2 applies unchanged. □

Lemma 10.4 (The swap sectors)**.** *Let X' be as in Lemma 10.3, write $\hat{\mathcal{O}}_s := \mathbf{P}\mathcal{O}_s^d\mathbf{P}$, and suppose that some outer layer contains no temporal plaquette of X'. Then*

$$\operatorname{Tr} \prod_s \hat{\mathcal{O}}_s = \operatorname{Tr}\Big(\hat{\mathcal{O}}_0 \prod_{s=1}^{t-1} (P_- \hat{\mathcal{O}}_s P_-)\, \hat{\mathcal{O}}_t \prod_{s=t+1}^{L_0-1} \hat{\mathcal{O}}_s\Big),$$

and $|\operatorname{Tr} \prod_s \hat{\mathcal{O}}_s| \le \prod_{s=1}^{t-1} \|P_-\hat{\mathcal{O}}_s P_-\| \prod_{s \notin \{1,\dots,t-1\}} \|\hat{\mathcal{O}}_s\|$. If some inner layer contains no temporal plaquette, the same holds with P_- at the slices $t+1, \dots, L_0-1$ instead of $1, \dots, t-1$.

Proof. S commutes with $\mathbf{P}$, with $G_g^{1/2}$, $\Pi_0 \otimes \Pi_0$ and the multiplications by g_p and $1+g_q$, which are invariant under the exchange of the copies; so $S\hat{\mathcal{O}}_s S = \hat{\mathcal{O}}_s$ for $s \notin \{0, t\}$, while $S\hat{\mathcal{O}}_0 S = -\hat{\mathcal{O}}_0$ and $S\hat{\mathcal{O}}_t S = -\hat{\mathcal{O}}_t$, because of the factors $A(U^{(0)}) - A(U'^{(0)})$ and $B(U^{(t)}) - B(U'^{(t)})$. Write $\operatorname{Tr} \prod_s \hat{\mathcal{O}}_s = \operatorname{Tr}(\hat{\mathcal{O}}_0 \mathcal{X} \hat{\mathcal{O}}_t \mathcal{Y})$ with $\mathcal{X} := \prod_{s=1}^{t-1} \hat{\mathcal{O}}_s$ and $\mathcal{Y} := \prod_{s=t+1}^{L_0-1} \hat{\mathcal{O}}_s$, and insert $1 = P_+ + P_-$ after $\hat{\mathcal{O}}_0$. As $\mathcal{X}$ commutes with S and $P_+\hat{\mathcal{O}}_t = \hat{\mathcal{O}}_t P_-$, the term with P_+ is $\operatorname{Tr}(\hat{\mathcal{O}}_0 \mathcal{X} \hat{\mathcal{O}}_t P_- \mathcal{Y})$. Let s_* be an outer layer without temporal plaquettes. Then $\tilde{\mathcal{K}}^{1/2}_{s_*}$ is the projection Π onto the constants, which is invariant under S, so that $P_-\Pi = \Pi P_- = 0$; it is the last factor of $\mathcal{O}^d_{s_*}$, which is $\mathcal{O}^d_t$ if $s_* = t$ and a factor of $\mathcal{Y}$ otherwise, and the factors between P_- and it commute with S. So the term with P_+ vanishes, and $P_-\mathcal{X} = P_-\mathcal{X}P_- = \prod_{s=1}^{t-1} P_-\hat{\mathcal{O}}_s P_-$. The bound follows by Hölder's inequality for traces, as in the proof of Lemma 9.2: the factor $\hat{\mathcal{O}}_{s_*+1} = \Pi \hat{\mathcal{O}}_{s_*+1}$, which is not among the $P_-\hat{\mathcal{O}}_s P_-$, has trace norm at most its norm. If an inner layer s_* is empty, insert $P_+ + P_-$ after $\hat{\mathcal{O}}_t$ instead; by cyclicity the term with P_+ is $\operatorname{Tr}(\hat{\mathcal{O}}_0 P_- \mathcal{X} \hat{\mathcal{O}}_t \mathcal{Y})$, which vanishes because Π is the first factor of $\mathcal{O}^d_{s_*+1}$. □

Lemma 10.5 (Shifts and jumps)**.** *Let J be a junction without support at a slice s whose clusters consist of two distinct plaquettes q, q' and whose front-pairs are $(\partial q, s-1)$ and $(\partial q', s)$ (a shift or a jump, Lemma 8.6(iii)). Then*

$$P^J O_J P^J = |\xi\rangle\langle\xi'|, \qquad \xi := \tilde{\mathcal{K}}^{1/2}_{J,s-1}\, g_q, \quad \xi' := \tilde{\mathcal{K}}^{1/2}_{J,s}\, g_{q'}.$$

The vectors ξ, ξ' are invariant under $S_{E(J)}$, so $P_-^{E(J)} P^J O_J P^J P_-^{E(J)} = 0$; and $\|P^J O_J P^J\| = \|\xi\|\|\xi'\| = 2\sigma_6 + \sigma_6^2$.

Proof. Here $O_J = \tilde{\mathcal{K}}^{1/2}_{J,s-1} M_{g_q g_{q'}} \tilde{\mathcal{K}}^{1/2}_{J,s}$, where $\tilde{\mathcal{K}}^{1/2}_{J,s-1}$ is $G_g^{1/2}$ on the links of ∂q and $\Pi_0 \otimes \Pi_0$ on those of $\partial q' \setminus \partial q$, and $\tilde{\mathcal{K}}^{1/2}_{J,s}$ vice versa. Let ϑ, ϑ' be in the range of P^J. The function $h := \tilde{\mathcal{K}}^{1/2}_{J,s-1}\vartheta$ does not depend on the links of $\partial q' \setminus \partial q$ and, as the $\tilde{\mathcal{K}}^{1/2}$ commute with gauge transformations, it is invariant under those at the corners of q; so it is a function of (U_q, U'_q) (proof of Lemma 9.3). Likewise $h' := \tilde{\mathcal{K}}^{1/2}_{J,s}\vartheta'$ is a function of $(U_{q'}, U'_{q'})$. The plaquettes $q \neq q'$ share at most one link, so integrating first over a link of q not in q' shows that (U_q, U'_q) is Haar distributed and independent of $(U_{q'}, U'_{q'})$. Hence $\langle\vartheta, O_J\vartheta'\rangle = \langle h, g_q g_{q'} h'\rangle = \langle h, g_q\rangle\langle g_{q'}, h'\rangle = \langle\vartheta, \xi\rangle\langle\xi', \vartheta'\rangle$, as g is real and the $\tilde{\mathcal{K}}^{1/2}$ are self-adjoint; and ξ, ξ' lie in the range of P^J. They are invariant

under the exchange of the copies because g and G_g are. Finally, in the notation of the proof of Lemma 9.3, $\xi = \sum_{(a,a')\neq(0,0)} \hat u^2_{aa'}\langle e_{aa'}, g\rangle e_{aa'}$ with $\langle e_{a0}, g\rangle = \langle e_{0a}, g\rangle = d_a u_a$ and $\langle e_{aa'}, g\rangle = d_a u_a d_{a'} u_{a'}$ for $a, a' \neq 0$, so $\|\xi\|^2 = 2\sigma_6 + \sigma_6^2$; the same holds for ξ'. □

Lemma 10.6 (The dressed tube)**.** *Suppose $u_* = u$, and let P^q be the gauge projection at the corners of q. Then $P^q D_q P^q \geq 0$ commutes with $S_{\partial q}$, $\|P^q D_q P^q\| \leq u^4(1+\gamma_g)$, and $\|P_-^{\partial q} P^q D_q P^q P_-^{\partial q}\| \leq u^4 \bar\Upsilon$, where*

$$n_B := \frac{u_2^2}{u^2}\,\epsilon + \sqrt{2-\iota}\,\frac{N}{u}\big(u_2^5(e^\beta - 1)\big)^{1/2},$$
$$\upsilon_1 := \Upsilon + \frac{n_B^2}{\Upsilon - (u_2/u)^4 e^\beta}, \qquad \bar\Upsilon := \upsilon_1 + \frac{2e^{2\beta}\sigma_6}{\upsilon_1 - u^4(1+\gamma_g)},$$

provided the two denominators are positive. The compression of $P^q D_q P^q$ to the span of $\frac{1}{\sqrt2}(e_{a0} - e_{0a})$, $a \in \{\rho, \bar\rho\}$, is $u^4\mathsf{V}$ (in the identification $\frac{1}{\sqrt2}(e_{a0} - e_{0a}) \leftrightarrow \chi_a$). As $\beta \downarrow 0$, $\bar\Upsilon = \Upsilon(\beta)(1 + O(\beta^6))$.

Proof. As in the proof of Lemma 9.3, $P^q D_q P^q$ is the operator $\Lambda_{\hat u} M_{1+g} \Lambda_{\hat u}$ on $\mathcal{H} = L^2_{\mathrm{cl}}(G) \otimes L^2_{\mathrm{cl}}(G)$, $\Lambda_{\hat u} e_{aa'} = \hat u^2_{aa'} e_{aa'}$. It is positive because $1 + g > 0$, it commutes with $S_{\partial q}$, which acts by $e_{aa'} \mapsto e_{a'a}$, and its norm is at most $u^4\|1+g\|_\infty \leq u^4(1+\gamma_g)$. Its matrix elements are $\hat u^2_{aa'} m_{ab} m_{a'b'} \hat u^2_{bb'}$, so, with the blocks of the proof of Lemma 9.3 and $\mathsf{m} = 1+\mathsf{B}$: $\mathsf{D}(1+\mathsf{B})\mathsf{D}$ on $\mathcal{H}_1$ and on $\mathcal{H}_2$; the rank-one operator $vv^{\mathsf T}$ between $\mathcal{H}_1$ and $\mathcal{H}_2$; $e_{bb'} \mapsto v_{b'} \sum_a (\mathsf{D}\mathsf{m}\mathsf{D})_{ab} e_{a0}$ from $\mathcal{H}_{12}$ to $\mathcal{H}_1$, and its mirror image to $\mathcal{H}_2$; and $(\mathsf{D}\otimes\mathsf{D})(\mathsf{m}\otimes\mathsf{m})(\mathsf{D}\otimes\mathsf{D})$ on $\mathcal{H}_{12}$, of norm at most $u^8(1+\gamma_g)$. The odd sector is $\mathcal{H}_1^- \oplus \mathcal{H}_{12}^-$, where $\mathcal{H}_1^-$ has the orthonormal basis $e_a^- := \frac{1}{\sqrt2}(e_{a0} - e_{0a})$, $a \neq 0$, and $\mathcal{H}_{12}^-$ consists of the antisymmetric vectors of $\mathcal{H}_{12}$. In the basis (e_a^-) the operator acts on $\mathcal{H}_1^-$ as $\mathsf{D}(1+\mathsf{B})\mathsf{D} - vv^{\mathsf T}$: the switch enters with the sign -1 on antisymmetric vectors. Its compression to the span of $e_\rho^-, e_{\bar\rho}^-$ is $u^4\big(P_\rho M_{1+f} P_\rho - |P_\rho f_\beta\rangle\langle P_\rho f_\beta|\big) = u^4\mathsf{V}$, since $P_\rho v = u^2 P_\rho f_\beta$ in the identification $e_a^- \leftrightarrow \chi_a$; its largest eigenvalue is $u^4\Upsilon$. Write $\mathsf{D} = u^2 P_\rho + \mathsf{D}'$ with $\|\mathsf{D}'\| \leq u_2^2$ and $P_\rho \mathsf{D}' = 0$. The block from the complement of $\rho, \bar\rho$ to $\rho, \bar\rho$ is $u^2 P_\rho \mathsf{B}\mathsf{D}' - (P_\rho v) v'^{\mathsf T}$, v' the part of v outside $\rho, \bar\rho$, of norm at most $u^4 n_B$, since $\|\mathsf{B}\| \leq \epsilon$, $\|P_\rho v\| = \sqrt{2-\iota}\,N u^3$ and $\|v'\|^2 \leq u_2^5 \sum_{a\neq 0} d_a^2 u_a \leq u_2^5(e^\beta - 1)$ (proof of Lemma 9.1(iv)); the block on the complement is at most $\mathsf{D}'(1+\mathsf{B})\mathsf{D}' \leq u_2^4 e^\beta$. For a self-adjoint operator with blocks $\left(\begin{smallmatrix} A & B \\ B^* & C\end{smallmatrix}\right)$, $A \leq c_1$, $C \leq c_2 < c_1$ and $\|B\| \leq n$, the largest eigenvalue is at most the largest eigenvalue of $\left(\begin{smallmatrix} c_1 & n \\ n & c_2\end{smallmatrix}\right)$, which is at most $c_1 + n^2/(c_1 - c_2)$. So the operator on $\mathcal{H}_1^-$ is at most $u^4\upsilon_1$. On $\mathcal{H}_{12}^-$ it is at most $u^8(1+\gamma_g)$, and the block from $\mathcal{H}_{12}^-$ to $\mathcal{H}_1^-$ maps an antisymmetric $y = \sum y_{bb'} e_{bb'}$ to $\sqrt2 \sum_a ((\mathsf{D}\mathsf{m}\mathsf{D}) Y v)_a e_a^-$, $Y := (y_{bb'})$, so its norm is at most $\sqrt2 u^4 e^\beta \sigma_6^{1/2}$ (as $\|Y\| \leq \|y\|$ and $\|v\|^2 = \sigma_6$). The same inequality gives $\bar\Upsilon$. As $\beta \downarrow 0$: $u_2 = O(\beta^2)$ and $u \geq c\beta$ (Lemma 9.1(iv), Lemma 5.1(iii)), so $n_B = O(\beta^3)$, $(u_2/u)^4 = O(\beta^4)$ and $\sigma_6 = O(\beta^6)$, while $\Upsilon = 1 + O(\beta)$. □

From now on the pieces at a slice are those of Lemma 9.2 for a set X' without caps, an untouched plaquette front ∂q at s carrying D_q (Lemma 10.3); for a piece π let Y_π be its operator with the gauge projection P^π, $\nu(\pi) := \|Y_\pi\|$ and $\nu^\pm(\pi) := \|P_\pm^{E(\pi)} Y_\pi P_\pm^{E(\pi)}\|$. Let $\varpi \in (0, 1]$ and

$$C_\varpi := 1 + \max\Big(\frac{1}{\varpi},\ \frac{1+\gamma_g}{\bar\Upsilon}\Big), \qquad w_\varpi(\pi) := \begin{cases} u^4\bar\Upsilon & \text{if } \pi \text{ is a dressed tube } D_q, \\ \varpi\,\nu(\pi) & \text{if } \pi \text{ is a junction as in Lemma 10.5}, \\ \nu(\pi) & \text{otherwise.}\end{cases}$$

Then $\nu(\pi) \leq C_\varpi w_\varpi(\pi)$ for every piece, by Lemma 10.6 for the dressed tubes.

Lemma 10.7 (A slice on the odd sector)**.** *Suppose $u_* = u$. Let $s \notin \{0, t\}$ be a slice, $\mathfrak{P}_s$ the set of pieces at s, and $\mu \in \mathfrak{P}_s$. Then*

$$\|P_- \mathbf{P} \mathcal{O}_s^d \mathbf{P} P_-\| \;\le\; w_\varpi(\mu) \prod_{\pi \in \mathfrak{P}_s \setminus \{\mu\}} C_\varpi\, \nu(\pi).$$

Proof. By Lemma 10.3, $\mathbf{P}\mathcal{O}_s^d\mathbf{P} = \bigotimes_\pi Y_\pi \otimes (\Pi_0 \otimes \Pi_0)^{\otimes \mathcal{R}}$ and $S = \bigotimes_\pi S_{E(\pi)} \otimes S_{\mathcal{R}}$, where $S_{\mathcal{R}}$ acts trivially on the range of $(\Pi_0 \otimes \Pi_0)^{\otimes \mathcal{R}}$, which consists of constants. Each Y_π commutes with $S_{E(\pi)}$, since no support lies at s. Hence $P_-\mathbf{P}\mathcal{O}_s^d\mathbf{P}P_-$ is the direct sum, over the sign patterns $(\varepsilon_\pi)_{\pi \in \mathfrak{P}_s}$ with $\prod_\pi \varepsilon_\pi = -1$, of the operators $\bigotimes_\pi P_{\varepsilon_\pi}^{E(\pi)} Y_\pi P_{\varepsilon_\pi}^{E(\pi)} \otimes (\Pi_0 \otimes \Pi_0)^{\otimes \mathcal{R}}$, and its norm is the largest of the products $\prod_\pi \nu^{\varepsilon_\pi}(\pi)$. In a pattern with $\varepsilon_\mu = +1$ some other piece exists. So the norm is at most $\max\big(\nu^-(\mu), [|\mathfrak{P}_s| \ge 2]\,\nu(\mu)\big) \prod_{\pi \ne \mu} \nu(\pi)$. If μ is a dressed tube, $\nu^-(\mu) \le u^4 \bar\Upsilon$ and $\nu(\mu) \le u^4(1+\gamma_g)$ (Lemma 10.6); if μ is as in Lemma 10.5, $\nu^-(\mu) = 0$ and $[|\mathfrak{P}_s| \ge 2] \le \varpi(1 + 1/\varpi)^{|\mathfrak{P}_s|-1}$; otherwise the maximum is at most $\nu(\mu)$. In the first case $\max(\cdot) \le u^4 \bar\Upsilon\,(1 + (1+\gamma_g)/\bar\Upsilon)^{|\mathfrak{P}_s|-1}$. □

Theorem 10.8 (The lower bound to second order)**.** *Suppose $u_* = u$ and that the two denominators in Lemma 10.6 are positive, and let $\varpi \in (0,1]$, $\omega := C_\varpi a^4(1+\gamma_g) < 1$ and $b_2 := e^{4(K+1)\theta_5}(2\sigma_6 + \sigma_6^2)/u^4$. In Definition 8.9 and Lemma 8.10 replace c_J by $e^{h_J}/(1-\omega)$ and the third condition of Lemma 8.10 by $C_\varpi k P_v Y_0 a^{k/2} e^{\xi_F k} \le h_J$ $(k \ge 4)$, and let Σ_4 be as there with these constants. Let $\Sigma_4^{(2)}$ be Σ_4 without its first term b and with its second term $(4P_v - 4 - K_S)b^2$ replaced by $(4P_v - 4 - K_S)\varpi \min(b^2, b_2)$. Let $\delta' > 0$ satisfy*

$$\Sigma_4^{(2)} < \bar\Upsilon\delta', \qquad P_v(g_C Y_1/5 + g_F Y_0) < F(\bar\Upsilon(1+\delta') - a),$$
$$\omega + \frac{C_\varpi a^4(2\bar\Upsilon(1+\delta') + \Sigma_4)}{1-\omega} \le 1, \qquad a^4\bar\Upsilon(1+\delta') \le \varpi.$$

Then for every periodic lattice with all spatial sides ≥ 5 there is $C_{A,B}$, independent of t and L_0, with

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A,B)| \le C_{A,B}\big(a^4\bar\Upsilon(1+\delta')\big)^t + \frac{2\|A\|_\infty \|B\|_\infty c^{|S|}}{1-\hat q}\,\hat q^{\,4L_0},$$

so $m_{\Lambda_s}(\beta) \ge 4\log(1/u) - 8(K+1)\theta_5 - \log\bar\Upsilon - \log(1+\delta')$; with $N_4^v = P_v$ and $N_5^v = 0$ the same holds for $m(\mu_\beta)$. As $\beta \downarrow 0$ the hypotheses hold with $\varpi := b$ and some $\delta' = O(\beta^3)$, so

$$m(\mu_\beta) \;\ge\; 4\log\frac{1}{u(\beta)} - \log\Upsilon(\beta) - O(\beta^3).$$

Proof. Reductions. As in the proof of Theorem 8.11, the sets with a temporal plaquette in every layer give the second term. The other sets X are summed over the caps (Lemma 10.3): it suffices to bound $\sum_{X'} r_{X'}^2 |\operatorname{Tr}\prod_s \hat{\mathcal{O}}_s|$ over the sets X' without caps, as in Lemma 10.3, that are in $\mathcal{X}_S$ with a component meeting $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Suppose X' has an empty outer layer (the other case is treated at the end of the proof). We use the representation of Definition 7.1 in which the slice of S_A is 0 and that of S_B is t, so that the inner slices are $1, \dots, t-1$. Choose the path $\pi(X') = (J_0, \sigma_1, J_1, \dots, \sigma_r, J_r)$, $J_0 = J_A$, $J_r = J_B$, and the forest as in the proof of Theorem 8.11; the dressed tubes are pieces of segments. The slices of a junction are its slice, those of a segment its interior slices; consecutive elements of $\pi(X')$ have adjacent sets of slices, so the slices of $J_0, \sigma_1, \dots, J_i$ form an interval $I_i \ni 0$, and every slice of $[0,t]$ is a slice of some element of $\pi(X')$ (Lemma 8.4). A slice s of an element other than J_0 is *new* for it if s exceeds every slice of the earlier elements. Every slice of $[1,t]$ is new for exactly one element, and no slice $s \le 0$ is new. At a new slice the piece of that element is the *main piece* $\mu(s)$; all other pieces are not main.

Weights. Put $z := 1/(a^4\bar\Upsilon(1+\delta'))$. By Lemma 10.4, Lemma 10.7 at the slices $1,\dots,t-1$ and Lemma 9.2 elsewhere,

$$z^t r_{X'}^2 \Big|\mathrm{Tr}\prod_s \hat{\mathcal{O}}_s\Big| \le 4\|A\|_\infty\|B\|_\infty e^{2\theta_5 J_S}\, C_\varpi\, W^{(2)}(X'),$$

where $W^{(2)}$ is the product of the following weights, each plaquette of X' carrying a factor $e^{2(K+1)\theta_5}$, distributed as in Section 8. A main piece carries $zw_\varpi(\mu)$ and every other piece $C_\varpi\nu(\pi)$, bounded as in Section 8 (an untouched front by $u^{|\varphi|}$, a dressed tube by $u^4(1+\gamma_g)$, a junction by $w(J)$ before the factors $e^{2(K+1)\theta_5}$). This uses $z^t = \prod_{s=1}^t z$, one factor for each slice of $[1,t]$; at the slice t, where Lemma 10.7 is not available, $\nu(\mu)\le C_\varpi w_\varpi(\mu)$, which gives the factor C_ϖ; and at a new slice $s>t$, $\nu(\mu)\le zw_\varpi(\mu)$: by the last hypothesis for a junction as in Lemma 10.5, by $a^4(1+\gamma_g)(1+\delta')\le 1$ (a consequence of the third hypothesis) for a dressed tube, and by $z\ge 1$ (a consequence of the last hypothesis) for the other pieces. With the factors $e^{2(K+1)\theta_5}$: a main dressed tube carries $za^4\bar\Upsilon = 1/(1+\delta')$, an untouched front of $k\ge 5$ edges za^k if main and $C_\varpi a^k$ otherwise, a dressed tube that is not main $C_\varpi a^4(1+\gamma_g)=\omega$, a junction as in Lemma 10.5 $z\varpi\min(b^2,b_2)$ if main and $C_\varpi b^2$ otherwise, each times the factors $a^{|p|/2}$ of its two front-pairs (with $\nu\le\min(b^2,b_2)u^4$ by Lemmas 9.2 and 10.5), and every other junction $zw(J)$ if main and $C_\varpi w(J)$ otherwise. The caps carry nothing: they are not in X', and $r_{X'}$ does not see them.

Decorations are bounded as in the proof of Theorem 8.11, as none of their pieces is main: the factor C_ϖ of a junction is covered by the modified third condition of Lemma 8.10, and that of the interior slices of a segment by the modified c_J, since a segment with an end at a front-pair p'' contributes at most $\sum_{\ell\ge1}\omega^{\ell-1}e^{h_J}$ (as $C_\varpi a^{|p''|}\le\omega$ for $|p''|\ge5$).

The path. For a front-pair p of a path junction that starts the next path segment, let $\Delta := \max I - s(p)\ge 0$, where $s(p)$ is the slice of p and I the interval of slices of the path up to this junction, and let $\mathcal{V}(p,\Delta)$ be the sum over the continuations of the path, as in the proof of Theorem 8.11, of the product of the weights of $W^{(2)}$ (with the front-pair factors $a^{|\sigma|/2}$ of the two ends of a path segment assigned to the segment). We show by induction on the number of segments of the continuations that $\mathcal{V}(p,\Delta)\le Mf(|p|)$, with f as in Definition 8.9 and

$$M := \max\Big(\frac{E_B(4)}{\bar\Upsilon\delta' - \Sigma_4^{(2)}},\ \frac{E_B}{F(\bar\Upsilon(1+\delta')-a) - P_v(g_C Y_1/5 + g_F Y_0)}\Big),$$

$E_B(k)\le E_B k^2 e^{\xi_F k}$ as there. (a) p upward, $\Delta=0$, $|p|=4$. A segment of length ℓ has $\ell-1$ interior slices, all new, with main dressed tubes, and its far end is new: it carries $(1+\delta')^{-(\ell-1)}za^4 = (1+\delta')^{-\ell}/\bar\Upsilon$, and $\sum_{\ell\ge1}$ gives $1/(\bar\Upsilon\delta')$. The far junction J' is main; it is not a cap, and if it is as in Lemma 10.5 it carries $\varpi\min(b^2,b_2)$ besides z, which is used for the far end. The exits of J' start with $\Delta=0$, and by the induction hypothesis and Lemma 8.10(ii) (whose bound for $k=4$ splits the junctions as in Definition 8.9) the sum over J' and the continuations is at most $E_B(4)+M\Sigma_4^{(2)}$. So $\mathcal{V}(p,0)\le(E_B(4)+M\Sigma_4^{(2)})/(\bar\Upsilon\delta')\le M$. (b) p upward, $\Delta=0$, $|p|=k\ge5$: $\sum_{\ell\ge1}(za^k)^\ell\le a^{k-4}/(\bar\Upsilon(1+\delta')-a)$, and J' carries at most $zw(J')$; as in the proof of Theorem 9.4, $\mathcal{V}(p,0)\le Mf(k)$. (c) p upward with $\Delta\ge1$, or downward. A segment that passes the first Δ interior slices upward (all old) and reaches new slices carries, at the old ones, at most ω each and no z, and then exactly what it carries in (a), (b); these continuations give at most ω^Δ times the bound of (a) or (b) for a segment starting at the slice $\max I$, hence at most $\omega Mf(|p|)$. In the remaining continuations (the segment ends at an old slice, or below 0) no slice of the segment or of J' is new: a segment of length ℓ carries at most $\omega^{\ell-1}a^{|p|}$, the junction J' at most $C_\varpi w(J')$, and their sum, with Lemma 8.10(ii), is at most $C_\varpi a^{|p|}(E_B(|p|)+M\mathcal{S})/(1-\omega)$, where $\mathcal{S}\le\Sigma_4$ if $|p|=4$ and $\mathcal{S}\le kP_v e^{\xi_F k}(g_CY_1+g_FkY_0)$ if $|p|=k$. By the definition of M, $E_B(4)\le M\bar\Upsilon\delta'$, $E_B\le MF\bar\Upsilon(1+\delta')$ and $P_v(g_CY_1/5+g_FY_0)\le F\bar\Upsilon(1+\delta')$; so in both cases $\mathcal{V}(p,\Delta)\le Mf(|p|)(\omega + C_\varpi a^4(2\bar\Upsilon(1+\delta')+\Sigma_4)/(1-\omega))\le Mf(|p|)$ by the third

hypothesis. The beginning of the path is treated as in the proof of Theorem 8.11, which gives $z^t \sum_{X'} W^{(2)}(X') \le MC_A$.

An empty inner layer. If X' has only empty inner layers, the same argument applies with the direction of time reversed: in the representation of Definition 7.1 in which the slices of S_A and S_B are L_0 and t, Lemma 10.4 gives P_- at the outer slices $t+1,\dots,L_0-1$; the path is followed from J_A downward, a slice is new if it lies below every slice of the earlier elements, the factor z is charged to the L_0-t slices $t,\dots,L_0-1$ (by $z^t \le z^{L_0-t}$, as $z \ge 1$), and the factor C_ϖ comes from the slice t.

Gaps: as in the proof of Theorem 7.8. The spatial sides ≥ 5 exclude fronts of four edges other than plaquette boundaries, which would carry $za^4 = 1/(\bar\Upsilon(1+\delta'))$ at every interior slice without being dressed; $4\log(1/a) = 4\log(1/u) - 8(K+1)\theta_5$.

Small β. Take $\varpi := b$, so that $C_\varpi = 1 + 1/b$ for small β; $h_J := 8P_v a^2$, $\xi_F := 2P_v b$, $\xi_C := 8P_v b$ and $F := 3P_v N_4^v b$. Then $C_\varpi Y_0 = 1 + O(\beta)$, so $C_\varpi k P_v Y_0 a^{k/2} e^{\xi_F k} \le h_J$; $\omega = O(\beta^3)$ and $c_J = 1 + O(\beta^2)$, so the conditions of Lemma 8.10 and the six inequalities of Definition 8.9 hold as in the proof of Theorem 9.4, and the second hypothesis holds since $\bar\Upsilon = 1 + O(\beta)$. In $\Sigma_4^{(2)}$ the term of the shifts and jumps is at most $(4P_v - 4 - K_S) b\, b_2 = O(\beta^3)$, since $b_2 = 2(2-\iota)N^2u^2(1+O(\beta))$ (Lemma 9.1(iv)), and the other terms are $O(\beta^3)$ as in the proof of Theorem 9.4. So every $\delta' > c\beta^3$ is admissible for a constant c; the third and fourth hypotheses hold as $C_\varpi a^4 = O(\beta^3)$ and $a^4/b = O(\beta^3)$. Finally $\log \bar\Upsilon = \log \Upsilon + O(\beta^6)$ (Lemma 10.6) and $\theta_5 = O(\beta^5)$. □

Corollary 10.9 (The second-order term)**.** *Let G be metrisable and ρ irreducible and non-trivial. As $\beta \downarrow 0$,*

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \log\Upsilon(\beta) + O(\beta^3),$$

and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the periodic spatial lattices with all sides ≥ 5. For the fundamental representations:

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} + \frac{\beta^2}{8} + O(\beta^3) \quad \textit{for } SU(2) \textit{ and for } U(1),$$

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \frac{\beta}{6} + \frac{\beta^2}{72} + O(\beta^3) \quad \textit{for } SU(3),$$

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \frac{\beta^2}{128} + O(\beta^3) \quad \textit{for } SU(4),$$

and $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta^3)$ for $SU(N)$, $N \ge 5$. For $SU(2)$, $u = I_2(\beta)/I_1(\beta) = \frac{\beta}{4}(1 - \frac{\beta^2}{24} + O(\beta^4))$, so $m(\mu_\beta) = 4\log(4/\beta) + \frac{7}{24}\beta^2 + O(\beta^3)$. (Corollary 11.6 improves all the remainders to $O(\beta^4)$.)

Proof. For small β, $u_* = u$ (proof of Corollary 6.5). The upper bound is Theorem 10.2, since $L_2 \to 1$; the lower bound is Theorem 10.8, in finite volume with the values of N_4^v, N_5^v of the lattice, which are bounded uniformly (Lemma 7.5). The values are Lemma 10.1(iii); for $SU(2)$, $I_n(\beta) = \frac{(\beta/2)^n}{n!}(1 + \frac{\beta^2}{4(n+1)} + O(\beta^4))$. □

Remark 10.10 (Meaning, numbers and checks). (i) Meaning (Heuristic). In the transfer-matrix picture the plaquette states $\chi_\rho(U_q)$, $\chi_{\bar\rho}(U_q)$ propagate with the amplitude u^4 per unit of time, the cap and the subtraction of the vacuum act on them through V, and a shift moves the tube from q to a neighbour q' with the amplitude of closing it on q and opening it on q', which the subtraction of the vacuum cancels; the band of plaquette states is flat to second order (in the exact spectra of (ii) its width is about $8u^4$ in $d = 3$, the state of momentum 0 being the lowest). For $SU(2)$ the gap is raised by $\beta^2/8$: the subtraction of the vacuum, $N^2u^2 = \beta^2/4 + O(\beta^4)$,

outweighs the second-order cap, $\kappa = 3u_1 = \beta^2/8 + O(\beta^4)$. For $U(1)$ the C-odd plaquette state is the lighter one ($+\beta^2/8$ against $+3\beta^2/8$); for $SU(3)$ the C-even one, by the first-order cap; for $SU(4)$ the C-even one is lowered by $\beta^2/128$, as four fundamental fluxes can meet at a link ($\langle\chi^4\rangle = 1$). Only the lowest mass is bounded here.

(ii) Checks (`theory/checks/second-order/`, a compass). Exact transfer-matrix spectra: for $\mathbb{Z}_n$ in $d = 3$, in the sector of winding zero ($\mathbb{Z}_2$ on the 4×4 torus; $\mathbb{Z}_3$, $\mathbb{Z}_4$, $\mathbb{Z}_5$ on 3×3), in both C-sectors, $m - 4\log(1/u) + \log\Upsilon_\pm \approx -8u^4$ for the lowest state; an independent referee, with separate code in dual plaquette variables (`zn_dual_spectra.py`), found the same residual $-8.0u^4$ at momentum 0 and $O(u^5)$ at momentum (π,π), also on a torus that satisfies the hypotheses of Theorem 10.8 ($\mathbb{Z}_2$ on 5×5; the 3×3 and 4×4 tori have sides < 5); on one square (class functions; $SU(2)$, S_3, the quaternion group) to all printed digits; and on two squares sharing an edge (the full link space with the gauge projection; S_3: 6^7, quaternion group: 8^7 states), where the tube can shift between the two plaquettes, the two lowest plaquette states agree with each other and with $-\log\Upsilon$ up to $3\cdot10^{-4}\beta^3$ at $\beta = 0.01$ and 0.04. Brute force on one cube ($\mathbb{Z}_3$, $\mathbb{Z}_5$, A_4) confirms (2) to machine precision; on the two-copy space of the four links of a square ($\mathbb{Z}_3$, $\mathbb{Z}_4$, S_3) the largest eigenvalue of the dressed tube on the odd sector lies between $u^4\Upsilon$ and $u^4\bar\Upsilon$, within a relative $3\cdot10^{-8}$ of $u^4\Upsilon$ at $\beta = 0.1$ (and equal to it for $\mathbb{Z}_3$, where charge conjugation decouples the other states); and for a shift and a jump ($\mathbb{Z}_2$; and a shift for $\mathbb{Z}_3$) the operator $P^J O_J P^J$ has rank one, with the singular value $2\sigma_6 + \sigma_6^2$ to ten digits, and vanishes on the odd sector (the other singular values are at the level of round-off). Weyl integration confirms Lemma 10.1(iii) for $U(1)$ and $SU(2), \dots, SU(5)$.

(iii) Numbers (`second_order_constants.py`; Numerical, an evaluation of the explicit constants). For $SU(2)$ the upper bound of Theorem 10.2 is $D_2 = \beta^2/8 + O(\beta^5)$ (for instance $1.25\cdot10^{-15}$ at $\beta = 10^{-7}$), and $D_2(\beta_0(4)) = 4.8\cdot10^{-4}$, where $\mathcal{T}_Q$ dominates. The lower bound of Theorem 10.8, with $\varpi = b$ and the smallest admissible δ', has $\delta'/\beta^3 \to 1.4\cdot10^6$ ($d = 4$) and $2.1\cdot10^5$ ($d = 3$); the largest terms are the crude bounds for the junctions with three or more front-pairs and for those with three plaquettes ($Z_{3,4}$, which the enumerations of Remark 8.12(iii) found to vanish). So the lower bound resolves the term $\beta^2/8$ only for very small β: at $\beta = 10^{-7}$ it gives $m(\mu_\beta) - 4\log(1/u) \geq 1.04\cdot10^{-15}$ for $d = 3$ (against $\beta^2/8 = 1.25\cdot10^{-15}$), the first lower bound above $4\log(1/u)$, and $\geq -1.4\cdot10^{-16}$ for $d = 4$. Near $\beta_0(d)$ it is slightly better than that of Theorem 9.4: $-1.25\cdot10^{-4}$ instead of $-1.28\cdot10^{-4}$ at $\beta_0(4)$, and $-2.38\cdot10^{-4}$ instead of $-2.44\cdot10^{-4}$ at $\beta_0(3)$. For $SU(3)$ at $\beta = 10^{-7}$ ($d = 4$) the two bounds give $-6.4\cdot10^{-16} \leq m(\mu_\beta) - 4\log(1/u) + \log\Upsilon \leq 5\cdot10^{-22}$, with $-\log\Upsilon = -1.6667\cdot10^{-8}$.

(iv) What remains: the higher-order terms (Section 11 shows that the third-order term is that of $-\log\Upsilon$); the analyticity of $m + 4\log u$, as in the classical results; the C-odd mass for $SU(3)$ ($-\log\Upsilon_- = +\beta/6 + O(\beta^2)$, Heuristic), which needs a lower bound restricted to C-odd observables; and, near $\beta_0(d)$, the crude counts of Remark 8.12(i) and $\mathcal{T}_Q$.

11 The third-order term

By Theorem 10.2 the upper bound is $4\log(1/u) - \log\Upsilon + O(\beta^5)$. In the lower bound of Theorem 10.8 three contributions to δ' are of order β^3, and none of them is a third-order effect. The shifts and jumps enter through the factor $\varpi = b$ of Lemma 10.7, which can be taken of order β^2. The junctions with three plaquettes and two front-pairs of four edges, counted by $Z_{3,4}$, do not exist (Lemma 11.2). The junctions with three or more front-pairs are bounded in Lemma 8.10 by tree sums that charge a further front-pair only the factor a^2 of its half; but such a junction has two or more plaquettes, which costs b^2a^2, or one plaquette and then either a front of at least five edges (if it has exactly three front-pairs, Lemma 11.3) or at least two further front-pairs, which costs ba^4. With these changes $\delta' = O(\beta^4)$ (Theorem 11.5),

and

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \log\Upsilon(\beta) + O(\beta^4)$$

(Corollary 11.6): the third-order term of the gap is that of $-\log\Upsilon$. With the fundamental representations it is $\beta^3/432$ for $SU(3)$, $-\beta^3/6000$ for $SU(5)$, where five fundamental fluxes can meet at a link, and 0 for $U(1)$, $SU(2)$, $SU(4)$ and $SU(N)$, $N \geq 6$ (Lemma 11.1). The constant in the $O(\beta^4)$ of the lower bound is very large (Remark 11.7(iii)); Section 13 reduces its limit by a factor of about $5 \cdot 10^8$ (Theorem 13.4).

Lemma 11.1 (The plaquette variance to third order). *(i) If $\rho(z) = -\mathbf{1}$ for a central element z of G, then Υ_+, Υ_- and Υ, defined by the formulas of Lemma 10.1 for all real β, are even functions of β. This applies to $U(1)$, $SU(2)$ and $SU(N)$ with N even ($z = -\mathbf{1}$), with the fundamental representations.*

(ii) For $SU(N)$, $N \geq 3$, with the fundamental representation, put $t := \beta/N$, $r_n := \langle(\operatorname{Re}\chi_\rho)^n\rangle$ and $m_3 := \langle(\operatorname{Re}\chi_\rho)^3(\operatorname{Im}\chi_\rho)^2\rangle$. Then

$$\Upsilon_+ = 1 + 2r_3 t + (r_4 - \tfrac{3}{4})t^2 + \tfrac{1}{3}(r_5 - 5r_3)t^3 + O(\beta^4),$$
$$\Upsilon_- = 1 - 2r_3 t + (\tfrac{3}{4} - r_4)t^2 + \tfrac{1}{3}(m_3 + r_3)t^3 + O(\beta^4),$$

with $r_3 = \frac{1}{4}[N=3]$, $r_4 = \frac{3}{4} + \frac{1}{8}[N=4]$, $r_5 = \frac{15}{16}[N=3] + \frac{1}{16}[N=5]$ and $m_3 = -\frac{3}{16}[N=3] - \frac{1}{16}[N=5]$.

(iii) Consequently: $-\log\Upsilon = -\frac{\beta}{6} + \frac{\beta^2}{72} + \frac{\beta^3}{432} + O(\beta^4)$ for $SU(3)$; $-\log\Upsilon = -\frac{\beta^3}{6000} + O(\beta^4)$ for $SU(5)$; and the term of order β^3 of $-\log\Upsilon$ vanishes for $U(1)$, $SU(2)$, $SU(4)$ and $SU(N)$, $N \geq 6$.

Proof. (i) Haar measure is invariant under $U \mapsto zU$, and $\chi_\rho(zU) = -\chi_\rho(U)$; so $w_{-\beta}(U) = w_\beta(zU)$, and every ψ in the span of $\chi_\rho, \chi_{\bar\rho}$ satisfies $\psi(zU) = -\psi(U)$. Hence $c_0(-\beta) = c_0(\beta)$, $\langle\psi^2 w_{-\beta}\rangle = \langle\psi^2 w_\beta\rangle$ and $\langle\psi w_{-\beta}\rangle = -\langle\psi w_\beta\rangle$, so the variance of ψ is the same at β and $-\beta$. For $SU(N)$, $-\mathbf{1}$ has determinant $(-1)^N = 1$ when N is even.

(ii) Write $R := \operatorname{Re}\chi_\rho$, $I := \operatorname{Im}\chi_\rho$ and $y = tR$, $|y| \leq \beta$. As in the proof of Lemma 10.1(ii), but to one more order: with $A_\pm := \langle e^y\psi_\pm^2\rangle$, $B_\pm := \langle e^y\psi_\pm\rangle$, $C := \langle e^y\rangle = c_0$ for $\psi_+ = \sqrt{2}R$, $\psi_- = \sqrt{2}I$, each expanded to order t^3 with remainder $O(\beta^4)$, $\Upsilon_\pm = A_\pm/C - B_\pm^2/C^2$. Here $r_1 = 0$ and $r_2 = \frac{1}{2}$ (as $\langle\chi_\rho^2\rangle = 0$ for $N \geq 3$), so $A_+ = 1 + 2r_3 t + r_4 t^2 + \frac{1}{3}r_5 t^3$, $B_+ = \sqrt{2}(\frac{1}{2}t + \frac{1}{2}r_3 t^2) + O(\beta^3)$ and $C = 1 + \frac{1}{4}t^2 + \frac{1}{6}r_3 t^3$, which gives Υ_+. Complex conjugation $U \mapsto \bar U$ is an automorphism of $SU(N)$, so it preserves Haar measure; it fixes R and changes the sign of I, so $B_- = 0$. Further $\langle R^k I^2\rangle = \langle R^k|\chi_\rho|^2\rangle - r_{k+2}$, with $\langle|\chi_\rho|^2\rangle = 1$, $\langle R|\chi_\rho|^2\rangle = 0$ and $\langle R^2|\chi_\rho|^2\rangle = 1$ (expanding in $\chi_\rho, \overline{\chi_\rho}$ and using the moments below). So $A_- = 1 - 2r_3 t + (1 - r_4)t^2 + \frac{1}{3}m_3 t^3$, which gives Υ_-. The moments: as in the proof of Lemma 10.1(iii), $\langle\chi^j\bar\chi^k\rangle = 0$ unless N divides $j - k$ ($\chi := \chi_\rho$), $\langle|\chi|^2\rangle = 1$, $\langle|\chi|^4\rangle = 2$ and $\langle\chi^N\rangle = 1$; for $j + k \leq 5$ the only other non-zero moments are $\langle\chi^4\bar\chi\rangle = \langle\chi\bar\chi^4\rangle$ for $N = 3$. This is the multiplicity of ρ in $\rho^{\otimes 4}$, i.e. the dimension of the space of vectors of weight e_1 in $(\mathbb{C}^3)^{\otimes 4}$ annihilated by the raising operators E_{12} and E_{23} of $\mathfrak{sl}(3)$ (by the theorem of the highest weight, a standard fact of Lie theory that we do not prove; the value obtained below is confirmed by the Weyl integration formula and, independently, by Bessel–Toeplitz determinants, Remark 11.7(ii)). Such a vector is $\sum_{a\neq b} c_{ab} w_{ab}$, where w_{ab} has e_3 in the tensor factor a, e_2 in the factor b and e_1 in the other two. E_{12} replaces an e_2 by e_1, and E_{23} an e_3 by e_2; so the conditions are $\sum_b c_{ab} = 0$ for every a and $c_{ab} + c_{ba} = 0$. The antisymmetric 4×4 matrices form a space of dimension 6, on which the row sums are 3 independent conditions, so $\langle\chi^4\bar\chi\rangle = 3$. Hence $r_3 = \frac{1}{8}(\langle\chi^3\rangle + \langle\bar\chi^3\rangle) = \frac{1}{4}[N=3]$; $r_4 = \frac{1}{16}(6\langle|\chi|^4\rangle + \langle\chi^4\rangle + \langle\bar\chi^4\rangle) = \frac{3}{4} + \frac{1}{8}[N=4]$; $r_5 = \frac{1}{32}(5\langle\chi^4\bar\chi\rangle + 5\langle\chi\bar\chi^4\rangle)[N=3] + \frac{1}{32}(\langle\chi^5\rangle + \langle\bar\chi^5\rangle)[N=5]$; and $m_3 = \frac{1}{8}(\langle\chi^4\bar\chi\rangle + \langle\chi\bar\chi^4\rangle)[N=3] - r_5$.

(iii) For $SU(3)$, $t = \beta/3$: $\Upsilon_+ = 1 + \frac{\beta}{6} - \frac{5\beta^3}{1296} + O(\beta^4)$ and $\Upsilon_- = 1 - \frac{\beta}{6} + O(\beta^3)$, so $-\log\Upsilon = -\log\Upsilon_+ = -\frac{\beta}{6} + \frac{\beta^2}{72} + \frac{\beta^3}{432} + O(\beta^4)$. For $SU(5)$, $t = \beta/5$: $\Upsilon_\pm = 1 \pm \frac{t^3}{48} + O(\beta^4) = 1 \pm \frac{\beta^3}{6000} + O(\beta^4)$. For $N \geq 6$, $r_3 = r_5 = m_3 = 0$ and $r_4 = \frac{3}{4}$, so $\Upsilon_\pm = 1 + O(\beta^4)$. For $U(1)$, $SU(2)$ and $SU(4)$ use (i). □

Lemma 11.2 (No junctions with three plaquettes and two four-edge fronts)**.** *Suppose all sides of Λ_s are ≥ 5. Then no junction without support has exactly two front-pairs, both with four edges, and exactly three plaquettes; that is, $Z_{3,4} = 0$ in Lemma 8.7.*

Proof. Let J be such a junction at a slice s, with $\Phi(J) = \{(\varphi,\lambda),(\varphi',\lambda')\}$. As the sides are ≥ 5, $\varphi = \partial q_1$ and $\varphi' = \partial q_2$ are plaquette boundaries (Lemma 7.5(ii)), and $q_1 \neq q_2$ by Lemma 8.6(iv). By Lemma 8.6(i), every plaquette P of J has at least two edges in $\varphi \cup \varphi'$, and every edge of P outside $\varphi \cup \varphi'$, an *outside edge*, lies in a second plaquette of J. If $P \notin \{q_1, q_2\}$, then P has at most one edge in each of φ and φ' (a plaquette with two edges in ∂q is q, proof of Lemma 8.6(iv)), hence exactly one edge in $\varphi \setminus \varphi'$, one in $\varphi' \setminus \varphi$, and two outside edges; we call such a plaquette a *bridge*. The plaquettes q_1, q_2 have no outside edges, and two distinct plaquettes share at most one link (proof of Lemma 5.4). If q_1 or q_2 is a plaquette of J, the other plaquettes of J are at most two bridges, and the two outside edges of a bridge would have to lie in the other bridge: impossible. So the three plaquettes are bridges, which cover at most three edges of $\varphi \setminus \varphi'$ and three of $\varphi' \setminus \varphi$. If $\lambda = \lambda'$, the fronts are vertex-disjoint and all eight edges of $\varphi \cup \varphi'$ lie in plaquettes of J (Lemma 8.6(i)): impossible. So $\lambda \neq \lambda'$, every edge of $\varphi \triangle \varphi'$ lies in a plaquette of J, and $|\varphi \triangle \varphi'| \leq 6$ forces $|\varphi \cap \varphi'| = 1$: q_1 and q_2 share a link $\{x, x + e_\mu\}$, with q_1 spanned at x by e_μ and a unit vector f_1, and q_2 by e_μ and a unit vector $f_2 \neq f_1$, both perpendicular to e_μ; so either $f_2 = -f_1$ (the two plaquettes are coplanar) or $f_2 \perp f_1$. The edges of $\varphi \setminus \varphi'$ are $\{x, x + f_1\}$, $\{x + e_\mu, x + e_\mu + f_1\}$ and $\{x + f_1, x + f_1 + e_\mu\}$, and those of $\varphi' \setminus \varphi$ are the same with f_2 in place of f_1. Two edges lie in a common plaquette only if they are perpendicular with a common vertex or parallel at distance one. If $f_2 = -f_1$, the pairs with a common vertex are collinear, and the only parallel pair, $\{x + f_1, x + f_1 + e_\mu\}$ and $\{x - f_1, x - f_1 + e_\mu\}$, is at distance two (the sides ≥ 5 are used here again: with a side 3 it would be at distance one); so there is no bridge. If $f_2 \perp f_1$, exactly two pairs exist, at x and at $x + e_\mu$: the bridges are the two plaquettes spanned by f_1, f_2 at x and at $x + e_\mu$. In either case there are fewer than three bridges, a contradiction. □

Lemma 11.3 (One plaquette, three front-pairs)**.** *Let J be a junction without support with exactly one plaquette c. Then every front-pair of J touches c; and if J has exactly three front-pairs, one of them has at least five edges.*

Proof. Every front-pair is touched by a marker of its junction (Lemma 8.2(i)), and c is the only marker of J. By Lemma 8.2(iii), an edge of a front of J of layer λ that lies in no front of J of the other layer lies in ∂c. Suppose all three fronts have four edges. Two of the front-pairs have the same layer λ; their fronts $\varphi_1 \neq \varphi_2$ are distinct components of T_λ, hence vertex-disjoint. If the third front-pair has layer λ too, each front is contained in ∂c and so equals ∂c, a contradiction. Otherwise its front φ_3 has the other layer, and $\varphi_1 \cup \varphi_2 \subseteq \varphi_3 \cup \partial c$. Since $|\varphi_1 \cup \varphi_2| = 8$, φ_3 and ∂c are disjoint with union $\varphi_1 \cup \varphi_2$. A cycle of four edges in the disjoint union of two cycles of four edges is one of them; so φ_3 is, say, φ_1, and $\partial c = \varphi_2$. But c touches φ_1, whereas $\partial c = \varphi_2$ shares no vertex with φ_1: a contradiction. □

The proof of the next lemma uses the following elementary bound: for any fixed numbers $W_J \geq 0$ and $0 < b \leq \bar{b}$,

$$\sum_{|J|_C \geq 2} b^{|J|_C} W_J \leq \left(\frac{b}{\bar{b}}\right)^2 \sum_{|J|_C \geq 2} \bar{b}^{|J|_C} W_J,$$

since $b^n \leq (b/\bar{b})^2 \bar{b}^n$ for $n \geq 2$.

Lemma 11.4 (Junction sums to third order)**.** *In the setting of Theorem 10.8, suppose all spatial sides are* ≥ 5*. Let* $\bar{b} \geq b$*, and let* $\bar{\xi}_F, \bar{\xi}_C$ *and* $\bar{g}_F, \bar{g}_C, \bar{g}'_F, \bar{g}'_C, \bar{\kappa}_F, \bar{\kappa}_C$ *be such that the series below converge and that they satisfy the first two conditions of Lemma 8.10 and the six inequalities of Definition 8.9 with* $\bar{b}$ *in place of* b *in* Y_j*. The resulting quantities are written* $\bar{Y}_j$*,* $\bar{A}_j$*,* $\bar{B}_j$*; they keep the* c_J *and the* f *of Theorem 10.8. Put*

$$\mathsf{F} := 8\sum_{k\geq 4} N_k^v f(k), \quad \mathsf{F}' := 8\sum_{k\geq 5} N_k^v f(k), \quad \mathsf{A} := 8e^{c_J}\sum_{k\geq 4} N_k^v a^{k/2}, \quad \mathsf{A}' := 8e^{c_J}\sum_{k\geq 5} N_k^v a^{k/2},$$

$$\begin{aligned}\Sigma_4^{(3)} :=\ & (4P_v - 4 - K_S)\,\varpi\min(b^2, b_2) + 2\sum_{n\geq 4}(n+1)(2d-3)\binom{4P_v + T_v n}{n} b^n \\ & + 2\sum_{n\geq 1}\binom{4P_v + T_v n}{n} b^n \sum_{k\geq 5} X_{n,k} f(k) + 4P_v b\big(\mathsf{F}'\mathsf{A} + \mathsf{F}\mathsf{A}' + \mathsf{F}(e^{\mathsf{A}} - 1 - \mathsf{A})\big) \\ & + \Big(\frac{b}{\bar{b}}\Big)^2 4P_v e^{4\bar{\xi}_F}\big(\bar{\kappa}_C \bar{Y}_2 + 4(\bar{g}_C\bar{g}'_F + \bar{g}'_C\bar{g}_F)\bar{Y}_1 + 16\bar{\kappa}_F\bar{Y}_0\big).\end{aligned}$$

Let p *be a front-pair candidate with four edges, and let* J *range over the junctions containing* p *that have no support and are not caps, a junction as in Lemma 10.5 carrying* $\varpi\min(b^2, b_2)$ *in place of* $b^{|J|_C}$*. Then the sum* $\mathcal{S}(p)$ *of Lemma 8.10(ii) is at most* $\Sigma_4^{(3)}$*.*

Proof. Split the pairs (J, p') as in the proof of Lemma 8.10(ii).

Two front-pairs, with $|p'| = 4$*.* With one plaquette J would be a cap; with two it is a shift or a jump (Lemma 8.6(iii)), of which there are at most $4(P_v - 1) - K_S$; there are none with three (Lemma 11.2); and those with $n \geq 4$ are counted as in Definition 8.9.

Two front-pairs, with $|p'| \geq 5$*:* as there.

Three or more front-pairs, one plaquette c*.* The plaquette c has a vertex on p ($\leq 4P_v$ choices). Every other front-pair of J touches c (Lemma 11.3), so it is (φ, λ) with φ a front through a vertex of c and $\lambda \in \{s-1, s\}$; there are at most $8N_k^v$ such candidates with k edges. Write $\Psi := \Phi(J) \setminus \{p, p'\} \neq \emptyset$. If $|\Psi| = 1$, then p' or the front-pair of Ψ has at least five edges (Lemma 11.3); these terms give at most $\mathsf{F}'\mathsf{A} + \mathsf{F}\mathsf{A}'$. The terms with $|\Psi| \geq 2$ give at most $\mathsf{F}\sum_{m\geq 2}\mathsf{A}^m/m!$.

Three or more front-pairs, two or more plaquettes. These terms are $\sum_{|J|_C\geq 2} b^{|J|_C} W_J$ with $W_J := \sum_{p'} f(|p'|)\prod_{\psi\neq p,p'} a^{|\psi|/2} e^{c_J}$, computed with the actual a, c_J and f. By the inequality before the lemma they are at most $(b/\bar{b})^2$ times the same sum with plaquette weight $\bar{b}$, and the doubly marked sums of the proof of Lemma 8.10(ii), with the constants at $\bar{b}$, bound that sum by the last factor. □

Theorem 11.5 (The lower bound to third order)**.** *Suppose all spatial sides are* ≥ 6*. Then Theorem 10.8 holds with the hypothesis* $\Sigma_4^{(2)} < \bar{\Upsilon}\delta'$ *replaced by* $\Sigma_4^{(3)} < \bar{\Upsilon}\delta'$*, for any* $\bar{b}$ *and constants as in Lemma 11.4. As* $\beta \downarrow 0$ *the hypotheses hold with* $\varpi := b^2$*, a fixed* $\bar{b} > 0$ *depending only on* d*, and some* $\delta' = O(\beta^4)$*; so*

$$m(\mu_\beta) \ \geq\ 4\log\frac{1}{u(\beta)} - \log\Upsilon(\beta) - O(\beta^4).$$

Proof. In case (a) of the proof of Theorem 10.8 the far junction J' is main and contains no cap. So Lemma 11.4, in place of Lemma 8.10(ii), bounds the sum over J' and the continuations by $E_B(4) + M\Sigma_4^{(3)}$, where the constant M of that proof is now defined with $\Sigma_4^{(3)}$ in place of $\Sigma_4^{(2)}$. Nothing else changes: case (c) uses only the full sum Σ_4, through the third hypothesis, where $C_\varpi a^4\Sigma_4 = O(a^4/b)$ is small.

Small β*.* Take $\varpi := b^2$; then $C_\varpi = 1 + 1/b^2$ for small β. Take $\xi_F := 2P_v b$, $\xi_C := 8P_v b$, $h_J := 8P_v a^2/b$ and $F := 3P_v N_4^v b$. Then:

- $C_\varpi Y_0 = (1 + O(\beta))/b$, so $C_\varpi k P_v Y_0 a^{k/2} e^{\xi_F k} \le h_J$;
- $h_J = O(\beta)$, $\omega = O(\beta^2)$ and $c_J = 1 + O(\beta)$, while $A_j = O(a^2)$, so the conditions of Lemma 8.10, the six inequalities and the second hypothesis hold as in the proof of Theorem 9.4;
- the third hypothesis holds as $C_\varpi a^4 = O(\beta^2)$, and the fourth as $a^4/b^2 = O(\beta^2)$.

Let $\bar b$ be so small, depending only on d, that the conditions at $\bar b$ hold with room when $a = 0$ and $F = 0$ (then $\bar A_j = 0$, $\bar B_0 = N_4^v e^{4\bar\xi_F}$ and $\bar B_1 = 4\bar B_0$); for small β they then hold for the actual a, c_J and F, as $\bar Y_j$ converges at $\bar b$ and $\bar A_j$, $\bar B_j$ converge uniformly as $a \to 0$ and $F \to 0$ ($N_k^v \le C_d^k$). We take for $\bar g$, $\bar\kappa$ the minimal solutions, i.e. the six inequalities with equality: a linear system which at $a = 0$, $F = 0$ is a contraction with room. Its sources for $\bar g'_F, \bar g'_C, \bar\kappa_F, \bar\kappa_C$ are $O(a^2)$, so $\bar g_F, \bar g_C = O(1)$ and $\bar g'_F, \bar g'_C, \bar\kappa_F, \bar\kappa_C = O(a^2)$. The terms of $\Sigma_4^{(3)}$ are then:

- the shifts and jumps, at most $(4P_v - 4 - K_S)b^4$;
- the terms with $n \ge 4$, $O(b^4)$;
- the exits with at least five edges, which have at least six ($N_5^v = 0$): $O(bf(6)) = O(b^2a^2)$;
- the one-plaquette term, $O(b(ba^4 + a^3 + a^4))$, as $\mathsf{F} = O(1)$, $\mathsf{F}' = O(ba^2)$, $\mathsf{A} = O(a^2)$ and $\mathsf{A}' = O(a^3)$;
- the last term, $O(b^2a^2)$.

So $\Sigma_4^{(3)} = O(\beta^4)$, and every $\delta' > c\beta^4$ is admissible for a constant c. Finally $\log \bar\Upsilon = \log \Upsilon + O(\beta^6)$ (Lemma 10.6) and $\theta_5 = O(\beta^5)$. □

Corollary 11.6 (The third-order term)**.** *Let G be metrisable and ρ irreducible and non-trivial. As $\beta \downarrow 0$,*

$$m(\mu_\beta) = 4 \log \frac{1}{u(\beta)} - \log \Upsilon(\beta) + O(\beta^4),$$

and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the periodic spatial lattices with all sides ≥ 6. For the fundamental representations:

$$m(\mu_\beta) = 4 \log \frac{1}{u(\beta)} + \frac{\beta^2}{8} + O(\beta^4) \quad \textit{for } SU(2) \textit{ and for } U(1),$$
$$m(\mu_\beta) = 4 \log \frac{1}{u(\beta)} - \frac{\beta}{6} + \frac{\beta^2}{72} + \frac{\beta^3}{432} + O(\beta^4) \quad \textit{for } SU(3),$$
$$m(\mu_\beta) = 4 \log \frac{1}{u(\beta)} - \frac{\beta^2}{128} + O(\beta^4) \quad \textit{for } SU(4),$$
$$m(\mu_\beta) = 4 \log \frac{1}{u(\beta)} - \frac{\beta^3}{6000} + O(\beta^4) \quad \textit{for } SU(5),$$

and $m(\mu_\beta) = 4 \log(1/u(\beta)) + O(\beta^4)$ for $SU(N)$, $N \ge 6$. For $SU(2)$, $m(\mu_\beta) = 4 \log(4/\beta) + \frac{7}{24}\beta^2 + O(\beta^4)$.

Proof. The upper bound is Theorem 10.2, whose error is $O(\beta^5)$; the lower bound is Theorem 11.5, in finite volume with the counts N_k^v of the lattice, which are bounded uniformly (Lemma 7.5). The values are Lemma 11.1, with the terms of first and second order from Lemma 10.1(iii). For $SU(2)$, $u = I_2(\beta)/I_1(\beta) = \frac{\beta}{4}(1 - \frac{\beta^2}{24} + O(\beta^4))$. □

Remark 11.7 (Meaning, checks and numbers). (i) Meaning (Heuristic). At third order nothing happens beyond the single plaquette: the cap and the subtraction of the vacuum are exact in Υ, and the shifts and jumps cancel. The next corrections are of order β^4:

- the hopping of the plaquette state through a junction that widens the tube to six edges and one that narrows it again, of order $u^2 \cdot a^2$ per hop in characters;
- in $d = 4$, the junctions with four plaquettes and two four-edge front-pairs (twelve, Lemma 13.1(i)); in $d = 3$ there are none;
- at order β^6, the vacuum.

(Numerical) In the exact $\mathbb{Z}_n$ spectra in $d = 3$ the residual $m - 4\log(1/u) + \log \Upsilon_C$ is about $-8u^4$ at momentum 0 and $O(u^5)$ at momentum (π, π) (Remark 10.10(ii)). (Theorem) By Theorem 10.2, whose error is $O(\beta^5)$, $\limsup_{\beta \downarrow 0} \beta^{-4}(m(\mu_\beta) - 4\log(1/u) + \log \Upsilon) \le 0$ for every G and ρ; Theorem 12.8 improves this to $\le -h_C u_1^4$.

The values of Lemma 11.1 follow a pattern. If $\iota = 0$, then exactly $\Upsilon_+ = 2\partial_t^2 K(t, 0)$ and $\Upsilon_- = 2\partial_s^2 K(t, 0)$ at $t = \beta/N$, where $K(t, s) := \log\langle \exp(t \operatorname{Re} \chi_\rho + s \operatorname{Im} \chi_\rho)\rangle$; so $\Upsilon_\pm - 1$ are given by the cumulants of $(\operatorname{Re} \chi_\rho, \operatorname{Im} \chi_\rho)$ of order at least three, and tilting a Gaussian changes no variance. For $SU(N)$ the moments $\langle \chi^j \bar{\chi}^k \rangle$ of degree $j + k < N$ are those of a standard complex Gaussian ($\langle |\chi|^{2j} \rangle = j!$ for $j \le N$ by Schur–Weyl duality, and $\langle \chi^j \bar{\chi}^k \rangle = 0$ for $j \ne k$), and at degree N the only excess is $\langle \chi^N \rangle = \langle \bar{\chi}^N \rangle = 1$. So $K = \frac{1}{4}(t^2 + s^2) + 2^{1-N} \operatorname{Re}((t + is)^N)/N! + O(|(t, s)|^{N+1})$ and $\Upsilon_\pm = 1 \pm 2^{2-N}(\beta/N)^{N-2}/(N-2)! + O(\beta^{N-1})$: for $SU(N)$ the C-even and C-odd plaquette states split first at order $N - 2$, when N fundamental fluxes meet at a link. For $N = 3, 4, 5$ this is part of Lemma 11.1; an independent referee's exact series confirm it for $N = 6, 7$ ($\mp\beta^4/497664$, $\mp\beta^5/64538880$), where it lies beyond the order controlled here.

(ii) Checks (`theory/checks/third-order/`, a compass).

- Lemmas 11.2 and 11.3, by exhaustive enumeration.
  - The first on spatial tori of dimensions 2, 3 and 4, with the plaquettes pre-filtered by Lemma 8.6(i): the junctions with two front-pairs of four edges and at most three plaquettes are exactly the cap and the $4(P_v - 1) - K_S$ shifts and jumps.
  - The second on dimensions 2 and 3, with the fronts of one layer restricted to four or six edges (those of the other layer arbitrary). There the junctions with one plaquette and three front-pairs merge two fronts into one of at least eight edges.
  - An independent referee, with separate code working from the definitions alone (all cycles of four edges, straight ones included, and no lemma of the paper), found the same on tori with sides 4 to 6 in spatial dimensions 2, 3 and 4, and in addition $Z_{4,4} = 0$ for $d = 3$ and $Z_{4,4} = 12$ for $d = 4$ (two jumps across a cube, eight turns through a cube, and two U-turns), as Lemma 13.1(i) now proves.
- Lemma 11.1, by Weyl integration (the moments exactly, and $-\log \Upsilon$ at small β). The referee, independently, from exact power series of Bessel–Toeplitz determinants.
- Exact transfer-matrix spectra of $\mathbb{Z}_3$ and $\mathbb{Z}_5$ on the 3×3 torus, in both C-sectors. This torus has sides $3 < 6$ (and girth 3), outside the hypotheses of Theorem 11.5. For these groups $-\log \Upsilon_C$ has a term $c_3\beta^3$; for $\mathbb{Z}_5$, $c_3 = \mp\frac{1}{48}$ for $C = \pm$.
  - After subtracting $-\log \Upsilon_C$, the residual divided by u^4 lies between -8.16 and -7.99 at $\beta = 0.04$, 0.02 and 0.01, tending to -8. The referee found the same on the 3×4 and 4×4 tori; fits give a β^3 coefficient of the residual below $5 \cdot 10^{-4}$ of c_3.

– After subtracting only its expansion to second order, the remainder divided by β^3 tends to c_3: for $\mathbb{Z}_5$, $C=+$, it is -0.042, -0.032 and -0.026 against $c_3=-0.021$, the difference being the term $-8u^4$.

(iii) Numbers (`third_order_constants.py`; Numerical). For $SU(2)$ the smallest admissible δ' of Theorem 11.5, with $\varpi=b^2$ and the best $\bar{b}$ among the powers of 2, has $\delta'/\beta^4\to 8.4\cdot10^{10}$ ($d=4$) and $3.4\cdot10^8$ ($d=3$). Almost all of it (91% and 97%) is the crude count of the junctions with four plaquettes and a four-edge exit, $2\cdot5\cdot(2d-3)\binom{4P_v+4T_v}{4}b^4$. Section 13 proves $Z_{4,4}\le 12(d-3)$ and counts the other junctions of order β^4; the limit then drops to 2 ($d=3$) and 166 ($d=4$) (Theorem 13.4, Remark 13.5(ii)). By Lemma 11.2 the term $Z_{3,4}b^3$ of Σ_4 (Definition 8.9) vanishes when all spatial sides are ≥ 5. The one-plaquette term reaches its limit slowly, because of the crude bound for N_7^v. For $SU(3)$ the limits are $8.2\cdot10^{10}$ and $3.1\cdot10^8$, so the lower bound resolves the term $\beta^3/432$ only for $\beta\lesssim 3\cdot10^{-14}$ ($d=4$) and $8\cdot10^{-12}$ ($d=3$): this part of the statement is asymptotic. For $SU(2)$ the two bounds now give $1.249966\cdot10^{-15}\le m(\mu_\beta)-4\log(1/u)\le 1.250000\cdot10^{-15}$ at $\beta=10^{-7}$ and $1.2465\cdot10^{-13}\le m(\mu_\beta)-4\log(1/u)\le 1.2500\cdot10^{-13}$ at $\beta=10^{-6}$ ($d=3$), and $1.2416\cdot10^{-15}\le m(\mu_\beta)-4\log(1/u)\le 1.2500\cdot10^{-15}$ at $\beta=10^{-7}$ ($d=4$): the second-order term $\beta^2/8$ is resolved there to a few parts in 10^5, 10^3 and 10^3.

(iv) What remains: the fourth-order term, of which Section 12 proves the upper half and Section 14 the lower half for $C=+$ (Corollary 14.3); the analyticity of $m+4\log u$; the C-odd mass; and, near $\beta_0(d)$, the crude counts and $\mathcal{T}_Q$.

12 The fourth-order term: the plaquette state moves

By Corollary 11.6, $m(\mu_\beta)=4\log(1/u)-\log\Upsilon+O(\beta^4)$: up to third order the plaquette state stays on its plaquette, dressed by the weight of that plaquette. At fourth order it moves. Three local processes change the tube, each with relative weight u^4 for every G and ρ (Remark 12.3):

- a *bulge*: for one layer the front ∂q of the tube widens to the six links of $\partial q\triangle\partial c$, where c shares a link with q, and then narrows back to ∂q; the tube acquires the unit cube $c\times[s,s+1]$ on one of its sides;
- a *hop*: the same widened layer, narrowed on the other side, so that the tube continues on c;
- a *turn* or a *jump* ($d\ge4$): at one slice, four faces of a unit cube of the spatial lattice carry the tube from a face q of the cube to an adjacent face (a turn) or to the opposite face (a jump).

They define a translation-invariant matrix H_C on the plaquettes of a time slice, for each sector $C=\pm$ of the plaquette function (Definition 12.1); its spectrum reaches up to $h_+=26d-70$ and $h_-=10d-22$ (Lemma 12.2). The prediction (Conjecture 12.4) is $m(\mu_\beta)=\min_C\big(4\log(1/u)-\log\Upsilon_C-h_Cu^4\big)+O(\beta^5)$; for $SU(2)$ in $d=4$ this is $m=-4\log u+2u^2-\frac{98}{3}u^4+O(u^5)$. We prove the upper half (Theorem 12.8; Section 14 proves the lower half for $C=+$, Corollary 14.3): for every finite set $\mathcal{W}$ of plaquettes of a slice and every $a\in\mathbb{R}^{\mathcal{W}}\setminus\{0\}$,

$$m(\mu_\beta)\;\le\;4\log\frac{1}{u}-\log\Upsilon_C-\log\Big(1+u^4\,\frac{a^\top H_{C,\mathcal{W}}\,a}{a^\top a}\Big)+O(\beta^5),$$

with a constant depending on $\mathcal{W}$ but not on a; hence $\limsup_{\beta\downarrow0}\beta^{-4}\big(m(\mu_\beta)-4\log(1/u)+\log\Upsilon_C\big)\le-h_Cu_1^4$, $u_1=\lim u/\beta$. The proof is that of Theorem 10.2 for a plaquette function spread over $\mathcal{W}$. Positivity (Lemma 5.2) is applied with one term for every four-edge front ∂q and every six-edge front $\partial q\triangle\partial c$ in the crossing layer. Each term is evaluated exactly

by characters on one unit cube or on two cubes glued along a face (Lemma 12.7), and two combinatorial lemmas (Lemmas 12.5 and 12.6) show that everything else is $O(\beta^5)$ relative to the tube. The matching lower bound is open (Remark 12.10(v)). The evidence for the conjecture is of two kinds (Remark 12.10). Exact transfer-matrix spectra of $\mathbb{Z}_2$ and $\mathbb{Z}_3$ in $d = 3$ and $d = 4$ follow every level of the predicted band, in both sectors. And the classical strong-coupling series of Münster and Seo for $SU(2)$ and $SU(3)$ in $d = 4$, as quoted in [LMP07], agree with it at order u^4 in four of five channels.

Definition 12.1 (The hopping matrix)**.** Let Λ_s be $\mathbb{Z}^{d-1}$ or a periodic spatial lattice with all sides ≥ 3. Give each plaquette q its standard orientation, let ∂q be its oriented boundary (a 1-chain), and write $q \sim q'$ if $q \neq q'$ share a link. For $q \sim c$ let $\sigma(q, c) \in \{\pm 1\}$ be the sign for which $\partial q + \sigma(q, c)\, \partial c$ has six links (the common link cancels). A *unit cube* of Λ_s is the set of the six plaquettes bounding a unit 3-cube; fix an orientation of the 3-cube, and for a face p let $\epsilon_\Gamma(p) = 1$ if the standard orientation of p is the induced boundary orientation and $\epsilon_\Gamma(p) = -1$ otherwise. For distinct faces q, q' of Γ put $\sigma_\Gamma(q, q') := -\epsilon_\Gamma(q)\epsilon_\Gamma(q')$, which does not depend on the orientation chosen. For $C \in \{+, -\}$ put $\mathsf{s}_C := [C = -]$ and let H_C be the matrix indexed by the plaquettes with

$$H_C(q, q) := K_S, \qquad H_C(q, q') := [q \sim q']\, \sigma(q, q')^{\mathsf{s}_C} + \sum_{\Gamma \ni q, q'} \sigma_\Gamma(q, q')^{\mathsf{s}_C} \qquad (q \neq q'),$$

the sum over the unit cubes having both q and q' as faces. Here $K_S = 4(2d - 5)$ is the number of plaquettes sharing a link with a given one. For a finite set $\mathcal{W}$ of plaquettes, $H_{C,\mathcal{W}} := (H_C(q, q'))_{q,q' \in \mathcal{W}}$.

In a hop, inserting c with the orientation $\sigma(q, c)$ at the lower slice turns the front ∂q into $\partial q + \sigma(q, c)\partial c = \partial(q + \sigma(q, c)c)$, and inserting q at the upper slice leaves $\sigma(q, c)\, \partial c$: a tube that carries ρ on q in its standard orientation continues on c in the orientation $\sigma(q, c)$, and likewise through a cube with σ_Γ. The function $\sqrt{2}\,\mathrm{Im}\,\chi_\rho$ changes sign with the orientation, $\sqrt{2}\,\mathrm{Re}\,\chi_\rho$ does not; this is the origin of the exponent s_C.

Lemma 12.2 (The hopping matrix)**.** *Let $\Lambda_s = \mathbb{Z}^{d-1}$.*

(i) Two distinct plaquettes are faces of at most one common unit cube, and of one if and only if they share a link and are perpendicular, or are translates of each other by a unit vector perpendicular to their plane. If $q \sim c$ are coplanar, $\sigma(q, c) = 1$. For opposite faces of a unit cube Γ, $\sigma_\Gamma(q, q') = 1$; for adjacent faces, $\sigma_\Gamma(q, q') = -\sigma(q, q')$.

(ii) Hence $H_+ = K_S\mathbf{1} + A_\parallel + 2A_\perp + A_J$ and $H_- = K_S\mathbf{1} + A_\parallel + A_J$, where $A_\parallel$, $A_\perp$, A_J are the adjacency matrices of the relations "coplanar and sharing a link" (4 per plaquette), "perpendicular and sharing a link" ($8(d-3)$ per plaquette) and "opposite faces of a unit cube" ($2(d-3)$ per plaquette). Every row of H_C sums to

$$h_+ := 26d - 70, \qquad h_- := 10d - 22.$$

(iii) H_C is a bounded self-adjoint operator on ℓ^2 of the plaquettes, and $\sup \mathrm{spec}\, H_C = h_C$. If N_L is the set of plaquettes of the box $[0, L]^{d-1}$, then $\mathbf{1}^\top H_{C,N_L}\mathbf{1}/|N_L| \to h_C$ as $L \to \infty$.

(iv) For $d = 4$, on the translation-invariant vectors, which are the functions of the three orientations of the plaquettes, H_+ has the eigenvalues 34 (the uniform vector) and 10 (twice), and H_- the eigenvalue 18 (three times). Under the cubic group these are the representations A_1, E and T_1.

Proof. (i) Let q be spanned at x by unit vectors e, e'. A unit cube having q as a face is spanned by e, e' and a unit vector $g \perp e, e'$ (on the side of g); there are $2(d-3)$ of them. Its faces other than q are the translate $q+g$ and the four plaquettes spanned by an edge of q and g. The plaquettes through an edge of q are q, its coplanar neighbour across that edge, and $2(d-3)$ perpendicular ones, one for each g. So the faces other than q of the cubes through q are exactly the $2(d-3)$ translates $q+g$ and the $8(d-3)$ perpendicular plaquettes sharing a link with q, and each lies in one such cube only, the one determined by g. This proves the first statement. If c is the coplanar neighbour of q across the edge $\{x+e, x+e+e'\}$, this edge is traversed in the direction e' by ∂q and in the direction $-e'$ by ∂c (both plaquettes have the orientation $e \wedge e'$); likewise for the other three edges, so $\sigma(q,c) = 1$. For a unit cube Γ, $\sum_{p \in \Gamma} \epsilon_\Gamma(p) \partial p = 0$ (the boundary of a boundary). Opposite faces have the same standard orientation and opposite outward normals, so $\epsilon_\Gamma(q) = -\epsilon_\Gamma(q')$ and $\sigma_\Gamma(q,q') = 1$. If q, q' are adjacent faces with common link l, no other face of Γ contains l, so the coefficients of l satisfy $\epsilon_\Gamma(q)\,\partial q(l) + \epsilon_\Gamma(q')\,\partial q'(l) = 0$, while $\partial q(l) + \sigma(q,q')\,\partial q'(l) = 0$ by definition; as all four numbers are ± 1, $\sigma(q,q') = \epsilon_\Gamma(q)\epsilon_\Gamma(q') = -\sigma_\Gamma(q,q')$.

(ii) By (i), for $C = +$ a perpendicular neighbour receives 1 from the hop and 1 from the turn through the unique common cube, a coplanar neighbour 1, a translate across a cube 1; for $C = -$ the hop and the turn cancel, and the other two terms are 1. There are 4 coplanar neighbours (one across each edge), $K_S - 4 = 8(d-3)$ perpendicular ones, and $2(d-3)$ translates. So the row sums are $K_S + 4 + 16(d-3) + 2(d-3) = 26d - 70$ and $K_S + 4 + 2(d-3) = 10d - 22$.

(iii) By (ii) H_C is symmetric with non-negative entries and constant row sums h_C, so $\|H_C\| \le h_C$ (Schur's test). Every plaquette of N_L at distance ≥ 2 from the boundary of the box has all its neighbours for H_C in N_L; these are all but $O(L^{d-2})$ of the $|N_L| \ge L^{d-1}$ plaquettes, so $\mathbf{1}^\top H_{C,N_L} \mathbf{1}/|N_L| \to h_C$, and $\sup \operatorname{spec} H_C \ge h_C$ by the variational principle.

(iv) By (ii), the plaquettes of the same orientation as q receive $K_S + 4 + 2 = 18$ in total from q, and those of each of the other two orientations $2 \cdot 4 = 8$ for $C = +$ (the 8 perpendicular neighbours of q split evenly between them) and 0 for $C = -$. The matrices $18 \cdot \mathbf{1} + 8(J - \mathbf{1})$ and $18 \cdot \mathbf{1}$ on $\mathbb{R}^3$ (J the matrix of ones) have the stated eigenvalues. The uniform vector spans A_1 and its orthogonal complement is E; the C-odd vectors change sign with the orientation of the plaquettes and transform as the components of an axial vector, T_1. □

Remark 12.3 (The three processes). (Heuristic.) Consider the tube on q, the four temporal plaquettes over ∂q in every layer, carrying ρ. A bulge replaces the temporal plaquette over the common link ℓ of q and c in one layer by the other five faces of the unit cube $c \times [s, s+1]$: four plaquettes more. A hop uses the six temporal plaquettes over $\partial q \triangle \partial c$ in one layer, with c at its lower slice and q at its upper one; this is again four plaquettes more than the tube. A turn or a jump uses the four faces of a unit cube Γ of a slice other than q and q', again four plaquettes more. In each case every link of the surface lies in exactly two of its plaquettes, and the surface is again a tube. By the character integrals of Lemma 12.7 it costs u^4 relative to the plain tube, for every G and ρ, with the orientation signs of Definition 12.1. Per unit of time a plaquette state thus has K_S bulges and moves to each of its neighbours in H_C with amplitude u^4, and the plaquette band is

$$e^{-m_C(k)} = u^4\, \Upsilon_C \left(1 + u^4 \lambda(\hat{H}_C(k)) + O(\beta^5)\right),$$

$\lambda(\hat{H}_C(k))$ the eigenvalues of the symbol of H_C at momentum k, a matrix over the $\binom{d-1}{2}$ orientations. Its bottom, the gap, is at $k = 0$, where the largest eigenvalue is h_C (Lemma 12.2(iii)). Nothing else should enter at relative order u^4. By Lemma 12.6, at one slice a four-edge front is carried to another four-edge front by fewer than five plaquettes only through a turn or a jump, and a six-edge front is closed by fewer than three only as in a bulge or a hop. The two junctions with four plaquettes "in the same layer" of the referee's enumeration (Remark 11.7(ii)) join two tubes, and enter the propagation of one plaquette state only at order u^8.

Conjecture 12.4 (The fourth-order term)**.** *Let G be metrisable and ρ irreducible and non-trivial. As $\beta \downarrow 0$,*

$$m(\mu_\beta) = \min_C \Big(4 \log \frac{1}{u(\beta)} - \log \Upsilon_C(\beta) - h_C\, u(\beta)^4 \Big) + O(\beta^5),$$

where C runs over $\{+,-\}$ if $\iota = 0$, and over $\{+\}$ with $\Upsilon_+ := \Upsilon$ if $\iota = 1$. For the fundamental representations in $d = 4$:

$$\begin{aligned} SU(2): \quad & m(\mu_\beta) = -4\log u + 2u^2 - \tfrac{98}{3}u^4 + O(u^5) = 4\log\tfrac{4}{\beta} + \tfrac{7}{24}\beta^2 - \tfrac{167}{1152}\beta^4 + O(\beta^5), \\ SU(3): \quad & m(\mu_\beta) = -4\log u - 3u + 9u^2 - \tfrac{27}{2}u^3 - 7u^4 + O(u^5), \\ U(1)\ (C=-): \quad & m(\mu_\beta) = -4\log u + \tfrac{1}{2}u^2 - \tfrac{425}{24}u^4 + O(u^5); \end{aligned}$$

in $d = 3$, $m + 4\log u$ is $2u^2 - \frac{20}{3}u^4$ ($SU(2)$), $-3u + 9u^2 - \frac{27}{2}u^3 + 19u^4$ ($SU(3)$) and $\frac{1}{2}u^2 - \frac{185}{24}u^4$ ($U(1)$), up to $O(u^5)$.

Lemma 12.5 (Six plaquettes)**.** *Let $\mathcal{Y}$ be a non-empty finite set of plaquettes of $\mathbb{Z}^n$, $n \geq 2$, such that every link of $E(\mathcal{Y})$ lies in at least two plaquettes of $\mathcal{Y}$. Then $|\mathcal{Y}| \geq 6$, with equality only if $\mathcal{Y}$ is a unit cube. The same holds on a periodic lattice with all sides ≥ 8.*

Proof. Let $p \in \mathcal{Y}$. Every other plaquette contains at most one link of p (two distinct plaquettes share at most one link, proof of Lemma 5.4), and every link of $E(\mathcal{Y} \setminus \{p\}) \setminus E(p)$ lies in two plaquettes of $\mathcal{Y} \setminus \{p\}$; the links of p lie in further plaquettes, so $\mathcal{Y} \setminus \{p\} \neq \emptyset$, and Lemma 5.4 with $\mathcal{L} = E(p)$ gives $|\mathcal{Y} \setminus \{p\}| \geq 5$.

Let $|\mathcal{Y}| = 6$, and choose a direction k such that some plaquette of $\mathcal{Y}$ is spanned by e_k and another unit vector. Let M and m be the largest and the smallest k-th coordinate of a vertex of $\mathcal{Y}$; then $m < M$. A plaquette with a vertex at height M lies either in the hyperplane $\{x_k = M\}$ (a *top* plaquette) or is spanned by e_k and another direction with its upper edge at height M (a *side* plaquette). A link at height M is the upper edge of exactly one plaquette spanned by e_k and its own direction, so it lies in at most one side plaquette; as it lies in at least two plaquettes of $\mathcal{Y}$ if it lies in one, it lies in a top plaquette. Some plaquette of $\mathcal{Y}$ has a link at height M, so the set $\mathcal{T}$ of top plaquettes is non-empty; let $t := |\mathcal{T}|$, and call a link of $E(\mathcal{T})$ *exposed* if it lies in exactly one plaquette of $\mathcal{T}$. Each exposed link is the upper edge of a side plaquette of $\mathcal{Y}$, distinct for distinct exposed links. By the same argument at height m, $\mathcal{Y}$ contains a bottom plaquette, which is neither a top nor a side plaquette. So $6 \geq t + e_{\mathcal{T}} + 1$, $e_{\mathcal{T}}$ the number of exposed links. If s_l is the number of plaquettes of $\mathcal{T}$ containing the link l, then $\sum_l s_l = 4t$ and $\sum_{s_l \geq 2} \binom{s_l}{2} \leq \binom{t}{2}$, so $\sum_{s_l \geq 2} s_l \leq t(t-1)$ and $e_{\mathcal{T}} \geq t(5-t)$. For $t = 2, 3, 4$ this gives $t + e_{\mathcal{T}} + 1 \geq 9$; for $t = 5$, $e_{\mathcal{T}} = 0$ would make $\mathcal{T}$ a set as in the lemma with five plaquettes, so $e_{\mathcal{T}} \geq 1$ and $t + e_{\mathcal{T}} + 1 \geq 7$; and $t \geq 6$ gives $t + 1 > 6$. Hence $t = 1$: the four links of the top plaquette τ are exposed and are the upper edges of four side plaquettes, and these, τ and the bottom plaquette b are all of $\mathcal{Y}$. At the bottom, the four links of b are the lower edges of four side plaquettes, which must be the same four. A side plaquette spans the heights $M - 1$ and M, so $m = M - 1$, and its lower edge is its upper edge translated by $-e_k$; so $b = \tau - e_k$, and $\mathcal{Y}$ is the unit cube spanned by τ and e_k.

On a periodic lattice with all sides ≥ 8, Lemma 5.4 applies as before. If $|\mathcal{Y}| = 6$, every link-connected component of $\mathcal{Y}$ satisfies the hypothesis, so $\mathcal{Y}$ is connected, and it lies in a box of side at most 7 that does not wrap around. It therefore lifts to $\mathbb{Z}^n$ with the hypothesis preserved. □

Lemma 12.6 (Configurations at a slice)**.** *Let Λ_s be $\mathbb{Z}^{d-1}$ or periodic with all sides ≥ 8, q a plaquette, and F either ∂q or $\partial q \triangle \partial c$ for some $c \sim q$, as a set of four, respectively six, links. Let q' be a plaquette and X a finite set of plaquettes with $q' \notin X$, such that every link e has*

$$n(e) := [e \in F] + [e \in E(q')] + \#\{p \in X : e \in E(p)\} \ \neq 1.$$

(i) If $F = \partial q$, then: $q' = q$ and $X = \emptyset$; or $q \notin X$, q and q' are distinct faces of a unit cube Γ, and $X = \Gamma \setminus \{q, q'\}$; or $|X| \geq 5$.

(ii) If $F = \partial q \triangle \partial c$, then: $q' \in \{q, c\}$ and $X = \{q, c\} \setminus \{q'\}$; or $|X| \geq 3$.

Proof. Distinct plaquettes share at most one link, so a plaquette other than q contains at most one link of q.

(i) If $q' = q$ (so $q \notin X$), every link of $E(X) \setminus E(q)$ lies in at least two plaquettes of X, so $X = \emptyset$ or $|X| \geq 5$ by Lemma 5.4 with $\mathcal{L} = E(q)$. If $q' \neq q$ and $q \notin X$, then $\mathcal{Y} := X \cup \{q, q'\}$ has $\#\{p \in \mathcal{Y} : e \in E(p)\} = n(e)$ for every link e, so it is as in Lemma 12.5: $|X| \geq 4$, with equality only if $\mathcal{Y}$ is a unit cube. If $q' \neq q$ and $q \in X$, the plaquettes of $Y := (X \setminus \{q\}) \cup \{q'\}$ differ from q, and every link $e \notin E(q)$ has $\#\{p \in Y : e \in E(p)\} = n(e)$; so $|X| = |Y| \geq 5$ by Lemma 5.4 with $\mathcal{L} = E(q)$.

(ii) Let ℓ be the common link of q and c. For $e \neq \ell$, $[e \in F] = [e \in E(q)] + [e \in E(c)]$, and $\ell \notin F$.

- $q' \notin \{q, c\}$ and $q, c \notin X$. Then $\mathcal{Y} := X \cup \{q', q, c\}$ is as in Lemma 12.5 (the link ℓ lies in q and c), so $|X| \geq 3$.
- $q' \notin \{q, c\}$ and X meets $\{q, c\}$. Every link of q' has $n \geq 2$, so it lies in $F \subseteq E(q) \cup E(c)$ or in a plaquette of X. As q' has at most one link in each plaquette boundary, $|X| \leq 2$ would leave at most three boundaries, those of q, c and at most one further plaquette, for its four links. So $|X| \geq 3$.
- $q' = q$ (the case $q' = c$ is symmetric), so $q \notin X$. The links of $E(c) \setminus \{\ell\}$ have $[e \in F] = 1$, and ℓ has $[\ell \in F] + [\ell \in E(q')] = 1$; so each of the four links of c lies in a plaquette of X. If $c \notin X$, these are four distinct plaquettes, so $|X| \geq 4$. If $c \in X$ and $X \neq \{c\}$, a plaquette $p \in X \setminus \{c\}$ has at most one link in each of $E(q)$ and $E(c)$. So it has a link outside $E(q) \cup E(c)$, which lies in a second plaquette of X other than c, and $|X| \geq 3$.

On a periodic lattice with sides ≥ 8 the lemmas used apply. □

Lemma 12.7 (One and two cubes)**.** *For a plaquette p let U_p be its holonomy in the standard orientation; for an irreducible r put $r^+ := r$ and $r^- := \bar{r}$. Let h_p be continuous class functions with $\sum_r d_r |\hat{h}_p(r)| < \infty$, where $\hat{h}_p(r) := \int h_p \overline{\chi_r}\, dU$.*

(i) Let Γ be the set of faces of a unit 3-cube of the lattice, oriented as its boundary (for an orientation of the 3-cube), and $\epsilon(p) = 1$ or -1 according as the standard orientation of $p \in \Gamma$ is or is not this orientation. Then

$$\int \prod_{p \in \Gamma} h_p(U_p)\, dU = \sum_r d_r^{-4} \prod_{p \in \Gamma} \hat{h}_p(r^{\epsilon(p)}).$$

(ii) Let Γ_1, Γ_2 be as in (i), with exactly one common face s, oriented so that s receives opposite orientations, let $\Sigma := (\Gamma_1 \cup \Gamma_2) \setminus \{s\}$, and let $\epsilon(p)$ be as in (i) for $p \in \Sigma$ as a face of Γ_1 or of Γ_2. Then

$$\int \prod_{p \in \Sigma} h_p(U_p)\, dU = \sum_r d_r^{-8} \prod_{p \in \Sigma} \hat{h}_p(r^{\epsilon(p)}).$$

In particular, if $h_p = f_\beta$ for all but two faces p_1, p_2, the two integrals are

$$\sum_{r \neq 0} u_r^k\, \hat{h}_{p_1}(r^{\epsilon(p_1)})\, \hat{h}_{p_2}(r^{\epsilon(p_2)}), \qquad k = 4 \text{ and } k = 8.$$

Proof. (i) Let V_p be the holonomy of p in its boundary orientation. If $\epsilon(p)=1$, $h_p(U_p)=\sum_r \hat h_p(r)\chi_r(V_p)$; if $\epsilon(p)=-1$, U_p is conjugate to V_p^{-1} and $h_p(U_p)=\sum_r \hat h_p(r)\chi_{\bar r}(V_p)=\sum_r \hat h_p(\bar r)\chi_r(V_p)$. So in both cases $h_p(U_p)=\sum_r \hat h_p(r^{\epsilon(p)})\chi_r(V_p)$. Insert these expansions. The six-fold series converges absolutely, as in the proof of Lemma 5.1(ii), and may be integrated term by term. By the identity quoted there, $\int\prod_p \chi_{r_p}(V_p)\,dU$ is d_r^{-4} if all the r_p equal the same r and 0 otherwise.

(ii) Two unit cubes with exactly one common face share exactly its four links. So we may fix the links of s and integrate separately over the other links of Γ_1 and of Γ_2, obtaining functions g_1 and g_2 of the links of s. A gauge transformation at the vertices of s acts on the links of s and on the links of Γ_k at these vertices; changing variables in the latter (Haar measure is invariant) shows that g_k is invariant under it. So $g_k=G_k(V_k)$ for a class function G_k of the holonomy V_k of s in its orientation in Γ_k, and V_2 is conjugate to V_1^{-1}. By (i), applied to Γ_k with the function $V_k\mapsto\overline{\chi_t(V_k)}=\chi_{\bar t}(V_k)$ on s, $\int G_k\overline{\chi_t}=d_t^{-4}c_k(t)$ with $c_k(t):=\prod_{p\in\Gamma_k\setminus\{s\}}\hat h_p(\bar t^{\,\epsilon(p)})$. The holonomy $V:=V_1$ of the four links of s is Haar distributed, so

$$\int g_1g_2=\int G_1(V)G_2(V^{-1})\,dV=\sum_{t,t'}d_t^{-4}d_{t'}^{-4}c_1(t)c_2(t')\int\chi_t(V)\overline{\chi_{t'}(V)}\,dV=\sum_t d_t^{-8}c_1(t)c_2(t),$$

which is the claim with $r=\bar t$. Finally $\hat f_\beta(r)=d_ru_r$ for $r\neq0$ and $\hat f_\beta(0)=0$, and $f_\beta(U^{-1})=f_\beta(U)$. □

Theorem 12.8 (The upper bound to fourth order)**.** *Let $d\geq3$, G metrisable and ρ irreducible and non-trivial. Let ψ be one of the real unit eigenvectors of $\mathsf V$ of Lemma 10.1(i) (χ_ρ if $\iota=1$; $\sqrt2\,\mathrm{Re}\,\chi_\rho$ or $\sqrt2\,\mathrm{Im}\,\chi_\rho$ if $\iota=0$), Υ_ψ its eigenvalue, and $C=+$ if $\psi(U^{-1})=\psi(U)$, $C=-$ if $\psi(U^{-1})=-\psi(U)$. Let $\mathcal W$ be a finite set of plaquettes of the time-zero slice of $\mathbb Z^d$ and $a\in\mathbb R^{\mathcal W}\setminus\{0\}$. Then, as $\beta\downarrow0$,*

$$m(\mu_\beta)\;\leq\;4\log\frac{1}{u(\beta)}-\log\Upsilon_\psi(\beta)-\log\Big(1+u(\beta)^4\,\frac{a^\top H_{C,\mathcal W}\,a}{a^\top a}\Big)+O(\beta^5),$$

with a constant depending on d, G, ρ and $\mathcal W$ but not on a. The same holds for $m_{\Lambda_s}(\beta)$, uniformly in the periodic spatial lattices whose sides all exceed by at least 12 the side of a box containing $\mathcal W$. Consequently

$$\limsup_{\beta\downarrow0}\;\beta^{-4}\Big(m(\mu_\beta)-4\log\frac{1}{u(\beta)}+\log\Upsilon_\psi(\beta)\Big)\;\leq\;-h_C\,u_1^4,\qquad u_1:=\lim_{\beta\downarrow0}\frac{u(\beta)}{\beta}=\frac{1+\iota}{2N^2}.$$

Proof. Write $f=f_\beta$, $u=u(\beta)$, $\Upsilon=\Upsilon_\psi$, $\mathsf s=\mathsf s_C$, $|a|^2=a^\top a$; constants may depend on $d,G,\rho,\mathcal W$ but not on a, since every remainder below is a multiple of $(\sum_q|a_q|)^2\leq|\mathcal W|\,|a|^2$. Let $\Lambda=(\mathbb Z/L_0\mathbb Z)\times\Lambda_s$ with $L_0\geq3$ and Λ_s as in the statement. Let B^+ be the box obtained by enlarging by 3 in every direction the smallest box containing the vertices of the plaquettes of $\mathcal W$, and E_0 the set of spatial links with both endpoints in B^+. All bounds below are uniform in L_0 and Λ_s, so, as in the proof of Theorem 5.6, they hold for the limits $L_0\to\infty$ and for μ_β.

Step 1: means, and the spectral inequality. In this proof $\langle\cdot\rangle_{\Lambda,\beta}$ is the Gibbs state and $\int\cdot\,dU$ the Haar integral. Put $m_w:=\int\psi(1+f)\,dU=\int f\psi\,dU$, $\tilde\psi:=\psi-m_w$ and $\tilde a:=\tilde\psi(1+f)$, a real class function with $\int\tilde a\,dU=0$ and absolutely summable character expansion. Since $\mathsf V\psi=P_\rho M_{1+f}\psi-(\int f\psi\,dU)P_\rho f$,

$$P_\rho\tilde a=P_\rho M_{1+f}\psi-m_wP_\rho f=\mathsf V\psi=\Upsilon\psi,$$

and $\|(1-P_\rho)\tilde a\|_2\leq\|\psi f\|_2+|m_w|(1+\|f\|_2)=O(\beta)$. Let $A_q:=\psi(U_q)$ for $q\in\mathcal W$ (time 0), $A_q(1)$ its translate to time 1, and $\Phi:=\sum_q a_qA_q$. By the proof of Theorem 10.2, which uses

only $\mathsf{V}\psi = \Upsilon\psi$, $\mathrm{Var}(A_q) \le \Upsilon + O(\beta^5)$ and $|\langle A_q\rangle_{\Lambda,\beta} - m_w| = O(\beta^5)$ for every q. As in the proof of Theorem 5.6, $m \le \log(\mathrm{Var}(\Phi)/\operatorname{Cov}(\Phi(0),\Phi(1)))$ once the covariance is positive.

Step 2: the variance. Let $q \neq q'$ in $\mathcal{W}$. Then $\operatorname{Cov}(A_q, A_{q'}) = \langle\tilde\psi(U_q)\tilde\psi(U_{q'})\rangle_{\Lambda,\beta} + O(\beta^{10})$. Let $\mathcal{C}$ be the smallest set of plaquettes of the slice that contains q, q' and every plaquette of the slice with at least two links in $S := E(\mathcal{C})$. It exists, and lies in the smallest box containing q and q', because a plaquette with two links in a set of links lies in every box containing them. A temporal plaquette contains one link of the slice, so every plaquette outside $\mathcal{C}$ contains at most one link of S. The derivation of (1) applies, with Lemma 5.4 for $X \setminus \mathcal{C}$ and Lemma 6.1 for the ratios (θ_5 in place of θ). It gives $\langle\tilde\psi(U_q)\tilde\psi(U_{q'})\rangle_{\Lambda,\beta} = R\,M + O(\beta^5)$, with $R = 1 + O(\beta^5)$ and

$$M = \int \tilde\psi(U_q)\tilde\psi(U_{q'}) \prod_{p\in\mathcal{C}} (1 + f(U_p))\, dU = \sum_X \int \tilde a(U_q)\tilde a(U_{q'}) \prod_{p\in X} f(U_p)\, dU,$$

the sum over $X \subseteq \mathcal{C} \setminus \{q, q'\}$. The functions $\tilde a$ and f have Haar mean zero, so a term vanishes if a link lies in exactly one of its factors. Otherwise $X \cup \{q, q'\}$ is as in Lemma 12.5, so $|X| \ge 5$ unless q, q' are faces of a unit cube Γ and $X = \Gamma \setminus \{q, q'\}$. The terms with $|X| \ge 5$ are at most $\epsilon^5 \|\tilde a\|_2^2$, because U_q and $U_{q'}$ are each Haar distributed. The term of the cube is, by Lemma 12.7(i), $\sum_{r\neq 0} u_r^4 \hat{\tilde a}(r^{\epsilon_\Gamma(q)})\hat{\tilde a}(r^{\epsilon_\Gamma(q')})$. Here $\hat{\tilde a}(r) = \Upsilon\hat\psi(r)$ for $r \in \{\rho, \bar\rho\}$; $\hat\psi(\bar r) = \overline{\hat\psi(r)}$; and $\sum_r \hat\psi(r)^2$ is 1 if $C = +$ and -1 if $C = -$ (for $\psi = \sqrt{2}\,\mathrm{Im}\,\chi_\rho$, $\hat\psi(\rho) = -i/\sqrt{2} = -\hat\psi(\bar\rho)$). Hence the terms $r \in \{\rho, \bar\rho\}$ give $(-\epsilon_\Gamma(q)\epsilon_\Gamma(q'))^{\mathsf{s}} u^4\Upsilon^2 = \sigma_\Gamma(q,q')^{\mathsf{s}} u^4\Upsilon^2$, and the others at most $\epsilon^4\|(1 - P_\rho)\tilde a\|_2^2 = O(\beta^6)$. So

$$\operatorname{Cov}(A_q, A_{q'}) = [\exists\,\Gamma \ni q, q']\,\sigma_\Gamma(q,q')^{\mathsf{s}} u^4\Upsilon^2 + O(\beta^5),$$
$$\mathrm{Var}(\Phi) \le \Upsilon|a|^2 + u^4\Upsilon^2\, a^\top C_{\mathcal{W}} a + O(\beta^5)|a|^2,$$

where $C_{\mathcal{W}}(q,q') := \sum_{\Gamma\ni q,q'} \sigma_\Gamma(q,q')^{\mathsf{s}}$ for $q \neq q'$ and $C_{\mathcal{W}}(q,q) := 0$.

Step 3: positivity with several fronts. For $\ell \in E_0$ let s_ℓ be the temporal plaquette of layer 0 based on ℓ. By Lemma 5.2, for every real b, $\langle\Delta\rangle_{\Lambda,\beta} = \sum_{F\subseteq E_0} T_F$ with $\Delta := (\Phi(0) - b)(\Phi(1) - b)$ and

$$T_F = \Big\langle \Delta \prod_{\ell\in F} \varphi_\beta(U_{s_\ell}) \prod_{\ell\in E_0\setminus F} (1 - \varphi_\beta(U_{s_\ell})) \Big\rangle_{\Lambda,\beta} \;\ge\; 0.$$

Take $b := m_w \sum_q a_q$; then $\langle\Delta\rangle_{\Lambda,\beta} = \operatorname{Cov}(\Phi(0),\Phi(1)) + (\langle\Phi\rangle_{\Lambda,\beta} - b)^2$ and $|\langle\Phi\rangle_{\Lambda,\beta} - b| = O(\beta^5)|a|$. Let $\mathcal{F}$ consist of the sets of links ∂q ($q \in \mathcal{W}$) and $\partial q \triangle \partial c$ ($q \in \mathcal{W}$, $c \sim q$). They lie in E_0 and are distinct: ∂q determines q, and q and c are the only plaquettes with three links in $\partial q \triangle \partial c$ (a plaquette with three links in a six-cycle contains three consecutive links of it, and only the three consecutive links of q and those of c lie in one plaquette). Hence

$$\operatorname{Cov}(\Phi(0),\Phi(1)) \;\ge\; \sum_{F\in\mathcal{F}} T_F - O(\beta^{10})|a|^2.$$

Step 4: localisation. Fix $F \in \mathcal{F}$. Since $w_\beta\varphi_\beta = c_0 f$, $w_\beta(1 - \varphi_\beta) = c_0$ and $w_\beta = c_0(1 + f)$, with $\mathcal{P}' := \mathcal{P} \setminus \{s_\ell : \ell \in E_0\}$,

$$T_F = \sum_{q',q''\in\mathcal{W}} a_{q'}a_{q''} T_F(q',q''), \qquad T_F(q',q'') := \frac{1}{\Xi(\mathcal{P})} \int H' \prod_{p\in\mathcal{P}'} (1 + f(U_p))\, dU,$$
$$H' := \tilde\psi(U_{q'})\,\tilde\psi(U_{q_1''}) \prod_{\ell\in F} f(U_{s_\ell}),$$

where q_1'' is the translate of q'' to time 1. Let $\mathcal{C}$ be the smallest set of spatial plaquettes of the slices 0 and 1 that contains q' and q_1'' and every spatial plaquette of these slices with at least two links in $S := E(\mathcal{C}) \cup \bigcup_{\ell\in F} E(s_\ell)$. Its links lie in the box spanned by $E(\mathcal{W})$ and F, so a temporal plaquette of layer 0 with two links in S is based on a link of E_0, hence not in $\mathcal{P}'$;

and the other temporal plaquettes contain at most one link of S. So every plaquette of $\mathcal{P}' \setminus \mathcal{C}$ contains at most one link of S. The proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)], with $\mathcal{P}'$ in place of $\mathcal{P}$ in the expansion of the product and S in place of E_C, writes $T_F(q', q'')$ as $\sum_X \Phi_{H'}(X)\, \Xi(\mathcal{P}' \setminus \overline{X})/\Xi(\mathcal{P})$. By Lemma 6.1 the ratio lies between $e^{\mp\theta_5(|E_0|+|\overline{X}|)}$, and the derivation of (1), with Lemma 5.4 for $X \setminus \mathcal{C}$, gives

$$T_F(q', q'') = R\, M + O\Big(\beta^5 \int |H'|\, dU\Big), \qquad M := \int H' \prod_{p \in \mathcal{C}} (1 + f(U_p))\, dU, \qquad R = 1 + O(\beta^5).$$

For fixed links of slice 0 and temporal links, the map from the links $(\ell, 1)$, $\ell \in F$, to the holonomies of the s_ℓ is a bijection preserving Haar measure, and likewise with the links $(\ell, 0)$. By the Cauchy–Schwarz inequality, as in the proof of Lemma 5.5, $\int |H'| \le \|\tilde{\psi}\|_2^2 (\int |f|)^{|F|} = O(\beta^{|F|})$.

Step 5: the local integrals. Since $q', q_1'' \in \mathcal{C}$,

$$M = \sum_{X_0, X_1} M(X_0, X_1), \qquad M(X_0, X_1) := \int \tilde{a}(U_{q'})\, \tilde{a}(U_{q_1''}) \prod_{\ell \in F} f(U_{s_\ell}) \prod_{p \in X_0 \cup X_1} f(U_p)\, dU,$$

over the sets X_0 and X_1 of plaquettes of $\mathcal{C}$ of the slices 0 and 1 other than q' and q_1''. All these functions have Haar mean zero, so $M(X_0, X_1) = 0$ if some link lies in exactly one of the plaquettes q', q_1'', s_ℓ ($\ell \in F$) and those of $X_0 \cup X_1$. A link of slice 0 lies in s_ℓ if and only if it is ℓ, and the vertical links lie only in the s_ℓ (two each, F being a cycle). So (q', X_0), and (q'', X_1) translated to time 0, satisfy the hypothesis of Lemma 12.6. Every term is at most $\epsilon^{|X_0|+|X_1|} \|\tilde{a}\|_2^2 (\int |f|)^{|F|}$, by the same change of variables.

(a) $F = \partial q$. By Lemma 12.6(i) the terms with $|X_0| + |X_1| \ge 5$ are $O(\beta^9)$, and the others are of two kinds.

- $q' = q'' = q$, $X_0 = X_1 = \emptyset$. The integral is over the unit cube $q \times [0, 1]$, and Lemma 12.7(i) gives $\sum_{r \ne 0} u_r^4 |\hat{\tilde{a}}(r)|^2 \ge u^4 \|P_\rho \tilde{a}\|_2^2 = u^4 \Upsilon^2$.
- $q'' = q$, $X_1 = \emptyset$, and q', q faces of a unit cube Γ of slice 0 with $X_0 = \Gamma \setminus \{q, q'\}$, or the same at time 1. This is $\sigma_\Gamma(q, q')^{\mathsf{s}} u^8 \Upsilon^2 + O(\beta^{10})$, by Lemma 12.7(ii) for Γ and $q \times [0, 1]$ and the signs below.

(b) $F = \partial q \triangle \partial c$. By Lemma 12.6(ii) the terms with $|X_0| \ge 3$ or $|X_1| \ge 3$ are $O(\beta^9)$. The others have $q', q'' \in \{q, c\}$, $X_0 = \{q, c\} \setminus \{q'\}$ and X_1 the translate of $\{q, c\} \setminus \{q''\}$. Their faces are those of the unit cubes $q \times [0, 1]$ and $c \times [0, 1]$ other than s_ℓ, ℓ the common link of q and c. By Lemma 12.7(ii) and the signs below they equal $\tau(q', q'')^{\mathsf{s}} u^8 \Upsilon^2 + O(\beta^{10})$, with $\tau(q, q) = \tau(c, c) = 1$ (bulges) and $\tau(q, c) = \tau(c, q) = \sigma(q, c)$ (hops).

Signs. In Lemma 12.7 let p_1, p_2 be the faces carrying $\tilde{a}$. As in Step 2, the terms $r \in \{\rho, \bar{\rho}\}$ give $(-\epsilon(p_1)\epsilon(p_2))^{\mathsf{s}} u^k \Upsilon^2$, and the others $O(\beta^{k+2})$. In a unit cube $p \times [0, 1]$ the bottom and the top have the same standard orientation and opposite outward normals, so $\epsilon(p_1) = -\epsilon(p)$ for the top p_1. For $q \times [0, 1]$ and $c \times [0, 1]$, the link $(\ell, 0)$ lies only in the faces q and c of Σ, which traverse it in opposite directions, so $\epsilon(q)\partial q(\ell) + \epsilon(c)\partial c(\ell) = 0$ and $\epsilon(c) = \sigma(q, c)\epsilon(q)$. A hop from q to c therefore has sign $-\epsilon(q) \cdot (-\epsilon(c)) = \sigma(q, c)$, and a bulge $-\epsilon(q) \cdot (-\epsilon(q)) = 1$. For Γ and $Q := q \times [0, 1]$, q receives opposite orientations, so in Q the top q_1 has $\epsilon(q_1) = \epsilon_\Gamma(q)$. A turn or a jump from q' to q has the sign $-\epsilon_\Gamma(q')\epsilon_\Gamma(q) = \sigma_\Gamma(q, q')$.

Step 6: assembly. By Step 4, $R\, M$ differs from M by $O(\beta^9)$, and there are finitely many terms. Summing (a) over $q \in \mathcal{W}$ gives at least $u^4 \Upsilon^2 |a|^2 + 2u^8 \Upsilon^2 a^\top C_{\mathcal{W}} a$. Each unordered pair $\{q, c\}$, $q \sim c$, with $q \in \mathcal{W}$ or $c \in \mathcal{W}$, occurs once in $\mathcal{F}$ and gives $u^8 \Upsilon^2 (a_q^2 + a_c^2 + 2\sigma(q, c)^{\mathsf{s}} a_q a_c)$, with $a_c := 0$ for $c \notin \mathcal{W}$. Every $q \in \mathcal{W}$ has K_S neighbours c, so (b) gives $u^8 \Upsilon^2 a^\top (K_S \mathbf{1} + A_{\mathcal{W}}) a$, where $A_{\mathcal{W}}(q, c) := [q \sim c]\sigma(q, c)^{\mathsf{s}}$. Hence

$$\mathrm{Cov}(\Phi(0), \Phi(1)) \;\ge\; u^4 \Upsilon^2 \Big(|a|^2 + u^4 a^\top (K_S \mathbf{1} + A_{\mathcal{W}} + 2C_{\mathcal{W}}) a \Big) - O(\beta^9) |a|^2.$$

With Step 2, $\Upsilon = 1 + O(\beta)$ and $K_S\mathbf{1} + A_{\mathcal{W}} + C_{\mathcal{W}} = H_{C,\mathcal{W}}$ (Definition 12.1),

$$\begin{aligned}\frac{\mathrm{Cov}(\Phi(0),\Phi(1))}{\mathrm{Var}(\Phi)} &\geq u^4\Upsilon\,\frac{1 + u^4 a^\top (K_S\mathbf{1} + A_{\mathcal{W}} + 2C_{\mathcal{W}})a/|a|^2 - O(\beta^5)}{1 + u^4 a^\top C_{\mathcal{W}} a/|a|^2 + O(\beta^5)} \\ &= u^4\Upsilon\Big(1 + u^4\,\frac{a^\top H_{C,\mathcal{W}} a}{|a|^2} - O(\beta^5)\Big),\end{aligned}$$

which is positive for small β, and Step 1 gives the bound. For the last statement take $\mathcal{W} = N_L$ and $a = \mathbf{1}$ (Lemma 12.2(iii)), use $u/\beta \to u_1$ (Lemma 5.1(iii)) and let $L \to \infty$. □

Corollary 12.9 (Beyond $-\log\Upsilon$ the gap drops at order β^4)**.** *With the fundamental representations, for $SU(N)$, $2 \leq N \leq 5$, and for $U(1)$,*

$$\limsup_{\beta\downarrow 0}\ \beta^{-4}\Big(m(\mu_\beta) - 4\log\frac{1}{u(\beta)} + \log\Upsilon(\beta)\Big)\ \leq\ -h_C u_1^4,$$

where $h_C u_1^4 = \frac{26d-70}{256}$ for $SU(2)$ ($\frac{17}{128}$ if $d = 4$, $\frac{1}{32}$ if $d = 3$), $\frac{26d-70}{16N^8}$ for $SU(N)$ with $3 \leq N \leq 5$, and $\frac{10d-22}{16}$ for $U(1)$. With Corollary 11.6, $-c\beta^4 \leq m(\mu_\beta) - 4\log(1/u) + \log\Upsilon \leq -(h_C u_1^4 - o(1))\beta^4$ for some c: the difference is negative and of exact order β^4, and Conjecture 12.4 identifies its leading term as $-h_C u^4$; Corollary 14.3 proves this for $SU(N)$, $2 \leq N \leq 5$, and for $U(1)$ in $d = 3$. For every $SU(N)$ the same bound holds with Υ_+ in place of Υ.

Proof. Apply Theorem 12.8 with ψ the eigenvector of Υ. For $SU(2)$, $\iota = 1$ and $\psi = \chi_\rho$, so $C = +$. For $SU(N)$ with $3 \leq N \leq 5$, $\Upsilon = \Upsilon_+ > \Upsilon_-$ for small β (Lemmas 10.1(iii) and 11.1), so $C = +$. For $U(1)$, $\Upsilon = \Upsilon_-$ (Lemma 10.1(iii)), so $C = -$. Finally $u_1 = \frac{1+\iota}{2N^2}$ is $\frac{1}{4}$, $\frac{1}{2N^2}$ and $\frac{1}{2}$ in the three cases. □

Remark 12.10 (Evidence, numbers and what remains). (i) Numbers. For $SU(2)$ in $d = 4$ the conjectured term of order β^4 of $m(\mu_\beta) - 4\log(4/\beta)$ is $-\frac{167}{1152} = -0.145$; the part $-\log\Upsilon$ and the expansion of $4\log(1/u)$ contribute $-\frac{7}{576}$, and the hopping $-\frac{34}{256}$. In terms of u the term is $-\frac{98}{3}u^4$, of which $\frac{4}{3}u^4$ comes from $-\log\Upsilon$. The lower bound of Theorem 11.5 still has a constant of about $8.4 \cdot 10^{10}$ at this order (Remark 11.7(iii)).

(ii) Exact spectra (Numerical; `theory/checks/fourth-order/`, a compass).

- $\mathbb{Z}_2$ in dual plaquette variables. After subtracting $4\log(1/u) - \log(1-u^2)$, every level m_i of the plaquette band, divided by u^4, tends to $-\log(1 + h_i u^4)/u^4 \approx -h_i$, where h_i are the eigenvalues of H for the same geometry, with relative corrections $O(u^2)$. This holds on the 3×3 and 4×4 tori ($d = 3$; on 4×4, $h_i = 8, 6^4, 4^6, 2^4, 0$: all 16 levels) and on blocks of $1 \times 1 \times 1$, $1 \times 1 \times 2$ and $1 \times 2 \times 2$ cubes with open boundary ($d = 4$; all 6, 11 and 20 levels), and for the four lowest levels of the $1 \times 2 \times 3$ block (2^{23} states). Without the turns and jumps the top value for one cube would be 8 instead of 13.

- $\mathbb{Z}_3$, both sectors. The same holds, with relative corrections $O(u)$, on the 3×3 torus and on the $1 \times 1 \times 1$ and $1 \times 1 \times 2$ blocks. On one cube the C-odd levels follow the signed matrix (eigenvalues $5, 5, 5, 3, 3, 3$) and the C-even ones the unsigned one $(13, 3, 3, 3, 1, 1)$.

- Lemmas 12.5 and 12.6, by exhaustive enumeration in spatial dimensions 2, 3 and 4; and Lemma 12.2(ii)–(iv), by building H_+ and H_- from the orientations on tori.

(iii) The classical series (Cited). Langelage, Münster and Philipsen [LMP07, eqs. (2.14)–(2.18)], "recognizing the first orders of the corresponding glueball masses" of [Mün81, Seo82],

give for $d = 4$

$$\begin{aligned} SU(2): \quad m(A_1^{++}) &= -4\log u + 2u^2 - \tfrac{98}{3}u^4 - \dots, \\ m(E^{++}) &= -4\log u + 2u^2 - \tfrac{26}{3}u^4 + \dots, \\ SU(3): \quad m(A_1^{++}) &= -4\log u - 3u + 9u^2 - \tfrac{27}{2}u^3 - 7u^4 - \dots, \\ m(E^{++}) &= -4\log u - 3u + 9u^2 - \tfrac{27}{2}u^3 + 17u^4 - \dots, \\ m(T_1^{+-}) &= -4\log u + 3u + \tfrac{9}{2}u^3 - \tfrac{98}{4}u^4 + \dots \end{aligned}$$

[LMP08, eq. (29)] repeats the $SU(2)$ series. At momentum 0 the conjecture predicts $-\log\Upsilon_C - hu^4$ with $h = 34$ (A_1^{++}), 10 (E^{++}) and 18 (T_1^{+-}, $C = -$) (Lemma 12.2(iv)). With the exact $-\log\Upsilon_C$ this gives $2u^2 - \frac{98}{3}u^4$ and $2u^2 - \frac{26}{3}u^4$ for $SU(2)$, and $-3u + 9u^2 - \frac{27}{2}u^3 - 7u^4$, the same with $+17u^4$, and $3u + \frac{9}{2}u^3 - \frac{99}{4}u^4$ for $SU(3)$ (`fourth_order_values.py`). The first four agree with [LMP07] term by term through u^4. For T_1^{+-} the terms through u^3 agree, and [LMP07] prints $-\frac{98}{4}u^4$ in both arXiv versions, a difference of $u^4/4$. It is very probably a misprint for $-\frac{99}{4}$, for three reasons.

- Of the coefficients printed in [LMP07, eqs. (2.14)–(2.18)], $\frac{98}{4}$ is the only one not in lowest terms.
- (Heuristic.) In the band picture of Remark 12.3, the coefficient of u^4 minus that of $-\log\Upsilon_C$ is minus an eigenvalue of an integer matrix at $k = 0$, and on T_1 the block at $k = 0$ is a multiple of the identity (cubic symmetry), so this difference is an integer. The four other channels give -34, -10, -34, -10; the printed T_1^{+-} value gives $-\frac{71}{4}$, while $-\frac{99}{4}$ gives -18.
- On our side, the series of $\Upsilon_\pm$ for $SU(3)$ agree with Weyl integration to about 10^{-16}, and on one cube of $\mathbb{Z}_3$ the C-odd levels follow the model.

We have not consulted [Mün81, Seo82]. Two further points of [LMP07] support the picture. First, its eqs. (2.6)–(2.9) write the tubes as $u^{4N_t}a^{N_t}$ ($SU(2)$) and $u^{4N_t}(b^{N_t} + c^{N_t})$ ($SU(3)$), with $a = 1 + 3v - 4u^2$, $b = 1 - 3u - 6v + 8w$ and $c = 1 + 3u + 6v + 8w - 18u^2$, where v, w are the expansion parameters of the next representations. These are exactly Υ for $SU(2)$ and Υ_-, Υ_+ for $SU(3)$: in Lemma 10.1, $\kappa = 3u_1$ for $SU(2)$, and $\alpha_\rho = 8u_8$, $\gamma_\rho = 3u + 6u_6$ for $SU(3)$ (from $3 \otimes \bar{3} = 1 \oplus 8$ and $3 \otimes 3 = \bar{3} \oplus 6$). The referee of this section checked these identities as exact series to order β^8; so the terms of orders u, u^2, u^3 of all five series are those of $-\log\Upsilon_C$, as Corollaries 10.9 and 11.6 and the C-odd first-order term of Remark 9.7(iii) predict. Second, the decorations of these tubes in the free energies [LMP07, eqs. (2.10)–(2.11)], [LMP08, eq. (23)] begin with $1 + 12N_t u^4$ in every sector, and $12 = K_S$ in $d = 4$: the bulges, with the sign $+$ in the C-odd sector as well. There is also a tension (Heuristic). In the band picture the coefficient of $N_t^2 u^8$ in that bracket is $\frac{1}{2}\sum_{q'} H_+(q, q')^2 = 91$. [LMP07, eq. (2.10)] prints 86 and [LMP08, eq. (23)] prints 83, which is what one obtains without the paths that hop to a perpendicular neighbour and turn back through the common cube, or the reverse. This bears on the band away from $k = 0$, not on the masses compared above.

(iv) The C-odd mass (Heuristic; proof sketched). For $SU(N)$, where $U \mapsto \bar{U}$ is an automorphism and the transfer matrix commutes with charge conjugation, the proof of Theorem 12.8 with $\psi = \sqrt{2}\,\mathrm{Im}\,\chi_\rho$, applied to the transfer matrix restricted to the C-odd sector, should bound the lowest mass of that sector by $4\log(1/u) - \log\Upsilon_- - h_- u^4 + o(\beta^4)$. We have not written out the spectral statement for that sector.

(v) What remains: the lower bound at order β^4. By Theorem 13.4, the constant δ' contains at order β^4 only the moves of H_C: the turns and jumps through a unit cube (Lemma 13.1(i)), at the sup norm b^4, and the bulges and hops (the term $bf(6)$: a widening to $\partial q \triangle \partial c$ and one of the two narrowings, Lemma 13.3), at b^2a^2. The shifts and jumps and the U-turns are of

order β^5 there. Section 14 evaluates these junctions by characters (Lemma 14.1): u for each widening or narrowing (so u^4 per bulge or hop) and u^4 per turn or jump. The recursion then sums the moves at a slice by the row sums of $|H_C|$, which are h_+. For $C = +$ this is sharp, because H_+ has non-negative entries (Theorem 14.2, Corollary 14.3). For the C-odd gap of $U(1)$ in $d \geq 4$ the signs must be kept: bounding the hop and the turn between perpendicular neighbours separately in absolute value gives H_+, with row sum $h_+ > h_-$, instead of H_-. Two terms of Theorem 11.5 are $O(\beta^5)$ by Lemma 13.1(iii),(iv): the b^2a^2 of the junctions with three or more front-pairs and two or more plaquettes, and the ba^3 of those with one plaquette. The terms of order u^6 (for $SU(2)$ in $d = 4$, $-\frac{20984}{405}u^6$ in [LMP07]) involve the vacuum cubes and pairs of moves.

13 Junctions of order β^4; the third-order term with an effective constant

In Theorem 11.5 the constant in $\delta' = O(\beta^4)$ is about $8.4 \cdot 10^{10}$ for $SU(2)$ in $d = 4$. Almost all of it is the crude count of the junctions with four plaquettes and a four-edge exit (Remark 11.7(iii)). With Lemmas 5.4 and 12.5 we determine all junctions that enter the junction sum at order β^4 (Lemma 13.1):

- the junctions with four plaquettes and two four-edge front-pairs are the turns, jumps and U-turns through a unit cube, so $Z_{4,4} \leq 12(d-3)$;
- a junction with one plaquette whose second front-pair has more than four edges widens the tube to the six-edge front $\partial q \triangle \partial c$, or to a front of seven or eight edges;
- one plaquette and three front-pairs force a front of at least eight edges, and two plaquettes and three front-pairs one of at least six.

A six-edge front $\partial q \triangle \partial c$, in turn, ends at leading order in one of exactly two junctions, which narrow it back to ∂q or to ∂c (Lemma 13.3). The junction sums then have explicit terms of order β^4 (Lemma 13.2). The shifts and jumps of Section 10 drop out with $\varpi = b^{5/2}$, and the U-turns because their exit runs back in time. So the third-order lower bound holds with $\delta' = (\delta'_4 + o(1))\beta^4$ (Theorem 13.4), where δ'_4 has two terms:

- the turns and jumps through a unit cube;
- the bulges and hops through six-edge fronts.

These are the moves of the plaquette state of Section 12, counted with the sup norm $b \approx 2\beta$ for each plaquette instead of in characters, where each move costs u^4. For $SU(2)$, $\delta'_4 = 2$ ($d = 3$) and 166 ($d = 4$), instead of $3.4 \cdot 10^8$ and $8.4 \cdot 10^{10}$. Evaluating the moves by characters is what the lower bound at fourth order needs (Remark 13.5). Throughout this section all spatial sides are ≥ 8, so that fronts with four edges are plaquette boundaries, there are no fronts with five edges, and fronts with six edges are cycles (Lemmas 7.5 and 8.5).

Lemma 13.1 (Junctions of order β^4)**.** *Suppose all spatial sides are ≥ 8. Let J be a junction without support at a slice s, and P the set of plaquettes of its clusters, so $|P| = |J|_C$.*

(i) If $\Phi(J) = \{p, p'\}$ with $|p| = |p'| = 4$ and $|P| = 4$, then $p = (\partial q, \lambda)$ and $p' = (\partial q', \lambda')$, where $q \neq q'$ are faces of a unit cube Γ of the slice, $P = \Gamma \setminus \{q, q'\}$, and q, q' are opposite faces if $\lambda = \lambda'$. Hence $Z_{4,4} \leq 12(d-3)$ in Lemma 8.7: at most $10(d-3)$ pairs with $\lambda' \neq \lambda$ (turns and jumps) and $2(d-3)$ with $\lambda' = \lambda$ (U-turns).

(ii) *If $P = \{c\}$ and $\Phi(J) = \{p, p'\}$ with $p = (\partial q, \lambda)$ and $|p'| \geq 5$, then $\lambda' \neq \lambda$ and $c \neq q$, and the front φ' of p' is one of the following: $\partial q \triangle \partial c$ (six edges) or $\partial q \cup \partial c$ (seven edges), where c shares a link with q; or $\partial q \cup \partial c$ (eight edges), where c shares exactly one vertex with q. So, given p, there are at most K_S such pairs (J, p') with $|p'| = 6$, K_S with $|p'| = 7$, $4(P_v - 1) - 2K_S$ with $|p'| = 8$, and none with $|p'| = 5$ or $|p'| \geq 9$.*

(iii) *If $P = \{c\}$ and $|\Phi(J)| = 3$, then one of the fronts has at least eight edges; if one front-pair has four edges, one of the other two has at least eight edges.*

(iv) *If $|P| = 2$, $|\Phi(J)| = 3$ and two front-pairs have four edges, then the third has at least six edges.*

Proof. Write $A := E_{s-1}(J)$ and $B := E_s(J)$, and $\pi(e)$ for the number of plaquettes of P containing e. By Lemma 8.2(iii), $[e \in A] + [e \in B] + \pi(e) \geq 2$ for every $e \in E(J)$. Hence:

(a) $A \triangle B \subseteq E(P)$;

(b) a link of $E(P)$ outside $A \cup B$ lies in two plaquettes of P.

Distinct fronts of one layer are distinct components of T_λ, hence vertex-disjoint, and every front-pair touches a plaquette of P (Lemma 8.2(i)). A front is connected, and each of its vertices has degree ≥ 2 in it (Lemma 7.2(i)); so a front contained in a union of vertex-disjoint fronts is contained in one of them, and a front contained in a cycle is that cycle. Two distinct plaquettes share at most one link (proof of Lemma 5.4), so a plaquette other than q has at most one link in ∂q.

(i) As the sides are ≥ 5, $p = (\partial q, \lambda)$ and $p' = (\partial q', \lambda')$. If $\lambda = \lambda'$, then $q \neq q'$ and the two fronts are vertex-disjoint. In both cases $[e \in A] + [e \in B] = [e \in \partial q] + [e \in \partial q']$, so every link e of $E(J) = E(P) \cup \partial q \cup \partial q'$ has $n(e) := [e \in \partial q] + [e \in \partial q'] + \pi(e) \geq 2$.

- $q = q'$ (so $\lambda \neq \lambda'$). The set $Y := P \setminus \{q\}$ has at least three plaquettes, each with at most one link in ∂q, and every link of $E(Y) \setminus \partial q$ has $n(e) = \pi(e) \geq 2$, so it lies in two plaquettes of Y. Lemma 5.4 with $\mathcal{L} = \partial q$ gives $|Y| \geq 5$, which is impossible.

- $q \neq q'$ and $q, q' \notin P$. Then $n(e)$ is the number of plaquettes of $Y := P \cup \{q, q'\}$ containing e, so every link of $E(Y)$ lies in at least two plaquettes of the six-element set Y, which is therefore a unit cube by Lemma 12.5. If $\lambda = \lambda'$, the faces q, q' share no vertex and are opposite.

- $q \in P$ and $q' \notin P$ (or the reverse). The plaquettes of $Y := (P \setminus \{q\}) \cup \{q'\}$ differ from q, and for a link e of $E(Y) \setminus \partial q$, $n(e)$ is the number of plaquettes of Y containing e. Lemma 5.4 gives $|P| = |Y| \geq 5$, which is impossible.

- $q, q' \in P$. Each of the other two plaquettes of P has at most one link in each of $\partial q, \partial q'$, so at least two links outside them, each of which lies in the other one. Two plaquettes share at most one link, so this is impossible.

Given p: Γ is one of the $2(d-3)$ unit cubes of the slice containing q; q' is one of its five other faces if $\lambda' \neq \lambda$, and the opposite face if $\lambda' = \lambda$; and then P is determined.

(ii) If $\lambda' = \lambda$, then $A \triangle B = \partial q \sqcup \varphi'$, and (a) would give $4 + |p'| \leq |\partial c| = 4$. So $\lambda' \neq \lambda$, and by (a) and (b), $\partial q \triangle \varphi' \subseteq \partial c \subseteq \partial q \cup \varphi'$. If $c = q$, then $\varphi' \subseteq \partial q$ and $|\varphi'| \leq 4$. So $c \neq q$, and $k := |\partial q \cap \partial c| \leq 1$. The inclusions give $\partial q \triangle \partial c \subseteq \varphi' \subseteq \partial q \cup \partial c$. If $k = 1$, then φ' is $\partial q \triangle \partial c$ or $\partial q \cup \partial c$. If $k = 0$, then $\varphi' = \partial q \cup \partial c$, and c shares exactly one vertex with q: at least one because φ' is connected, and at most one because two distinct plaquettes with two common vertices share the link joining them (two diagonal vertices of a plaquette determine it). The plaquettes sharing exactly one vertex with q number $4(P_v - 1) - 2K_S$: there are $4(P_v - 1)$

pairs (vertex of q, plaquette $\neq q$ at it), and each of the K_S plaquettes sharing a link with q occurs in two of them.

(iii) By (a), $|A\triangle B| \le 4$, and every front touches c (Lemma 11.3). If no front has four edges, all three have at least six. Two of them lie in one layer, so $|A| \ge 12$ for that layer, and $|A\triangle B| \ge 12 - |\psi|$ if the third, ψ, lies in the other layer, while $|A\triangle B| \ge 18$ if it does not; so $|\psi| \ge 8$. Now let $\varphi = \partial q$ be a four-edge front and ψ_1, ψ_2 the other two, and suppose $|\psi_1|, |\psi_2| \le 7$.

- All three in one layer: $|A\triangle B| \ge 12$.
- φ alone in its layer: $|A\triangle B| \ge |\psi_1| + |\psi_2| - 4 \ge 4$, with equality only if $|\psi_1| = |\psi_2| = 4$ and $\varphi \subseteq \psi_1 \sqcup \psi_2$, i.e. φ is one of them, say ψ_1. Then $A\triangle B = \psi_2 \subseteq \partial c$, so $\partial c = \psi_2$, and c does not touch φ.
- φ and ψ_1 in one layer, ψ_2 in the other. Then

$$|A\triangle B| = 4 + |\psi_1| + |\psi_2| - 2|(\varphi \sqcup \psi_1) \cap \psi_2| \le 4.$$

  - If $\psi_2 \subseteq \varphi \sqcup \psi_1$, then $\psi_2 \subseteq \varphi$ or $\psi_2 \subseteq \psi_1$. If $\psi_2 \subseteq \varphi$, then $\psi_2 = \varphi$, so $A\triangle B = \psi_1 \subseteq \partial c$, $\partial c = \psi_1$, and c does not touch φ. If $\psi_2 \subseteq \psi_1$, then $\varphi \subseteq A\triangle B \subseteq \partial c$, so $c = q$, which does not touch ψ_1.
  - Otherwise, ψ_2 has $|\psi_2| - |(\varphi \sqcup \psi_1) \cap \psi_2| \le (|\psi_2| - |\psi_1|)/2 < 2$ links outside $\varphi \sqcup \psi_1$, so exactly one, e_0; then $|\psi_2| \ge |\psi_1| + 2$, so $|\psi_1| = 4$ and $|\psi_2| \in \{6, 7\}$. If ψ_2 had no link in ψ_1, it would lie in $\varphi \cup \{e_0\}$, so $|\psi_2| \le 5$; likewise it has a link in φ. As ψ_2 is connected and φ, ψ_1 are vertex-disjoint, e_0 joins a vertex of φ to one of ψ_1. In $\psi_2 \cap \varphi$, a non-empty subgraph of the 4-cycle φ, every vertex other than the end of e_0 has degree ≥ 2, so $\psi_2 \cap \varphi = \varphi$; likewise $\psi_2 \cap \psi_1 = \psi_1$. Then $|\psi_2| = 9$, a contradiction.

(iv) Let F_1, F_2, F_3 be the three fronts. The third has four edges if it has fewer than six, so all three would be plaquette boundaries $F_j = \partial q_j$. By (a), $|A\triangle B| \le |E(P)| \le 8$. All three cannot lie in one layer, since then $|A\triangle B| = 12$. So let F_1, F_2 lie in one layer (they are vertex-disjoint) and F_3 in the other.

- If $F_3 \notin \{F_1, F_2\}$, it shares at most one link with each, so $|A\triangle B| = 12 - 2|F_3 \cap (F_1 \sqcup F_2)| \ge 8$, and equality holds throughout: the two plaquettes share no link, so $E(P) \subseteq A \cup B$ by (b) and $E(P) = A\triangle B$; and F_3 shares exactly one link with each of F_1, F_2. A plaquette of P has its four links in $F_1 \cup F_2 \cup F_3$ and at most one in each F_j other than its own boundary, so it is one of q_1, q_2, q_3. But every F_j contains a link of $F_3 \cap (F_1 \sqcup F_2) \subseteq A \cap B$, which is not in $A\triangle B = E(P)$: a contradiction.
- If $F_3 = F_1$ (or F_2), then $A\triangle B = F_2$, and by (b) every link of $E(P)$ lies in $F_1 \cup F_2$ or is the common link of the two plaquettes. A plaquette other than q_1, q_2 has at most three links there, so $P = \{q_1, q_2\}$. These are vertex-disjoint, and no front of J meets both, so J is not connected: a contradiction. □

Lemma 13.2 (Junction sums with the junctions of order β^4)**.** *In the setting of Lemma 11.4, suppose all spatial sides are ≥ 8, and let $\bar{b} \ge b$ and $\bar{a} \ge a$. Let $\bar{\xi}_F, \bar{\xi}_C$ and $\bar{g}_F, \dots, \bar{\kappa}_C$ be such that the barred series below converge and that they satisfy the first two conditions of Lemma 8.10 and the six inequalities of Definition 8.9, with $\bar{b}$ in place of b in Y_j, with $\bar{a}$ in place of a in the front weights $a^{k/2}$ of A_j, and with $\bar{f}(4) := 1$, $\bar{f}(k) := f(k)(\bar{a}/a)^{k-4}$ ($k \ge 5$) in place of f in B_j, where f and c_J are those of the bound at hand (Theorem 10.8, or Theorem 13.4 with its $f(6)$). Let $\eta \in [0, 1]$, and let $\mathcal{S}_\eta(p)$ be the sum $\mathcal{S}(p)$ of Lemma 11.4 in*

which the weight $f(|p'|)$ is multiplied by η when p' has the layer of p. Put $\mathsf{F}_8 := 8\sum_{k\ge 8} N_k^v f(k)$, $\mathsf{A}_8 := 8e^{c_J}\sum_{k\ge 8} N_k^v a^{k/2}$,

$$M_4 := \Big(\Big(\frac{b}{\bar b}\Big)^3\Big(\frac{a}{\bar a}\Big)^2 + \Big(\frac{b}{\bar b}\Big)^2\Big(\frac{a}{\bar a}\Big)^3\Big)\,4P_v e^{4\bar\xi_F}\,(\bar\kappa_C\bar Y_2 + 4(\bar g_C\bar g'_F + \bar g'_C\bar g_F)\bar Y_1 + 16\bar\kappa_F\bar Y_0),$$

let M_3 be the last term of $\Sigma_4^{(3)}$ (for some $\bar b$ and constants as in Lemma 11.4), and

$$\begin{aligned}\Sigma_4^{(4)} :=\ & (4P_v - 4 - K_S)\,\varpi\min(b^2, b_2) + (10+2\eta)(d-3)\,b^4\\ & + 2\sum_{n\ge 5}(n+1)(2d-3)\binom{4P_v+T_vn}{n}b^n\\ & + K_S b\big(f(6)+f(7)\big) + \big(4(P_v-1)-2K_S\big)bf(8) + 2\sum_{n\ge 2}\binom{4P_v+T_vn}{n}b^n\sum_{k\ge 5}X_{n,k}f(k)\\ & + 4P_v b\big(\mathsf{F}_8\mathsf{A} + \mathsf{F}\mathsf{A}_8 + \mathsf{F}(e^{\mathsf{A}}-1-\mathsf{A})\big) + \min(M_3, M_4).\end{aligned}$$

Then $\mathcal{S}_\eta(p) \le \Sigma_4^{(4)}$; in particular $\mathcal{S}(p) = \mathcal{S}_1(p)$ is at most $\Sigma_4^{(4)}$ with $\eta = 1$.

Proof. Split the pairs (J, p') as in the proof of Lemma 11.4; as $\eta \le 1$, the factor η may be dropped except for the U-turns.

- *Two front-pairs, $|p'| = 4$.* The shifts and jumps are as there, and there are none with three plaquettes. Those with four plaquettes number at most $12(d-3)$, and only the $2(d-3)$ U-turns have p' in the layer of p (Lemma 13.1(i)). Those with $n \ge 5$ are counted as there.

- *Two front-pairs, $|p'| \ge 5$.* With one plaquette, Lemma 13.1(ii) gives the stated numbers, each pair with the weight $bf(|p'|)$. With $n \ge 2$ plaquettes the count is as there.

- *Three or more front-pairs, one plaquette.* As there, but if there are exactly three front-pairs, one of p' and the third has at least eight edges (Lemma 13.1(iii)).

- *Three or more front-pairs, $n \ge 2$ plaquettes.* The bound M_3 is the one of Lemma 11.4. For M_4: the weight $f(|p'|)\prod_\psi a^{|\psi|/2}e^{c_J}$ of a pair (J, p') in the proof of Lemma 11.4 is κa^E, where $E := (|p'|-4) + \sum_\psi |\psi|/2$, the sum over the ≥ 1 further front-pairs ψ, and $\kappa := f(|p'|)a^{4-|p'|}\prod_\psi e^{c_J}$; with the barred weights it is $\kappa\bar a^E$, with the same κ. So $E \ge 2$. If $n = 2$, then $E \ge 3$: with exactly three front-pairs, either $|p'| = 4$ and the third has at least six edges (Lemma 13.1(iv)), or $|p'| \ge 6$ and $E \ge 4$; with more front-pairs, $E \ge 4$. Hence either $n \ge 3$ and $E \ge 2$, or $n = 2$ and $E \ge 3$, and in both cases, as $b \le \bar b$ and $a \le \bar a$,

  $$b^n a^E \le \Big(\Big(\frac{b}{\bar b}\Big)^3\Big(\frac{a}{\bar a}\Big)^2 + \Big(\frac{b}{\bar b}\Big)^2\Big(\frac{a}{\bar a}\Big)^3\Big)\bar b^n\bar a^E.$$

  The sum of the terms $\kappa\bar b^n\bar a^E$ is the sum of the proof of Lemma 8.10(ii) with $\bar b$, $\bar a$ and $\bar f$ in place of b, a and f, and the doubly marked tree sums with the barred constants bound it by the last factor of M_4. □

Lemma 13.3 (Junction sums at a six-edge front)**.** *In the setting of Lemma 11.4, suppose all spatial sides are ≥ 8, and let $f \ge 0$ with $f(4) = 1$ (in Theorem 13.4, the f with F_6). Let p^* be a front-pair candidate with six edges, and $\mathcal{S}(p^*)$ the sum of Lemma 8.10(ii) at p^*, over the junctions without support containing p^*. Put $X_{n,k}^{(6)} := \min\{\binom{4n+6}{k}, (4n+6)N_k^v\}$ for $k \ge 6$, and*

$$\begin{aligned}\Sigma^{(6)} :=\ & 2b + \sum_{n\ge 2}(2d-3)(2n+3)\binom{6P_v+T_vn}{n}b^n + 2\sum_{n\ge 1}\binom{6P_v+T_vn}{n}b^n\sum_{k\ge 6}X_{n,k}^{(6)}f(k)\\ & + 6P_v e^{6\xi_F}\,(\kappa_C Y_2 + 6(g_C g'_F + g'_C g_F)Y_1 + 36\kappa_F Y_0).\end{aligned}$$

Then $\mathcal{S}(p^*) \le \Sigma^{(6)}$. The term $2b$ counts the two narrowings*: if the front of p^* is $\partial q \triangle \partial c$ with $c \sim q$, they are the junctions with the one plaquette c (resp. q) and the second front-pair $(\partial q, \lambda')$ (resp. $(\partial c, \lambda')$), λ' the other layer.*

Proof. Split the pairs (J, p') as in the proof of Lemma 8.10(ii), and let φ be the front of p^*.

- *Two front-pairs, one plaquette c, $|p'| = 4$.* By Lemma 13.1(ii), with the roles of the two front-pairs exchanged, $p' = (\partial q, \lambda')$ with λ' the other layer, and $\varphi = \partial q \triangle \partial c$, where c shares a link e_0 with q. Write $e_0 = \{x, x + e_\mu\}$, and let q, c be spanned at x by e_μ and unit vectors $g \ne g'$ perpendicular to e_μ ($g' = -g$ for a flat domino). As all sides are ≥ 4, the only links joining two of the six vertices x, $x+e_\mu$, $x+g$, $x+e_\mu+g$, $x+g'$, $x+e_\mu+g'$ of φ are the six links of φ and e_0. So φ determines e_0, and with it the unordered pair $\{q, c\}$: the two paths of φ between the ends of e_0, each closed by e_0, are ∂q and ∂c. Hence there are at most two such pairs (J, p'), each of weight $bf(4) = b$.

- *Two front-pairs, the other cases.* As in the proof of Lemma 8.7: by Lemma 8.6(i), whose proof does not use $|\varphi| = 4$, the n plaquettes of J together with φ form a connected set in the graph in which φ is joined to the at most $6P_v$ plaquettes with a vertex in $V(\varphi)$; so there are at most $\binom{6P_v+T_v n}{n}$ of them. Given them, p' has one of two layers, and its front lies in $F := E_M(J) \cup \varphi$, with $|F| \le 4n + 6$. Fronts with four edges in F are plaquette boundaries (the sides are ≥ 8), at most $2(d-2)|F|/4 \le (2d-3)(2n+3)/2$ of them; for $n = 1$ these are the narrowings, counted above. Fronts with $k \ge 6$ edges in F number at most $X^{(6)}_{n,k}$, and there are none with five.

- *Three or more front-pairs.* As in the proof of Lemma 8.10(ii), with $|p^*| = 6$ in place of 4. The doubly marked sums over the at most $6P_v\binom{K_S n}{n-1}$ clusters C_0 with n plaquettes touching p^* give the last term. □

Theorem 13.4 (The third-order term with an effective constant)**.** *Suppose all spatial sides are ≥ 8. In Theorem 10.8, make four changes:*

- *give $f(6)$ a constant of its own, $f(6) := 36F_6 e^{6\xi_F} a^2$ with $F_6 > 0$ (the other values of f as in Definition 8.9);*

- *replace the hypothesis $\Sigma_4^{(2)} < \bar\Upsilon\delta'$ by $\Sigma_4^{(4)} < \bar\Upsilon\delta'$, with $\Sigma_4^{(4)}$ taken at $\eta := C_\varpi a^4(2\bar\Upsilon(1+\delta') + \Sigma_4)/(1-\omega)$;*

- *replace 5 by 7 in the second hypothesis;*

- *add the hypothesis $\Sigma^{(6)} < 36F_6 e^{6\xi_F}(\bar\Upsilon(1+\delta') - a)$.*

All quantities are computed with this f, and $\bar b, \bar a$ and the barred constants are as in Lemma 13.2. Then the conclusions of Theorem 10.8 hold. As $\beta \downarrow 0$ the hypotheses hold with $\varpi := b^{5/2}$, fixed $\bar b, \bar a > 0$ depending only on d, and $\delta' := (\delta'_4 + \varepsilon)\beta^4$ for every $\varepsilon > 0$, where

$$\delta'_4 := 160\,(d-3) + 8K_S u_1^2, \qquad u_1 = \frac{1+\iota}{2N^2}.$$

Hence $m(\mu_\beta) \ge 4\log(1/u) - \log\Upsilon - (\delta'_4 + o(1))\beta^4$, and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the periodic spatial lattices with all sides ≥ 8. For the fundamental representations, $\delta'_4 = 2$ ($d = 3$) and 166 ($d = 4$) for $SU(2)$, 0.099 and 160.3 for $SU(3)$, and 8 and 184 for $U(1)$.

Proof. The bound. We follow the proof of Theorem 11.5, which is that of Theorem 10.8, with Lemma 13.2 in place of Lemma 11.4 for the far junctions of front-pairs with four edges. Lemmas 7.7, 8.8 and 8.10 hold for every $f \ge 0$. Three things change.

- *Six-edge front-pairs.* For a path front-pair p with $|p| = 6$, in the steps (b) and (c) of the proof of Theorem 10.8, the sum over the far junction J' and the continuations is at most $E_B(6) + M\Sigma^{(6)}$ by Lemma 13.3, instead of $E_B(6) + M \cdot 6P_v e^{6\xi_F}(g_C Y_1 + 6g_F Y_0)$. Add $36E_B e^{6\xi_F}/(36F_6 e^{6\xi_F}(\bar{\Upsilon}(1+\delta') - a) - \Sigma^{(6)})$ to the maximum that defines M. Then in step (b), as $E_B(6) \le 36E_B e^{6\xi_F}$,

$$\mathcal{V}(p,0) \le \frac{a^2(E_B(6) + M\Sigma^{(6)})}{\bar{\Upsilon}(1+\delta') - a} \le 36MF_6 e^{6\xi_F} a^2 = Mf(6),$$

  and step (c) goes through as there, since $\Sigma^{(6)} \le 36F_6 e^{6\xi_F}\bar{\Upsilon}(1+\delta')$ and $E_B \le MF_6\bar{\Upsilon}(1+\delta')$.

- *Larger front-pairs.* For $|p| = k \ge 7$ the steps are as there, and $(g_C Y_1 + g_F k Y_0)/k \le g_C Y_1/7 + g_F Y_0$; there are no fronts with five edges. So 7 replaces 5 in the second entry of M and in the inequality used in step (c).

- *Downward exits.* Call a front-pair of a junction at slice s' *downward* if its segment lies on the side of the earlier slices: its layer is $s'-1$, or s' when the direction of time is reversed (for an empty inner layer). For a downward p the first segment of every continuation runs through old slices, so step (c) gives the stronger bound $\mathcal{V}(p,\Delta) \le \eta Mf(|p|)$; it uses only the bound $\mathcal{V} \le Mf$ for shorter continuations, and we include it in the induction. In step (a) the far junction J', at a new slice, is reached through a downward front-pair p^*, and its exits in the layer of p^* are downward. So the sum over J' and the continuations is at most $E_B(4) + M\mathcal{S}_\eta(p^*) \le E_B(4) + M\Sigma_4^{(4)}$ (Lemma 13.2).

In all,

$$M := \max\Big(\frac{E_B(4)}{\bar{\Upsilon}\delta' - \Sigma_4^{(4)}},\ \frac{E_B}{F(\bar{\Upsilon}(1+\delta') - a) - P_v(g_C Y_1/7 + g_F Y_0)},\ \frac{36E_B e^{6\xi_F}}{36F_6 e^{6\xi_F}(\bar{\Upsilon}(1+\delta') - a) - \Sigma^{(6)}}\Big),$$

and the denominators are positive by the hypotheses; the third hypothesis gives $\eta \le 1$.

Small β. Take $\varpi := b^{5/2}$; then $C_\varpi = 1 + b^{-5/2}$ for small β, so that $C_\varpi a^4$, ω and η are $O(\beta^{3/2})$. Take $\xi_F := 2P_v b$, $\xi_C := 8P_v b$ and $h_J := \max_{k\ge 4} C_\varpi k P_v Y_0 a^{k/2} e^{\xi_F k}$. The maximum is at $k = 4$ for small β, and $h_J = 4P_v a^2 b^{-3/2}(1 + O(\beta)) = O(\beta^{1/2})$, so $c_J \to 1$ (here $b = 2\beta(1+O(\beta))$ is used from both sides). So the conditions of Lemma 8.10 and the six inequalities hold as in the proof of Theorem 11.5, the third hypothesis as $\omega + \eta = O(\beta^{3/2})$, and the fourth as $a^4/b^{5/2} = O(\beta^{3/2})$. Choose $\bar{b}, \bar{a} > 0$, depending only on d, so small that the conditions at $(\bar{b}, \bar{a})$ hold with room when $c_J = 1$ and $F = F_6 = 0$. For small β they then hold for the actual c_J, F and F_6: these tend to 1, 0 and 0, $\bar{Y}_j$ converges at $\bar{b}$, and $\bar{A}_j$, $\bar{B}_j$ depend continuously on c_J, F and F_6. Take for $\bar{g}$, $\bar{\kappa}$ the minimal solutions. The last factor of M_4 then stays bounded, so $M_4 = O(b^3a^2 + b^2a^3) = O(\beta^5)$.

Take $F := (1+\varepsilon')P_v(g_C Y_1/7 + g_F Y_0)/(\bar{\Upsilon}(1+\delta') - a)$ with $\varepsilon' > 0$; this is consistent for small β, since g_F and g_C depend on F only through terms $O(Fa^3)$. As $\beta \downarrow 0$, $Y_0, Y_1 \sim b$, $g_F \to N_4^v = P_v$ and $g_C \to 8P_v$, so $F = O(b)$. Take $F_6 := (1+\varepsilon')\Sigma^{(6)}/(36e^{6\xi_F}(\bar{\Upsilon}(1+\delta') - a))$. In $\Sigma^{(6)}$ the terms after $2b$ are $O(b^2)$, $O(bf(6) + bf(7)) = O(\beta^3)$ and $O(a^2 b)$, so $\Sigma^{(6)} = 2b(1 + O(\beta))$ and $F_6 \sim (1+\varepsilon')b/18$. This is consistent, since $\Sigma^{(6)}$ depends on F_6 only through terms $O(bF_6a^2)$. With $b = 2\beta(1 + O(\beta))$ and $a = u_1\beta(1 + O(\beta))$, the terms of $\Sigma_4^{(4)}$ are:

- $\varpi \min(b^2, b_2) \le b^{5/2} b_2 = O(\beta^{9/2})$, as $b_2 = 2(2-\iota)N^2u^2(1 + O(\beta))$ (by Lemma 9.1(iv), $\sigma_6/u^4 = (2-\iota)N^2u^2 + O(\beta^7)$);

- $(10+2\eta)(d-3)b^4 = 160(d-3)\beta^4(1+O(\beta))$;
- the terms with $n \geq 5$ in the first line, and with $n \geq 2$ in the second, $O(\beta^5)$;
- $K_S bf(6) = 36K_S bF_6 e^{6\xi_F} a^2 \sim (1+\varepsilon')\, 2K_S b^2 a^2 \sim (1+\varepsilon')\, 8K_S u_1^2 \beta^4$, while $bf(7)$ and $bf(8)$ are $O(\beta^5)$;
- the one-plaquette term, $O(ba^4)$, as $\mathsf{F} = O(1)$, $\mathsf{F}_8 = O(Fa^4)$, $\mathsf{A} = O(a^2)$ and $\mathsf{A}_8 = O(a^4)$;
- $\min(M_3, M_4) \leq M_4 = O(\beta^5)$.

So $\Sigma_4^{(4)} \leq (\delta_4' + \varepsilon'')\beta^4$ for small β, with $\varepsilon'' \to 0$ as $\varepsilon' \to 0$, and $\bar{\Upsilon} \to 1$; with $\delta' := (\delta_4' + \varepsilon)\beta^4$ and ε' small, all hypotheses hold. Finally $\log \bar{\Upsilon} = \log \Upsilon + O(\beta^6)$ (Lemma 10.6), $\theta_5 = O(\beta^5)$, and the counts N_k^v of a periodic lattice are bounded uniformly (Lemma 7.5). The values follow from $K_S = 4(2d-5)$ and $u_1 = \frac{1}{4}, \frac{1}{18}, \frac{1}{2}$. □

Remark 13.5 (Meaning, window, numbers and checks). (i) Meaning (Heuristic). The two terms of $\delta_4' = 10(d-3)\cdot 16 + 2K_S \cdot 4u_1^2$ count the moves of the plaquette state of Section 12:

- the $10(d-3)$ turns and jumps through a unit cube (Lemma 13.1(i));
- the K_S bulges and K_S hops: a widening to $\partial q \triangle \partial c$ followed by one of the two narrowings, by c (a bulge) or by q (a hop).

These are the moves counted by the hopping matrix, whose top is $h_+ = 2K_S + 10(d-3)$ (Lemma 12.2). The bound charges them at the sup norm, $b^4 \approx 16\beta^4$ and $b \cdot a^2 \cdot b \approx 4u_1^2\beta^4$, whereas in characters each costs $u^4 = u_1^4\beta^4$; for $SU(2)$ these are factors 4096 and 64. The other junctions of order β^4 drop out of the bound:

- the shifts and jumps, through the choice $\varpi = b^{5/2}$: their operators vanish on the odd sector (Lemma 10.5), which Lemma 10.7 turns into the factor ϖ;
- the U-turns, whose exit runs downward and so costs $\eta = O(\beta^{3/2})$. They join two tubes and are not moves of H_C (Remark 12.3).

Section 14 evaluates the junctions of these moves by characters, u^4 per move in place of b^4 and b^2a^2. Then the moves at a slice are summed by the row sums of $|H_C|$, and the lower bound to fourth order follows (Theorem 14.2).

(ii) Window (Theorem). By Theorem 13.4 and Corollary 12.9, for $SU(2)$ in $d = 4$,

$$-166 \leq \liminf_{\beta\downarrow 0} R(\beta) \leq \limsup_{\beta\downarrow 0} R(\beta) \leq -\frac{17}{128}, \qquad R(\beta) := \beta^{-4}\Big(m(\mu_\beta) - 4\log\frac{1}{u} + \log\Upsilon\Big),$$

and in $d = 3$ both lie in $[-2, -\frac{1}{32}]$. Theorem 14.2 raises the lower end to the upper one, so the limit exists: $-\frac{17}{128}$ and $-\frac{1}{32}$ (Corollary 14.3).

(iii) Numbers (Numerical; script `third_order_effective.py`, in the directory of (iv)). The table gives, for $SU(2)$, the smallest admissible δ' of Theorem 13.4, over $\varpi \in \{b^2, b^{5/2}, b^3\}$, $\bar{b} \in \{2^{-3}, \dots, 2^{-17}\}$ and $\bar{a} \in \{2^{-2}, \dots, 2^{-25}\}$, and that of Theorem 11.5.

	$d=3$				$d=4$			
δ'/β^4 at $\beta =$	10^{-5}	10^{-7}	10^{-9}	10^{-12}	10^{-5}	10^{-7}	10^{-9}	10^{-12}
Theorem 11.5	$3.4\cdot 10^8$	$3.4\cdot 10^8$	$3.4\cdot 10^8$	$3.4\cdot 10^8$	$8.7\cdot 10^{10}$	$8.4\cdot 10^{10}$	$8.4\cdot 10^{10}$	$8.4\cdot 10^{10}$
Theorem 13.4	$4.1\cdot 10^6$	$4.1\cdot 10^4$	412	2.4	$1.9\cdot 10^9$	$1.9\cdot 10^7$	$1.9\cdot 10^5$	353

With $\varpi = b^2$ the limit is $\delta_4' + 8(4P_v - 4 - K_S)(2-\iota)N^2u_1^2$, i.e. 18 and 230 for $SU(2)$. The choices $\varpi = b^{5/2}$ and b^3 have the smaller limit δ_4'; $b^{5/2}$, for which $c_J \to 1$, also has smaller terms of order β^5 than b^3. It gives the smallest δ' at $\beta = 10^{-12}$ (2.4 against 18.4 for b^2 in $d = 3$, and 353 against 417 in $d = 4$) and at 10^{-9} in $d = 3$; at the other entries of the table b^2 is as good or slightly better. The loss in $R(\beta)$ also contains $8(K+1)\theta_5/\beta^4 \approx 8(K+1)(2e^2K)^5\beta$, from Lemma 6.1: at $\beta = 10^{-12}$ it is 18 ($d = 3$) and 379 ($d = 4$), so there the lower bound for $R(\beta)$ is about -21 ($d = 3$) and $-7.3 \cdot 10^2$ ($d = 4$). Among the terms of δ', those of order β^5 with crude counts dominate at practical β:

- the junctions with five or more plaquettes;
- the majorant M_4. Its tree sum includes trees with a single plaquette, which do not occur among the junctions it bounds. It is dominated by the crude counts N_k^v, $k \geq 7$: for instance $N_7^v \leq 3.9 \cdot 10^8$ in $d = 4$, whereas $N_7^v = \frac{3}{4}P_vK_S = 108$ for sides ≥ 8. (Closed walks of length ≤ 7 lift to $\mathbb{Z}^{d-1}$, so a front with seven edges is bipartite. A bipartite graph on at most five vertices has at most six edges, and a 7-cycle is odd. So the front has six vertices and is a theta graph whose three paths have odd lengths, $(1, 3, 3)$: the union $\partial q \cup \partial c$ of the boundaries of two plaquettes sharing a link. There are $\frac{P_v}{4} \cdot \frac{K_S}{2}$ such pairs per site, each with six vertices.)

At larger β the minimum in $\min(M_3, M_4)$ is M_3, the majorant of Section 11; even there the bound improves on Theorem 11.5, because $Z_{4,4}$ and the one-plaquette exits are now counted. For $SU(2)$ in $d = 4$ at $\beta = 10^{-7}$ the two bounds give $1.2499943 \cdot 10^{-15} \leq m(\mu_\beta) - 4\log(1/u) \leq 1.2500000 \cdot 10^{-15}$ (Theorem 11.5: $1.2416 \cdot 10^{-15}$ on the left), so the second-order term $\beta^2/8$ is resolved to 5 parts in 10^6.

(iv) Checks (a compass; in `theory/checks/third-order/`).

- Our enumeration (`junction_lemmas_abcd.py`), in spatial dimensions 2 and 3, with fronts of four and six edges:
  - (i): exactly the 10 turns and jumps and the 2 U-turns in $d = 4$, and none in $d = 3$;
  - (ii), for $|p'| \leq 6$: exactly the cap and the K_S six-edge exits;
  - (iii), for fronts of 4, 4 and at most 6 edges: none;
  - (iv): none.
- An independent referee (scripts in `referee-section13/`) used code written from the definitions alone. Fronts there are all connected edge sets of minimum degree ≥ 2, found by a Redelmeier search. The referee confirmed the full statements:
  - (i), in spatial dimensions 2, 3, 4: 0, 12, 24 junctions, namely $8(d-3)$ turns and $2(d-3)$ jumps in the other layer and $2(d-3)$ U-turns in the same layer;
  - (ii), for second fronts with up to 10 edges ($d = 3$) and 9 edges ($d = 4$): exactly the cap and K_S, K_S, $4(P_v - 1) - 2K_S$ exits with 6, 7, 8 edges, and none with 5, 9 or 10;
  - (iii), for all further fronts with up to 8 edges ($d = 3$, where the only junctions have fronts of 4, 4 and 8 edges) and 7 edges ($d = 4$);
  - (iv), in spatial dimensions 2, 3, 4.

  The referee also found $N_6^v = 12$, 132, 456 and $N_7^v = 12$, 108, 360 for $d = 3, 4, 5$.
- Lemma 13.3, by both codes (`six_edge_narrowings.py` and the referee's `check_six.py`): of the 132 six-cycles through a vertex of $\mathbb{Z}^3$, the 108 domino boundaries have exactly two narrowings each and the 24 skew hexagons none; in $\mathbb{Z}^2$ all 12 are dominos, with two each.

14 The fourth-order term: the lower bound

In Section 13 the junctions of order β^4 are charged at the sup norm, $b \approx 2\beta$ for each plaquette. The moves of the plaquette state go through junctions whose plaquettes, together with the fronts on either side, form a closed surface in which every link lies in exactly two faces:

- a widening or a narrowing through $\partial q \triangle \partial c$;
- a turn or a jump through a unit cube.

On such a junction the transfer operator is diagonal in the representations carried by the fronts, and its norm is exactly u per plaquette (Lemma 14.1). So each factor b of these junctions becomes $a \approx u$, and the constant of Theorem 13.4 becomes the row sum of the hopping matrix, $\delta_4' = h_+ u_1^4$ (Theorem 14.2). With the upper bound of Section 12 this gives the fourth-order term for $SU(N)$, $2 \le N \le 5$, in every dimension, and for $U(1)$ in $d = 3$: there Conjecture 12.4 holds with $o(\beta^4)$ in place of $O(\beta^5)$ (Corollary 14.3). For $U(1)$ in $d \ge 4$ the two bounds differ. The lower bound counts the hops and turns between perpendicular neighbours in absolute value, $h_+ = 26d - 70$, while in the C-odd sector, which carries the gap of $U(1)$, they cancel, $h_- = 10d - 22$.

Lemma 14.1 (The moves by characters)**.** *Let J be a junction without support at a slice s, with $\Phi(J) = \{(\varphi_-, s-1), (\varphi_+, s)\}$ and plaquettes P, of one of two kinds:*

(i) $P = \{c\}$, and $\{\varphi_-, \varphi_+\} = \{\partial q, \partial q \triangle \partial c\}$ for a plaquette $q \ne c$ sharing a link with c (a widening, or a narrowing);

(ii) $P = \Gamma \setminus \{q, q'\}$ for a unit cube Γ of the slice and two of its faces $q \ne q'$, and $\{\varphi_-, \varphi_+\} = \{\partial q, \partial q'\}$ (a turn, or a jump).

Then $\|P^J O_J P^J\| = u_*^{|P| + (|\varphi_-| + |\varphi_+|)/2}$, that is u_*^6 in case (i) and u_*^8 in case (ii).

Proof. The two spaces. Here $O_J = \tilde{\mathcal{K}}_{J,s-1}^{1/2} M_J \tilde{\mathcal{K}}_{J,s}^{1/2}$ and M_J is multiplication by $\prod_{p \in P} g_p$. The range of $\tilde{\mathcal{K}}_{J,s}^{1/2} P^J$ consists of functions of the links of φ_+, in both copies, that are invariant under the gauge transformations at its vertices; gauge transformations at the other vertices of $E(J)$ act trivially on them. As φ_+ is a cycle, these are the class functions of its holonomy U_+ in each copy, with orthonormal basis $e^+_{bb'} := \chi_b(U_+)\chi_{b'}(U_+')$; likewise for φ_-, with $e^-_{aa'}$. The factors of O_J commute with P^J. As in the proof of Lemma 9.3, $(G_g^{1/2})^{\otimes \varphi_\pm}$ multiplies $e^\pm_{aa'}$ by $\hat{u}_{aa'}^{|\varphi_\pm|/2}$, where $\hat{u}_{a0} = u_a$, $\hat{u}_{0a'} = u_{a'}$, $\hat{u}_{aa'} = u_a u_{a'}$ for $a, a' \ne 0$, and $\hat{u}_{00} = 0$. So

$$P^J O_J P^J = \sum_{a,a',b,b'} \hat{u}_{aa'}^{|\varphi_-|/2}\, \hat{u}_{bb'}^{|\varphi_+|/2}\, \langle e^-_{aa'}, M_J e^+_{bb'} \rangle\, |e^-_{aa'}\rangle\langle e^+_{bb'}|.$$

The matrix elements. Expanding $g_p = f_\beta(U_p) + f_\beta(U_p') + f_\beta(U_p) f_\beta(U_p')$, the integral over the two copies factorises:

$$\langle e^-_{aa'}, M_J e^+_{bb'} \rangle = \sum_{S_1 \cup S_2 = P} I(a, b, S_1)\, I(a', b', S_2),$$

where

$$I(a, b, S) := \int \overline{\chi_a(U_-)}\, \chi_b(U_+) \prod_{p \in S} f_\beta(U_p)\, dU,$$

the integral over the links of $E(J)$. We claim that $I(a, b, S)$ is

- 1 if $a = b = 0$ and $S = \emptyset$;

- $u_a^{|P|}$ if $a \neq 0$, $S = P$ and $b = \sigma(a)$;
- 0 otherwise.

Here σ is a bijection of the non-trivial irreducible representations, $\sigma(a) \in \{a, \bar{a}\}$, fixed by the orientations of $\varphi_\pm$.

In case (i) let $e_0 = \{x, y\}$ be the common link of q and c, V its variable from x to y, and W_q, W_c the products along the other three links of q, resp. c, from y back to x. Then $U_q := VW_q$ and $U_c := VW_c$ are independent and Haar distributed, and with suitable orientations the holonomy of $\partial q \triangle \partial c$ is conjugate to $U_q U_c^{-1}$. Let $\varphi_- = \partial q$ (the other case is the same with a and b exchanged), and let $h := f_\beta$ if $c \in S$, $h := 1$ otherwise. Then, by $\int \overline{\chi_a(X)} \chi_b(XY)\, dX = \delta_{ab}\chi_a(Y)/d_a$ and $\int f_\beta \overline{\chi_a} = d_a u_a$ for $a \neq 0$,

$$I(a,b,S) = \iint \overline{\chi_a(U_q)}\, h(U_c)\, \chi_b(U_q U_c^{-1})\, dU_q\, dU_c = \frac{\delta_{ab}}{d_a} \int h(U_c)\, \overline{\chi_a(U_c)}\, dU_c,$$

which is the claim with $\sigma = \mathrm{id}$. In case (ii) apply Lemma 12.7(i) to the faces of Γ, with $h_q := \overline{\chi_a}$, $h_{q'} := \chi_b$, $h_p := f_\beta$ for $p \in S$ and $h_p := 1$ for $p \in P \setminus S$. As $\hat{h}_p(r) = d_r u_r$ for $p \in S$, $\hat{h}_p(r) = \delta_{r0}$ for $p \in P \setminus S$, and $\hat{f}_\beta(0) = 0$:

- if $S \neq P$, only $r = 0$ contributes, and then $S = \emptyset$ and $a = b = 0$;
- if $S = P$, only one $r \neq 0$ contributes, the one determined by a; it requires $b = \sigma(a)$ and gives $d_r^{-4}(d_r u_r)^4 = u_a^4$.

The norm. Hence the terms of the sum over (S_1, S_2) vanish unless $S_1, S_2 \in \{\emptyset, P\}$, not both empty, and $\langle e^-_{aa'}, M_J e^+_{bb'}\rangle = \hat{u}_{aa'}^{|P|}$ if $(a,a') \neq (0,0)$ and $(b,b') = (\sigma(a), \sigma(a'))$ (with $\sigma(0) := 0$), and 0 otherwise. As $\hat{u}_{\sigma(a)\sigma(a')} = \hat{u}_{aa'}$,

$$P^J O_J P^J = \sum_{(a,a') \neq (0,0)} \hat{u}_{aa'}^{|P| + (|\varphi_-| + |\varphi_+|)/2}\, |e^-_{aa'}\rangle\langle e^+_{\sigma(a)\sigma(a')}|,$$

an operator that maps an orthonormal family bijectively onto another. Its norm is

$$\sup_{(a,a') \neq (0,0)} \hat{u}_{aa'}^{|P| + (|\varphi_-| + |\varphi_+|)/2} = u_*^{|P| + (|\varphi_-| + |\varphi_+|)/2},$$

since $\hat{u}_{a0} = u_a$, $\hat{u}_{aa'} = u_a u_{a'} \leq u_*$ and $u_* \leq 1$. □

Theorem 14.2 (The lower bound to fourth order)**.** *Suppose all spatial sides are ≥ 8. Then Theorem 13.4 holds with three terms of the junction sums reduced, b being replaced by a for the junctions of Lemma 14.1:*

- *in $\Sigma_4^{(4)}$, the term $(10 + 2\eta)(d-3)\, b^4$ by $(10a^4 + 2\eta b^4)(d-3)$;*
- *in $\Sigma_4^{(4)}$, the term $K_S b\, f(6)$ by $K_S a\, f(6)$;*
- *in $\Sigma^{(6)}$, the term $2b$ by $2a$.*

As $\beta \downarrow 0$ the hypotheses hold with $\varpi := b^{5/2}$, fixed $\bar{b}, \bar{a} > 0$ depending only on d, and $\delta' := (h_+ u_1^4 + \varepsilon)\beta^4$ for every $\varepsilon > 0$, where $h_+ = 26d - 70 = 10(d-3) + 2K_S$ is the top of the spectrum of the hopping matrix H_+ (Lemma 12.2). Hence

$$\liminf_{\beta \downarrow 0}\ \beta^{-4}\Big(m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} + \log \Upsilon(\beta)\Big)\ \geq\ -h_+ u_1^4,$$

and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the periodic spatial lattices with all sides ≥ 8.

Proof. The bound. In the proof of Theorem 10.8 a main junction J carries $z\,\nu(J)$, $\nu(J) = \|P^J O_J P^J\|$, which there is bounded by $w(J)$ before the factors $e^{2(K+1)\theta_5}$. For the junctions of Lemma 14.1, $\nu(J) = u^{|P|}\prod_{p\in\Phi(J)} u^{|p|/2}$ as $u_* = u$. With the factors $e^{2(K+1)\theta_5}$ such a junction may therefore carry $a^{|P|}$ in place of $b^{|P|}$, times the factors $a^{|p|/2}$ of its front-pairs. These junctions are the following:

- the widenings: the K_S pairs (J, p') with one plaquette and $|p'| = 6$ of Lemma 13.1(ii);
- the turns and jumps: the $10(d-3)$ pairs with four plaquettes and p' in the other layer, of Lemma 13.1(i);
- the narrowings: the two pairs with $|p'| = 4$ in the first item of the proof of Lemma 13.3.

The proofs of Lemmas 13.2 and 13.3, and that of Theorem 13.4, then go through with the reduced terms.

Small β. As in the proof of Theorem 13.4: now $\Sigma^{(6)} = 2a(1+O(\beta))$, so $F_6 \sim (1+\varepsilon')a/18$ and $K_S a\, f(6) = 36K_S a F_6 e^{6\xi_F} a^2 \sim (1+\varepsilon')\,2K_S a^4 \sim (1+\varepsilon')\,2K_S u_1^4\beta^4$; and $(10a^4 + 2\eta b^4)(d-3) = 10(d-3)u_1^4\beta^4(1+O(\beta))$, as $\eta = O(\beta^{3/2})$. The other terms of $\Sigma_4^{(4)}$ are $o(\beta^4)$, as there. So $\Sigma_4^{(4)} \le (h_+ u_1^4 + \varepsilon'')\beta^4$ for small β, with $\varepsilon'' \to 0$ as $\varepsilon' \to 0$, since $10(d-3) + 2K_S = 10d - 30 + 16d - 40 = 26d - 70$. The rest is as there. □

Corollary 14.3 (The fourth-order term)**.** *With the fundamental representations:*

(i) for $SU(N)$, $2 \le N \le 5$, and $d \ge 3$,

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \log\Upsilon(\beta) - (26d-70)\,u(\beta)^4 + o(\beta^4);$$

(ii) for $U(1)$ and $d = 3$, $m(\mu_\beta) = 4\log(1/u) - \log\Upsilon - 8u^4 + o(\beta^4)$;

(iii) for $U(1)$ and $d \ge 4$, with $R(\beta)$ as in Remark 13.5(ii),

$$-\frac{26d-70}{16} \le \liminf_{\beta\downarrow 0} R(\beta) \le \limsup_{\beta\downarrow 0} R(\beta) \le -\frac{10d-22}{16}.$$

So Conjecture 12.4 holds for $SU(N)$, $2 \le N \le 5$, in every $d \ge 3$, and for $U(1)$ in $d = 3$, with $o(\beta^4)$ in place of $O(\beta^5)$, including the values listed there for $SU(2)$, $SU(3)$ and, in $d = 3$, $U(1)$. For $SU(2)$:

- $m(\mu_\beta) = -4\log u + 2u^2 - \frac{98}{3}u^4 + o(u^4) = 4\log\frac{4}{\beta} + \frac{7}{24}\beta^2 - \frac{167}{1152}\beta^4 + o(\beta^4)$ *in $d = 4$;*
- $m(\mu_\beta) = -4\log u + 2u^2 - \frac{20}{3}u^4 + o(u^4)$ *in $d = 3$.*

Proof. The lower bounds are Theorem 14.2, and the upper bounds are Corollary 12.9: for $SU(N)$, $2 \le N \le 5$, it gives $h_C u_1^4 = h_+ u_1^4$; for $U(1)$ it gives $h_- u_1^4 = (10d-22)/16$, and $h_- = h_+ = 8$ when $d = 3$. The values are those of Remark 12.10(i) and of Conjecture 12.4: for $SU(2)$, $-\log\Upsilon = 2u^2 + \frac{4}{3}u^4 + O(u^6)$, and $\frac{4}{3} - 34 = -\frac{98}{3}$, $\frac{4}{3} - 8 = -\frac{20}{3}$. □

Remark 14.4 (Meaning and what remains). (i) Meaning (Heuristic). The recursion of the lower bound follows the moving tube automatically. After a move, the bound $\mathcal{V} \le Mf(4)$ for the new four-edge front is the same as for the old one, by translation invariance, so the moves at a slice are summed by the row sums of $|H_C|$, which are h_+. For $C = +$ the row sum is the top of the spectrum, as H_+ has non-negative entries, so nothing is lost. Each move costs u^4, as in the upper bound. The bulges and hops cost $u \cdot u^2 \cdot u$: a widening, a layer of the six-edge front, and a narrowing. The turns and jumps cost u^4: the four plaquettes of the cube.

(ii) The C-odd sector. For $U(1)$ in $d \geq 4$ the gap lies in the C-odd sector, where the hop and the turn between perpendicular neighbours cancel (H_- has no $A_\perp$). Bounding them separately, in norm, gives h_+. A lower bound with h_- would have to keep the signs, that is, use the odd and even sectors of charge conjugation in the recursion as the swap is used in Section 10.

(iii) What remains:

- $U(1)$ in $d \geq 4$, as in (ii);
- $SU(N)$ with $N \geq 6$, where $\Upsilon_\pm$ split only at order β^{N-2} (Remark 11.7(i)) and the sector of the gap is not yet identified at order β^4;
- a remainder $O(\beta^5)$ in place of $o(\beta^4)$, which needs a quantitative version of the limit over $\mathcal{W}$ in Theorem 12.8. The lower bound has a remainder $O(\beta^{9/2})$ with $\varpi = b^{5/2}$, and $O(\beta^5)$ with $\varpi = b^3$, for which c_J tends to $e^{P_v u_1^2}$ instead of 1;
- the term of order u^6;
- effective constants. At practical β the terms of order β^5 with crude counts still dominate (Remark 13.5(iii)), so the fourth-order term is resolved only asymptotically. With the reduced sums and $\varpi = b^{5/2}$ (Numerical; `third_order_effective.py`, option `chars`), the admissible δ'/β^4 for $SU(2)$ is 0.44 at $\beta = 10^{-12}$ and 0.035 at $\beta = 10^{-14}$ in $d = 3$, where the limit is $\frac{1}{32} = 0.031$; in $d = 4$ it is 187 and 2.0, where the limit is $\frac{17}{128} = 0.133$. The loss in $R(\beta)$ also contains $8(K+1)\theta_5/\beta^4$, which is 0.18 at $\beta = 10^{-14}$ in $d = 3$.

15 Two dimensions; discussion

Proposition 15.1 (Two dimensions)**.** *Let $d = 2$ and $0 < \beta < \beta_0(2)$, so that $K = 4$ and $q(\beta) < 0.909$ (Lemma 4.2).*

(i) For every $L_1 \geq 3$, the spatial lattice $\Lambda_s = \mathbb{Z}/L_1\mathbb{Z}$ has $m_{\Lambda_s}(\beta) \geq L_1 \log(1/q(\beta))$.

(ii) In infinite volume, $\mathrm{Cov}_{\mu_\beta}(A, B) = 0$ for all A, B as in Theorem 1.2(iii) (time ranges with $a_2 < b_1$); hence $m(\mu_\beta) = \infty$.

Proof. By Lemma 3.1, $g = L_1$, and Theorem 4.1 gives the bound with $q^{L_1\tau}$. (i) follows as in the proof of Theorem 1.2(i), with L_1 in place of 4. For (ii), the bound for the periodised observables on tori with $L_0, L_1 \to \infty$ tends to 0, since $\tau \geq 1$ and $g = L_1 \to \infty$, and the torus covariances converge to $\mathrm{Cov}_{\mu_\beta}(A, B)$ as in the proof of Theorem 1.2(iii) (note $\beta_0(2) < \beta_*(2)$, Lemma 4.2); then [II, Theorem `iv:thm:OS`](iii) applies with every $m > 0$. □

Remark 15.2 (Consistency with the exact solution). In $d = 2$ the transfer matrix on a spatial circle of length L_1 is diagonal in the characters of the holonomy, with eigenvalue ratios $(c_r/(d_r c_0))^{L_1}$ (`lattice/ymlattice/transfer2d.py`, tested against the Monte Carlo); so the exact finite-volume gap is $L_1 \log(1/u(\beta))$ for $SU(2)$, linear in L_1, and the infinite-volume theory has no propagating states. Proposition 15.1 reproduces both features (with q in place of u). It is also consistent with [II, Proposition 'Gauge invariance of the ground-state chain' (`iv:prop:phys`)]: in $d = 2$ the spatial links of a time slice form a line, gauge-invariant functions of finitely many of them are constant, and P acts trivially on the physical sector. The difference between $d = 2$ and $d \geq 3$ is the girth of the spatial graph (L_1 versus 4): in $d \geq 3$ a closed surface can turn a corner in every time layer with four plaquettes, the minimal glueball.

Remark 15.3 (The strong-coupling test, sharpness, and the $U(1)$ test). (i) Principle 3 of the research program requires the correct asymptotics $m(\beta) \sim 4\log(1/u(\beta))$. Theorem 1.2 gives

the lower half, $m \geq 4\log(1/\beta) - O(1)$ for all $\beta < \beta_0(d)$, and Theorem 5.6 the upper half, $m \leq 4\log(1/u(\beta)) + O(\beta)$, on the same range when $N^4 \leq 1200\tau_0^2$ and for sufficiently small β in general; Corollary 5.8 combines them. The upper bound uses a single correlator at time separations 0 and 1; by the positivity of P ([II, Remark 'Consequences for the numerics' (`iv:rem:numerics`)]) any larger pair of separations could be used instead. The additive constants of the two halves differ ($-4\log(4e^3K)$ versus $+4\log\frac{2N^2}{1+\iota}$); the strong-coupling expansion predicts $m = 4\log(1/u) + O(u)$, i.e. the upper one. (ii) The constant $-4\log(4e^3K)$ of Theorem 1.2 reflects the crude counting of [III] and the sup-norm bound $\|g_p\|_\infty \leq e^{4\beta} - 1$ on every plaquette. Section 6 removes the second loss for the temporal plaquettes, which gives the constant $-4\log(eK)$ relative to $4\log(1/u)$ as $\beta \downarrow 0$ (Corollary 6.5); counting by fronts (Section 7) removes the counting loss as well, so that the lower bound is $4\log(1/u_*) - O(\beta)$, i.e. $4\log(1/u) - O(\beta)$ when $u_* = u$, and grouping by junctions (Section 8) brings the $O(\beta)$ to $2\beta(1 + O(\beta))$ when ρ has no trivial summand. Evaluating the cap by characters (Section 9) brings it to $\kappa(\beta) + O(\beta^2)$, which is also the first-order term of the upper bound: $m = 4\log(1/u) - \kappa + O(\beta^2)$. At second order (Section 10) the two bounds meet again: $m = 4\log(1/u) - \log\Upsilon + O(\beta^3)$, and at third order (Section 11) $m = 4\log(1/u) - \log\Upsilon + O(\beta^4)$. At fourth order (Section 12) the upper bound is $4\log(1/u) - \log\Upsilon - h_C u^4 + o(\beta^4)$. Counted at the sup norm, the lower bound is $4\log(1/u) - \log\Upsilon - (\delta_4' + o(1))\beta^4$ (Section 13), with $\delta_4' = 166$ for $SU(2)$ in $d = 4$. Evaluating the moves by characters (Section 14) brings it to $-h_+u^4$, so the two bounds meet once more for $C = +$: $m = 4\log(1/u) - \log\Upsilon - (26d - 70)u^4 + o(\beta^4)$ for $SU(N)$, $2 \leq N \leq 5$. (iii) The hypothesis $L_i \geq 4$ cannot be dropped from the sharper finite-volume bounds: if a spatial side equals 3, Theorem 4.1 holds only with $g = 3$, and closed strips of three plaquettes per layer around that side (torelons, carrying the flux of a Polyakov loop) give states of energy $\approx 3\log(1/u(\beta))$. For $\mathbb{Z}_2$ in $d = 3$ on a 3×3 spatial torus the transfer matrix preserves the winding class, and a high-precision computation of the top eigenvalues of the sectors of winding $(0,0)$ and $(1,0)$ gives a torelon energy $43.526 = 3\log(1/u)$ at $\beta = 5 \cdot 10^{-7}$, below the bound $4\log(1/q_u) = 44.086$ of Theorem 6.4(i) (Numerical; also $46.27 < 47.76$ at $\beta = 2 \cdot 10^{-7}$; `theory/checks/lower-bound/z2_torelon_mp.py`). So Theorem 6.4(i) fails on such lattices. For the weaker bound $4\log(1/q)$ of Theorem 1.2(i) the crossing is expected only near $\beta \approx 10^{-12}$. (iv) Nothing here is non-abelian: the argument applies verbatim to $G = U(1)$, whose $d = 4$ theory is expected to be massless at large β (proved in [FS82] for the Villain action and asserted there for Wilson's; [III, Remark 'The $U(1)$ test' (`al:rem:u1test`)]). The result therefore concerns the strong-coupling endpoint only; it shows that the endpoint is controlled with the right leading behaviour (Corollary 5.8), which is what a renormalisation-group argument would need to reach.

Summary of results

Statement	Where	Status
Duplicated system; loop expansion with Ξ^2; cancellation of non-connecting terms	Lem. 2.2, 2.4, 2.5	Theorem
Surfaces without free edges have $\geq g$ temporal plaquettes per crossed layer ($g = 4$ for $d \geq 3$, spatial sides ≥ 4)	Lem. 3.1, 3.2, 3.3, Cor. 3.4	Theorem
Covariance bound $q^{g\tau}$ in finite and infinite volume	Thm. 4.1, 1.2(iii)	Theorem
$m(\mu_\beta) \geq 4\log(1/q(\beta)) = 4\log(1/\beta) - O(1)$, and $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$ for spatial sides ≥ 4; $d \geq 3$, $\beta < \beta_0(d)$	Thm. 1.2	Theorem
$d = 2$: gap $\geq L_1 \log(1/q)$; infinite gap in infinite volume	Prop. 15.1	Theorem
Positivity: $\mathrm{Cov}(A(0), A(1))$ dominates the expectation in which the four sides of the cube carry φ_β (every β)	Lem. 5.2	Theorem
Cube identity $\sum_{r\neq 0} \hat{a}_{0,r}\hat{a}_{1,r}u_r^4$; five-plaquette lemma	Lem. 5.1, 5.4	Theorem
Matching upper bound $m \leq 4\log(1/u(\beta)) + D(\beta)$, $D = O(\beta)$ (infinite volume; finite volume, sides ≥ 3): for all $\beta < \beta_0(d)$ if $N^4 \leq 1200\tau_0^2$ ($D < 0.32$ if $N \leq 3$), for small β in general	Thm. 5.6, Lem. 5.7	Theorem
$4\log(1/q) \leq m(\mu_\beta) \leq 4\log(1/u) + D$ for all $\beta < \beta_0(d)$ ($N^4 \leq 1200\tau_0^2$; also finite volume, sides ≥ 4); $m = 4\log(1/\beta) + O(1)$	Cor. 5.8	Theorem
Ratio bound with $\theta_5 = x^5/(1-x)$ (only closed polymers count); $\|g_p\|_\infty \leq e^{2\beta} - 1 + \beta^2 e^\beta$ if ρ has no trivial summand	Lem. 6.1, 2.2(d)	Theorem
Transfer bound: the temporal plaquettes of all but one layer weigh $u_* = \max_{r\neq 0} u_r$, not $e^{4\beta} - 1$	Lem. 6.2, 6.3	Theorem
$m \geq 4\log(1/q_u)$ for all $\beta < \beta_0(d)$ (infinite volume; finite volume, sides ≥ 4), $= 4\log(1/u) - 4\log(eK) - O(\beta)$ for $\dim \rho \leq 49$; window $4\log(eK) + O(\beta)$ when also $N^4 \leq 1200\tau_0^2$	Thm. 6.4, Cor. 6.5	Theorem
Fronts and segments; incidence graph; fronts with 4 and 5 edges; local counts at vertices; tree sums	Lem. 7.2–7.7	Theorem
Sharp constant on the whole range: $m \geq 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1+\delta)$, $\delta = O(\beta)$, hypotheses verified for all $\beta < \beta_0(d)$; $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$ (also finite volume, sides ≥ 4)	Thm. 7.8, Lem. 7.9, Cor. 7.11	Theorem

Statement (continued)	Where	Status
$SU(2)$, all $\beta < \beta_0(d)$: $-1.38 \le m - 4\log(1/u) \le 0.0014$ ($d = 4$), $-0.63 \le m - 4\log(1/u) \le 0.005$ ($d = 3$)	Rem. 7.10	Numerical
Junctions: locality and local admissibility; junction graph; fronts with six edges; junctions with two front-pairs (cap, shift, jump); marked tree sums	Lem. 8.2–8.10	Theorem
First-order constant: $m \ge 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1+\delta_J)$, $\delta_J = b(1+O(\beta))$, so $m(\mu_\beta) \ge 4\log(1/u(\beta)) - 2\beta + O(\beta^2)$ (ρ without trivial summand)	Thm. 8.11, Cor. 8.13	Theorem
$SU(2)$, all $\beta < \beta_0(d)$: $-5.0\cdot 10^{-4} \le m - 4\log(1/u) \le 1.4\cdot 10^{-3}$ ($d = 4$), $-1.1\cdot 10^{-3} \le m - 4\log(1/u) \le 4.9\cdot 10^{-3}$ ($d = 3$)	Rem. 8.12	Numerical
Cap matrix element: $\kappa = \|P_\rho M_f P_\rho\| = \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3 + O(\beta^2)$; $O(\beta^2)$ if the centre acts on ρ by a scalar that is not a cube root of unity ($U(1)$, $SU(N)$ with $N \ne 3$, fundamental); $\frac{\beta}{6} + O(\beta^2)$ for $SU(3)$	Lem. 9.1	Theorem
Transfer bound by pieces (junctions with their gauge projections); the cap by characters, $\|P^J O_J P^J\| \le u^4(\kappa + (2-\iota)N^2u^2 + O(\beta^3))$	Lem. 9.2, 9.3	Theorem
First-order term: $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ (also finite volume, sides ≥ 4): $4\log(1/u) + O(\beta^2)$ for $U(1)$ and $SU(N)$, $N \ne 3$, and $4\log(1/u) - \beta/6 + O(\beta^2)$ for $SU(3)$ (fundamental)	Thm. 9.4, 9.5, Cor. 9.6	Theorem
$SU(2)$, all $\beta < \beta_0(d)$: $-1.3\cdot 10^{-4} \le m - 4\log(1/u) \le 4.8\cdot 10^{-4}$ ($d = 4$), $-2.5\cdot 10^{-4} \le m - 4\log(1/u) \le 2.4\cdot 10^{-3}$ ($d = 3$); as $\beta \downarrow 0$, between $-128.4\beta^2$ and $0.375\beta^2$ ($d = 4$)	Rem. 9.7	Numerical
$\mathbb{Z}_3$ (3×3 torus, winding zero): C-even and C-odd plaquette states at $4\log(1/u) \mp u(1+O(u))$ ($\kappa = u$; also on the 4×4 torus); $\mathbb{Z}_2$: no first-order term	Rem. 9.7(iv)	Numerical

Statement (continued)	Where	Status
Plaquette variance $\Upsilon(\beta)$, the largest eigenvalue of $\mathsf{V} = P_\rho(M_{1+f} - \lvert f\rangle\langle f\rvert)P_\rho$; its expansion; $-\log\Upsilon = \beta^2/8$ ($SU(2)$, $U(1)$), $-\beta/6 + \beta^2/72$ ($SU(3)$), $-\beta^2/128$ ($SU(4)$), 0 ($SU(N)$, $N \geq 5$), up to $O(\beta^3)$	Lem. 10.1	Theorem
Cube identity for real class functions; upper bound to second order, $D_2 = -\log\Upsilon + O(\beta^5)$	Eq. (2), Thm. 10.2	Theorem
Caps summed into the dressed tube; the swap-odd sector; shifts and jumps of rank one and even; the dressed tube on the odd sector, $\leq u^4\Upsilon(1 + O(\beta^6))$; a slice on the odd sector	Lem. 10.3–10.7	Theorem
Second-order term: $m = 4\log(1/u) - \log\Upsilon + O(\beta^3)$ (infinite volume; finite volume, sides ≥ 5); for $SU(2)$, $m(\mu_\beta) = 4\log(4/\beta) + \frac{7}{24}\beta^2 + O(\beta^3)$	Thm. 10.8, Cor. 10.9	Theorem
$SU(2)$: $\delta'/\beta^3 \to 1.4{\cdot}10^6$ ($d = 4$), $2.1{\cdot}10^5$ ($d = 3$); at $\beta = 10^{-7}$, $d = 3$: $1.04 \cdot 10^{-15} \leq m - 4\log(1/u) \leq 1.25 \cdot 10^{-15}$; exact spectra ($\mathbb{Z}_n$ tori, one and two squares) and brute-force checks of the lemmas	Rem. 10.10	Numerical
Sides ≥ 4 needed for Thm. 6.4(i): torelon below the bound on a 3×3 torus ($\mathbb{Z}_2$)	Rem. 15.3(iii)	Numerical
Plaquette variance to third order: β^3 terms $\frac{1}{432}$ ($SU(3)$), $-\frac{1}{6000}$ ($SU(5)$), 0 ($U(1)$, $SU(2)$, $SU(4)$, $SU(N \geq 6)$); no junctions with three plaquettes and two four-edge fronts ($Z_{3,4} = 0$); one plaquette and three front-pairs force a front of ≥ 5 edges; junction sums to third order	Lem. 11.1–11.4	Theorem
Third-order term: $m = 4\log(1/u) - \log\Upsilon + O(\beta^4)$ (infinite volume; finite volume, sides ≥ 6); for $SU(2)$, $m(\mu_\beta) = 4\log(4/\beta) + \frac{7}{24}\beta^2 + O(\beta^4)$	Thm. 11.5, Cor. 11.6	Theorem
$SU(2)$: $\delta'/\beta^4 \to 8.4 \cdot 10^{10}$ ($d = 4$), $3.4 \cdot 10^8$ ($d = 3$); at $\beta = 10^{-7}$, $d = 3$: $1.249966 \cdot 10^{-15} \leq m - 4\log(1/u) \leq 1.25 \cdot 10^{-15}$; enumerations, Weyl integration, exact $\mathbb{Z}_3$, $\mathbb{Z}_5$ spectra on tori outside the hypotheses (consistent with the third-order claim, residual $\approx -8u^4$)	Rem. 11.7	Numerical

Statement (continued)	Where	Status
Hopping matrix H_C and its signs: $H_+ = K_S\mathbf{1}+A_\parallel+2A_\perp+A_J$, $H_- = K_S\mathbf{1}+A_\parallel+A_J$; $\sup\operatorname{spec} H_+ = 26d-70$, $\sup\operatorname{spec} H_- = 10d-22$; at $k=0$ in $d=4$: 34 (A_1), 10 (E), 18 (T_1)	Def. 12.1, Lem. 12.2	Theorem
A set of plaquettes in which no link lies in exactly one of them has ≥ 6 plaquettes, 6 only for a unit cube; configurations at a slice; one and two cubes by characters	Lem. 12.5–12.7	Theorem
Upper bound to fourth order: $m \leq 4\log(1/u) - \log\Upsilon_C - \log(1 + u^4 a^\top H_{C,\mathcal{W}} a/a^\top a) + O(\beta^5)$ for every finite $\mathcal{W}$; $\limsup \beta^{-4}(m - 4\log(1/u) + \log\Upsilon) \leq -h_C u_1^4$, e.g. $-\frac{17}{128}$ for $SU(2)$, $d=4$	Thm. 12.8, Cor. 12.9	Theorem
Fourth-order term: $m = 4\log(1/u) - \log\Upsilon - h_C u^4 + O(\beta^5)$; for $SU(2)$, $d=4$: $m = -4\log u + 2u^2 - \frac{98}{3}u^4 + O(u^5)$; proved, with $o(\beta^4)$, for $SU(N)$, $N \leq 5$, and for $U(1)$, $d=3$ (Cor. 14.3)	Conj. 12.4, Rem. 12.3	Conjecture
Exact $\mathbb{Z}_2$ and $\mathbb{Z}_3$ spectra ($d=3$ tori, $d=4$ blocks of cubes, both sectors): every level of the plaquette band follows H_C; exhaustive checks of the lemmas	Rem. 12.10(ii)	Numerical
Classical series of [Mün81, Seo82] as quoted in [LMP07]: the terms of order u^4 agree in four of five channels; for T_1^{+-} of $SU(3)$, $-\frac{99}{4}u^4$ predicted, $-\frac{98}{4}u^4$ printed	Rem. 12.10(iii)	Cited
Junctions of order β^4: with four plaquettes and two four-edge fronts only through a unit cube ($Z_{4,4} \leq 12(d-3)$); one plaquette widens a four-edge front only to $\partial q \triangle \partial c$ or $\partial q \cup \partial c$; three front-pairs force a front of ≥ 8 edges (one plaquette) or ≥ 6 (two); a six-edge front ends at leading order in one of two narrowings	Lem. 13.1, 13.3	Theorem
Third-order lower bound with an effective constant: $m \geq 4\log(1/u) - \log\Upsilon - (\delta_4' + o(1))\beta^4$, $\delta_4' = 160(d-3) + 8K_S u_1^2$, the moves of H_C at the sup norm (the shifts and jumps and the U-turns drop out); for $SU(2)$, 2 ($d=3$), 166 ($d=4$). With Cor. 12.9: lim inf and lim sup of $\beta^{-4}(m - 4\log(1/u) + \log\Upsilon)$ lie in $[-166, -\frac{17}{128}]$ ($SU(2)$, $d=4$)	Lem. 13.2, 13.3, Thm. 13.4, Rem. 13.5(ii)	Theorem
$SU(2)$: admissible δ'/β^4 at $\beta = 10^{-9}$: $1.9 \cdot 10^5$ ($d=4$), 412 ($d=3$) (Theorem 11.5: $8.4 \cdot 10^{10}$, $3.35 \cdot 10^8$); enumerations of the junction lemmas and the narrowings, with two independent codes	Rem. 13.5(iii),(iv)	Numerical
Widenings, narrowings, turns and jumps by characters: diagonal, with norm u_* per plaquette and per half-link of the fronts	Lem. 14.1	Theorem
Lower bound to fourth order: $\liminf \beta^{-4}(m - 4\log(1/u) + \log\Upsilon) \geq -h_+ u_1^4$, $h_+ = 26d - 70$	Thm. 14.2	Theorem
The fourth-order term: $m = 4\log(1/u) - \log\Upsilon - (26d-70)u^4 + o(\beta^4)$ for $SU(N)$, $2 \leq N \leq 5$, $d \geq 3$, and $-8u^4$ for $U(1)$, $d=3$; for $SU(2)$, $d=4$: $m = -4\log u + 2u^2 - \frac{98}{3}u^4 + o(u^4)$; for $U(1)$, $d \geq 4$: between $-\frac{26d-70}{16}\beta^4$ and $-\frac{10d-22}{16}\beta^4$	Cor. 14.3	Theorem
Analyticity of $m(\beta) + 4\log u(\beta)$, as in the classical results; the lower bound at fourth order in the C-odd sector ($U(1)$, $d \geq 4$), and for $SU(N \geq 6)$; the C-odd mass; the term of order u^6	Rem. 12.10(iv),(v), Rem. 13.5	open

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